

Project Euler — All Problems (1-974)

Problem 1: Multiples of 3 or 5

[Link](#)

Tags: Number Theory

If we list all the natural numbers below 10 that are multiples of 3 or 5, we get 3, 5, 6 and 9. The sum of these multiples is 23.

Find the sum of all the multiples of 3 or 5 below 1000.

Problem 2: Even Fibonacci Numbers

[Link](#)

Tags: Sequences

Each new term in the Fibonacci sequence is generated by adding the previous two terms. By starting with 1 and 2, the first 10 terms will be:

$$1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

By considering the terms in the Fibonacci sequence whose values do not exceed four million, find the sum of the even-valued terms.

Problem 3: Largest Prime Factor

[Link](#)

Tags: Number Theory

The prime factors of 13195 are 5, 7, 13 and 29.

What is the largest prime factor of the number 600851475143?

Problem 4: Largest Palindrome Product

[Link](#)

Tags: Number Theory

A palindromic number reads the same both ways. The largest palindrome made from the product of two 2-digit numbers is $9009 = 91 \times 99$.

Find the largest palindrome made from the product of two 3-digit numbers.

Problem 5: Smallest Multiple

Link

Tags: Number Theory

2520 is the smallest number that can be divided by each of the numbers from 1 to 10 without any remainder.

What is the smallest positive number that is evenly divisible by all of the numbers from 1 to 20?

Problem 6: Sum Square Difference

Link

Tags: Number Theory

The sum of the squares of the first ten natural numbers is,

$$1^2 + 2^2 + \dots + 10^2 = 385.$$

The square of the sum of the first ten natural numbers is,

$$(1 + 2 + \dots + 10)^2 = 55^2 = 3025.$$

Hence the difference between the sum of the squares of the first ten natural numbers and the square of the sum is $3025 - 385 = 2640$.

Find the difference between the sum of the squares of the first one hundred natural numbers and the square of the sum.

Problem 7: 10 001st Prime

Link

Tags: Number Theory

By listing the first six prime numbers: 2, 3, 5, 7, 11, and 13, we can see that the 6th prime is 13.

What is the 10 001st prime number?

Problem 8: Largest Product in a Series

Link

Tags: Number Theory

The four adjacent digits in the 1000-digit number that have the greatest product are $9 \times 9 \times 8 \times 9 = 5832$.

73167176531330624919225119674426574742355349194934
96983520312774506326239578318016984801869478851843
85861560789112949495459501737958331952853208805511
12540698747158523863050715693290963295227443043557
66896648950445244523161731856403098711121722383113
62229893423380308135336276614282806444486645238749
30358907296290491560440772390713810515859307960866
70172427121883998797908792274921901699720888093776
65727333001053367881220235421809751254540594752243
52584907711670556013604839586446706324415722155397
53697817977846174064955149290862569321978468622482
83972241375657056057490261407972968652414535100474
82166370484403199890008895243450658541227588666881
16427171479924442928230863465674813919123162824586
17866458359124566529476545682848912883142607690042
24219022671055626321111109370544217506941658960408
07198403850962455444362981230987879927244284909188
84580156166097919133875499200524063689912560717606
05886116467109405077541002256983155200055935729725
71636269561882670428252483600823257530420752963450

Find the thirteen adjacent digits in the 1000-digit number that have the greatest product. What is the value of this product?

Problem 9: Special Pythagorean Triplet

Link

Tags: Geometry

A Pythagorean triplet is a set of three natural numbers, abc , for which,

$$a^2 + b^2 = c^2.$$

For example, $3^2 + 4^2 = 9 + 16 = 25 = 5^2$.

There exists exactly one Pythagorean triplet for which $a + b + c = 1000$. Find the product abc .

Problem 10: Summation of Primes

Link

Tags: Number Theory

The sum of the primes below 10 is $2 + 3 + 5 + 7 = 17$.

Find the sum of all the primes below two million.

Problem 11: Largest Product in a Grid

Link

Tags: Combinatorics

In the 20×20 grid below, four numbers along a diagonal line have been marked in red.

08	02	22	97	38	15	00	40	00	75	04	05	07	78	52	12	50	77	91	08
49	49	99	40	17	81	18	57	60	87	17	40	98	43	69	48	04	56	62	00
81	49	31	73	55	79	14	29	93	71	40	67	53	88	30	03	49	13	36	65
52	70	95	23	04	60	11	42	69	24	68	56	01	32	56	71	37	02	36	91
22	31	16	71	51	67	63	89	41	92	36	54	22	40	40	28	66	33	13	80
24	47	32	60	99	03	45	02	44	75	33	53	78	36	84	20	35	17	12	50
32	98	81	28	64	23	67	10	26	38	40	67	59	54	70	66	18	38	64	70
67	26	20	68	02	62	12	20	95	63	94	39	63	08	40	91	66	49	94	21
24	55	58	05	66	73	99	26	97	17	78	78	96	83	14	88	34	89	63	72

21 36 23 09 75 00 76 44 20 45 35 14 00 61 33 97 34 31 33 95
 78 17 53 28 22 75 31 67 15 94 03 80 04 62 16 14 09 53 56 92
 16 39 05 42 96 35 31 47 55 58 88 24 00 17 54 24 36 29 85 57
 86 56 00 48 35 71 89 07 05 44 44 37 44 60 21 58 51 54 17 58
 19 80 81 68 05 94 47 69 28 73 92 13 86 52 17 77 04 89 55 40
 04 52 08 83 97 35 99 16 07 97 57 32 16 26 26 79 33 27 98 66
 88 36 68 87 57 62 20 72 03 46 33 67 46 55 12 32 63 93 53 69
 04 42 16 73 38 25 39 11 24 94 72 18 08 46 29 32 40 62 76 36
 20 69 36 41 72 30 23 88 34 62 99 69 82 67 59 85 74 04 36 16
 20 73 35 29 78 31 90 01 74 31 49 71 48 86 81 16 23 57 05 54
 01 70 54 71 83 51 54 69 16 92 33 48 61 43 52 01 89 19 67 48

The product of these numbers is $26 \times 63 \times 78 \times 14 = 1788696$.

What is the greatest product of four adjacent numbers in the same direction (up, down, left, right, or diagonally) in the 20×20 grid?

Problem 12: Highly Divisible Triangular Number

Link

Tags: Geometry, Number Theory, Sequences

The sequence of triangle numbers is generated by adding the natural numbers. So the 7th triangle number would be $1 + 2 + 3 + 4 + 5 + 6 + 7 = 28$. The first ten terms would be:

1, 3, 6, 10, 15, 21, 28, 36, 45, 55, ...

Let us list the factors of the first seven triangle numbers:

1: 1

3: 1, 3

6: 1, 2, 3, 6

10: 1, 2, 5, 10

15: 1, 3, 5, 15

21: 1, 3, 7, 21

28: 1, 2, 4, 7, 14, 28

We can see that 28 is the first triangle number to have over five divisors.

What is the value of the first triangle number to have over five hundred divisors?

Problem 13: Large Sum

Link

Tags: Game Theory, Number Theory

Work out the first ten digits of the sum of the following one-hundred 50-digit numbers.

37107287533902102798797998220837590246510135740250
46376937677490009712648124896970078050417018260538
74324986199524741059474233309513058123726617309629
91942213363574161572522430563301811072406154908250
23067588207539346171171980310421047513778063246676
89261670696623633820136378418383684178734361726757
28112879812849979408065481931592621691275889832738
44274228917432520321923589422876796487670272189318
47451445736001306439091167216856844588711603153276
70386486105843025439939619828917593665686757934951
62176457141856560629502157223196586755079324193331
64906352462741904929101432445813822663347944758178
92575867718337217661963751590579239728245598838407
58203565325359399008402633568948830189458628227828
80181199384826282014278194139940567587151170094390
35398664372827112653829987240784473053190104293586
86515506006295864861532075273371959191420517255829
71693888707715466499115593487603532921714970056938
54370070576826684624621495650076471787294438377604
53282654108756828443191190634694037855217779295145
36123272525000296071075082563815656710885258350721
45876576172410976447339110607218265236877223636045
17423706905851860660448207621209813287860733969412

81142660418086830619328460811191061556940512689692
51934325451728388641918047049293215058642563049483
62467221648435076201727918039944693004732956340691
15732444386908125794514089057706229429197107928209
55037687525678773091862540744969844508330393682126
18336384825330154686196124348767681297534375946515
80386287592878490201521685554828717201219257766954
78182833757993103614740356856449095527097864797581
16726320100436897842553539920931837441497806860984
48403098129077791799088218795327364475675590848030
87086987551392711854517078544161852424320693150332
59959406895756536782107074926966537676326235447210
69793950679652694742597709739166693763042633987085
41052684708299085211399427365734116182760315001271
65378607361501080857009149939512557028198746004375
35829035317434717326932123578154982629742552737307
94953759765105305946966067683156574377167401875275
88902802571733229619176668713819931811048770190271
25267680276078003013678680992525463401061632866526
36270218540497705585629946580636237993140746255962
24074486908231174977792365466257246923322810917141
91430288197103288597806669760892938638285025333403
34413065578016127815921815005561868836468420090470
23053081172816430487623791969842487255036638784583
11487696932154902810424020138335124462181441773470
63783299490636259666498587618221225225512486764533
67720186971698544312419572409913959008952310058822
95548255300263520781532296796249481641953868218774
76085327132285723110424803456124867697064507995236
37774242535411291684276865538926205024910326572967
23701913275725675285653248258265463092207058596522

29798860272258331913126375147341994889534765745501
18495701454879288984856827726077713721403798879715
38298203783031473527721580348144513491373226651381
34829543829199918180278916522431027392251122869539
40957953066405232632538044100059654939159879593635
29746152185502371307642255121183693803580388584903
41698116222072977186158236678424689157993532961922
62467957194401269043877107275048102390895523597457
23189706772547915061505504953922979530901129967519
86188088225875314529584099251203829009407770775672
11306739708304724483816533873502340845647058077308
82959174767140363198008187129011875491310547126581
97623331044818386269515456334926366572897563400500
42846280183517070527831839425882145521227251250327
55121603546981200581762165212827652751691296897789
32238195734329339946437501907836945765883352399886
75506164965184775180738168837861091527357929701337
62177842752192623401942399639168044983993173312731
32924185707147349566916674687634660915035914677504
99518671430235219628894890102423325116913619626622
73267460800591547471830798392868535206946944540724
76841822524674417161514036427982273348055556214818
97142617910342598647204516893989422179826088076852
87783646182799346313767754307809363333018982642090
10848802521674670883215120185883543223812876952786
71329612474782464538636993009049310363619763878039
62184073572399794223406235393808339651327408011116
66627891981488087797941876876144230030984490851411
60661826293682836764744779239180335110989069790714
85786944089552990653640447425576083659976645795096
66024396409905389607120198219976047599490197230297

64913982680032973156037120041377903785566085089252
16730939319872750275468906903707539413042652315011
94809377245048795150954100921645863754710598436791
78639167021187492431995700641917969777599028300699
15368713711936614952811305876380278410754449733078
40789923115535562561142322423255033685442488917353
44889911501440648020369068063960672322193204149535
41503128880339536053299340368006977710650566631954
81234880673210146739058568557934581403627822703280
82616570773948327592232845941706525094512325230608
22918802058777319719839450180888072429661980811197
77158542502016545090413245809786882778948721859617
72107838435069186155435662884062257473692284509516
20849603980134001723930671666823555245252804609722
53503534226472524250874054075591789781264330331690

Problem 14: Longest Collatz Sequence

Link

Tags: Game Theory, Sequences

The following iterative sequence is defined for the set of positive integers:

- $n \rightarrow n/2$ (n is even)
- $n \rightarrow 3n + 1$ (n is odd)

Using the rule above and starting with 13, we generate the following sequence:

$$13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

It can be seen that this sequence (starting at 13 and finishing at 1) contains 10 terms. Although it has not been proved yet (Collatz Problem), it is thought that all starting numbers finish at 1.

Which starting number, under one million, produces the longest chain?

NOTE: Once the chain starts the terms are allowed to go above one million.

Problem 15: Lattice Paths

Link

Tags: Combinatorics

Starting in the top left corner of a 2×2 grid, and only being able to move to the right and down, there are exactly 6 routes to the bottom right corner.

Problem 16: Power Digit Sum

Link

Tags: Number Theory

$2^{15} = 32768$ and the sum of its digits is $3 + 2 + 7 + 6 + 8 = 26$.

What is the sum of the digits of the number 2^{1000} ?

Problem 17: Number Letter Counts

Link

Tags: Number Theory

If the numbers 1 to 5 are written out in words: one, two, three, four, five, then there are $3 + 3 + 5 + 4 + 4 = 19$ letters used in total.

If all the numbers from 1 to 1000 (one thousand) inclusive were written out in words, how many letters would be used?

NOTE: Do not count spaces or hyphens. For example, 342 (three hundred and forty-two) contains 23 letters and 115 (one hundred and fifteen) contains 20 letters. The use of “and” when writing out numbers is in compliance with British usage.

Problem 18: Maximum Path Sum I

Link

Tags: Geometry

By starting at the top of the triangle below and moving to adjacent numbers on the row below, the maximum total from top to bottom is 23.

3 7 4

2 4 6

8 5 9 3

That is, $3 + 7 + 4 + 9 = 23$.

Find the maximum total from top to bottom of the triangle below:

75

95 64

17 47 82

18 35 87 10

20 04 82 47 65

19 01 23 75 03 34

88 02 77 73 07 63 67

99 65 04 28 06 16 70 92

41 41 26 56 83 40 80 70 33

41 48 72 33 47 32 37 16 94 29

53 71 44 65 25 43 91 52 97 51 14

70 11 33 28 77 73 17 78 39 68 17 57

91 71 52 38 17 14 91 43 58 50 27 29 48

63 66 04 68 89 53 67 30 73 16 69 87 40 31

04 62 98 27 23 09 70 98 73 93 38 53 60 04 23

NOTE: As there are only 16384 routes, it is possible to solve this problem by trying every route. However, Problem 67, is the same challenge with a triangle containing one-hundred rows; it cannot be solved by brute force, and requires a clever method! ;o)

Problem 19: Counting Sundays

Link

Tags: Game Theory

You are given the following information, but you may prefer to do some research for yourself.

- 1 Jan 1900 was a Monday.
- Thirty days has September,

April, June and November.

All the rest have thirty-one,

Saving February alone,

Which has twenty-eight, rain or shine.

And on leap years, twenty-nine.

- A leap year occurs on any year evenly divisible by 4, but not on a century unless it is divisible by 400.

How many Sundays fell on the first of the month during the twentieth century (1 Jan 1901 to 31 Dec 2000)?

Problem 20: Factorial Digit Sum

[Link](#)

Tags: Number Theory

$n!$ means $n \times (n - 1) \times \cdots \times 3 \times 2 \times 1$.

For example, $10! = 10 \times 9 \times \cdots \times 3 \times 2 \times 1 = 3628800$, and the sum of the digits in the number $10!$ is $3 + 6 + 2 + 8 + 8 + 0 + 0 = 27$.

Find the sum of the digits in the number $100!$.

Problem 21: Amicable Numbers

[Link](#)

Tags: Number Theory

Let $d(n)$ be defined as the sum of proper divisors of n (numbers less than n which divide evenly into n).

If $d(a) = b$ and $d(b) = a$, where $a \neq b$, then a and b are an amicable pair and each of a and b are called amicable numbers.

For example, the proper divisors of 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55 and 110; therefore $d(220) = 284$. The proper divisors of 284 are 1, 2, 4, 71 and 142; so $d(284) = 220$.

Evaluate the sum of all the amicable numbers under 10000.

Problem 22: Names Scores

Link

Tags: Number Theory

Using names.txt (right click and ‘Save Link/Target As...’), a 46K text file containing over five-thousand first names, begin by sorting it into alphabetical order. Then working out the alphabetical value for each name, multiply this value by its alphabetical position in the list to obtain a name score.

For example, when the list is sorted into alphabetical order, COLIN, which is worth $3 + 15 + 12 + 9 + 14 = 53$, is the 938th name in the list. So, COLIN would obtain a score of $938 \times 53 = 49714$.

What is the total of all the name scores in the file?

Problem 23: Non-Abundant Sums

Link

Tags: Number Theory

A perfect number is a number for which the sum of its proper divisors is exactly equal to the number. For example, the sum of the proper divisors of 28 would be $1 + 2 + 4 + 7 + 14 = 28$, which means that 28 is a perfect number.

A number n is called deficient if the sum of its proper divisors is less than n and it is called abundant if this sum exceeds n .

As 12 is the smallest abundant number, $1 + 2 + 3 + 4 + 6 = 16$, the smallest number that can be written as the sum of two abundant numbers is 24. By mathematical analysis, it can be shown that all integers greater than 28123 can be written as the sum of two abundant numbers. However, this upper limit cannot be reduced any further by analysis even though it is known that the greatest number that cannot be expressed as the sum of two abundant numbers is less than this limit.

Find the sum of all the positive integers which cannot be written as the sum of two abundant numbers.

Problem 24: Lexicographic Permutations

Link

Tags: Combinatorics, Graph Theory, Number Theory

A permutation is an ordered arrangement of objects. For example, 3124 is one possible permutation of the digits 1, 2, 3 and 4. If all of the permutations

are listed numerically or alphabetically, we call it lexicographic order. The lexicographic permutations of 0, 1 and 2 are:

012 021 102 120 201 210

What is the millionth lexicographic permutation of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9?

Problem 25: 1000-digit Fibonacci Number

[Link](#)

Tags: Number Theory, Sequences

The Fibonacci sequence is defined by the recurrence relation:

$F_n = F_{n-1} + F_{n-2}$, where $F_1 = 1$ and $F_2 = 1$.

Hence the first 12 terms will be:

$$F_1 = 1$$

$$F_2 = 1$$

$$F_3 = 2$$

$$F_4 = 3$$

$$F_5 = 5$$

$$F_6 = 8$$

$$F_7 = 13$$

$$F_8 = 21$$

$$F_9 = 34$$

$$F_{10} = 55$$

$$F_{11} = 89$$

$$F_{12} = 144$$

The 12th term, F_{12} , is the first term to contain three digits.

What is the index of the first term in the Fibonacci sequence to contain 1000 digits?

Problem 26: Reciprocal Cycles

[Link](#)

Tags: Number Theory

A unit fraction contains 1 in the numerator. The decimal representation of the unit fractions with denominators 2 to 10 are given:

$$\begin{aligned} 1/2 &= 0.5 \\ 1/3 &= 0.(3) \\ 1/4 &= 0.25 \\ 1/5 &= 0.2 \\ 1/6 &= 0.1(6) \\ 1/7 &= 0.(142857) \\ 1/8 &= 0.125 \\ 1/9 &= 0.(1) \\ 1/10 &= 0.1 \end{aligned}$$

Where $0.1(6)$ means $0.166666\cdots$, and has a 1-digit recurring cycle. It can be seen that $1/7$ has a 6-digit recurring cycle.

Find the value of $d1000$ for which $1/d$ contains the longest recurring cycle in its decimal fraction part.

Problem 27: Quadratic Primes

Link

Tags: Algebra, Number Theory

Euler discovered the remarkable quadratic formula:

$$n^2 + n + 41$$

It turns out that the formula will produce 40 primes for the consecutive integer values $0 \leq n \leq 39$. However, when $n = 40$, $40^2 + 40 + 41 = 40(40 + 1) + 41$ is divisible by 41, and certainly when $n = 41$, $41^2 + 41 + 41$ is clearly divisible by 41.

The incredible formula $n^2 - 79n + 1601$ was discovered, which produces 80 primes for the consecutive values $0 \leq n \leq 79$. The product of the coefficients, -79 and 1601 , is -126479 .

Considering quadratics of the form:

$$n^2 + an + b, \text{ where } |a| < 1000 \text{ and } |b| \leq 1000$$

where $|n|$ is the modulus/absolute value of n e.g. $|11| = 11$ and $|-4| = 4$

Problem 28: Number Spiral Diagonals

Link

Tags: Number Theory

Starting with the number 1 and moving to the right in a clockwise direction a 5 by 5 spiral is formed as follows:

21 22 23 24 25

20 7 8 9 10

19 6 1 2 11

18 5 4 3 12 17 16 15 14 13

It can be verified that the sum of the numbers on the diagonals is 101.

What is the sum of the numbers on the diagonals in a 1001 by 1001 spiral formed in the same way?

Problem 29: Distinct Powers

Link

Tags: Combinatorics, Game Theory, Sequences

Consider all integer combinations of a^b for $2 \leq a \leq 5$ and $2 \leq b \leq 5$:

$$\begin{array}{cccc} 2^2 = 4, & 2^3 = 8, & 2^4 = 16, & 2^5 = 32 \\ 3^2 = 9, & 3^3 = 27, & 3^4 = 81, & 3^5 = 243 \\ 4^2 = 16, & 4^3 = 64, & 4^4 = 256, & 4^5 = 1024 \\ 5^2 = 25, & 5^3 = 125, & 5^4 = 625, & 5^5 = 3125 \end{array}$$

If they are then placed in numerical order, with any repeats removed, we get the following sequence of 15 distinct terms:

$$4, 8, 9, 16, 25, 27, 32, 64, 81, 125, 243, 256, 625, 1024, 3125.$$

How many distinct terms are in the sequence generated by a^b for $2 \leq a \leq 100$ and $2 \leq b \leq 100$?

Problem 30: Digit Fifth Powers

Link

Tags: Number Theory

Surprisingly there are only three numbers that can be written as the sum of fourth powers of their digits:

$$1634 = 1^4 + 6^4 + 3^4 + 4^4$$

$$8208 = 8^4 + 2^4 + 0^4 + 8^4$$

$$9474 = 9^4 + 4^4 + 7^4 + 4^4$$

As $1 = 1^4$ is not a sum it is not included.

The sum of these numbers is $1634 + 8208 + 9474 = 19316$.

Find the sum of all the numbers that can be written as the sum of fifth powers of their digits.

Problem 31: Coin Sums

Link

Tags: Game Theory, Probability

In the United Kingdom the currency is made up of pound (£) and pence (p). There are eight coins in general circulation:

1p, 2p, 5p, 10p, 20p, 50p, £1 (100p), and £2 (200p).

It is possible to make £2 in the following way:

$$1 \times £1 + 1 \times 50p + 2 \times 20p + 1 \times 5p + 1 \times 2p + 3 \times 1p$$

How many different ways can £2 be made using any number of coins?

Problem 32: Pandigital Products

Link

Tags: Number Theory

We shall say that an n -digit number is pandigital if it makes use of all the digits 1 to n exactly once; for example, the 5-digit number, 15234, is 1 through 5 pandigital.

The product 7254 is unusual, as the identity, $39 \times 186 = 7254$, containing multiplicand, multiplier, and product is 1 through 9 pandigital.

Find the sum of all products whose multiplicand/multiplier/product identity can be written as a 1 through 9 pandigital.

HINT: Some products can be obtained in more than one way so be sure to only include it once in your sum.

Problem 33: Digit Cancelling Fractions

Link

Tags: Number Theory

The fraction $49/98$ is a curious fraction, as an inexperienced mathematician in attempting to simplify it may incorrectly believe that $49/98 = 4/8$, which is correct, is obtained by cancelling the 9s.

We shall consider fractions like, $30/50 = 3/5$, to be trivial examples.

There are exactly four non-trivial examples of this type of fraction, less than one in value, and containing two digits in the numerator and denominator.

If the product of these four fractions is given in its lowest common terms, find the value of the denominator.

Problem 34: Digit Factorials

Link

Tags: Number Theory

145 is a curious number, as $1! + 4! + 5! = 1 + 24 + 120 = 145$.

Find the sum of all numbers which are equal to the sum of the factorial of their digits.

Note: As $1! = 1$ and $2! = 2$ are not sums they are not included.

Problem 35: Circular Primes

Link

Tags: Number Theory

The number, 197, is called a circular prime because all rotations of the digits: 197, 971, and 719, are themselves prime.

There are thirteen such primes below 100: 2, 3, 5, 7, 11, 13, 17, 31, 37, 71, 73, 79, and 97.

How many circular primes are there below one million?

Problem 36: Double-base Palindromes

[Link](#)

Tags: Number Theory

The decimal number, $585 = 1001001001_2$ (binary), is palindromic in both bases.

Find the sum of all numbers, less than one million, which are palindromic in base 10 and base 2.

(Please note that the palindromic number, in either base, may not include leading zeros.)

Problem 37: Truncatable Primes

[Link](#)

Tags: Number Theory

The number 3797 has an interesting property. Being prime itself, it is possible to continuously remove digits from left to right, and remain prime at each stage: 3797, 797, 97, and 7. Similarly we can work from right to left: 3797, 379, 37, and 3.

Find the sum of the only eleven primes that are both truncatable from left to right and right to left.

NOTE: 2, 3, 5, and 7 are not considered to be truncatable primes.

Problem 38: Pandigital Multiples

[Link](#)

Tags: Number Theory

Take the number 192 and multiply it by each of 1, 2, and 3:

$$192 \times 1 = 192$$

$$192 \times 2 = 384$$

$$192 \times 3 = 576$$

By concatenating each product we get the 1 to 9 pandigital, 192384576. We will call 192384576 the concatenated product of 192 and (1, 2, 3).

The same can be achieved by starting with 9 and multiplying by 1, 2, 3, 4, and 5, giving the pandigital, 918273645, which is the concatenated product of 9 and (1, 2, 3, 4, 5).

What is the largest 1 to 9 pandigital 9-digit number that can be formed as the concatenated product of an integer with $(1, 2, \dots, n)$ where $n \geq 1$?

Problem 39: Integer Right Triangles

Link

Tags: Geometry

If p is the perimeter of a right angle triangle with integral length sides, $\{a, b, c\}$, there are exactly three solutions for $p = 120$.

$\{20, 48, 52\}$, $\{24, 45, 51\}$, $\{30, 40, 50\}$

For which value of $p \leq 1000$, is the number of solutions maximised?

Problem 40: Champernowne's Constant

Link

Tags: Game Theory, Number Theory

An irrational decimal fraction is created by concatenating the positive integers:

0.123456789101112131415161718192021...

It can be seen that the 12th digit of the fractional part is 1.

If d_n represents the n th digit of the fractional part, find the value of the following expression.

$$d_1 \times d_{10} \times d_{100} \times d_{1000} \times d_{10000} \times d_{100000} \times d_{1000000}$$

Problem 41: Pandigital Prime

Link

Tags: Number Theory

We shall say that an n -digit number is pandigital if it makes use of all the digits 1 to n exactly once. For example, 2143 is a 4-digit pandigital and is also prime.

What is the largest n -digit pandigital prime that exists?

Problem 42: Coded Triangle Numbers

Link

Tags: Geometry, Sequences

The n th term of the sequence of triangle numbers is given by, $t_n = \frac{1}{2}n(n+1)$; so the first ten triangle numbers are:

$$1, 3, 6, 10, 15, 21, 28, 36, 45, 55, \dots$$

By converting each letter in a word to a number corresponding to its alphabetical position and adding these values we form a word value. For example, the word value for SKY is $19 + 11 + 25 = 55 = t_{10}$. If the word value is a triangle number then we shall call the word a triangle word.

Using words.txt (right click and ‘Save Link/Target As...’), a 16K text file containing nearly two-thousand common English words, how many are triangle words?

Problem 43: Sub-string Divisibility

Link

Tags: Game Theory, Number Theory

The number, 1406357289, is a 0 to 9 pandigital number because it is made up of each of the digits 0 to 9 in some order, but it also has a rather interesting sub-string divisibility property.

Let d_1 be the 1st digit, d_2 be the 2nd digit, and so on. In this way, we note the following:

- $d_2d_3d_4 = 406$ is divisible by 2
- $d_3d_4d_5 = 063$ is divisible by 3
- $d_4d_5d_6 = 635$ is divisible by 5
- $d_5d_6d_7 = 357$ is divisible by 7
- $d_6d_7d_8 = 572$ is divisible by 11
- $d_7d_8d_9 = 728$ is divisible by 13
- $d_8d_9d_{10} = 289$ is divisible by 17

Find the sum of all 0 to 9 pandigital numbers with this property.

Problem 44: Pentagon Numbers

Link

Tags: Game Theory

Pentagonal numbers are generated by the formula, $P_n = n(3n - 1)/2$. The first ten pentagonal numbers are:

$$1, 5, 12, 22, 35, 51, 70, 92, 117, 145, \dots$$

It can be seen that $P_4 + P_7 = 22 + 70 = 92 = P_8$. However, their difference, $70 - 22 = 48$, is not pentagonal.

Find the pair of pentagonal numbers, P_j and P_k , for which their sum and difference are pentagonal and $D = |P_k - P_j|$ is minimised; what is the value of D ?

Problem 45: Triangular, Pentagonal, and Hexagonal

Link

Tags: Game Theory, Geometry

Triangle, pentagonal, and hexagonal numbers are generated by the following formulae:

$$\begin{array}{lll} \text{Triangle} & T_n = n(n+1)/2 & 1, 3, 6, 10, 15, \dots \\ \text{Pentagonal} & P_n = n(3n-1)/2 & 1, 5, 12, 22, 35, \dots \\ \text{Hexagonal} & H_n = n(2n-1) & 1, 6, 15, 28, 45, \dots \end{array}$$

It can be verified that $T_{285} = P_{165} = H_{143} = 40755$.

Find the next triangle number that is also pentagonal and hexagonal.

Problem 46: Goldbach's Other Conjecture

Link

Tags: Number Theory

It was proposed by Christian Goldbach that every odd composite number can be written as the sum of a prime and twice a square.

$$\begin{aligned}
9 &= 7 + 2 \times 1^2 \\
15 &= 7 + 2 \times 2^2 \\
21 &= 3 + 2 \times 3^2 \\
25 &= 7 + 2 \times 3^2 \\
27 &= 19 + 2 \times 2^2 \\
33 &= 31 + 2 \times 1^2
\end{aligned}$$

It turns out that the conjecture was false.

What is the smallest odd composite that cannot be written as the sum of a prime and twice a square?

Problem 47: Distinct Primes Factors

Link

Tags: Number Theory

The first two consecutive numbers to have two distinct prime factors are:

$$\begin{aligned}
14 &= 2 \times 7 \\
15 &= 3 \times 5.
\end{aligned}$$

The first three consecutive numbers to have three distinct prime factors are:

$$\begin{aligned}
644 &= 2^2 \times 7 \times 23 \\
645 &= 3 \times 5 \times 43 \\
646 &= 2 \times 17 \times 19.
\end{aligned}$$

Find the first four consecutive integers to have four distinct prime factors each. What is the first of these numbers?

Problem 48: Self Powers

Link

Tags: Number Theory, Sequences

The series, $1^1 + 2^2 + 3^3 + \dots + 10^{10} = 10405071317$.

Find the last ten digits of the series, $1^1 + 2^2 + 3^3 + \dots + 1000^{1000}$.

Problem 49: Prime Permutations

[Link](#)

Tags: Combinatorics, Number Theory, Sequences

The arithmetic sequence, 1487, 4817, 8147, in which each of the terms increases by 3330, is unusual in two ways: (i) each of the three terms are prime, and, (ii) each of the 4-digit numbers are permutations of one another.

There are no arithmetic sequences made up of three 1-, 2-, or 3-digit primes, exhibiting this property, but there is one other 4-digit increasing sequence.

What 12-digit number do you form by concatenating the three terms in this sequence?

Problem 50: Consecutive Prime Sum

[Link](#)

Tags: Number Theory

The prime 41, can be written as the sum of six consecutive primes:

$$41 = 2 + 3 + 5 + 7 + 11 + 13.$$

This is the longest sum of consecutive primes that adds to a prime below one-hundred.

The longest sum of consecutive primes below one-thousand that adds to a prime, contains 21 terms, and is equal to 953.

Which prime, below one-million, can be written as the sum of the most consecutive primes?

Problem 51: Prime Digit Replacements

[Link](#)

Tags: Number Theory

By replacing the 1st digit of the 2-digit number *3, it turns out that six of the nine possible values: 13, 23, 43, 53, 73, and 83, are all prime.

By replacing the 3rd and 4th digits of 56**3 with the same digit, this 5-digit number is the first example having seven primes among the ten generated numbers, yielding the family: 56003, 56113, 56333, 56443, 56663, 56773, and

56993. Consequently 56003, being the first member of this family, is the smallest prime with this property.

Find the smallest prime which, by replacing part of the number (not necessarily adjacent digits) with the same digit, is part of an eight prime value family.

Problem 52: Permuted Multiples

[Link](#)

Tags: Number Theory

It can be seen that the number, 125874, and its double, 251748, contain exactly the same digits, but in a different order.

Find the smallest positive integer, x , such that $2x$, $3x$, $4x$, $5x$, and $6x$, contain the same digits.

Problem 53: Combinatoric Selections

[Link](#)

Tags: Combinatorics

There are exactly ten ways of selecting three from five, 12345:

123, 124, 125, 134, 135, 145, 234, 235, 245, and 345

In combinatorics, we use the notation, $\binom{5}{3} = 10$.

In general, $\binom{n}{r} = \frac{n!}{r!(n-r)!}$, where $r \leq n$, $n! = n \times (n-1) \times \dots \times 3 \times 2 \times 1$, and $0! = 1$.

It is not until $n = 23$, that a value exceeds one-million: $\binom{23}{10} = 1144066$.

How many, not necessarily distinct, values of $\binom{n}{r}$ for $1 \leq n \leq 100$, are greater than one-million?

Problem 54: Poker Hands

[Link](#)

Tags: Game Theory

In the card game poker, a hand consists of five cards and are ranked, from lowest to highest, in the following way:

- High Card: Highest value card.
- One Pair: Two cards of the same value.
- Two Pairs: Two different pairs.
- Three of a Kind: Three cards of the same value.
- Straight: All cards are consecutive values.
- Flush: All cards of the same suit.
- Full House: Three of a kind and a pair.
- Four of a Kind: Four cards of the same value.
- Straight Flush: All cards are consecutive values of same suit.
- Royal Flush: Ten, Jack, Queen, King, Ace, in same suit.

The cards are valued in the order: 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King, Ace.

If two players have the same ranked hands then the rank made up of the highest value wins; for example, a pair of eights beats a pair of fives (see example 1 below). But if two ranks tie, for example, both players have a pair of queens, then highest cards in each hand are compared (see example 4 below); if the highest cards tie then the next highest cards are compared, and so on.

Consider the following five hands dealt to two players:

Hand	Player 1	Player 2	Winner
1	5H 5C 6S 7S KD		

Pair of Fives

Problem 55: Lychrel Numbers

Link

Tags: Number Theory

If we take 47, reverse and add, $47 + 74 = 121$, which is palindromic.

Not all numbers produce palindromes so quickly. For example,

$$\begin{aligned} 349 + 943 &= 1292 \\ 1292 + 2921 &= 4213 \\ 4213 + 3124 &= 7337 \end{aligned}$$

That is, 349 took three iterations to arrive at a palindrome.

Although no one has proved it yet, it is thought that some numbers, like 196, never produce a palindrome. A number that never forms a palindrome through the reverse and add process is called a Lychrel number. Due to the theoretical nature of these numbers, and for the purpose of this problem, we shall assume that a number is Lychrel until proven otherwise. In addition you are given that for every number below ten-thousand, it will either (i) become a palindrome in less than fifty iterations, or, (ii) no one, with all the computing power that exists, has managed so far to map it to a palindrome. In fact, 10677 is the first number to be shown to require over fifty iterations before producing a palindrome: 4668731596684224866951378664 (53 iterations, 28-digits).

Surprisingly, there are palindromic numbers that are themselves Lychrel numbers; the first example is 4994.

How many Lychrel numbers are there below ten-thousand?

NOTE: Wording was modified slightly on 24 April 2007 to emphasise the theoretical nature of Lychrel numbers.

Problem 56: Powerful Digit Sum

Link

Tags: Game Theory, Number Theory

A googol (10^{100}) is a massive number: one followed by one-hundred zeros; 100^{100} is almost unimaginably large: one followed by two-hundred zeros. Despite their size, the sum of the digits in each number is only 1.

Considering natural numbers of the form, a^b , where $a, b \leq 100$, what is the maximum digital sum?

Problem 57: Square Root Convergents

Link

Tags: Number Theory, Sequences

It is possible to show that the square root of two can be expressed as an infinite continued fraction.

$$\sqrt{2} = 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots}}}$$

By expanding this for the first four iterations, we get:

$$1 + \frac{1}{2} = \frac{3}{2} = 1.5$$

$$1 + \frac{1}{2 + \frac{1}{2}} = \frac{7}{5} = 1.4$$

$$1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}} = \frac{17}{12} = 1.41666\dots$$

$$1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}} = \frac{41}{29} = 1.41379\dots$$

The next three expansions are $\frac{99}{70}$, $\frac{239}{169}$, and $\frac{577}{408}$, but the eighth expansion, $\frac{1393}{985}$, is the first example where the number of digits in the numerator exceeds the number of digits in the denominator.

In the first one-thousand expansions, how many fractions contain a numerator with more digits than the denominator?

Problem 58: Spiral Primes

Link

Tags: Game Theory, Number Theory

Starting with 1 and spiralling anticlockwise in the following way, a square spiral with side length 7 is formed.

37 36 35 34 33 32 31

38 17 16 15 14 13 30

39 18 5 4 3 12 29

40 19 6 1 2 11 28

41 20 7 8 9 10 27

42 21 22 23 24 25 26 43 44 45 46 47 48 49

It is interesting to note that the odd squares lie along the bottom right diagonal, but what is more interesting is that 8 out of the 13 numbers lying along both diagonals are prime; that is, a ratio of $8/13 \approx 62\%$.

If one complete new layer is wrapped around the spiral above, a square spiral with side length 9 will be formed. If this process is continued, what is the side length of the square spiral for which the ratio of primes along both diagonals first falls below 10%?

Problem 59: XOR Decryption

Link

Tags: Graph Theory, Probability

Each character on a computer is assigned a unique code and the preferred standard is ASCII (American Standard Code for Information Interchange). For example, uppercase A = 65, asterisk (*) = 42, and lowercase k = 107.

A modern encryption method is to take a text file, convert the bytes to ASCII, then XOR each byte with a given value, taken from a secret key. The advantage with the XOR function is that using the same encryption key on the cipher text, restores the plain text; for example, $65 \text{ XOR } 42 = 107$, then $107 \text{ XOR } 42 = 65$.

For unbreakable encryption, the key is the same length as the plain text message, and the key is made up of random bytes. The user would keep the encrypted message and the encryption key in different locations, and without both “halves”, it is impossible to decrypt the message.

Unfortunately, this method is impractical for most users, so the modified method is to use a password as a key. If the password is shorter than the message, which is likely, the key is repeated cyclically throughout the message. The balance for this method is using a sufficiently long password key for security, but short enough to be memorable.

Your task has been made easy, as the encryption key consists of three lower case characters. Using 0059_cipher.txt (right click and ‘Save Link/Target As...’), a file containing the encrypted ASCII codes, and the knowledge that the plain text must contain common English words, decrypt the message and find the sum of the ASCII values in the original text.

Problem 60: Prime Pair Sets

[Link](#)

Tags: Number Theory

The primes 3, 7, 109, and 673, are quite remarkable. By taking any two primes and concatenating them in any order the result will always be prime. For example, taking 7 and 109, both 7109 and 1097 are prime. The sum of these four primes, 792, represents the lowest sum for a set of four primes with this property.

Find the lowest sum for a set of five primes for which any two primes concatenate to produce another prime.

Problem 61: Cyclical Figurate Numbers

[Link](#)

Tags: Game Theory, Geometry, Number Theory, Sequences

Triangle, square, pentagonal, hexagonal, heptagonal, and octagonal numbers are all figurate (polygonal) numbers and are generated by the following formulae:

Triangle $P_{3,n} = n(n+1)/2$ 1, 3, 6, 10, 15, ... Square $P_{4,n} = n^2$ 1, 4, 9, 16, 25, ...
 Pentagonal $P_{5,n} = n(3n-1)/2$ 1, 5, 12, 22, 35, ... Hexagonal $P_{6,n} = n(2n-1)$
 1, 6, 15, 28, 45, ... Heptagonal $P_{7,n} = n(5n-3)/2$ 1, 7, 18, 34, 55, ... Octagonal
 $P_{8,n} = n(3n-2)$ 1, 8, 21, 40, 65, ...

The ordered set of three 4-digit numbers: 8128, 2882, 8281, has three interesting properties.

- The set is cyclic, in that the last two digits of each number is the first two digits of the next number (including the last number with the first).
- Each polygonal type: triangle ($P_{3,127} = 8128$), square ($P_{4,91} = 8281$), and pentagonal ($P_{5,44} = 2882$), is represented by a different number in the set.
- This is the only set of 4-digit numbers with this property.

Find the sum of the only ordered set of six cyclic 4-digit numbers for which each polygonal type: triangle, square, pentagonal, hexagonal, heptagonal, and octagonal, is represented by a different number in the set.

Problem 62: Cubic Permutations

[Link](#)

Tags: Combinatorics, Number Theory

The cube, 41063625 (345^3), can be permuted to produce two other cubes: 56623104 (384^3) and 66430125 (405^3). In fact, 41063625 is the smallest cube which has exactly three permutations of its digits which are also cube.

Find the smallest cube for which exactly five permutations of its digits are cube.

Problem 63: Powerful Digit Counts

[Link](#)

Tags: Number Theory

The 5-digit number, $16807 = 7^5$, is also a fifth power. Similarly, the 9-digit number, $134217728 = 8^9$, is a ninth power.

How many n -digit positive integers exist which are also an n th power?

Problem 64: Odd Period Square Roots

Link

Tags: Algebra, Game Theory, Sequences

All square roots are periodic when written as continued fractions and can be written in the form:

$$\sqrt{N} = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots}}}$$

For example, let us consider $\sqrt{23}$:

$$\sqrt{23} = 4 + \sqrt{23} - 4 = 4 + \frac{1}{\frac{1}{\sqrt{23} - 4}} = 4 + \frac{1}{1 + \frac{\sqrt{23} - 3}{7}}$$

If we continue we would get the following expansion:

$$\sqrt{23} = 4 + \frac{1}{1 + \frac{1}{3 + \frac{1}{1 + \frac{1}{8 + \dots}}}}$$

The process can be summarised as follows:

$$\begin{aligned}
a_0 &= 4, \frac{1}{\sqrt{23}-4} = \frac{\sqrt{23}+4}{7} = 1 + \frac{\sqrt{23}-3}{7} \\
a_1 &= 1, \frac{7}{\sqrt{23}-3} = \frac{7(\sqrt{23}+3)}{14} = 3 + \frac{\sqrt{23}-3}{2} \\
a_2 &= 3, \frac{2}{\sqrt{23}-3} = \frac{2(\sqrt{23}+3)}{14} = 1 + \frac{\sqrt{23}-4}{7} \\
a_3 &= 1, \frac{7}{\sqrt{23}-4} = \frac{7(\sqrt{23}+4)}{7} = 8 + \sqrt{23}-4 \\
a_4 &= 8, \frac{1}{\sqrt{23}-4} = \frac{\sqrt{23}+4}{7} = 1 + \frac{\sqrt{23}-3}{7} \\
a_5 &= 1, \frac{7}{\sqrt{23}-3} = \frac{7(\sqrt{23}+3)}{14} = 3 + \frac{\sqrt{23}-3}{2} \\
a_6 &= 3, \frac{2}{\sqrt{23}-3} = \frac{2(\sqrt{23}+3)}{14} = 1 + \frac{\sqrt{23}-4}{7} \\
a_7 &= 1, \frac{7}{\sqrt{23}-4} = \frac{7(\sqrt{23}+4)}{7} = 8 + \sqrt{23}-4
\end{aligned}$$

It can be seen that the sequence is repeating. For conciseness, we use the notation $\sqrt{23} = [4; (1, 3, 1, 8)]$, to indicate that the block (1,3,1,8) repeats indefinitely.

The first ten continued fraction representations of (irrational) square roots are:

$$\begin{aligned}
\sqrt{2} &= [1; (2)], \text{ period}=1 \\
\sqrt{3} &= [1; (1, 2)], \text{ period}=2 \\
\sqrt{5} &= [2; (4)], \text{ period}=1 \\
\sqrt{6} &= [2; (2, 4)], \text{ period}=2 \\
\sqrt{7} &= [2; (1, 1, 1, 4)], \text{ period}=4 \\
\sqrt{8} &= [2; (1, 4)], \text{ period}=2 \\
\sqrt{10} &= [3; (6)], \text{ period}=1 \\
\sqrt{11} &= [3; (3, 6)], \text{ period}=2 \\
\sqrt{12} &= [3; (2, 6)], \text{ period}=2 \\
\sqrt{13} &= [3; (1, 1, 1, 1, 6)], \text{ period}=5
\end{aligned}$$

Exactly four continued fractions, for $N \leq 13$, have an odd period.

How many continued fractions for $N \leq 10\,000$ have an odd period?

Problem 65: Convergents of e

Link

Tags: Algebra, Number Theory, Sequences

The square root of 2 can be written as an infinite continued fraction.

$$\sqrt{2} = 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots}}}}$$

The infinite continued fraction can be written, $\sqrt{2} = [1; (2)]$, (2) indicates that 2 repeats ad infinitum. In a similar way, $\sqrt{23} = [4; (1, 3, 1, 8)]$.

It turns out that the sequence of partial values of continued fractions for square roots provide the best rational approximations. Let us consider the convergents for $\sqrt{2}$.

$$\begin{aligned} 1 + \frac{1}{2} &= \frac{3}{2} \\ 1 + \frac{1}{2 + \frac{1}{2}} &= \frac{7}{5} \\ 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}} &= \frac{17}{12} \\ 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}} &= \frac{41}{29} \end{aligned}$$

Hence the sequence of the first ten convergents for $\sqrt{2}$ are:

$$1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \frac{99}{70}, \frac{239}{169}, \frac{577}{408}, \frac{1393}{985}, \frac{3363}{2378}, \dots$$

What is most surprising is that the important mathematical constant,

$$e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, \dots, 1, 2k, 1, \dots]$$

The first ten terms in the sequence of convergents for e are:

$$2, 3, \frac{8}{3}, \frac{11}{4}, \frac{19}{7}, \frac{87}{32}, \frac{106}{39}, \frac{193}{71}, \frac{1264}{465}, \frac{1457}{536}, \dots$$

The sum of digits in the numerator of the 10th convergent is $1 + 4 + 5 + 7 = 17$.

Find the sum of digits in the numerator of the 100th convergent of the continued fraction for e .

Problem 66: Diophantine Equation

Link

Tags: Algebra, Game Theory

Consider quadratic Diophantine equations of the form:

$$x^2 - Dy^2 = 1$$

For example, when $D = 13$, the minimal solution in x is $649^2 - 13 \times 180^2 = 1$.

It can be assumed that there are no solutions in positive integers when D is square.

By finding minimal solutions in x for $D = \{2, 3, 5, 6, 7\}$, we obtain the following:

$$3^2 - 2 \times 2^2 = 1$$

$$2^2 - 3 \times 1^2 = 1$$

$$\textcolor{red}{9}^2 - 5 \times 4^2 = 1$$

$$5^2 - 6 \times 2^2 = 1$$

$$8^2 - 7 \times 3^2 = 1$$

Hence, by considering minimal solutions in x for $D \leq 7$, the largest x is obtained when $D = 5$.

Find the value of $D \leq 1000$ in minimal solutions of x for which the largest value of x is obtained.

Problem 67: Maximum Path Sum II

Link

Tags: Geometry

By starting at the top of the triangle below and moving to adjacent numbers on the row below, the maximum total from top to bottom is 23.

3 7 4

2 4 6

8 5 9 3

That is, $3 + 7 + 4 + 9 = 23$.

Find the maximum total from top to bottom in triangle.txt (right click and ‘Save Link/Target As...’), a 15K text file containing a triangle with one-hundred rows.

NOTE: This is a much more difficult version of Problem 18. It is not possible to try every route to solve this problem, as there are 2^{99} altogether! If you could check one trillion (10^{12}) routes every second it would take over twenty billion years to check them all. There is an efficient algorithm to solve it. ;o)

Problem 68: Magic 5-gon Ring

Link

Tags: Game Theory

Consider the following “magic” 3-gon ring, filled with the numbers 1 to 6, and each line adding to nine.

Problem 69: Totient Maximum

Link

Tags: Number Theory

Euler’s totient function, $\phi(n)$ [sometimes called the phi function], is defined as the number of positive integers not exceeding n which are relatively prime to n . For example, as 1, 2, 4, 5, 7, and 8, are all less than or equal to nine and relatively prime to nine, $\phi(9) = 6$.

n Relatively Prime $\phi(n)$ $n/\phi(n)$ 2 1 1 2 3 1,2 2 1.5 4 1,3 2 2 5 1,2,3,4 4 1.25 6 1,5 2 3 7 1,2,3,4,5,6 6 1.1666... 8 1,3,5,7 4 2 9 1,2,4,5,7,8 6 1.5 10 1,3,7,9 4 2.5

Problem 70: Totient Permutation

Link

Tags: Combinatorics, Game Theory, Number Theory

Euler’s totient function, $\phi(n)$ [sometimes called the phi function], is used to determine the number of positive numbers less than or equal to n which are

relatively prime to n . For example, as 1, 2, 4, 5, 7, and 8, are all less than nine and relatively prime to nine, $\phi(9) = 6$. The number 1 is considered to be relatively prime to every positive number, so $\phi(1) = 1$.

Interestingly, $\phi(87109) = 79180$, and it can be seen that 87109 is a permutation of 79180.

Find the value of n , $1n10^7$, for which $\phi(n)$ is a permutation of n and the ratio $n/\phi(n)$ produces a minimum.

Problem 71: Ordered Fractions

Link

Tags: Number Theory

Consider the fraction, $\frac{n}{d}$, where n and d are positive integers. If nd and $\text{HCF}(n, d) = 1$, it is called a reduced proper fraction.

If we list the set of reduced proper fractions for $d \leq 8$ in ascending order of size, we get:

$$\frac{1}{8}, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}, \frac{4}{7}, \frac{3}{5}, \frac{5}{8}, \frac{2}{3}, \frac{5}{7}, \frac{3}{4}, \frac{5}{5}, \frac{6}{7}, \frac{7}{8}$$

It can be seen that $\frac{2}{5}$ is the fraction immediately to the left of $\frac{3}{7}$.

By listing the set of reduced proper fractions for $d \leq 1\,000\,000$ in ascending order of size, find the numerator of the fraction immediately to the left of $\frac{3}{7}$.

Problem 72: Counting Fractions

Link

Tags: Number Theory

Consider the fraction, $\frac{n}{d}$, where n and d are positive integers. If nd and $\text{HCF}(n, d) = 1$, it is called a reduced proper fraction.

If we list the set of reduced proper fractions for $d \leq 8$ in ascending order of size, we get:

$$\frac{1}{8}, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}, \frac{4}{7}, \frac{3}{5}, \frac{5}{8}, \frac{2}{3}, \frac{5}{7}, \frac{3}{4}, \frac{5}{5}, \frac{6}{7}, \frac{7}{8}$$

It can be seen that there are 21 elements in this set.

How many elements would be contained in the set of reduced proper fractions for $d \leq 1\,000\,000$?

Problem 73: Counting Fractions in a Range

Link

Tags: Number Theory

Consider the fraction, $\frac{n}{d}$, where n and d are positive integers. If nd and $\text{HCF}(n, d) = 1$, it is called a reduced proper fraction.

If we list the set of reduced proper fractions for $d \leq 8$ in ascending order of size, we get:

$$\frac{1}{8}, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}, \frac{4}{7}, \frac{3}{5}, \frac{5}{8}, \frac{2}{3}, \frac{5}{7}, \frac{3}{4}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}$$

It can be seen that there are 3 fractions between $\frac{1}{3}$ and $\frac{1}{2}$.

How many fractions lie between $\frac{1}{3}$ and $\frac{1}{2}$ in the sorted set of reduced proper fractions for $d \leq 12\,000$?

Problem 74: Digit Factorial Chains

Link

Tags: Number Theory

The number 145 is well known for the property that the sum of the factorial of its digits is equal to 145:

$$1! + 4! + 5! = 1 + 24 + 120 = 145.$$

Perhaps less well known is 169, in that it produces the longest chain of numbers that link back to 169; it turns out that there are only three such loops that exist:

$$\begin{aligned} 169 &\rightarrow 363601 \rightarrow 1454 \rightarrow 169 \\ 871 &\rightarrow 45361 \rightarrow 871 \\ 872 &\rightarrow 45362 \rightarrow 872 \end{aligned}$$

It is not difficult to prove that EVERY starting number will eventually get stuck in a loop. For example,

$$\begin{aligned}
&69 \rightarrow 363600 \rightarrow 1454 \rightarrow 169 \rightarrow 363601(\rightarrow 1454) \\
&78 \rightarrow 45360 \rightarrow 871 \rightarrow 45361(\rightarrow 871) \\
&540 \rightarrow 145(\rightarrow 145)
\end{aligned}$$

Starting with 69 produces a chain of five non-repeating terms, but the longest non-repeating chain with a starting number below one million is sixty terms.

How many chains, with a starting number below one million, contain exactly sixty non-repeating terms?

Problem 75: Singular Integer Right Triangles

Link

Tags: Geometry

It turns out that 12cm is the smallest length of wire that can be bent to form an integer sided right angle triangle in exactly one way, but there are many more examples.

- **12cm:** (3, 4, 5)
- **24cm:** (6, 8, 10)
- **30cm:** (5, 12, 13)
- **36cm:** (9, 12, 15)
- **40cm:** (8, 15, 17)
- **48cm:** (12, 16, 20)

In contrast, some lengths of wire, like 20cm , cannot be bent to form an integer sided right angle triangle, and other lengths allow more than one solution to be found; for example, using 120cm it is possible to form exactly three different integer sided right angle triangles.

- **120cm:** (30, 40, 50), (20, 48, 52), (24, 45, 51)

Given that L is the length of the wire, for how many values of $L \leq 1\,500\,000$ can exactly one integer sided right angle triangle be formed?

Problem 76: Counting Summations

Link

Tags: Number Theory

It is possible to write five as a sum in exactly six different ways:

$4 + 1$
 $3 + 2$
 $3 + 1 + 1$
 $2 + 2 + 1$
 $2 + 1 + 1 + 1$
 $1 + 1 + 1 + 1 + 1$

How many different ways can one hundred be written as a sum of at least two positive integers?

Problem 77: Prime Summations

Link

Tags: Number Theory

It is possible to write ten as the sum of primes in exactly five different ways:

$7 + 3$
 $5 + 5$
 $5 + 3 + 2$
 $3 + 3 + 2 + 2$
 $2 + 2 + 2 + 2 + 2$

What is the first value which can be written as the sum of primes in over five thousand different ways?

Problem 78: Coin Partitions

Link

Tags: Probability

Let $p(n)$ represent the number of different ways in which n coins can be separated into piles. For example, five coins can be separated into piles in exactly seven different ways, so $p(5) = 7$.

OOOOO
 OOOO O
 OOO OO
 OOO O O

OO OO O
OO O O O
O O O O O

Problem 79: Passcode Derivation

[Link](#)

Tags: Probability

A common security method used for online banking is to ask the user for three random characters from a passcode. For example, if the passcode was 531278, they may ask for the 2nd, 3rd, and 5th characters; the expected reply would be: 317.

The text file, `keylog.txt`, contains fifty successful login attempts.

Given that the three characters are always asked for in order, analyse the file so as to determine the shortest possible secret passcode of unknown length.

Problem 80: Square Root Digital Expansion

[Link](#)

Tags: Algebra, Number Theory

It is well known that if the square root of a natural number is not an integer, then it is irrational. The decimal expansion of such square roots is infinite without any repeating pattern at all.

The square root of two is $1.41421356237309504880\dots$, and the digital sum of the first one hundred decimal digits is 475.

For the first one hundred natural numbers, find the total of the digital sums of the first one hundred decimal digits for all the irrational square roots.

Problem 81: Path Sum: Two Ways

[Link](#)

Tags: Algebra, Combinatorics, Game Theory

In the 5 by 5 matrix below, the minimal path sum from the top left to the bottom right, by only moving to the right and down, is indicated in bold red and is equal to 2427.

$$\begin{pmatrix} \mathbf{131} & 673 & 234 & 103 & 18 \\ \mathbf{201} & \mathbf{96} & \mathbf{342} & 965 & 150 \\ 630 & 803 & \mathbf{746} & \mathbf{422} & 111 \\ 537 & 699 & 497 & \mathbf{121} & 956 \\ 805 & 732 & 524 & \mathbf{37} & \mathbf{331} \end{pmatrix}$$

Problem 82: Path Sum: Three Ways

Link

Tags: Algebra, Combinatorics, Game Theory

NOTE: This problem is a more challenging version of Problem 81.

The minimal path sum in the 5 by 5 matrix below, by starting in any cell in the left column and finishing in any cell in the right column, and only moving up, down, and right, is indicated in red and bold; the sum is equal to 994.

$$\begin{pmatrix} 131 & 673 & \mathbf{234} & \mathbf{103} & \mathbf{18} \\ \mathbf{201} & \mathbf{96} & \mathbf{342} & 965 & 150 \\ 630 & 803 & 746 & 422 & 111 \\ 537 & 699 & 497 & 121 & 956 \\ 805 & 732 & 524 & 37 & 331 \end{pmatrix}$$

Problem 83: Path Sum: Four Ways

Link

Tags: Algebra, Combinatorics, Game Theory

NOTE: This problem is a significantly more challenging version of Problem 81.

In the 5 by 5 matrix below, the minimal path sum from the top left to the bottom right, by moving left, right, up, and down, is indicated in bold red and is equal to 2297.

$$\begin{pmatrix} \mathbf{131} & 673 & \mathbf{234} & \mathbf{103} & \mathbf{18} \\ \mathbf{201} & \mathbf{96} & \mathbf{342} & 965 & \mathbf{150} \\ 630 & 803 & 746 & \mathbf{422} & \mathbf{111} \\ 537 & 699 & 497 & \mathbf{121} & 956 \\ 805 & 732 & 524 & \mathbf{37} & \mathbf{331} \end{pmatrix}$$

Problem 84: Monopoly Odds

[Link](#)

Tags: Game Theory

In the game, Monopoly, the standard board is set up in the following way:

Problem 85: Counting Rectangles

[Link](#)

Tags: Combinatorics, Geometry

By counting carefully it can be seen that a rectangular grid measuring 3 by 2 contains eighteen rectangles:

Problem 86: Cuboid Route

[Link](#)

Tags: Combinatorics, Geometry

A spider, S, sits in one corner of a cuboid room, measuring 6 by 5 by 3, and a fly, F, sits in the opposite corner. By travelling on the surfaces of the room the shortest “straight line” distance from S to F is 10 and the path is shown on the diagram.

Problem 87: Prime Power Triples

[Link](#)

Tags: Number Theory

The smallest number expressible as the sum of a prime square, prime cube, and prime fourth power is 28. In fact, there are exactly four numbers below fifty that can be expressed in such a way:

$$28 = 2^2 + 2^3 + 2^4$$

$$33 = 3^2 + 2^3 + 2^4$$

$$49 = 5^2 + 2^3 + 2^4$$

$$47 = 2^2 + 3^3 + 2^4$$

How many numbers below fifty million can be expressed as the sum of a prime square, prime cube, and prime fourth power?

Problem 88: Product-sum Numbers

Link

Tags: Game Theory

A natural number, N , that can be written as the sum and product of a given set of at least two natural numbers, $\{a_1, a_2, \dots, a_k\}$ is called a product-sum number: $N = a_1 + a_2 + \dots + a_k = a_1 \times a_2 \times \dots \times a_k$.

For example, $6 = 1 + 2 + 3 = 1 \times 2 \times 3$.

For a given set of size, k , we shall call the smallest N with this property a minimal product-sum number. The minimal product-sum numbers for sets of size, $k = 2, 3, 4, 5$, and 6 are as follows.

- $k = 2$: $4 = 2 \times 2 = 2 + 2$
- $k = 3$: $6 = 1 \times 2 \times 3 = 1 + 2 + 3$
- $k = 4$: $8 = 1 \times 1 \times 2 \times 4 = 1 + 1 + 2 + 4$
- $k = 5$: $8 = 1 \times 1 \times 2 \times 2 \times 2 = 1 + 1 + 2 + 2 + 2$
- $k = 6$: $12 = 1 \times 1 \times 1 \times 1 \times 2 \times 6 = 1 + 1 + 1 + 1 + 2 + 6$

Hence for $2 \leq k \leq 6$, the sum of all the minimal product-sum numbers is $4 + 6 + 8 + 12 = 30$; note that 8 is only counted once in the sum.

In fact, as the complete set of minimal product-sum numbers for $2 \leq k \leq 12$ is $\{4, 6, 8, 12, 15, 16\}$, the sum is 61 .

What is the sum of all the minimal product-sum numbers for $2 \leq k \leq 12000$?

Problem 89: Roman Numerals

Link

Tags: Game Theory

For a number written in Roman numerals to be considered valid there are basic rules which must be followed. Even though the rules allow some numbers to be expressed in more than one way there is always a “best” way of writing a particular number.

For example, it would appear that there are at least six ways of writing the number sixteen:

IIIIIIIIIIIIIIII

VIIIIIIIIIIII

VIIIIII

XIIIIII

VVVI

XVI

However, according to the rules only XIIIIII and XVI are valid, and the last example is considered to be the most efficient, as it uses the least number of numerals.

The 11K text file, roman.txt (right click and ‘Save Link/Target As...’), contains one thousand numbers written in valid, but not necessarily minimal, Roman numerals; see About... Roman Numerals for the definitive rules for this problem.

Find the number of characters saved by writing each of these in their minimal form.

Note: You can assume that all the Roman numerals in the file contain no more than four consecutive identical units.

Problem 90: Cube Digit Pairs

[Link](#)

Tags: Number Theory

Each of the six faces on a cube has a different digit (0 to 9) written on it; the same is done to a second cube. By placing the two cubes side-by-side in different positions we can form a variety of 2-digit numbers.

For example, the square number 64 could be formed:

Problem 91: Right Triangles with Integer Coordinates

[Link](#)

Tags: Geometry

The points $P(x_1, y_1)$ and $Q(x_2, y_2)$ are plotted at integer co-ordinates and are joined to the origin, $O(0, 0)$, to form $\triangle OPQ$.

Problem 92: Square Digit Chains

[Link](#)

Tags: Number Theory

A number chain is created by continuously adding the square of the digits in a number to form a new number until it has been seen before.

For example,

$$44 \rightarrow 32 \rightarrow 13 \rightarrow 10 \rightarrow \mathbf{1} \rightarrow \mathbf{1}$$

$$85 \rightarrow \mathbf{89} \rightarrow 145 \rightarrow 42 \rightarrow 20 \rightarrow 4 \rightarrow 16 \rightarrow 37 \rightarrow 58 \rightarrow \mathbf{89}$$

Therefore any chain that arrives at 1 or 89 will become stuck in an endless loop. What is most amazing is that EVERY starting number will eventually arrive at 1 or 89.

How many starting numbers below ten million will arrive at 89?

Problem 93: Arithmetic Expressions

Link

Tags: Number Theory

By using each of the digits from the set, $\{1, 2, 3, 4\}$, exactly once, and making use of the four arithmetic operations $(+, -, \times, /)$ and brackets/parentheses, it is possible to form different positive integer targets.

For example,

$$8 = (4 \times (1 + 3))/2$$

$$14 = 4 \times (3 + 1/2)$$

$$19 = 4 \times (2 + 3) - 1$$

$$36 = 3 \times 4 \times (2 + 1)$$

Note that concatenations of the digits, like $12 + 34$, are not allowed.

Using the set, $\{1, 2, 3, 4\}$, it is possible to obtain thirty-one different target numbers of which 36 is the maximum, and each of the numbers 1 to 28 can be obtained before encountering the first non-expressible number.

Find the set of four distinct digits, $abcd$, for which the longest set of consecutive positive integers, 1 to n , can be obtained, giving your answer as a string: $abcd$.

Problem 94: Almost Equilateral Triangles

Link

Tags: Geometry

It is easily proved that no equilateral triangle exists with integral length sides and integral area. However, the almost equilateral triangle 5-5-6 has an area of 12 square units.

We shall define an almost equilateral triangle to be a triangle for which two sides are equal and the third differs by no more than one unit.

Find the sum of the perimeters of all almost equilateral triangles with integral side lengths and area and whose perimeters do not exceed one billion (1 000 000 000).

Problem 95: Amicable Chains

[Link](#)

Tags: Number Theory

The proper divisors of a number are all the divisors excluding the number itself. For example, the proper divisors of 28 are 1, 2, 4, 7, and 14. As the sum of these divisors is equal to 28, we call it a perfect number.

Interestingly the sum of the proper divisors of 220 is 284 and the sum of the proper divisors of 284 is 220, forming a chain of two numbers. For this reason, 220 and 284 are called an amicable pair.

Perhaps less well known are longer chains. For example, starting with 12496, we form a chain of five numbers:

$$12496 \rightarrow 14288 \rightarrow 15472 \rightarrow 14536 \rightarrow 14264 (\rightarrow 12496 \rightarrow \dots)$$

Since this chain returns to its starting point, it is called an amicable chain.

Find the smallest member of the longest amicable chain with no element exceeding one million.

Problem 96: Su Doku

[Link](#)

Tags: Combinatorics, Number Theory

Su Doku (Japanese meaning number place) is the name given to a popular puzzle concept. Its origin is unclear, but credit must be attributed to Leonhard Euler who invented a similar, and much more difficult, puzzle idea called Latin Squares. The objective of Su Doku puzzles, however, is to replace the blanks (or zeros) in a 9 by 9 grid in such that each row, column, and 3 by 3 box contains each of the digits 1 to 9. Below is an example of a typical starting puzzle grid and its solution grid.

Problem 97: Large Non-Mersenne Prime

[Link](#)

Tags: Number Theory

The first known prime found to exceed one million digits was discovered in 1999, and is a Mersenne prime of the form $2^{6972593} - 1$; it contains exactly 2 098 960 digits. Subsequently other Mersenne primes, of the form $2^p - 1$, have been found which contain more digits.

However, in 2004 there was found a massive non-Mersenne prime which contains 2 357 207 digits: $28433 \times 2^{7830457} + 1$.

Find the last ten digits of this prime number.

Problem 98: Anagramic Squares

[Link](#)

Tags: Number Theory

By replacing each of the letters in the word CARE with 1, 2, 9, and 6 respectively, we form a square number: $1296 = 36^2$. What is remarkable is that, by using the same digital substitutions, the anagram, RACE, also forms a square number: $9216 = 96^2$. We shall call CARE (and RACE) a square anagram word pair and specify further that leading zeroes are not permitted, neither may a different letter have the same digital value as another letter.

Using words.txt (right click and ‘Save Link/Target As...’), a 16K text file containing nearly two-thousand common English words, find all the square anagram word pairs (a palindromic word is NOT considered to be an anagram of itself).

What is the largest square number formed by any member of such a pair?

NOTE: All anagrams formed must be contained in the given text file.

Problem 99: Largest Exponential

[Link](#)

Tags: Number Theory

Comparing two numbers written in index form like 2^{11} and 3^7 is not difficult, as any calculator would confirm that $2^{11} = 2048$ and $3^7 = 2187$.

However, confirming that $632382^{518061} 519432^{525806}$ would be much more difficult, as both numbers contain over three million digits.

Using base_exp.txt (right click and ‘Save Link/Target As...’), a 22K text file containing one thousand lines with a base/exponent pair on each line, determine which line number has the greatest numerical value.

NOTE: The first two lines in the file represent the numbers in the example given above.

Problem 100: Arranged Probability

Link

Tags: Combinatorics, Probability

If a box contains twenty-one coloured discs, composed of fifteen blue discs and six red discs, and two discs were taken at random, it can be seen that the probability of taking two blue discs, $P(BB) = (15/21) \times (14/20) = 1/2$.

The next such arrangement, for which there is exactly 50% chance of taking two blue discs at random, is a box containing eighty-five blue discs and thirty-five red discs.

By finding the first arrangement to contain over $10^{12} = 1\,000\,000\,000\,000$ discs in total, determine the number of blue discs that the box would contain.

Problem 101: Optimum Polynomial

Link

Tags: Algebra, Game Theory, Sequences

If we are presented with the first k terms of a sequence it is impossible to say with certainty the value of the next term, as there are infinitely many polynomial functions that can model the sequence.

As an example, let us consider the sequence of cube numbers. This is defined by the generating function, $u_n = n^3$: 1, 8, 27, 64, 125, 216, ...

Suppose we were only given the first two terms of this sequence. Working on the principle that “simple is best” we should assume a linear relationship and predict the next term to be 15 (common difference 7). Even if we were presented with the first three terms, by the same principle of simplicity, a quadratic relationship should be assumed.

We shall define $OP(k, n)$ to be the n th term of the optimum polynomial generating function for the first k terms of a sequence. It should be clear that $OP(k, n)$ will accurately generate the terms of the sequence for $n \leq k$, and potentially the first incorrect term (FIT) will be $OP(k, k + 1)$; in which case we shall call it a bad OP (BOP).

As a basis, if we were only given the first term of sequence, it would be most sensible to assume constancy; that is, for $n \geq 2$, $OP(1, n) = u_1$.

Hence we obtain the following OPs for the cubic sequence:

$$OP(1, n) = 1 \quad 1, \mathbf{1}, 1, 1, \dots \quad OP(2, n) = 7n - 6 \quad 1, 8, \mathbf{15}, \dots \quad OP(3, n) = 6n^2 - 11n + 6 \quad 1, 8, 27, \mathbf{58}, \dots \quad OP(4, n) = n^3 \quad 1, 8, 27, 64, 125, \dots$$

Problem 102: Triangle Containment

Link

Tags: Game Theory, Geometry, Probability

Three distinct points are plotted at random on a Cartesian plane, for which $-1000 \leq x, y \leq 1000$, such that a triangle is formed.

Consider the following two triangles:

It can be verified that triangle ABC contains the origin, whereas triangle XYZ does not.

Using triangles.txt (right click and ‘Save Link/Target As...’), a 27K text file containing the co-ordinates of one thousand “random” triangles, find the number of triangles for which the interior contains the origin.

NOTE: The first two examples in the file represent the triangles in the example given above.

Problem 103: Special Subset Sums: Optimum

Link

Tags: Combinatorics, Game Theory

Let $S(A)$ represent the sum of elements in set A of size n . We shall call it a special sum set if for any two non-empty disjoint subsets, B and C , the following properties are true:

- $S(B) \neq S(C)$; that is, sums of subsets cannot be equal.
- If B contains more elements than C then $S(B) > S(C)$.

If $S(A)$ is minimised for a given n , we shall call it an optimum special sum set. The first five optimum special sum sets are given below.

- $n = 1$: $\{1\}$

- $n = 2$: $\{1, 2\}$
- $n = 3$: $\{2, 3, 4\}$
- $n = 4$: $\{3, 5, 6, 7\}$
- $n = 5$: $\{6, 9, 11, 12, 13\}$

It seems that for a given optimum set, $A = \{a_1, a_2, \dots, a_n\}$, the next optimum set is of the form $B = \{b, a_1 + b, a_2 + b, \dots, a_n + b\}$, where b is the “middle” element on the previous row.

By applying this “rule” we would expect the optimum set for $n = 6$ to be $A = \{11, 17, 20, 22, 23, 24\}$, with $S(A) = 117$. However, this is not the optimum set, as we have merely applied an algorithm to provide a near optimum set. The optimum set for $n = 6$ is $A = \{11, 18, 19, 20, 22, 25\}$, with $S(A) = 115$ and corresponding set string: 111819202225.

Given that A is an optimum special sum set for $n = 7$, find its set string.

NOTE: This problem is related to Problem 105 and Problem 106.

Problem 104: Pandigital Fibonacci Ends

[Link](#)

Tags: Number Theory, Sequences

The Fibonacci sequence is defined by the recurrence relation:

$$F_n = F_{n-1} + F_{n-2}, \text{ where } F_1 = 1 \text{ and } F_2 = 1.$$

It turns out that F_{541} , which contains 113 digits, is the first Fibonacci number for which the last nine digits are 1-9 pandigital (contain all the digits 1 to 9, but not necessarily in order). And F_{2749} , which contains 575 digits, is the first Fibonacci number for which the first nine digits are 1-9 pandigital.

Given that F_k is the first Fibonacci number for which the first nine digits AND the last nine digits are 1-9 pandigital, find k .

Problem 105: Special Subset Sums: Testing

[Link](#)

Tags: Combinatorics, Game Theory

Let $S(A)$ represent the sum of elements in set A of size n . We shall call it a special sum set if for any two non-empty disjoint subsets, B and C , the following properties are true:

- $S(B) \neq S(C)$; that is, sums of subsets cannot be equal.
- If B contains more elements than C then $S(B)S(C)$.

For example, $\{81, 88, 75, 42, 87, 84, 86, 65\}$ is not a special sum set because $65 + 87 + 88 = 75 + 81 + 84$, whereas $\{157, 150, 164, 119, 79, 159, 161, 139, 158\}$ satisfies both rules for all possible subset pair combinations and $S(A) = 1286$.

Using sets.txt (right click and “Save Link/Target As...”), a 4K text file with one-hundred sets containing seven to twelve elements (the two examples given above are the first two sets in the file), identify all the special sum sets, $A_1, A_2, \dots, A_k$, and find the value of $S(A_1) + S(A_2) + \dots + S(A_k)$.

NOTE: This problem is related to Problem 103 and Problem 106.

Problem 106: Special Subset Sums: Meta-testing

Link

Tags: Combinatorics, Game Theory

Let $S(A)$ represent the sum of elements in set A of size n . We shall call it a special sum set if for any two non-empty disjoint subsets, B and C , the following properties are true:

- $S(B) \neq S(C)$; that is, sums of subsets cannot be equal.
- If B contains more elements than C then $S(B)S(C)$.

For this problem we shall assume that a given set contains n strictly increasing elements and it already satisfies the second rule.

Surprisingly, out of the 25 possible subset pairs that can be obtained from a set for which $n = 4$, only 1 of these pairs need to be tested for equality (first rule). Similarly, when $n = 7$, only 70 out of the 966 subset pairs need to be tested.

For $n = 12$, how many of the 261625 subset pairs that can be obtained need to be tested for equality?

NOTE: This problem is related to Problem 103 and Problem 105.

Problem 107: Minimal Network

Link

Tags: Game Theory, Graph Theory

The following undirected network consists of seven vertices and twelve edges with a total weight of 243.

Problem 108: Diophantine Reciprocals I

Link

Tags: Algebra, Game Theory

In the following equation x , y , and n are positive integers.

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

For $n = 4$ there are exactly three distinct solutions:

$$\begin{aligned}\frac{1}{5} + \frac{1}{20} &= \frac{1}{4} \\ \frac{1}{6} + \frac{1}{12} &= \frac{1}{4} \\ \frac{1}{8} + \frac{1}{8} &= \frac{1}{4}\end{aligned}$$

What is the least value of n for which the number of distinct solutions exceeds one-thousand?

NOTE: This problem is an easier version of Problem 110; it is strongly advised that you solve this one first.

Problem 109: Darts

Link

Tags: Game Theory

In the game of darts a player throws three darts at a target board which is split into twenty equal sized sections numbered one to twenty.

Problem 110: Diophantine Reciprocals II

Link

Tags: Algebra, Game Theory

In the following equation x , y , and n are positive integers.

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

It can be verified that when $n = 1260$ there are 113 distinct solutions and this is the least value of n for which the total number of distinct solutions exceeds one hundred.

What is the least value of n for which the number of distinct solutions exceeds four million?

NOTE: This problem is a much more difficult version of Problem 108 and as it is well beyond the limitations of a brute force approach it requires a clever implementation.

Problem 111: Primes with Runs

[Link](#)

Tags: Game Theory, Number Theory

Considering 4-digit primes containing repeated digits it is clear that they cannot all be the same: 1111 is divisible by 11, 2222 is divisible by 22, and so on. But there are nine 4-digit primes containing three ones:

1117, 1151, 1171, 1181, 1511, 1811, 2111, 4111, 8111.

We shall say that $M(n, d)$ represents the maximum number of repeated digits for an n -digit prime where d is the repeated digit, $N(n, d)$ represents the number of such primes, and $S(n, d)$ represents the sum of these primes.

So $M(4, 1) = 3$ is the maximum number of repeated digits for a 4-digit prime where one is the repeated digit, there are $N(4, 1) = 9$ such primes, and the sum of these primes is $S(4, 1) = 22275$. It turns out that for $d = 0$, it is only possible to have $M(4, 0) = 2$ repeated digits, but there are $N(4, 0) = 13$ such cases.

In the same way we obtain the following results for 4-digit primes.

Digit, d	M(4, d)	N(4, d)	S(4, d)
0	2	13	67061
1	3	9	22275
2	3	1	2221
3	3	12	46214
4	3	2	8888
5	3	1	5557
6	3	1	6661
7	3	9	57863
8	3	1	8887
9	3	7	48073

Problem 112: Bouncy Numbers

[Link](#)

Tags: Number Theory

Working from left-to-right if no digit is exceeded by the digit to its left it is called an increasing number; for example, 134468.

Similarly if no digit is exceeded by the digit to its right it is called a decreasing number; for example, 66420.

We shall call a positive integer that is neither increasing nor decreasing a “bouncy” number; for example, 155349.

Clearly there cannot be any bouncy numbers below one-hundred, but just over half of the numbers below one-thousand (525) are bouncy. In fact, the least number for which the proportion of bouncy numbers first reaches 50% is 538.

Surprisingly, bouncy numbers become more and more common and by the time we reach 21780 the proportion of bouncy numbers is equal to 90%.

Find the least number for which the proportion of bouncy numbers is exactly 99%.

Problem 113: Non-bouncy Numbers

Link

Tags: Number Theory

Working from left-to-right if no digit is exceeded by the digit to its left it is called an increasing number; for example, 134468.

Similarly if no digit is exceeded by the digit to its right it is called a decreasing number; for example, 66420.

We shall call a positive integer that is neither increasing nor decreasing a “bouncy” number; for example, 155349.

As n increases, the proportion of bouncy numbers below n increases such that there are only 12951 numbers below one-million that are not bouncy and only 277032 non-bouncy numbers below 10^{10} .

How many numbers below a googol (10^{100}) are not bouncy?

Problem 114: Counting Block Combinations I

Link

Tags: Game Theory

A row measuring seven units in length has red blocks with a minimum length of three units placed on it, such that any two red blocks (which are allowed to be different lengths) are separated by at least one grey square. There are exactly seventeen ways of doing this.

Problem 115: Counting Block Combinations II

[Link](#)

Tags: Game Theory

NOTE: This is a more difficult version of Problem 114.

A row measuring n units in length has red blocks with a minimum length of m units placed on it, such that any two red blocks (which are allowed to be different lengths) are separated by at least one black square.

Let the fill-count function, $F(m, n)$, represent the number of ways that a row can be filled.

For example, $F(3, 29) = 673135$ and $F(3, 30) = 1089155$.

That is, for $m = 3$, it can be seen that $n = 30$ is the smallest value for which the fill-count function first exceeds one million.

In the same way, for $m = 10$, it can be verified that $F(10, 56) = 880711$ and $F(10, 57) = 1148904$, so $n = 57$ is the least value for which the fill-count function first exceeds one million.

For $m = 50$, find the least value of n for which the fill-count function first exceeds one million.

Problem 116: Red, Green or Blue Tiles

[Link](#)

Tags: Number Theory

A row of five grey square tiles is to have a number of its tiles replaced with coloured oblong tiles chosen from red (length two), green (length three), or blue (length four).

If red tiles are chosen there are exactly seven ways this can be done.

Problem 117: Red, Green, and Blue Tiles

[Link](#)

Tags: Combinatorics

Using a combination of grey square tiles and oblong tiles chosen from: red tiles (measuring two units), green tiles (measuring three units), and blue tiles (measuring four units), it is possible to tile a row measuring five units in length in exactly fifteen different ways.

Problem 118: Pandigital Prime Sets

Link

Tags: Number Theory

Using all of the digits 1 through 9 and concatenating them freely to form decimal integers, different sets can be formed. Interestingly with the set $\{2, 5, 47, 89, 631\}$, all of the elements belonging to it are prime.

How many distinct sets containing each of the digits one through nine exactly once contain only prime elements?

Problem 119: Digit Power Sum

Link

Tags: Number Theory, Sequences

The number 512 is interesting because it is equal to the sum of its digits raised to some power: $5 + 1 + 2 = 8$, and $8^3 = 512$. Another example of a number with this property is $614656 = 28^4$.

We shall define a_n to be the n th term of this sequence and insist that a number must contain at least two digits to have a sum.

You are given that $a_2 = 512$ and $a_{10} = 614656$.

Find a_{30} .

Problem 120: Square Remainders

Link

Tags: Number Theory

Let r be the remainder when $(a - 1)^n + (a + 1)^n$ is divided by a^2 .

For example, if $a = 7$ and $n = 3$, then $r = 42$: $6^3 + 8^3 = 728 \equiv 42 \pmod{49}$. And as n varies, so too will r , but for $a = 7$ it turns out that $r_{\max} = 42$.

For $3 \leq a \leq 1000$, find $\sum r_{\max}$.

Problem 121: Disc Game Prize Fund

Link

Tags: Game Theory, Probability

A bag contains one red disc and one blue disc. In a game of chance a player takes a disc at random and its colour is noted. After each turn the disc is returned to the bag, an extra red disc is added, and another disc is taken at random.

The player pays £1 to play and wins if they have taken more blue discs than red discs at the end of the game.

If the game is played for four turns, the probability of a player winning is exactly $11/120$, and so the maximum prize fund the banker should allocate for winning in this game would be £10 before they would expect to incur a loss. Note that any payout will be a whole number of pounds and also includes the original £1 paid to play the game, so in the example given the player actually wins £9.

Find the maximum prize fund that should be allocated to a single game in which fifteen turns are played.

Problem 122: Efficient Exponentiation

Link

Tags: Game Theory

The most naive way of computing n^{15} requires fourteen multiplications:

$$n \times n \times \cdots \times n = n^{15}.$$

But using a “binary” method you can compute it in six multiplications:

$$\begin{aligned} n \times n &= n^2 \\ n^2 \times n^2 &= n^4 \\ n^4 \times n^4 &= n^8 \\ n^8 \times n^4 &= n^{12} \\ n^{12} \times n^2 &= n^{14} \\ n^{14} \times n &= n^{15} \end{aligned}$$

However it is yet possible to compute it in only five multiplications:

$$\begin{aligned}
n \times n &= n^2 \\
n^2 \times n &= n^3 \\
n^3 \times n^3 &= n^6 \\
n^6 \times n^6 &= n^{12} \\
n^{12} \times n^3 &= n^{15}
\end{aligned}$$

We shall define $m(k)$ to be the minimum number of multiplications to compute n^k ; for example $m(15) = 5$.

Find $\sum_{k=1}^{200} m(k)$.

Problem 123: Prime Square Remainders

[Link](#)

Tags: Number Theory

Let p_n be the n th prime: $2, 3, 5, 7, 11, \dots$, and let r be the remainder when $(p_n - 1)^n + (p_n + 1)^n$ is divided by p_n^2 .

For example, when $n = 3$, $p_3 = 5$, and $4^3 + 6^3 = 280 \equiv 5 \pmod{25}$.

The least value of n for which the remainder first exceeds 10^9 is 7037.

Find the least value of n for which the remainder first exceeds 10^{10} .

Problem 124: Ordered Radicals

[Link](#)

Tags: Number Theory

The radical of n , $\text{rad}(n)$, is the product of the distinct prime factors of n . For example, $504 = 2^3 \times 3^2 \times 7$, so $\text{rad}(504) = 2 \times 3 \times 7 = 42$.

If we calculate $\text{rad}(n)$ for $1 \leq n \leq 10$, then sort them on $\text{rad}(n)$, and sorting on n if the radical values are equal, we get:

Unsorted Sorted

n $\text{rad}(n)$ n $\text{rad}(n)$ k

11 111

22 222

33 423

42 824

55 335

66 936

77 557

82 668

93 779

1010 101010

Let $E(k)$ be the k -th element in the sorted n column; for example, $E(4) = 8$ and $E(6) = 9$.

If $\text{rad}(n)$ is sorted for $1 \leq n \leq 100000$, find $E(10000)$.

Problem 125: Palindromic Sums

[Link](#)

Tags: Number Theory

The palindromic number 595 is interesting because it can be written as the sum of consecutive squares: $6^2 + 7^2 + 8^2 + 9^2 + 10^2 + 11^2 + 12^2$.

There are exactly eleven palindromes below one-thousand that can be written as consecutive square sums, and the sum of these palindromes is 4164. Note that $1 = 0^2 + 1^2$ has not been included as this problem is concerned with the squares of positive integers.

Find the sum of all the numbers less than 10^8 that are both palindromic and can be written as the sum of consecutive squares.

Problem 126: Cuboid Layers

[Link](#)

Tags: Game Theory

The minimum number of cubes to cover every visible face on a cuboid measuring $3 \times 2 \times 1$ is twenty-two.

Problem 127: abc-hits

Link

Tags: Number Theory

The radical of n , $\text{rad}(n)$, is the product of distinct prime factors of n . For example, $504 = 2^3 \times 3^2 \times 7$, so $\text{rad}(504) = 2 \times 3 \times 7 = 42$.

We shall define the triplet of positive integers (a, b, c) to be an abc-hit if:

- $\gcd(a, b) = \gcd(a, c) = \gcd(b, c) = 1$
- ab
- $a + b = c$
- $\text{rad}(abc)c$

For example, $(5, 27, 32)$ is an abc-hit, because:

- $\gcd(5, 27) = \gcd(5, 32) = \gcd(27, 32) = 1$
- 527
- $5 + 27 = 32$
- $\text{rad}(4320) = 3032$

It turns out that abc-hits are quite rare and there are only thirty-one abc-hits for $c \leq 1000$, with $\sum c = 12523$.

Find $\sum c$ for $c \leq 120000$.

Problem 128: Hexagonal Tile Differences

Link

Tags: Number Theory

A hexagonal tile with number 1 is surrounded by a ring of six hexagonal tiles, starting at “12 o’clock” and numbering the tiles 2 to 7 in an anti-clockwise direction.

New rings are added in the same fashion, with the next rings being numbered 8 to 19, 20 to 37, 38 to 61, and so on. The diagram below shows the first three rings.

Problem 129: Repunit Divisibility

[Link](#)

Tags: Number Theory

A number consisting entirely of ones is called a repunit. We shall define $R(k)$ to be a repunit of length k ; for example, $R(6) = 111111$.

Given that n is a positive integer and $\gcd(n, 10) = 1$, it can be shown that there always exists a value, k , for which $R(k)$ is divisible by n , and let $A(n)$ be the least such value of k ; for example, $A(7) = 6$ and $A(41) = 5$.

The least value of n for which $A(n)$ first exceeds ten is 17.

Find the least value of n for which $A(n)$ first exceeds one-million.

Problem 130: Composites with Prime Repunit Property

[Link](#)

Tags: Number Theory

A number consisting entirely of ones is called a repunit. We shall define $R(k)$ to be a repunit of length k ; for example, $R(6) = 111111$.

Given that n is a positive integer and $\gcd(n, 10) = 1$, it can be shown that there always exists a value, k , for which $R(k)$ is divisible by n , and let $A(n)$ be the least such value of k ; for example, $A(7) = 6$ and $A(41) = 5$.

You are given that for all primes, $p \neq 5$, that $p - 1$ is divisible by $A(p)$. For example, when $p = 41$, $A(41) = 5$, and 40 is divisible by 5.

However, there are rare composite values for which this is also true; the first five examples being 91, 259, 451, 481, and 703.

Find the sum of the first twenty-five composite values of n for which $\gcd(n, 10) = 1$ and $n - 1$ is divisible by $A(n)$.

Problem 131: Prime Cube Partnership

[Link](#)

Tags: Number Theory

There are some prime values, p , for which there exists a positive integer, n , such that the expression $n^3 + n^2p$ is a perfect cube.

For example, when $p = 19$, $8^3 + 8^2 \times 19 = 12^3$.

What is perhaps most surprising is that for each prime with this property the value of n is unique, and there are only four such primes below one-hundred.

How many primes below one million have this remarkable property?

Problem 132: Large Repunit Factors

[Link](#)

Tags: Number Theory

A number consisting entirely of ones is called a repunit. We shall define $R(k)$ to be a repunit of length k .

For example, $R(10) = 1111111111 = 11 \times 41 \times 271 \times 9091$, and the sum of these prime factors is 9414.

Find the sum of the first forty prime factors of $R(10^9)$.

Problem 133: Repunit Nonfactors

[Link](#)

Tags: Number Theory

A number consisting entirely of ones is called a repunit. We shall define $R(k)$ to be a repunit of length k ; for example, $R(6) = 111111$.

Let us consider repunits of the form $R(10^n)$.

Although $R(10)$, $R(100)$, or $R(1000)$ are not divisible by 17, $R(10000)$ is divisible by 17. Yet there is no value of n for which $R(10^n)$ will divide by 19. In fact, it is remarkable that 11, 17, 41, and 73 are the only four primes below one-hundred that can be a factor of $R(10^n)$.

Find the sum of all the primes below one-hundred thousand that will never be a factor of $R(10^n)$.

Problem 134: Prime Pair Connection

[Link](#)

Tags: Number Theory

Consider the consecutive primes $p_1 = 19$ and $p_2 = 23$. It can be verified that 1219 is the smallest number such that the last digits are formed by p_1 whilst also being divisible by p_2 .

In fact, with the exception of $p_1 = 3$ and $p_2 = 5$, for every pair of consecutive primes, $p_2 p_1$, there exist values of n for which the last digits are formed by p_1 and n is divisible by p_2 . Let S be the smallest of these values of n .

Find $\sum S$ for every pair of consecutive primes with $5 \leq p_1 \leq 1000000$.

Problem 135: Same Differences

Link

Tags: Algebra

Given the positive integers, x , y , and z , are consecutive terms of an arithmetic progression, the least value of the positive integer, n , for which the equation, $x^2 - y^2 - z^2 = n$, has exactly two solutions is $n = 27$:

$$34^2 - 27^2 - 20^2 = 12^2 - 9^2 - 6^2 = 27.$$

It turns out that $n = 1155$ is the least value which has exactly ten solutions.

How many values of n less than one million have exactly ten distinct solutions?

Problem 136: Singleton Difference

Link

Tags: Algebra

The positive integers, x , y , and z , are consecutive terms of an arithmetic progression. Given that n is a positive integer, the equation, $x^2 - y^2 - z^2 = n$, has exactly one solution when $n = 20$:

$$13^2 - 10^2 - 7^2 = 20.$$

In fact there are twenty-five values of n below one hundred for which the equation has a unique solution.

How many values of n less than fifty million have exactly one solution?

Problem 137: Fibonacci Golden Nuggets

Link

Tags: Algebra, Sequences

Consider the infinite polynomial series $A_F(x) = xF_1 + x^2F_2 + x^3F_3 + \dots$, where F_k is the k th term in the Fibonacci sequence: 1, 1, 2, 3, 5, 8, $\dots$; that is, $F_k = F_{k-1} + F_{k-2}$, $F_1 = 1$ and $F_2 = 1$.

For this problem we shall be interested in values of x for which $A_F(x)$ is a positive integer.

Surprisingly

The corresponding values of x for the first five natural numbers are shown below.

$$xA_F(x) \quad \sqrt{2} - 11 \quad \frac{1}{2}2 \quad \frac{\sqrt{13}-2}{3}3 \quad \frac{\sqrt{89}-5}{8}4 \quad \frac{\sqrt{34}-3}{5}5$$

Problem 138: Special Isosceles Triangles

Link

Tags: Geometry

Consider the isosceles triangle with base length, $b = 16$, and legs, $L = 17$.

Problem 139: Pythagorean Tiles

Link

Tags: Geometry

Let (a, b, c) represent the three sides of a right angle triangle with integral length sides. It is possible to place four such triangles together to form a square with length c .

For example, $(3, 4, 5)$ triangles can be placed together to form a 5 by 5 square with a 1 by 1 hole in the middle and it can be seen that the 5 by 5 square can be tiled with twenty-five 1 by 1 squares.

Problem 140: Modified Fibonacci Golden Nuggets

Link

Tags: Algebra, Sequences

Consider the infinite polynomial series $A_G(x) = xG_1 + x^2G_2 + x^3G_3 + \dots$, where G_k is the k th term of the second order recurrence relation $G_k = G_{k-1} + G_{k-2}$, $G_1 = 1$ and $G_2 = 4$; that is, 1, 4, 5, 9, 14, 23, $\dots$

For this problem we shall be concerned with values of x for which $A_G(x)$ is a positive integer.

The corresponding values of x for the first five natural numbers are shown below.

$$x A_G(x) \quad \frac{\sqrt{5}-1}{4} 1 \quad \frac{2}{5} 2 \quad \frac{\sqrt{22}-2}{6} 3 \quad \frac{\sqrt{137}-5}{14} 4 \quad \frac{1}{2} 5$$

Problem 141: Square Progressive Numbers

Link

Tags: Sequences

A positive integer, n , is divided by d and the quotient and remainder are q and r respectively. In addition d , q , and r are consecutive positive integer terms in a geometric sequence, but not necessarily in that order.

For example, 58 divided by 6 has quotient 9 and remainder 4. It can also be seen that 4, 6, 9 are consecutive terms in a geometric sequence (common ratio $3/2$).

We will call such numbers, n , progressive.

Some progressive numbers, such as 9 and $10404 = 102^2$, happen to also be perfect squares. The sum of all progressive perfect squares below one hundred thousand is 124657.

Find the sum of all progressive perfect squares below one trillion (10^{12}).

Problem 142: Perfect Square Collection

Link

Tags: Number Theory

Find the smallest $x + y + z$ with integers $xyz \neq 0$ such that $x + y$, $x - y$, $x + z$, $x - z$, $y + z$, $y - z$ are all perfect squares.

Problem 143: Torricelli Triangles

Link

Tags: Game Theory, Geometry

Let ABC be a triangle with all interior angles being less than 120 degrees. Let X be any point inside the triangle and let $XA = p$, $XC = q$, and $XB = r$.

Fermat challenged Torricelli to find the position of X such that $p + q + r$ was minimised.

Torricelli was able to prove that if equilateral triangles AOB , BNC and AMC are constructed on each side of triangle ABC , the circumscribed circles of AOB ,

BNC , and AMC will intersect at a single point, T , inside the triangle. Moreover he proved that T , called the Torricelli/Fermat point, minimises $p + q + r$. Even more remarkable, it can be shown that when the sum is minimised, $AN = BM = CO = p + q + r$ and that AN , BM and CO also intersect at T .

Problem 144: Laser Beam Reflections

Link

Tags: Algebra, Game Theory, Geometry

In laser physics, a “white cell” is a mirror system that acts as a delay line for the laser beam. The beam enters the cell, bounces around on the mirrors, and eventually works its way back out.

The specific white cell we will be considering is an ellipse with the equation $4x^2 + y^2 = 100$.

The section corresponding to $-0.01 \leq x \leq +0.01$ at the top is missing, allowing the light to enter and exit through the hole.

Problem 145: Reversible Numbers

Link

Tags: Number Theory

Some positive integers n have the property that the sum $[n + \text{reverse}(n)]$ consists entirely of odd (decimal) digits. For instance, $36 + 63 = 99$ and $409 + 904 = 1313$. We will call such numbers reversible; so 36, 63, 409, and 904 are reversible. Leading zeroes are not allowed in either n or $\text{reverse}(n)$.

There are 120 reversible numbers below one-thousand.

How many reversible numbers are there below one-billion (10^9)?

Problem 146: Investigating a Prime Pattern

Link

Tags: Number Theory

The smallest positive integer n for which the numbers $n^2 + 1$, $n^2 + 3$, $n^2 + 7$, $n^2 + 9$, $n^2 + 13$, and $n^2 + 27$ are consecutive primes is 10. The sum of all such integers n below one-million is 1242490.

What is the sum of all such integers n below 150 million?

Problem 147: Rectangles in Cross-hatched Grids

[Link](#)

Tags: Combinatorics, Geometry

In a 3×2 cross-hatched grid, a total of 37 different rectangles could be situated within that grid as indicated in the sketch.

Problem 148: Exploring Pascal's Triangle

[Link](#)

Tags: Geometry

We can easily verify that none of the entries in the first seven rows of Pascal's triangle are divisible by 7:

1
1
1
1
2
1
1
3
3
1
1
4
6
4
1
1
5
10

10
5
1
1
6
15
20
15
6
1

However, if we check the first one hundred rows, we will find that only 2361 of the 5050 entries are not divisible by 7.

Find the number of entries which are not divisible by 7 in the first one billion (10^9) rows of Pascal's triangle.

Problem 149: Maximum-sum Subsequence

[Link](#)

Tags: Number Theory

Looking at the table below, it is easy to verify that the maximum possible sum of adjacent numbers in any direction (horizontal, vertical, diagonal or anti-diagonal) is 16 ($= 8 + 7 + 1$).

−25329−6513273−18−48

Problem 150: Sub-triangle Sums

[Link](#)

Tags: Geometry

In a triangular array of positive and negative integers, we wish to find a sub-triangle such that the sum of the numbers it contains is the smallest possible.

In the example below, it can be easily verified that the marked triangle satisfies this condition having a sum of 42.

Problem 151: A Preference for A5

[Link](#)

Tags: Number Theory

A printing shop runs 16 batches (jobs) every week and each batch requires a sheet of special colour-proofing paper of size A5.

Every Monday morning, the supervisor opens a new envelope, containing a large sheet of the special paper with size A1.

The supervisor proceeds to cut it in half, thus getting two sheets of size A2. Then one of the sheets is cut in half to get two sheets of size A3 and so on until an A5-size sheet is obtained, which is needed for the first batch of the week.

All the unused sheets are placed back in the envelope.

Problem 152: Sums of Square Reciprocals

[Link](#)

Tags: Number Theory

There are several ways to write the number $\frac{1}{2}$ as a sum of square reciprocals using distinct integers.

For instance, the numbers $\{2, 3, 4, 5, 7, 12, 15, 20, 28, 35\}$ can be used:

$$\begin{aligned}\frac{1}{2} = & \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \\ & \frac{1}{7^2} + \frac{1}{12^2} + \frac{1}{15^2} + \frac{1}{20^2} + \\ & \frac{1}{28^2} + \frac{1}{35^2}\end{aligned}$$

In fact, only using integers between 2 and 45 inclusive, there are exactly three ways to do it, the remaining two being: $\{2, 3, 4, 6, 7, 9, 10, 20, 28, 35, 36, 45\}$ and $\{2, 3, 4, 6, 7, 9, 12, 15, 28, 30, 35, 36, 45\}$.

How many ways are there to write $\frac{1}{2}$ as a sum of reciprocals of squares using distinct integers between 2 and 80 inclusive?

Problem 153: Investigating Gaussian Integers

[Link](#)

Tags: Algebra, Game Theory, Number Theory

As we all know the equation $x^2 = -1$ has no solutions for real x .

If we however introduce the imaginary number i this equation has two solutions: $x = i$ and $x = -i$.

If we go a step further the equation $(x - 3)^2 = -4$ has two complex solutions: $x = 3 + 2i$ and $x = 3 - 2i$.

$x = 3 + 2i$ and $x = 3 - 2i$ are called each others' complex conjugate.

Numbers of the form $a + bi$ are called complex numbers.

In general $a + bi$ and $a - bi$ are each other's complex conjugate.

A Gaussian Integer is a complex number $a + bi$ such that both a and b are integers.

The regular integers are also Gaussian integers (with $b = 0$).

To distinguish them from Gaussian integers with $b \neq 0$ we call such integers “rational integers.”

A Gaussian integer $a + bi$ is called a divisor of a rational integer n if the result $\frac{n}{a + bi}$ is also a Gaussian integer.

If for example we divide 5 by $1 + 2i$ we can simplify $\frac{5}{1 + 2i}$ in the following manner:

Multiply numerator and denominator by the complex conjugate of $1 + 2i$: $1 - 2i$.

The result is $\frac{5}{1+2i} = \frac{5}{1+2i} \frac{1-2i}{1-2i} = \frac{5(1-2i)}{1-(2i)^2} = \frac{5(1-2i)}{1-(-4)} = \frac{5(1-2i)}{5} = 1-2i$.

So $1 + 2i$ is a divisor of 5.

Note that $1+i$ is not a divisor of 5 because $\frac{5}{1+i} = \frac{5}{2} - \frac{5}{2}i$.

Note also that if the Gaussian Integer $(a + bi)$ is a divisor of a rational integer n , then its complex conjugate $(a - bi)$ is also a divisor of n .

In fact, 5 has six divisors such that the real part is positive: $\{1, 1 + 2i, 1 - 2i, 2 + i, 2 - i, 5\}$.

The following is a table of all of the divisors for the first five positive rational integers:

 n Gaussian integer divisors

with positive real partSum $s(n)$ of these divisors111 $21, 1 + i, 1 - i, 25$ 31, 34
41, $1 + i, 1 - i, 2, 2 + 2i, 2 - 2i, 413$ 51, $1 + 2i, 1 - 2i, 2 + i, 2 - i, 512$

For divisors with positive real parts, then, we have: $\sum_{n=1}^5 s(n) = 35$.

$$\sum_{n=1}^{10^5} s(n) = 17924657155.$$

What is $\sum_{n=1}^{10^8} s(n)$?

Problem 154: Exploring Pascal's Pyramid

Link

Tags: Number Theory

A triangular pyramid is constructed using spherical balls so that each ball rests on exactly three balls of the next lower level.

Problem 155: Counting Capacitor Circuits

Link

Tags: Game Theory, Sequences

An electric circuit uses exclusively identical capacitors of the same value C .

The capacitors can be connected in series or in parallel to form sub-units, which can then be connected in series or in parallel with other capacitors or other sub-units to form larger sub-units, and so on up to a final circuit.

Using this simple procedure and up to n identical capacitors, we can make circuits having a range of different total capacitances. For example, using up to $n = 3$ capacitors of $60\mu F$ each, we can obtain the following 7 distinct total capacitance values:

Problem 156: Counting Digits

Link

Tags: Algebra, Number Theory

Starting from zero the natural numbers are written down in base 10 like this:

0 1 2 3 4 5 6 7 8 9 10 11 12 $\dots$

Consider the digit $d = 1$. After we write down each number n , we will update the number of ones that have occurred and call this number $f(n, 1)$. The first values for $f(n, 1)$, then, are as follows:

n	$f(n, 1)$
0	0
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1
10	2
11	4
12	5

Note that $f(n, 1)$ never equals 3.

So the first two solutions of the equation $f(n, 1) = n$ are $n = 0$ and $n = 1$. The next solution is $n = 199981$.

In the same manner the function $f(n, d)$ gives the total number of digits d that have been written down after the number n has been written.

In fact, for every digit $d \neq 0$, 0 is the first solution of the equation $f(n, d) = n$.

Let $s(d)$ be the sum of all the solutions for which $f(n, d) = n$.

You are given that $s(1) = 22786974071$.

Find $\sum s(d)$ for $1 \leq d \leq 9$.

Note: if, for some n , $f(n, d) = n$ for more than one value of d this value of n is counted again for every value of d for which $f(n, d) = n$.

Problem 157: Base-10 Diophantine Reciprocal

Link

Tags: Algebra

Consider the diophantine equation $\frac{1}{a} + \frac{1}{b} = \frac{p}{10^n}$ with a, b, p, n positive integers and $a \leq b$.

For $n = 1$ this equation has 20 solutions that are listed below:

$$\begin{array}{cccccc}
 \frac{1}{1} + \frac{1}{1} = \frac{20}{10} & \frac{1}{1} + \frac{1}{2} = \frac{15}{10} & \frac{1}{1} + \frac{1}{5} = \frac{12}{10} & \frac{1}{1} + \frac{1}{10} = \frac{11}{10} & \frac{1}{2} + \frac{1}{2} = \frac{10}{10} \\
 \frac{1}{2} + \frac{1}{5} = \frac{7}{10} & \frac{1}{2} + \frac{1}{10} = \frac{6}{10} & \frac{1}{3} + \frac{1}{6} = \frac{5}{10} & \frac{1}{3} + \frac{1}{15} = \frac{4}{10} & \frac{1}{4} + \frac{1}{4} = \frac{5}{10} \\
 \frac{1}{4} + \frac{1}{20} = \frac{3}{10} & \frac{1}{5} + \frac{1}{5} = \frac{4}{10} & \frac{1}{5} + \frac{1}{10} = \frac{3}{10} & \frac{1}{6} + \frac{1}{30} = \frac{2}{10} & \frac{1}{10} + \frac{1}{10} = \frac{2}{10} \\
 \frac{1}{11} + \frac{1}{110} = \frac{1}{10} & \frac{1}{12} + \frac{1}{60} = \frac{1}{10} & \frac{1}{14} + \frac{1}{35} = \frac{1}{10} & \frac{1}{15} + \frac{1}{30} = \frac{1}{10} & \frac{1}{20} + \frac{1}{20} = \frac{1}{10}
 \end{array}$$

How many solutions has this equation for $1 \leq n \leq 9$?

Problem 158: Lexicographical Neighbours

Link

Tags: Graph Theory

Taking three different letters from the 26 letters of the alphabet, character strings of length three can be formed.

Examples are ‘abc’, ‘hat’ and ‘zyx’.

When we study these three examples we see that for ‘abc’ two characters come lexicographically after its neighbour to the left.

For ‘hat’ there is exactly one character that comes lexicographically after its neighbour to the left. For ‘zyx’ there are zero characters that come lexicographically after its neighbour to the left.

In all there are 10400 strings of length 3 for which exactly one character comes lexicographically after its neighbour to the left.

We now consider strings of $n \leq 26$ different characters from the alphabet.

For every n , $p(n)$ is the number of strings of length n for which exactly one character comes lexicographically after its neighbour to the left.

What is the maximum value of $p(n)$?

Problem 159: Digital Root Sums of Factorisations

Link

Tags: Algebra, Number Theory

A composite number can be factored many different ways.

For instance, not including multiplication by one, 24 can be factored in 7 distinct ways:

$$24 = 2 \times 2 \times 2 \times 3$$

$$24 = 2 \times 3 \times 4$$

$$24 = 2 \times 2 \times 6$$

$$24 = 4 \times 6$$

$$24 = 3 \times 8$$

$$24 = 2 \times 12$$

$$24 = 24$$

Recall that the digital root of a number, in base 10, is found by adding together the digits of that number, and repeating that process until a number is arrived at that is less than 10.

Thus the digital root of 467 is 8.

We shall call a Digital Root Sum (DRS) the sum of the digital roots of the individual factors of our number.

The chart below demonstrates all of the DRS values for 24.

Factorisation	Digital Root Sum	$2 \times 2 \times 2 \times 3$	$2 \times 3 \times 4$	$2 \times 2 \times 6$	4×6	3×8	2×12	24
		39	49	610	610	811	125	246

The maximum Digital Root Sum of 24 is 11.

The function $\text{mdrs}(n)$ gives the maximum Digital Root Sum of n . So $\text{mdrs}(24) = 11$.

Find $\sum \text{mdrs}(n)$ for $1 \leq n \leq 1\,000\,000$.

Problem 160: Factorial Trailing Digits

Link

Tags: Number Theory

For any N , let $f(N)$ be the last five digits before the trailing zeroes in $N!$.

For example,

- $9! = 362880$ so $f(9) = 36288$
- $10! = 3628800$ so $f(10) = 36288$
- $20! = 2432902008176640000$ so $f(20) = 17664$

Find $f(1\,000\,000\,000\,000)$.

Problem 161: Triominoes

Link

Tags: Combinatorics, Graph Theory

A triomino is a shape consisting of three squares joined via the edges. There are two basic forms:

If all possible orientations are taken into account there are six:

Any n by m grid for which $n \times m$ is divisible by 3 can be tiled with triominoes.

If we consider tilings that can be obtained by reflection or rotation from another tiling as different there are 41 ways a 2 by 9 grid can be tiled with triominoes:

In how many ways can a 9 by 12 grid be tiled in this way by triominoes?

Problem 162: Hexadecimal Numbers

Link

Tags: Number Theory

In the hexadecimal number system numbers are represented using 16 different digits:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.

The hexadecimal number AF when written in the decimal number system equals $10 \times 16 + 15 = 175$.

In the 3-digit hexadecimal numbers 10A, 1A0, A10, and A01 the digits 0, 1 and A are all present.

Like numbers written in base ten we write hexadecimal numbers without leading zeroes.

How many hexadecimal numbers containing at most sixteen hexadecimal digits exist with all of the digits 0, 1, and A present at least once?

Give your answer as a hexadecimal number.

(A, B, C, D, E and F in upper case, without any leading or trailing code that marks the number as hexadecimal and without leading zeroes, e.g. 1A3F and not: 1a3f and not 0x1a3f and not \$1A3F and not #1A3F and not 0000001A3F)

Problem 163: Cross-hatched Triangles

[Link](#)

Tags: Geometry, Graph Theory

Consider an equilateral triangle in which straight lines are drawn from each vertex to the middle of the opposite side, such as in the size 1 triangle in the sketch below.

Problem 164: Three Consecutive Digital Sum Limit

[Link](#)

Tags: Number Theory

How many 20 digit numbers n (without any leading zero) exist such that no three consecutive digits of n have a sum greater than 9?

Problem 165: Intersections

[Link](#)

Tags: Geometry, Probability

A segment is uniquely defined by its two endpoints. By considering two line segments in plane geometry there are three possibilities:

the segments have zero points, one point, or infinitely many points in common.

Moreover when two segments have exactly one point in common it might be the case that that common point is an endpoint of either one of the segments or of both. If a common point of two segments is not an endpoint of either of the segments it is an interior point of both segments.

We will call a common point T of two segments L_1 and L_2 a true intersection point of L_1 and L_2 if T is the only common point of L_1 and L_2 and T is an interior point of both segments.

Consider the three segments L_1 , L_2 , and L_3 :

- L_1 : (27, 44) to (12, 32)
- L_2 : (46, 53) to (17, 62)
- L_3 : (46, 70) to (22, 40)

It can be verified that line segments L_2 and L_3 have a true intersection point. We note that as the one of the end points of L_3 : (22, 40) lies on L_1 this is not

considered to be a true point of intersection. L_1 and L_2 have no common point. So among the three line segments, we find one true intersection point.

Now let us do the same for 5000 line segments. To this end, we generate 20000 numbers using the so-called “Blum Blum Shub” pseudo-random number generator.

$$\begin{aligned}s_0 &= 290797 \\ s_{n+1} &= s_n \times s_n \pmod{50515093} \\ t_n &= s_n \pmod{500}\end{aligned}$$

To create each line segment, we use four consecutive numbers t_n . That is, the first line segment is given by:

(t_1, t_2) to (t_3, t_4) .

The first four numbers computed according to the above generator should be: 27, 144, 12 and 232. The first segment would thus be (27, 144) to (12, 232).

How many distinct true intersection points are found among the 5000 line segments?

Problem 166: Criss Cross

[Link](#)

Tags: Algebra, Combinatorics, Number Theory

A 4×4 grid is filled with digits d , $0 \leq d \leq 9$.

It can be seen that in the grid

$$\begin{array}{cccc}6 & 3 & 3 & 0 \\ 5 & 0 & 4 & 3 \\ 0 & 7 & 1 & 4 \\ 1 & 2 & 4 & 5\end{array}$$

the sum of each row and each column has the value 12. Moreover the sum of each diagonal is also 12.

In how many ways can you fill a 4×4 grid with the digits d , $0 \leq d \leq 9$ so that each row, each column, and both diagonals have the same sum?

Problem 167: Investigating Ulam Sequences

[Link](#)

Tags: Sequences

For two positive integers a and b , the Ulam sequence $U(a, b)$ is defined by $U(a, b)_1 = a$, $U(a, b)_2 = b$ and for $k \geq 2$, $U(a, b)_k$ is the smallest integer greater than $U(a, b)_{k-1}$ which can be written in exactly one way as the sum of two distinct previous members of $U(a, b)$.

For example, the sequence $U(1, 2)$ begins with

1, 2, 3 = 1 + 2, 4 = 1 + 3, 6 = 2 + 4, 8 = 2 + 6, 11 = 3 + 8;

5 does not belong to it because $5 = 1 + 4 = 2 + 3$ has two representations as the sum of two previous members, likewise $7 = 1 + 6 = 3 + 4$.

Find $\sum_{n=2}^{10} U(2, 2n+1)_k$, where $k = 10^{11}$.

Problem 168: Number Rotations

Link

Tags: Number Theory

Consider the number 142857. We can right-rotate this number by moving the last digit (7) to the front of it, giving us 714285.

It can be verified that $714285 = 5 \times 142857$.

This demonstrates an unusual property of 142857: it is a divisor of its right-rotation.

Find the last 5 digits of the sum of all integers n , $10n10^{100}$, that have this property.

Problem 169: Sums of Powers of Two

Link

Tags: Number Theory

Define $f(0) = 1$ and $f(n)$ to be the number of different ways n can be expressed as a sum of integer powers of 2 using each power no more than twice.

For example, $f(10) = 5$ since there are five different ways to express 10:

$$\begin{aligned}
&1 + 1 + 8 \\
&1 + 1 + 4 + 4 \\
&1 + 1 + 2 + 2 + 4 \\
&2 + 4 + 4 \\
&2 + 8
\end{aligned}$$

What is $f(10^{25})$?

Problem 170: Pandigital Concatenating Products

[Link](#)

Tags: Number Theory

Take the number 6 and multiply it by each of 1273 and 9854:

$$\begin{aligned}
6 \times 1273 &= 7638 \\
6 \times 9854 &= 59124
\end{aligned}$$

By concatenating these products we get the 1 to 9 pandigital 763859124. We will call 763859124 the “concatenated product of 6 and (1273, 9854)”. Notice too, that the concatenation of the input numbers, 612739854, is also 1 to 9 pandigital.

The same can be done for 0 to 9 pandigital numbers.

What is the largest 0 to 9 pandigital 10-digit concatenated product of an integer with two or more other integers, such that the concatenation of the input numbers is also a 0 to 9 pandigital 10-digit number?

Problem 171: Square Sum of the Digital Squares

[Link](#)

Tags: Number Theory

For a positive integer n , let $f(n)$ be the sum of the squares of the digits (in base 10) of n , e.g.

$$\begin{aligned}
f(3) &= 3^2 = 9, \\
f(25) &= 2^2 + 5^2 = 4 + 25 = 29, \\
f(442) &= 4^2 + 4^2 + 2^2 = 16 + 16 + 4 = 36
\end{aligned}$$

Find the last nine digits of the sum of all n , $0 \leq n < 10^{20}$, such that $f(n)$ is a perfect square.

Problem 172: Few Repeated Digits

[Link](#)

Tags: Number Theory

How many 18-digit numbers n (without leading zeros) are there such that no digit occurs more than three times in n ?

Problem 173: Hollow Square Laminae I

[Link](#)

Tags: Number Theory

We shall define a square lamina to be a square outline with a square “hole” so that the shape possesses vertical and horizontal symmetry. For example, using exactly thirty-two square tiles we can form two different square laminae:

Problem 174: Hollow Square Laminae II

[Link](#)

Tags: Number Theory

We shall define a square lamina to be a square outline with a square “hole” so that the shape possesses vertical and horizontal symmetry.

Given eight tiles it is possible to form a lamina in only one way: 3×3 square with a 1×1 hole in the middle. However, using thirty-two tiles it is possible to form two distinct laminae.

Problem 175: Fractions and Sum of Powers of Two

[Link](#)

Tags: Number Theory

Define $f(0) = 1$ and $f(n)$ to be the number of ways to write n as a sum of powers of 2 where no power occurs more than twice.

For example, $f(10) = 5$ since there are five different ways to express 10: $10 = 8 + 2 = 8 + 1 + 1 = 4 + 4 + 2 = 4 + 2 + 2 + 1 + 1 = 4 + 4 + 1 + 1$.

It can be shown that for every fraction p/q ($p > 0, q > 0$) there exists at least one integer n such that $f(n)/f(n-1) = p/q$.

For instance, the smallest n for which $f(n)/f(n-1) = 13/17$ is 241.

The binary expansion of 241 is 11110001.

Reading this binary number from the most significant bit to the least significant bit there are 4 one's, 3 zeroes and 1 one. We shall call the string 4,3,1 the Shortened Binary Expansion of 241.

Find the Shortened Binary Expansion of the smallest n for which $f(n)/f(n-1) = 123456789/987654321$.

Give your answer as comma separated integers, without any whitespaces.

Problem 176: Common Cathetus Right-angled Triangles

[Link](#)

Tags: Geometry

The four right-angled triangles with sides $(9, 12, 15)$, $(12, 16, 20)$, $(5, 12, 13)$ and $(12, 35, 37)$ all have one of the shorter sides (catheti) equal to 12. It can be shown that no other integer sided right-angled triangle exists with one of the catheti equal to 12.

Find the smallest integer that can be the length of a cathetus of exactly 47547 different integer sided right-angled triangles.

Problem 177: Integer Angled Quadrilaterals

[Link](#)

Tags: Geometry, Graph Theory

Let $ABCD$ be a convex quadrilateral, with diagonals AC and BD . At each vertex the diagonal makes an angle with each of the two sides, creating eight corner angles.

For example, at vertex A , the two angles are CAD, CAB .

We call such a quadrilateral for which all eight corner angles have integer values when measured in degrees an "integer angled quadrilateral". An example of an integer angled quadrilateral is a square, where all eight corner angles are

45° . Another example is given by $DAC = 20^\circ$, $BAC = 60^\circ$, $ABD = 50^\circ$, $CBD = 30^\circ$, $BCA = 40^\circ$, $DCA = 30^\circ$, $CDB = 80^\circ$, $ADB = 50^\circ$.

What is the total number of non-similar integer angled quadrilaterals?

Note: In your calculations you may assume that a calculated angle is integral if it is within a tolerance of 10^{-9} of an integer value.

Problem 178: Step Numbers

Link

Tags: Number Theory

Consider the number 45656.

It can be seen that each pair of consecutive digits of 45656 has a difference of one.

A number for which every pair of consecutive digits has a difference of one is called a step number.

A pandigital number contains every decimal digit from 0 to 9 at least once.

How many pandigital step numbers less than 10^{40} are there?

Problem 179: Consecutive Positive Divisors

Link

Tags: Number Theory

Find the number of integers $1 \leq n \leq 10^7$, for which n and $n + 1$ have the same number of positive divisors. For example, 14 has the positive divisors 1, 2, 7, 14 while 15 has 1, 3, 5, 15.

Problem 180: Golden Triplets

Link

Tags: Combinatorics

For any integer n , consider the three functions

$$\begin{aligned} f_{1,n}(x, y, z) &= x^{n+1} + y^{n+1} - z^{n+1} \\ f_{2,n}(x, y, z) &= (xy + yz + zx) \cdot (x^{n-1} + y^{n-1} - z^{n-1}) \\ f_{3,n}(x, y, z) &= xyz \cdot (x^{n-2} + y^{n-2} - z^{n-2}) \end{aligned}$$

and their combination

$$f_n(x, y, z) = f_{1,n}(x, y, z) + f_{2,n}(x, y, z) - f_{3,n}(x, y, z).$$

We call (x, y, z) a golden triple of order k if x , y , and z are all rational numbers of the form a/b with $0 < a, b \leq k$ and there is (at least) one integer n , so that $f_n(x, y, z) = 0$.

Let $s(x, y, z) = x + y + z$.

Let $t = u/v$ be the sum of all distinct $s(x, y, z)$ for all golden triples (x, y, z) of order 35. All the $s(x, y, z)$ and t must be in reduced form.

Find $u + v$.

Problem 181: Grouping Two Different Coloured Objects

Link

Tags: Number Theory

Having three black objects B and one white object W they can be grouped in 7 ways like this:

(BBBW)(B,BBW)(B,B,BW)(B,B,B,W) (B,BB,W)(BBB,W)(BB,BW)

In how many ways can sixty black objects B and forty white objects W be thus grouped?

Problem 182: RSA Encryption

Link

Tags: Combinatorics, Game Theory, Number Theory

The RSA encryption is based on the following procedure:

Generate two distinct primes p and q . Compute $n = pq$ and $\phi = (p - 1)(q - 1)$.

Find an integer e , $1 < e < \phi$, such that $\gcd(e, \phi) = 1$.

A message in this system is a number in the interval $[0, n - 1]$.

A text to be encrypted is then somehow converted to messages (numbers in the interval $[0, n - 1]$).

To encrypt the text, for each message, m , $c = m^e \bmod n$ is calculated.

To decrypt the text, the following procedure is needed: calculate d such that $ed = 1 \bmod \phi$, then for each encrypted message, c , calculate $m = c^d \bmod n$.

There exist values of e and m such that $m^e \bmod n = m$. We call messages m for which $m^e \bmod n = m$ unconcealed messages.

An issue when choosing e is that there should not be too many unconcealed messages. For instance, let $p = 19$ and $q = 37$.

Then $n = 19 \cdot 37 = 703$ and $\phi = 18 \cdot 36 = 648$.

If we choose $e = 181$, then, although $\gcd(181, 648) = 1$ it turns out that all possible messages m ($0 \leq m \leq n - 1$) are unconcealed when calculating $m^e \bmod n$.

For any valid choice of e there exist some unconcealed messages.

It's important that the number of unconcealed messages is at a minimum.

Choose $p = 1009$ and $q = 3643$.

Find the sum of all values of e , $1 \leq e \leq \phi(1009, 3643)$ and $\gcd(e, \phi) = 1$, so that the number of unconcealed messages for this value of e is at a minimum.

Problem 183: Maximum Product of Parts

Link

Tags: Number Theory

Let N be a positive integer and let N be split into k equal parts, $r = N/k$, so that $N = r + r + \dots + r$.

Let P be the product of these parts, $P = r \times r \times \dots \times r = r^k$.

For example, if 11 is split into five equal parts, $11 = 2.2 + 2.2 + 2.2 + 2.2 + 2.2$, then $P = 2.2^5 = 51.53632$.

Let $M(N) = P_{\max}$ for a given value of N .

It turns out that the maximum for $N = 11$ is found by splitting eleven into four equal parts which leads to $P_{\max} = (11/4)^4$; that is, $M(11) = 14641/256 = 57.19140625$, which is a terminating decimal.

However, for $N = 8$ the maximum is achieved by splitting it into three equal parts, so $M(8) = 512/27$, which is a non-terminating decimal.

Let $D(N) = N$ if $M(N)$ is a non-terminating decimal and $D(N) = -N$ if $M(N)$ is a terminating decimal.

For example, $\sum_{N=5}^{100} D(N)$ is 2438.

Find $\sum_{N=5}^{10000} D(N)$.

Problem 184: Triangles Containing the Origin

[Link](#)

Tags: Geometry

Consider the set I_r of points (x, y) with integer co-ordinates in the interior of the circle with radius r , centered at the origin, i.e. $x^2 + y^2 < r^2$.

For a radius of 2, I_2 contains the nine points $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$, $(-1, 1)$, $(-1, 0)$, $(-1, -1)$, $(0, -1)$ and $(1, -1)$. There are eight triangles having all three vertices in I_2 which contain the origin in the interior. Two of them are shown below, the others are obtained from these by rotation.

For a radius of 3, there are 360 triangles containing the origin in the interior and having all vertices in I_3 and for I_5 the number is 10600.

How many triangles are there containing the origin in the interior and having all three vertices in I_{105} ?

Problem 185: Number Mind

[Link](#)

Tags: Game Theory, Number Theory, Sequences

The game Number Mind is a variant of the well known game Master Mind.

Instead of coloured pegs, you have to guess a secret sequence of digits. After each guess you're only told in how many places you've guessed the correct digit. So, if the sequence was 1234 and you guessed 2036, you'd be told that you have one correct digit; however, you would NOT be told that you also have another digit in the wrong place.

For instance, given the following guesses for a 5-digit secret sequence,

90342 ;2 correct

70794 ;0 correct

39458 ;2 correct

34109 ;1 correct

51545 ;2 correct

12531 ;1 correct

The correct sequence 39542 is unique.

Based on the following guesses,

5616185650518293 ;2 correct

3847439647293047 ;1 correct
 5855462940810587 ;3 correct
 9742855507068353 ;3 correct
 4296849643607543 ;3 correct
 3174248439465858 ;1 correct
 4513559094146117 ;2 correct
 7890971548908067 ;3 correct
 8157356344118483 ;1 correct
 2615250744386899 ;2 correct
 8690095851526254 ;3 correct
 6375711915077050 ;1 correct
 6913859173121360 ;1 correct
 6442889055042768 ;2 correct
 2321386104303845 ;0 correct
 2326509471271448 ;2 correct
 5251583379644322 ;2 correct
 1748270476758276 ;3 correct
 4895722652190306 ;1 correct
 3041631117224635 ;3 correct
 1841236454324589 ;3 correct
 2659862637316867 ;2 correct

Find the unique 16-digit secret sequence.

Problem 186: Connectedness of a Network

Link

Tags: Number Theory

Here are the records from a busy telephone system with one million users:

RecNrCallerCalled 120000710005326001835004393600863701497.....

Problem 187: Semiprimes

Link

Tags: Number Theory

A composite is a number containing at least two prime factors. For example, $15 = 3 \times 5$; $9 = 3 \times 3$; $12 = 2 \times 2 \times 3$.

There are ten composites below thirty containing precisely two, not necessarily distinct, prime factors: 4, 6, 9, 10, 14, 15, 21, 22, 25, 26.

How many composite integers, $n10^8$, have precisely two, not necessarily distinct, prime factors?

Problem 188: Hyperexponentiation

Link

Tags: Number Theory

The hyperexponentiation or tetration of a number a by a positive integer b , denoted by $a \uparrow\uparrow b$ or ${}^b a$, is recursively defined by:

$$a \uparrow\uparrow 1 = a,$$

$$a \uparrow\uparrow (k+1) = a^{(a \uparrow\uparrow k)}.$$

Thus we have e.g. $3 \uparrow\uparrow 2 = 3^3 = 27$, hence $3 \uparrow\uparrow 3 = 3^{27} = 7625597484987$ and $3 \uparrow\uparrow 4$ is roughly $10^{3.6383346400240996 \cdot 10^{12}}$.

Find the last 8 digits of $1777 \uparrow\uparrow 1855$.

Problem 189: Tri-colouring a Triangular Grid

Link

Tags: Game Theory, Geometry

Consider the following configuration of 64 triangles:

Problem 190: Maximising a Weighted Product

Link

Tags: Number Theory

Let $S_m = (x_1, x_2, \dots, x_m)$ be the m -tuple of positive real numbers with $x_1 + x_2 + \dots + x_m = m$ for which $P_m = x_1 \cdot x_2^2 \cdot \dots \cdot x_m^m$ is maximised.

For example, it can be verified that $\lfloor P_{10} \rfloor = 4112$ ($\lfloor \cdot \rfloor$ is the integer part function).

Find $\sum_{m=2}^{15} \lfloor P_m \rfloor$.

Problem 191: Prize Strings

Link

Tags: Number Theory

A particular school offers cash rewards to children with good attendance and punctuality. If they are absent for three consecutive days or late on more than one occasion then they forfeit their prize.

During an n -day period a trinary string is formed for each child consisting of L's (late), O's (on time), and A's (absent).

Although there are eighty-one trinary strings for a 4-day period that can be formed, exactly forty-three strings would lead to a prize:

OOOO OOOA OOOL OOAO OOAA OOAL OOLO OOLA OAOO OAOA
 OAOL OAAO OAAL OALO OALA OLOO OLOA OLAO OLAA AOOO
 AOOA AOOL AOAO AOAA AOAL AOLO AOLA AAOO AAOA AAOL
 AALO AALA ALOO ALOA ALAO ALAA LOOO LOOA LOAO LOAA
 LAOO LAOA LAAO

How many "prize" strings exist over a 30-day period?

Problem 192: Best Approximations

Link

Tags: Game Theory

Let x be a real number.

A best approximation to x for the denominator bound d is a rational number $\frac{p}{q}$ in reduced form, with $q \leq d$, such that any rational number which is closer to x than $\frac{p}{q}$ has a denominator larger than d :

$$\left| \frac{p}{q} - x \right| < \left| \frac{r}{s} - x \right| \Rightarrow q > d$$

Problem 193: Squarefree Numbers

Link

Tags: Number Theory

A positive integer n is called squarefree, if no square of a prime divides n , thus 1, 2, 3, 5, 6, 7, 10, 11 are squarefree, but not 4, 8, 9, 12.

How many squarefree numbers are there below 2^{50} ?

Problem 194: Coloured Configurations

Link

Tags: Graph Theory, Number Theory

Consider graphs built with the units A : and B : , where the units are glued along the vertical edges as in the graph .

A configuration of type (a, b, c) is a graph thus built of a units A and b units B , where the graph's vertices are coloured using up to c colours, so that no two adjacent vertices have the same colour.

The compound graph above is an example of a configuration of type $(2, 2, 6)$, in fact of type $(2, 2, c)$ for all $c \geq 4$.

Let $N(a, b, c)$ be the number of configurations of type (a, b, c) .

For example, $N(1, 0, 3) = 24$, $N(0, 2, 4) = 92928$ and $N(2, 2, 3) = 20736$.

Find the last 8 digits of $N(25, 75, 1984)$.

Problem 195: 60-degree Triangle Inscribed Circles

Link

Tags: Geometry

Let's call an integer sided triangle with exactly one angle of 60 degrees a 60-degree triangle.

Let r be the radius of the inscribed circle of such a 60-degree triangle.

There are 1234 60-degree triangles for which $r \leq 100$.

Let $T(n)$ be the number of 60-degree triangles for which $r \leq n$, so

$T(100) = 1234$, $T(1000) = 22767$, and $T(10000) = 359912$.

Find $T(1053779)$.

Problem 196: Prime Triplets

Link

Tags: Game Theory, Geometry, Number Theory

Build a triangle from all positive integers in the following way:

1
2 3
4 5 6
7 8 9 10 11 12 13 14 15
16 17 18 19 20 21
22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45
46 47 48 49 50 51 52 53 54 55
56 57 58 59 60 61 62 63 64 65 66
...

Each positive integer has up to eight neighbours in the triangle.

A set of three primes is called a prime triplet if one of the three primes has the other two as neighbours in the triangle.

For example, in the second row, the prime numbers 2 and 3 are elements of some prime triplet.

If row 8 is considered, it contains two primes which are elements of some prime triplet, i.e. 29 and 31.

If row 9 is considered, it contains only one prime which is an element of some prime triplet: 37.

Define $S(n)$ as the sum of the primes in row n which are elements of any prime triplet.

Then $S(8) = 60$ and $S(9) = 37$.

You are given that $S(10000) = 950007619$.

Find $S(5678027) + S(7208785)$.

Problem 197: A Recursively Defined Sequence

Link

Tags: Number Theory, Sequences

Given is the function $f(x) = \lfloor 2^{30.403243784-x^2} \rfloor \times 10^{-9}$ ($\lfloor \cdot \rfloor$ is the floor-function), the sequence u_n is defined by $u_0 = -1$ and $u_{n+1} = f(u_n)$.

Find $u_n + u_{n+1}$ for $n = 10^{12}$.

Give your answer with 9 digits after the decimal point.

Problem 198: Ambiguous Numbers

Link

Tags: Game Theory

A best approximation to a real number x for the denominator bound d is a rational number $\frac{r}{s}$ (in reduced form) with $s \leq d$, so that any rational number $\frac{p}{q}$ which is closer to x than $\frac{r}{s}$ has qd .

Usually the best approximation to a real number is uniquely determined for all denominator bounds. However, there are some exceptions, e.g. $\frac{9}{40}$ has the two best approximations $\frac{1}{4}$ and $\frac{1}{5}$ for the denominator bound 6. We shall call a real number x ambiguous, if there is at least one denominator bound for which x possesses two best approximations. Clearly, an ambiguous number is necessarily rational.

How many ambiguous numbers $x = \frac{p}{q}, 0 < x < \frac{1}{100}$, are there whose denominator q does not exceed 10^8 ?

Problem 199: Iterative Circle Packing

Link

Tags: Geometry

Three circles of equal radius are placed inside a larger circle such that each pair of circles is tangent to one another and the inner circles do not overlap. There are four uncovered “gaps” which are to be filled iteratively with more tangent circles.

Problem 200: Prime-proof Squbes

Link

Tags: Number Theory

We shall define a sqube to be a number of the form, p^2q^3 , where p and q are distinct primes.

For example, $200 = 5^22^3$ or $120072949 = 23^261^3$.

The first five squbes are 72, 108, 200, 392, and 500.

Interestingly, 200 is also the first number for which you cannot change any single digit to make a prime; we shall call such numbers, prime-proof. The next prime-proof sqube which contains the contiguous sub-string “200” is 1992008.

Find the 200th prime-proof sqube containing the contiguous sub-string “200”.

Problem 201: Subsets with a Unique Sum

Link

Tags: Combinatorics

For any set A of numbers, let $\text{sum}(A)$ be the sum of the elements of A .

Consider the set $B = \{1, 3, 6, 8, 10, 11\}$. There are 20 subsets of B containing three elements, and their sums are:

$$\begin{aligned}
\text{sum}(\{1, 3, 6\}) &= 10, \\
\text{sum}(\{1, 3, 8\}) &= 12, \\
\text{sum}(\{1, 3, 10\}) &= 14, \\
\text{sum}(\{1, 3, 11\}) &= 15, \\
\text{sum}(\{1, 6, 8\}) &= 15, \\
\text{sum}(\{1, 6, 10\}) &= 17, \\
\text{sum}(\{1, 6, 11\}) &= 18, \\
\text{sum}(\{1, 8, 10\}) &= 19, \\
\text{sum}(\{1, 8, 11\}) &= 20, \\
\text{sum}(\{1, 10, 11\}) &= 22, \\
\text{sum}(\{3, 6, 8\}) &= 17, \\
\text{sum}(\{3, 6, 10\}) &= 19, \\
\text{sum}(\{3, 6, 11\}) &= 20, \\
\text{sum}(\{3, 8, 10\}) &= 21, \\
\text{sum}(\{3, 8, 11\}) &= 22, \\
\text{sum}(\{3, 10, 11\}) &= 24, \\
\text{sum}(\{6, 8, 10\}) &= 24, \\
\text{sum}(\{6, 8, 11\}) &= 25, \\
\text{sum}(\{6, 10, 11\}) &= 27, \\
\text{sum}(\{8, 10, 11\}) &= 29.
\end{aligned}$$

Some of these sums occur more than once, others are unique.

For a set A , let $U(A, k)$ be the set of unique sums of k -element subsets of A , in our example we find $U(B, 3) = \{10, 12, 14, 18, 21, 25, 27, 29\}$ and $\text{sum}(U(B, 3)) = 156$.

Now consider the 100-element set $S = \{1^2, 2^2, \dots, 100^2\}$.

S has 100891344545564193334812497256 50-element subsets.

Determine the sum of all integers which are the sum of exactly one of the 50-element subsets of S , i.e. find $\text{sum}(U(S, 50))$.

Problem 202: Laserbeam

Link

Tags: Combinatorics, Geometry, Graph Theory

Three mirrors are arranged in the shape of an equilateral triangle, with their reflective surfaces pointing inwards. There is an infinitesimal gap at each vertex of the triangle through which a laser beam may pass.

Label the vertices A , B and C . There are 2 ways in which a laser beam may enter vertex C , bounce off 11 surfaces, then exit through the same vertex: one way is shown below; the other is the reverse of that.

Problem 203: Squarefree Binomial Coefficients

[Link](#)

Tags: Combinatorics, Geometry

The binomial coefficients $\binom{n}{k}$ can be arranged in triangular form, Pascal's triangle, like this:

111121133114641151010511615201561172135352171

Problem 204: Generalised Hamming Numbers

[Link](#)

Tags: Number Theory

A Hamming number is a positive number which has no prime factor larger than 5.

So the first few Hamming numbers are 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15.

There are 1105 Hamming numbers not exceeding 10^8 .

We will call a positive number a generalised Hamming number of type n , if it has no prime factor larger than n .

Hence the Hamming numbers are the generalised Hamming numbers of type 5.

How many generalised Hamming numbers of type 100 are there which don't exceed 10^9 ?

Problem 205: Dice Game

[Link](#)

Tags: Game Theory, Probability

Peter has nine four-sided (pyramidal) dice, each with faces numbered 1, 2, 3, 4.

Colin has six six-sided (cubic) dice, each with faces numbered 1, 2, 3, 4, 5, 6.

Peter and Colin roll their dice and compare totals: the highest total wins. The result is a draw if the totals are equal.

What is the probability that Pyramidal Peter beats Cubic Colin? Give your answer rounded to seven decimal places in the form 0.abcdefg.

Problem 206: Concealed Square

Link

Tags: Number Theory

Find the unique positive integer whose square has the form $1_2_3_4_5_6_7_8_9_0$, where each “ $_$ ” is a single digit.

Problem 207: Integer Partition Equations

Link

Tags: Combinatorics, Game Theory

For some positive integers k , there exists an integer partition of the form $4^t = 2^t + k$,

where 4^t , 2^t , and k are all positive integers and t is a real number.

The first two such partitions are $4^1 = 2^1 + 2$ and $4^{1.5849625\cdots} = 2^{1.5849625\cdots} + 6$.

Partitions where t is also an integer are called perfect.

For any $m \geq 1$ let $P(m)$ be the proportion of such partitions that are perfect with $k \leq m$.

Thus $P(6) = 1/2$.

In the following table are listed some values of $P(m)$.

$P(5) = 1/1$
$P(10) = 1/2$
$P(15) = 2/3$
$P(20) = 1/2$
$P(25) = 1/2$
$P(30) = 2/5$
$\dots$
$P(180) = 1/4$
$P(185) = 3/13$

Find the smallest m for which $P(m)1/12345$.

Problem 208: Robot Walks

[Link](#)

Tags: Combinatorics, Game Theory, Sequences

A robot moves in a series of one-fifth circular arcs (72°), with a free choice of a clockwise or an anticlockwise arc for each step, but no turning on the spot.

One of 70932 possible closed paths of 25 arcs starting northward is

Problem 209: Circular Logic

[Link](#)

Tags: Number Theory

A k -input binary truth table is a map from k input bits (binary digits, 0 [false] or 1 [true]) to 1 output bit. For example, the 2-input binary truth tables for the logical AND and XOR functions are:

$x \ y \ x \text{ AND } y$ 000010100111

Problem 210: Obtuse Angled Triangles

[Link](#)

Tags: Geometry

Consider the set $S(r)$ of points (x, y) with integer coordinates satisfying $|x| + |y| \leq r$.

Let O be the point $(0, 0)$ and C the point $(r/4, r/4)$.

Let $N(r)$ be the number of points B in $S(r)$, so that the triangle OBC has an obtuse angle, i.e. the largest angle α satisfies $90^\circ < \alpha < 180^\circ$.

So, for example, $N(4) = 24$ and $N(8) = 100$.

What is $N(1\,000\,000\,000)$?

Problem 211: Divisor Square Sum

[Link](#)

Tags: Number Theory

For a positive integer n , let $\sigma_2(n)$ be the sum of the squares of its divisors. For example,

$$\sigma_2(10) = 1 + 4 + 25 + 100 = 130.$$

Find the sum of all n , $0 \leq n < 10^9$ such that $\sigma_2(n)$ is a perfect square.

Problem 212: Combined Volume of Cuboids

[Link](#)

Tags: Geometry, Sequences

An axis-aligned cuboid, specified by parameters $\{(x_0, y_0, z_0), (dx, dy, dz)\}$, consists of all points (X, Y, Z) such that $x_0 \leq X \leq x_0 + dx$, $y_0 \leq Y \leq y_0 + dy$ and $z_0 \leq Z \leq z_0 + dz$. The volume of the cuboid is the product, $dx \times dy \times dz$. The combined volume of a collection of cuboids is the volume of their union and will be less than the sum of the individual volumes if any cuboids overlap.

Let $C_1, \dots, C_{50000}$ be a collection of 50000 axis-aligned cuboids such that C_n has parameters

$$\begin{aligned} x_0 &= S_{6n-5} \bmod 10000 \\ y_0 &= S_{6n-4} \bmod 10000 \\ z_0 &= S_{6n-3} \bmod 10000 \\ dx &= 1 + (S_{6n-2} \bmod 399) \\ dy &= 1 + (S_{6n-1} \bmod 399) \\ dz &= 1 + (S_{6n} \bmod 399) \end{aligned}$$

where $S_1, \dots, S_{300000}$ come from the “Lagged Fibonacci Generator”:

- For $1 \leq k \leq 55$, $S_k = [100003 - 200003k + 300007k^3] \pmod{1000000}$.
- For $56 \leq k$, $S_k = [S_{k-24} + S_{k-55}] \pmod{1000000}$.

Thus, C_1 has parameters $\{(7, 53, 183), (94, 369, 56)\}$, C_2 has parameters $\{(2383, 3563, 5079), (42, 212, 344)\}$, and so on.

The combined volume of the first 100 cuboids, $C_1, \dots, C_{100}$, is 723581599.

What is the combined volume of all 50000 cuboids, $C_1, \dots, C_{50000}$?

Problem 213: Flea Circus

[Link](#)

Tags: Combinatorics, Graph Theory, Probability

A 30×30 grid of squares contains 900 fleas, initially one flea per square.

When a bell is rung, each flea jumps to an adjacent square at random (usually 4 possibilities, except for fleas on the edge of the grid or at the corners).

What is the expected number of unoccupied squares after 50 rings of the bell?
Give your answer rounded to six decimal places.

Problem 214: Totient Chains

[Link](#)

Tags: Number Theory, Sequences

Let ϕ be Euler's totient function, i.e. for a natural number n , $\phi(n)$ is the number of k , $1 \leq k \leq n$, for which $\gcd(k, n) = 1$.

By iterating ϕ , each positive integer generates a decreasing chain of numbers ending in 1.

E.g. if we start with 5 the sequence 5, 4, 2, 1 is generated.

Here is a listing of all chains with length 4:

5, 4, 2, 1
7, 6, 2, 1
8, 4, 2, 1
9, 6, 2, 1
10, 4, 2, 1
12, 4, 2, 1
14, 6, 2, 1
18, 6, 2, 1

Only two of these chains start with a prime, their sum is 12.

What is the sum of all primes less than 40000000 which generate a chain of length 25?

Problem 215: Crack-free Walls

[Link](#)

Tags: Game Theory

Consider the problem of building a wall out of 2×1 and 3×1 bricks (horizontal $\times$ vertical dimensions) such that, for extra strength, the gaps between horizontally-adjacent bricks never line up in consecutive layers, i.e. never form a “running crack”.

For example, the following 9×3 wall is not acceptable due to the running crack shown in red:

Problem 216: The Primality of $2n^2 - 1$

Link

Tags: Number Theory

Consider numbers $t(n)$ of the form $t(n) = 2n^2 - 1$ with $n \geq 1$.

The first such numbers are 7, 17, 31, 49, 71, 97, 127 and 161.

It turns out that only $49 = 7 \cdot 7$ and $161 = 7 \cdot 23$ are not prime.

For $n \leq 10000$ there are 2202 numbers $t(n)$ that are prime.

How many numbers $t(n)$ are prime for $n \leq 50\,000\,000$?

Problem 217: Balanced Numbers

Link

Tags: Number Theory

A positive integer with k (decimal) digits is called balanced if its first $\lceil k/2 \rceil$ digits sum to the same value as its last $\lceil k/2 \rceil$ digits, where $\lceil x \rceil$, pronounced ceiling of x , is the smallest integer $\geq x$, thus $\lceil \pi \rceil = 4$ and $\lceil 5 \rceil = 5$.

So, for example, all palindromes are balanced, as is 13722.

Let $T(n)$ be the sum of all balanced numbers less than 10^n .

Thus: $T(1) = 45$, $T(2) = 540$ and $T(5) = 334795890$.

Find $T(47) \bmod 3^{15}$.

Problem 218: Perfect Right-angled Triangles

Link

Tags: Geometry, Number Theory

Consider the right angled triangle with sides $a = 7$, $b = 24$ and $c = 25$. The area of this triangle is 84, which is divisible by the perfect numbers 6 and 28.

Moreover it is a primitive right angled triangle as $\gcd(a, b) = 1$ and $\gcd(b, c) = 1$.

Also c is a perfect square.

We will call a right angled triangle perfect if

-it is a primitive right angled triangle

-its hypotenuse is a perfect square.

We will call a right angled triangle super-perfect if

-it is a perfect right angled triangle and

-its area is a multiple of the perfect numbers 6 and 28.

How many perfect right-angled triangles with $c \leq 10^{16}$ exist that are not super-perfect?

Problem 219: Skew-cost Coding

Link

Tags: Sequences

Let A and B be bit strings (sequences of 0's and 1's).

If A is equal to the leftmost $\text{length}(A)$ bits of B , then A is said to be a prefix of B .

For example, 00110 is a prefix of 001101001, but not of 00111 or 100110.

A prefix-free code of size n is a collection of n distinct bit strings such that no string is a prefix of any other. For example, this is a prefix-free code of size 6:

0000, 0001, 001, 01, 10, 11

Now suppose that it costs one penny to transmit a '0' bit, but four pence to transmit a '1'.

Then the total cost of the prefix-free code shown above is 35 pence, which happens to be the cheapest possible for the skewed pricing scheme in question.

In short, we write $\text{Cost}(6) = 35$.

What is $\text{Cost}(109)$?

Problem 220: Highway Dragon

Link

Tags: Combinatorics, Game Theory, Graph Theory

Let D_0 be the two-letter string “Fa”. For $n \geq 1$, derive D_n from D_{n-1} by the string-rewriting rules:

“a” $\rightarrow$ “aRbFR”

“b” $\rightarrow$ “LFaLb”

Thus, $D_0 = \text{“Fa”}$, $D_1 = \text{“FaRbFR”}$, $D_2 = \text{“FaRbFRRLFaLbFR”}$, and so on.

These strings can be interpreted as instructions to a computer graphics program, with “F” meaning “draw forward one unit”, “L” meaning “turn left 90 degrees”, “R” meaning “turn right 90 degrees”, and “a” and “b” being ignored. The initial position of the computer cursor is $(0, 0)$, pointing up towards $(0, 1)$.

Then D_n is an exotic drawing known as the Highway Dragon of order n . For example, D_{10} is shown below; counting each “F” as one step, the highlighted spot at $(18, 16)$ is the position reached after 500 steps.

Problem 221: Alexandrian Integers

Link

Tags: Number Theory

We shall call a positive integer A an “Alexandrian integer”, if there exist integers p, q, r such that:

$$A = p \cdot q \cdot r$$

and

$$\frac{1}{A} = \frac{1}{p} + \frac{1}{q} + \frac{1}{r}.$$

For example, 630 is an Alexandrian integer ($p = 5, q = -7, r = -18$). In fact, 630 is the 6th Alexandrian integer, the first 6 Alexandrian integers being: 6, 42, 120, 156, 420, and 630.

Find the 150000th Alexandrian integer.

Problem 222: Sphere Packing

[Link](#)

Tags: Number Theory

What is the length of the shortest pipe, of internal radius $50mm$, that can fully contain 21 balls of radii $30mm, 31mm, \dots, 50mm$?

Give your answer in micrometres ($10^{-6}m$) rounded to the nearest integer.

Problem 223: Almost Right-angled Triangles I

[Link](#)

Tags: Geometry

Let us call an integer sided triangle with sides $a \leq b \leq c$ barely acute if the sides satisfy $a^2 + b^2 = c^2 + 1$.

How many barely acute triangles are there with perimeter $\leq 25\,000\,000$?

Problem 224: Almost Right-angled Triangles II

[Link](#)

Tags: Geometry

Let us call an integer sided triangle with sides $a \leq b \leq c$ barely obtuse if the sides satisfy $a^2 + b^2 = c^2 - 1$.

How many barely obtuse triangles are there with perimeter $\leq 75\,000\,000$?

Problem 225: Tribonacci Non-divisors

[Link](#)

Tags: Sequences

The sequence $1, 1, 1, 3, 5, 9, 17, 31, 57, 105, 193, 355, 653, 1201, \dots$

is defined by $T_1 = T_2 = T_3 = 1$ and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$.

It can be shown that 27 does not divide any terms of this sequence. In fact, 27 is the first odd number with this property.

Find the 124th odd number that does not divide any terms of the above sequence.

Problem 226: A Scoop of Blancmange

[Link](#)

Tags: Geometry

The blancmange curve is the set of points (x, y) such that $0 \leq x \leq 1$ and $y = \sum_{n=0}^{\infty} \frac{s(2^n x)}{2^n}$, where $s(x)$ is the distance from x to the nearest integer.

The area under the blancmange curve is equal to $\frac{1}{2}$, shown in pink in the diagram below.



Problem 227: The Chase

[Link](#)

Tags: Game Theory, Number Theory, Probability

The Chase is a game played with two dice and an even number of players.

The players sit around a table and the game begins with two opposite players having one die each. On each turn, the two players with a die roll it.

If the player rolls 1, then the die passes to the neighbour on the left.

If the player rolls 6, then the die passes to the neighbour on the right.

Otherwise, the player keeps the die for the next turn.

The game ends when one player has both dice after they have been rolled and passed; that player has then lost.

In a game with 100 players, what is the expected number of turns the game lasts?

Give your answer rounded to ten significant digits.



Problem 228: Minkowski Sums

[Link](#)

Tags: Geometry

Let S_n be the regular n -sided polygon – or shape – whose vertices

v_k ($k = 1, 2, \dots, n$) have coordinates:

$$\begin{aligned}x_k &= \cos((2k-1)/n \times 180^\circ) \\y_k &= \sin((2k-1)/n \times 180^\circ)\end{aligned}$$

Each S_n is to be interpreted as a filled shape consisting of all points on the perimeter and in the interior.

The Minkowski sum, $S + T$, of two shapes S and T is the result of adding every point in S to every point in T , where point addition is performed coordinate-wise: $(u, v) + (x, y) = (u + x, v + y)$.

For example, the sum of S_3 and S_4 is the six-sided shape shown in pink below:

Problem 229: Four Representations Using Squares

Link

Tags: Game Theory

Consider the number 3600. It is very special, because

$$\begin{aligned} 3600 &= 48^2 + 36^2 \\ 3600 &= 20^2 + 2 \times 40^2 \\ 3600 &= 30^2 + 3 \times 30^2 \\ 3600 &= 45^2 + 7 \times 15^2 \end{aligned}$$

Similarly, we find that $88201 = 99^2 + 280^2 = 287^2 + 2 \times 54^2 = 283^2 + 3 \times 52^2 = 197^2 + 7 \times 84^2$.

In 1747, Euler proved which numbers are representable as a sum of two squares. We are interested in the numbers n which admit representations of all of the following four types:

$$\begin{aligned} n &= a_1^2 + b_1^2 \\ n &= a_2^2 + 2 b_2^2 \\ n &= a_3^2 + 3 b_3^2 \\ n &= a_7^2 + 7 b_7^2, \end{aligned}$$

where the a_k and b_k are positive integers.

There are 75373 such numbers that do not exceed 10^7 .

How many such numbers are there that do not exceed 2×10^9 ?

Problem 230: Fibonacci Words

Link

Tags: Number Theory, Sequences

For any two strings of digits, A and B , we define $F_{A,B}$ to be the sequence $(A, B, AB, BAB, ABBAB, \dots)$ in which each term is the concatenation of the previous two.

Further, we define $D_{A,B}(n)$ to be the n th digit in the first term of $F_{A,B}$ that contains at least n digits.

Example:

Let $A = 1415926535$, $B = 8979323846$. We wish to find $D_{A,B}(35)$, say.

The first few terms of $F_{A,B}$ are:

1415926535

8979323846

14159265358979323846

897932384614159265358979323846

14159265358979323846897932384614159265358979323846

Then $D_{A,B}(35)$ is the 35th digit in the fifth term, which is 9.

Now we use for A the first 100 digits of π behind the decimal point:

14159265358979323846264338327950288419716939937510

58209749445923078164062862089986280348253421170679

and for B the next hundred digits:

82148086513282306647093844609550582231725359408128

48111745028410270193852110555964462294895493038196.

Find $\sum_{n=0}^{17} 10^n \times D_{A,B}((127 + 19n) \times 7^n)$.

Problem 231: Prime Factorisation of Binomial Coefficients

Link

Tags: Combinatorics, Number Theory

The binomial coefficient $\binom{10}{3} = 120$.

$120 = 2^3 \times 3 \times 5 = 2 \times 2 \times 2 \times 3 \times 5$, and $2 + 2 + 2 + 3 + 5 = 14$.

So the sum of the terms in the prime factorisation of $\binom{10}{3}$ is 14.

Find the sum of the terms in the prime factorisation of $\binom{20\,000\,000}{15\,000\,000}$.

Problem 232: The Race

[Link](#)

Tags: Game Theory, Probability

Two players share an unbiased coin and take it in turns to play The Race.

On Player 1's turn, the coin is tossed once. If it comes up Heads, then Player 1 scores one point; if it comes up Tails, then no points are scored.

On Player 2's turn, a positive integer, T , is chosen by Player 2 and the coin is tossed T times. If it comes up all Heads, then Player 2 scores 2^{T-1} points; otherwise, no points are scored.

Player 1 goes first and the winner is the first to 100 or more points.

Player 2 will always select the number, T , of coin tosses that maximises the probability of winning.

What is the probability that Player 2 wins?

Give your answer rounded to eight decimal places in the form 0.abcdefgh.

Problem 233: Lattice Points on a Circle

[Link](#)

Tags: Geometry

Let $f(N)$ be the number of points with integer coordinates that are on a circle passing through $(0, 0)$, $(N, 0)$, $(0, N)$, and (N, N) .

It can be shown that $f(10000) = 36$.

What is the sum of all positive integers $N \leq 10^{11}$ such that $f(N) = 420$?

Problem 234: Semidivisible Numbers

[Link](#)

Tags: Number Theory

For an integer $n \geq 4$, we define the lower prime square root of n , denoted by $\text{lps}(n)$, as the largest prime $\leq \sqrt{n}$ and the upper prime square root of n , $\text{ups}(n)$, as the smallest prime $\geq \sqrt{n}$.

So, for example, $\text{lps}(4) = 2 = \text{ups}(4)$, $\text{lps}(1000) = 31$, $\text{ups}(1000) = 37$.

Let us call an integer $n \geq 4$ semidivisible, if one of $\text{lps}(n)$ and $\text{ups}(n)$ divides n , but not both.

The sum of the semidivisible numbers not exceeding 15 is 30, the numbers are 8, 10 and 12. 15 is not semidivisible because it is a multiple of both $\text{lps}(15) = 3$ and $\text{ups}(15) = 5$.

As a further example, the sum of the 92 semidivisible numbers up to 1000 is 34825.

What is the sum of all semidivisible numbers not exceeding 999966663333?

Problem 235: An Arithmetic Geometric Sequence

Link

Tags: Sequences

Given is the arithmetic-geometric sequence $u(k) = (900 - 3k)r^{k-1}$.

Let $s(n) = \sum_{k=1}^n u(k)$.

Find the value of r for which $s(5000) = -600\,000\,000\,000$.

Give your answer rounded to 12 places behind the decimal point.

Problem 236: Luxury Hampers

Link

Tags: Game Theory, Number Theory

Suppliers 'A' and 'B' provided the following numbers of products for the luxury hamper market:

Product	'A'	'B'	Beluga Caviar	5248640	Christmas Cake	13121888	Gammon Joint	26243776	Vintage Port	57603776	Champagne Truffles	39365664
---------	-----	-----	---------------	---------	----------------	----------	--------------	----------	--------------	----------	--------------------	----------

Although the suppliers try very hard to ship their goods in perfect condition, there is inevitably some spoilage - i.e. products gone bad.

The suppliers compare their performance using two types of statistic:

- The five per-product spoilage rates for each supplier are equal to the number of products gone bad divided by the number of products supplied, for each of the five products in turn.
 - The overall spoilage rate for each supplier is equal to the total number of products gone bad divided by the total number of products provided by that supplier.

To their surprise, the suppliers found that each of the five per-product spoilage rates was worse (higher) for ‘B’ than for ‘A’ by the same factor (ratio of spoilage rates), $m > 1$; and yet, paradoxically, the overall spoilage rate was worse for ‘A’ than for ‘B’, also by a factor of m .

There are thirty-five $m > 1$ for which this surprising result could have occurred, the smallest of which is $1476/1475$.

What’s the largest possible value of m ?

Give your answer as a fraction reduced to its lowest terms, in the form u/v .

Problem 237: Tours on a $4 \times N$ Playing Board

Link

Tags: Number Theory

Let $T(n)$ be the number of tours over a $4 \times n$ playing board such that:

- The tour starts in the top left corner.
- The tour consists of moves that are up, down, left, or right one square.
- The tour visits each square exactly once.
- The tour ends in the bottom left corner.

The diagram shows one tour over a 4×10 board:

Problem 238: Infinite String Tour

Link

Tags: Number Theory, Probability, Sequences

Create a sequence of numbers using the “Blum Blum Shub” pseudo-random number generator:

$$s_0 = 14025256$$

$$s_{n+1} = s_n^2 \bmod 20300713$$

Concatenate these numbers $s_0s_1s_2\cdots$ to create a string w of infinite length.

Then, $w = 14025256741014958470038053646\cdots$

For a positive integer k , if no substring of w exists with a sum of digits equal to k , $p(k)$ is defined to be zero. If at least one substring of w exists with a sum of digits equal to k , we define $p(k) = z$, where z is the starting position of the earliest such substring.

For instance:

The substrings $1, 14, 1402, \dots$

with respective sums of digits equal to $1, 5, 7, \dots$

start at position **1**, hence $p(1) = p(5) = p(7) = \cdots = \mathbf{1}$.

The substrings $4, 402, 4025, \dots$

with respective sums of digits equal to $4, 6, 11, \dots$

start at position **2**, hence $p(4) = p(6) = p(11) = \cdots = \mathbf{2}$.

The substrings $02, 0252, \dots$

with respective sums of digits equal to $2, 9, \dots$

start at position **3**, hence $p(2) = p(9) = \cdots = \mathbf{3}$.

Note that substring 025 starting at position **3**, has a sum of digits equal to 7, but there was an earlier substring (starting at position **1**) with a sum of digits equal to 7, so $p(7) = 1$, not 3.

We can verify that, for $0k \leq 10^3$, $\sum p(k) = 4742$.

Find $\sum p(k)$, for $0k \leq 2 \times 10^{15}$.

Problem 239: Twenty-two Foolish Primes

Link

Tags: Number Theory, Probability

A set of disks numbered 1 through 100 are placed in a line in random order.

What is the probability that we have a partial derangement such that exactly 22 prime number discs are found away from their natural positions?

(Any number of non-prime disks may also be found in or out of their natural positions.)

Give your answer rounded to 12 places behind the decimal point in the form 0.abcdefghijkl.

Problem 240: Top Dice

[Link](#)

Tags: Probability

There are 1111 ways in which five 6-sided dice (sides numbered 1 to 6) can be rolled so that the top three sum to 15. Some examples are:

$$D_1, D_2, D_3, D_4, D_5 = 4, 3, 6, 3, 5$$

$$D_1, D_2, D_3, D_4, D_5 = 4, 3, 3, 5, 6$$

$$D_1, D_2, D_3, D_4, D_5 = 3, 3, 3, 6, 6$$

$$D_1, D_2, D_3, D_4, D_5 = 6, 6, 3, 3, 3$$

In how many ways can twenty 12-sided dice (sides numbered 1 to 12) be rolled so that the top ten sum to 70?

Problem 241: Perfection Quotients

[Link](#)

Tags: Number Theory

For a positive integer n , let $\sigma(n)$ be the sum of all divisors of n . For example, $\sigma(6) = 1 + 2 + 3 + 6 = 12$.

A perfect number, as you probably know, is a number with $\sigma(n) = 2n$.

Let us define the perfection quotient of a positive integer as $p(n) = \frac{\sigma(n)}{n}$.

Find the sum of all positive integers $n \leq 10^{18}$ for which $p(n)$ has the form $k + \frac{1}{2}$, where k is an integer.

Problem 242: Odd Triplets

[Link](#)

Tags: Combinatorics

Given the set $\{1, 2, \dots, n\}$, we define $f(n, k)$ as the number of its k -element subsets with an odd sum of elements. For example, $f(5, 3) = 4$, since the set $\{1, 2, 3, 4, 5\}$ has four 3-element subsets having an odd sum of elements, i.e.: $\{1, 2, 4\}$, $\{1, 3, 5\}$, $\{2, 3, 4\}$ and $\{2, 4, 5\}$.

When all three values n , k and $f(n, k)$ are odd, we say that they make an odd-triplet $[n, k, f(n, k)]$.

There are exactly five odd-triplets with $n \leq 10$, namely:

$[1, 1, f(1, 1) = 1]$, $[5, 1, f(5, 1) = 3]$, $[5, 5, f(5, 5) = 1]$, $[9, 1, f(9, 1) = 5]$ and $[9, 9, f(9, 9) = 1]$.

How many odd-triplets are there with $n \leq 10^{12}$?

Problem 243: Resilience

[Link](#)

Tags: Number Theory

A positive fraction whose numerator is less than its denominator is called a proper fraction.

For any denominator, d , there will be $d - 1$ proper fractions; for example, with $d = 12$: $1/12, 2/12, 3/12, 4/12, 5/12, 6/12, 7/12, 8/12, 9/12, 10/12, 11/12$.

We shall call a fraction that cannot be cancelled down a resilient fraction.

Furthermore we shall define the resilience of a denominator, $R(d)$, to be the ratio of its proper fractions that are resilient; for example, $R(12) = 4/11$.

In fact, $d = 12$ is the smallest denominator having a resilience $R(d) > 4/10$.

Find the smallest denominator d , having a resilience $R(d) > 15499/94744$.

Problem 244: Sliders

[Link](#)

Tags: Game Theory, Sequences

You probably know the game Fifteen Puzzle. Here, instead of numbered tiles, we have seven red tiles and eight blue tiles.

A move is denoted by the uppercase initial of the direction (Left, Right, Up, Down) in which the tile is slid, e.g. starting from configuration (S), by the sequence LULUR we reach the configuration (E):

(S), (E)

Problem 245: Coresilience

[Link](#)

Tags: Number Theory

We shall call a fraction that cannot be cancelled down a resilient fraction. Furthermore we shall define the resilience of a denominator, $R(d)$, to be the ratio of its proper fractions that are resilient; for example, $R(12) = \frac{4}{11}$.

The resilience of a number d is then $\frac{\varphi(d)}{d-1}$, where φ is Euler's totient function.

We further define the coresilience of a number n as $C(n) = \frac{n - \varphi(n)}{n-1}$.

The coresilience of a prime p is $C(p) = \frac{1}{p-1}$.

Find the sum of all composite integers $1n \leq 2 \times 10^{11}$, for which $C(n)$ is a unit fraction.

Problem 246: Tangents to an Ellipse

Link

Tags: Geometry

A definition for an ellipse is:

Given a circle c with centre M and radius r and a point G such that $d(G, M) > r$, the locus of the points that are equidistant from c and G form an ellipse.

The construction of the points of the ellipse is shown below.

Problem 247: Squares Under a Hyperbola

Link

Tags: Geometry

Consider the region constrained by $1 \leq x$ and $0 \leq y \leq 1/x$.

Let S_1 be the largest square that can fit under the curve.

Let S_2 be the largest square that fits in the remaining area, and so on.

Let the index of S_n be the pair (left, below) indicating the number of squares to the left of S_n and the number of squares below S_n .

Problem 248: Euler's Totient Function Equals 13!

Link

Tags: Number Theory

The first number n for which $\phi(n) = 13!$ is 6227180929.

Find the 150 000th such number.

Problem 249: Prime Subset Sums

Link

Tags: Combinatorics, Number Theory

Let $S = \{2, 3, 5, \dots, 4999\}$ be the set of prime numbers less than 5000.

Find the number of subsets of S , the sum of whose elements is a prime number.

Enter the rightmost 16 digits as your answer.

Problem 250: 250250

Link

Tags: Combinatorics, Number Theory

Find the number of non-empty subsets of $\{1^1, 2^2, 3^3, \dots, 250250^{250250}\}$, the sum of whose elements is divisible by 250. Enter the rightmost 16 digits as your answer.

Problem 251: Cardano Triplets

Link

Tags: Number Theory

A triplet of positive integers (a, b, c) is called a Cardano Triplet if it satisfies the condition:

$$\sqrt[3]{a + b\sqrt{c}} + \sqrt[3]{a - b\sqrt{c}} = 1$$

For example, $(2, 1, 5)$ is a Cardano Triplet.

There exist 149 Cardano Triplets for which $a + b + c \leq 1000$.

Find how many Cardano Triplets exist such that $a + b + c \leq 110\,000\,000$.

Problem 252: Convex Holes

[Link](#)

Tags: Geometry

Given a set of points on a plane, we define a convex hole to be a convex polygon having as vertices any of the given points and not containing any of the given points in its interior (in addition to the vertices, other given points may lie on the perimeter of the polygon).

As an example, the image below shows a set of twenty points and a few such convex holes. The convex hole shown as a red heptagon has an area equal to 1049694.5 square units, which is the highest possible area for a convex hole on the given set of points.

Problem 253: Tidying Up A

[Link](#)

Tags: Probability

A small child has a “number caterpillar” consisting of forty jigsaw pieces, each with one number on it, which, when connected together in a line, reveal the numbers 1 to 40 in order.

Every night, the child’s father has to pick up the pieces of the caterpillar that have been scattered across the play room. He picks up the pieces at random and places them in the correct order. As the caterpillar is built up in this way, it forms distinct segments that gradually merge together. The number of segments starts at zero (no pieces placed), generally increases up to about eleven or twelve, then tends to drop again before finishing at a single segment (all pieces placed).

For example:

Piece Placed Segments So Far 121422936434554354.

Problem 254: Sums of Digit Factorials

[Link](#)

Tags: Number Theory

Define $f(n)$ as the sum of the factorials of the digits of n . For example, $f(342) = 3! + 4! + 2! = 32$.

Define $sf(n)$ as the sum of the digits of $f(n)$. So $sf(342) = 3 + 2 = 5$.

Define $g(i)$ to be the smallest positive integer n such that $sf(n) = i$. Though $sf(342)$ is 5, $sf(25)$ is also 5, and it can be verified that $g(5)$ is 25.

Define $sg(i)$ as the sum of the digits of $g(i)$. So $sg(5) = 2 + 5 = 7$.

Further, it can be verified that $g(20)$ is 267 and $\sum sg(i)$ for $1 \leq i \leq 20$ is 156.

What is $\sum sg(i)$ for $1 \leq i \leq 150$?

Problem 255: Rounded Square Roots

Link

Tags: Game Theory, Number Theory

We define the rounded-square-root of a positive integer n as the square root of n rounded to the nearest integer.

The following procedure (essentially Heron's method adapted to integer arithmetic) finds the rounded-square-root of n :

Let d be the number of digits of the number n .

If d is odd, set $x_0 = 2 \times 10^{(d-1)/2}$.

If d is even, set $x_0 = 7 \times 10^{(d-2)/2}$.

Repeat:

$$x_{k+1} = \left\lfloor \frac{x_k + \lceil n/x_k \rceil}{2} \right\rfloor$$

until $x_{k+1} = x_k$.

As an example, let us find the rounded-square-root of $n = 4321$. n has 4 digits, so $x_0 = 7 \times 10^{(4-2)/2} = 70$.

$$x_1 = \left\lfloor \frac{70 + \lceil 4321/70 \rceil}{2} \right\rfloor = 66$$

$$x_2 = \left\lfloor \frac{66 + \lceil 4321/66 \rceil}{2} \right\rfloor = 66$$

Since $x_2 = x_1$, we stop here.

So, after just two iterations, we have found that the rounded-square-root of 4321 is 66 (the actual square root is $65.7343137\dots$).

The number of iterations required when using this method is surprisingly low.

For example, we can find the rounded-square-root of a 5-digit integer ($10\,000 \leq n \leq 99\,999$) with an average of 3.210288889 iterations (the average value was rounded to 10 decimal places).

Using the procedure described above, what is the average number of iterations required to find the rounded-square-root of a 14-digit number ($10^{13} \leq n < 10^{14}$)?

Give your answer rounded to 10 decimal places.

Note: The symbols $\lfloor x \rfloor$ and $\lceil x \rceil$ represent the floor function and ceiling function respectively.

Problem 256: Tatami-Free Rooms

Link

Tags: Combinatorics, Geometry

Tatami are rectangular mats, used to completely cover the floor of a room, without overlap.

Assuming that the only type of available tatami has dimensions 1×2 , there are obviously some limitations for the shape and size of the rooms that can be covered.

For this problem, we consider only rectangular rooms with integer dimensions a, b and even size $s = a \cdot b$.

We use the term ‘size’ to denote the floor surface area of the room, and — without loss of generality — we add the condition $a \leq b$.

There is one rule to follow when laying out tatami: there must be no points where corners of four different mats meet.

For example, consider the two arrangements below for a 4×4 room:

Problem 257: Angular Bisectors

Link

Tags: Geometry

Given is an integer sided triangle ABC with sides $a \leq b \leq c$. ($AB = c$, $BC = a$ and $AC = b$.)

The angular bisectors of the triangle intersect the sides at points E , F and G (see picture below).

Problem 258: A Lagged Fibonacci Sequence

Link

Tags: Sequences

A sequence is defined as:

- $g_k = 1$, for $0 \leq k \leq 1999$
- $g_k = g_{k-2000} + g_{k-1999}$, for $k \geq 2000$.

Find $g_k \bmod 20092010$ for $k = 10^{18}$.

Problem 259: Reachable Numbers

Link

Tags: Game Theory, Number Theory

A positive integer will be called reachable if it can result from an arithmetic expression obeying the following rules:

- Uses the digits 1 through 9, in that order and exactly once each.
- Any successive digits can be concatenated (for example, using the digits 2, 3 and 4 we obtain the number 234).
- Only the four usual binary arithmetic operations (addition, subtraction, multiplication and division) are allowed.
- Each operation can be used any number of times, or not at all.
- Unary minus is not allowed.
- Any number of (possibly nested) parentheses may be used to define the order of operations.

For example, 42 is reachable, since $(1/23) \times ((4 \times 5) - 6) \times (78 - 9) = 42$.

What is the sum of all positive reachable integers?

Problem 260: Stone Game

Link

Tags: Combinatorics, Game Theory

A game is played with three piles of stones and two players.

On each player's turn, the player may remove one or more stones from the piles. However, if the player takes stones from more than one pile, then the same number of stones must be removed from each of the selected piles.

In other words, the player chooses some N and removes:

- N stones from any single pile; or
- N stones from each of any two piles ($2N$ total); or
- N stones from each of the three piles ($3N$ total).

The player taking the last stone(s) wins the game.

A winning configuration is one where the first player can force a win.

For example, $(0, 0, 13)$, $(0, 11, 11)$, and $(5, 5, 5)$ are winning configurations because the first player can immediately remove all stones.

A losing configuration is one where the second player can force a win, no matter what the first player does.

For example, $(0, 1, 2)$ and $(1, 3, 3)$ are losing configurations: any legal move leaves a winning configuration for the second player.

Consider all losing configurations (x_i, y_i, z_i) where $x_i \leq y_i \leq z_i \leq 100$.

We can verify that $\sum(x_i + y_i + z_i) = 173895$ for these.

Find $\sum(x_i + y_i + z_i)$ where (x_i, y_i, z_i) ranges over the losing configurations with $x_i \leq y_i \leq z_i \leq 1000$.

Problem 261: Pivotal Square Sums

Link

Tags: Number Theory

Let us call a positive integer k a square-pivot, if there is a pair of integers m and $n \geq k$, such that the sum of the $(m + 1)$ consecutive squares up to k equals the sum of the m consecutive squares from $(n + 1)$ on:

$$(k - m)^2 + \cdots + k^2 = (n + 1)^2 + \cdots + (n + m)^2.$$

Some small square-pivots are

- **4:** $3^2 + 4^2 = 5^2$
- **21:** $20^2 + 21^2 = 29^2$
- **24:** $21^2 + 22^2 + 23^2 + 24^2 = 25^2 + 26^2 + 27^2$
- **110:** $108^2 + 109^2 + 110^2 = 133^2 + 134^2$

Find the sum of all distinct square-pivots $\leq 10^{10}$.

Problem 262: Mountain Range

[Link](#)

Tags: Algebra, Combinatorics, Game Theory, Geometry, Graph Theory, Number Theory

The following equation represents the continuous topography of a mountainous region, giving the elevation h at any point (x, y) :

$$h = \left(5000 - \frac{x^2 + y^2 + xy}{200} + \frac{25(x + y)}{2} \right) \cdot e^{-\left| \frac{x^2 + y^2}{1000000} - \frac{3(x + y)}{2000} + \frac{7}{10} \right|}.$$

A mosquito intends to fly from $A(200, 200)$ to $B(1400, 1400)$, without leaving the area given by $0 \leq x, y \leq 1600$.

Because of the intervening mountains, it first rises straight up to a point A' , having elevation f . Then, while remaining at the same elevation f , it flies around any obstacles until it arrives at a point B' directly above B .

First, determine $f_{\min}$ which is the minimum constant elevation allowing such a trip from A to B , while remaining in the specified area.

Then, find the length of the shortest path between A' and B' , while flying at that constant elevation $f_{\min}$.

Give that length as your answer, rounded to three decimal places.

Note: For convenience, the elevation function shown above is repeated below, in a form suitable for most programming languages:

$$h = \left(5000 - 0.005(x^2 + y^2 + xy) + 12.5(x + y) \right) \cdot \exp\left(-\text{abs}(0.000001(x^2 + y^2) - 0.0015(x + y) + 0.7) \right)$$

Problem 263: An Engineers' Dream Come True

[Link](#)

Tags: Number Theory

Consider the number 6. The divisors of 6 are: 1, 2, 3 and 6.

Every number from 1 up to and including 6 can be written as a sum of distinct divisors of 6:

$$1 = 1, 2 = 2, 3 = 1 + 2, 4 = 1 + 3, 5 = 2 + 3, 6 = 6.$$

A number n is called a practical number if every number from 1 up to and including n can be expressed as a sum of distinct divisors of n .

A pair of consecutive prime numbers with a difference of six is called a sexy pair (since “sex” is the Latin word for “six”). The first sexy pair is (23, 29).

We may occasionally find a triple-pair, which means three consecutive sexy prime pairs, such that the second member of each pair is the first member of the next pair.

We shall call a number n such that :

- $(n - 9, n - 3)$, $(n - 3, n + 3)$, $(n + 3, n + 9)$ form a triple-pair, and
- the numbers $n - 8$, $n - 4$, n , $n + 4$ and $n + 8$ are all practical,

an engineers’ paradise.

Find the sum of the first four engineers’ paradises.

Problem 264: Triangle Centres

Link

Tags: Combinatorics, Geometry

Consider all the triangles having:

- All their vertices on lattice points.
- Circumcentre at the origin O .
- Orthocentre at the point $H(5, 0)$.

There are nine such triangles having a perimeter ≤ 50 .

Listed and shown in ascending order of their perimeter, they are:

$A(-4, 3)$, $B(5, 0)$, $C(4, -3)$

$A(4, 3)$, $B(5, 0)$, $C(-4, -3)$

$A(-3, 4)$, $B(5, 0)$, $C(3, -4)$

$A(3, 4)$, $B(5, 0)$, $C(-3, -4)$

$A(0, 5)$, $B(5, 0)$, $C(0, -5)$

$A(1, 8)$, $B(8, -1)$, $C(-4, -7)$

$A(8, 1)$, $B(1, -8)$, $C(-4, 7)$

$A(2, 9)$, $B(9, -2)$, $C(-6, -7)$

$A(9, 2)$, $B(2, -9)$, $C(-6, 7)$

The sum of their perimeters, rounded to four decimal places, is 291.0089.

Find all such triangles with a perimeter $\leq 10^5$.

Enter as your answer the sum of their perimeters rounded to four decimal places.

Problem 265: Binary Circles

[Link](#)

Tags: Combinatorics, Geometry, Number Theory, Sequences

2^N binary digits can be placed in a circle so that all the N -digit clockwise subsequences are distinct.

For $N = 3$, two such circular arrangements are possible, ignoring rotations:

Problem 266: Pseudo Square Root

[Link](#)

Tags: Number Theory

The divisors of 12 are: 1, 2, 3, 4, 6 and 12.

The largest divisor of 12 that does not exceed the square root of 12 is 3.

We shall call the largest divisor of an integer n that does not exceed the square root of n the pseudo square root (PSR) of n .

It can be seen that $\text{PSR}(3102) = 47$.

Let p be the product of the primes below 190.

Find $\text{PSR}(p) \bmod 10^{16}$.

Problem 267: Billionaire

[Link](#)

Tags: Combinatorics, Game Theory, Number Theory, Probability

You are given a unique investment opportunity.

Starting with £1 of capital, you can choose a fixed proportion, f , of your capital to bet on a fair coin toss repeatedly for 1000 tosses.

Your return is double your bet for heads and you lose your bet for tails.

For example, if $f=1/4$, for the first toss you bet £0.25, and if heads comes up you win £0.5 and so then have £1.5. You then bet £0.375 and if the second toss is tails, you have £1.125.

Choosing f to maximize your chances of having at least £1,000,000,000 after 1,000 flips, what is the chance that you become a billionaire?

All computations are assumed to be exact (no rounding), but give your answer rounded to 12 digits behind the decimal point in the form 0.abcdefghijkl.

Problem 268: At Least Four Distinct Prime Factors Less Than 100

[Link](#)

Tags: Number Theory

It can be verified that there are 23 positive integers less than 1000 that are divisible by at least four distinct primes less than 100.

Find how many positive integers less than 10^{16} are divisible by at least four distinct primes less than 100.

Problem 269: Polynomials with at Least One Integer Root

[Link](#)

Tags: Algebra, Number Theory

A root or zero of a polynomial $P(x)$ is a solution to the equation $P(x) = 0$.

Define P_n as the polynomial whose coefficients are the digits of n .

For example, $P_{5703}(x) = 5x^3 + 7x^2 + 3$.

We can see that:

- $P_n(0)$ is the last digit of n ,
- $P_n(1)$ is the sum of the digits of n ,
- $P_n(10)$ is n itself.

Define $Z(k)$ as the number of positive integers, n , not exceeding k for which the polynomial P_n has at least one integer root.

It can be verified that $Z(100\,000)$ is 14696.

What is $Z(10^{16})$?

Problem 270: Cutting Squares

Link

Tags: Combinatorics, Game Theory

A square piece of paper with integer dimensions $N \times N$ is placed with a corner at the origin and two of its sides along the x - and y -axes. Then, we cut it up respecting the following rules:

- We only make straight cuts between two points lying on different sides of the square, and having integer coordinates.
- Two cuts cannot cross, but several cuts can meet at the same border point.
- Proceed until no more legal cuts can be made.

Counting any reflections or rotations as distinct, we call $C(N)$ the number of ways to cut an $N \times N$ square. For example, $C(1) = 2$ and $C(2) = 30$ (shown below).

Problem 271: Modular Cubes, Part 1

Link

Tags: Number Theory

For a positive number n , define $S(n)$ as the sum of the integers x , for which $1 \leq x \leq n$ and $x^3 \equiv 1 \pmod{n}$.

When $n = 91$, there are 8 possible values for x , namely: 9, 16, 22, 29, 53, 74, 79, 81.

Thus, $S(91) = 9 + 16 + 22 + 29 + 53 + 74 + 79 + 81 = 363$.

Find $S(13082761331670030)$.

Problem 272: Modular Cubes, Part 2

Link

Tags: Number Theory

For a positive number n , define $C(n)$ as the number of the integers x , for which $1 \leq x \leq n$ and $x^3 \equiv 1 \pmod{n}$.

When $n = 91$, there are 8 possible values for x , namely: 9, 16, 22, 29, 53, 74, 79, 81.

Thus, $C(91) = 8$.

Find the sum of the positive numbers $n \leq 10^{11}$ for which $C(n) = 242$.

Problem 273: Sum of Squares

[Link](#)

Tags: Algebra, Number Theory

Consider equations of the form: $a^2 + b^2 = N$, $0 \leq a \leq b$, a , b and N integer.

For $N = 65$ there are two solutions:

$a = 1$, $b = 8$ and $a = 4$, $b = 7$.

We call $S(N)$ the sum of the values of a of all solutions of $a^2 + b^2 = N$, $0 \leq a \leq b$, a , b and N integer.

Thus $S(65) = 1 + 4 = 5$.

Find $\sum S(N)$, for all squarefree N only divisible by primes of the form $4k + 1$ with $4k + 1150$.

Problem 274: Divisibility Multipliers

[Link](#)

Tags: Game Theory, Number Theory

For each integer p coprime to 10 there is a positive divisibility multiplier mp which preserves divisibility by p for the following function on any positive integer, n :

$f(n) = (\text{all but the last digit of } n) + (\text{the last digit of } n) \cdot m$.

That is, if m is the divisibility multiplier for p , then $f(n)$ is divisible by p if and only if n is divisible by p .

(When n is much larger than p , $f(n)$ will be less than n and repeated application of f provides a multiplicative divisibility test for p .)

For example, the divisibility multiplier for 113 is 34.

$f(76275) = 7627 + 5 \cdot 34 = 7797$: 76275 and 7797 are both divisible by 113.

$f(12345) = 1234 + 5 \cdot 34 = 1404$: 12345 and 1404 are both not divisible by 113.

The sum of the divisibility multipliers for the primes that are coprime to 10 and less than 1000 is 39517. What is the sum of the divisibility multipliers for the primes that are coprime to 10 and less than 10^7 ?

Problem 275: Balanced Sculptures

[Link](#)

Tags: Combinatorics

Let us define a balanced sculpture of order n as follows:

- A polyomino made up of $n + 1$ tiles known as the blocks (n tiles) and the plinth (remaining tile);
- the plinth has its centre at position $(x = 0, y = 0)$;
- the blocks have y -coordinates greater than zero (so the plinth is the unique lowest tile);
- the centre of mass of all the blocks, combined, has x -coordinate equal to zero.

When counting the sculptures, any arrangements which are simply reflections about the y -axis, are not counted as distinct. For example, the 18 balanced sculptures of order 6 are shown below; note that each pair of mirror images (about the y -axis) is counted as one sculpture:

Problem 276: Primitive Triangles

Link

Tags: Geometry, Number Theory

Consider the triangles with integer sides a , b and c with $a \leq b \leq c$.

An integer sided triangle (a, b, c) is called primitive if $\gcd(a, b, c) = 1$.

How many primitive integer sided triangles exist with a perimeter not exceeding 10 000 000?

Problem 277: A Modified Collatz Sequence

Link

Tags: Game Theory, Sequences

A modified Collatz sequence of integers is obtained from a starting value a_1 in the following way:

$a_{n+1} = \frac{a_n}{3}$ if a_n is divisible by 3. We shall denote this as a large downward step, “D”.

$a_{n+1} = \frac{4a_n+2}{3}$ if a_n divided by 3 gives a remainder of 1. We shall denote this as an upward step, “U”.

$a_{n+1} = \frac{2a_n-1}{3}$ if a_n divided by 3 gives a remainder of 2. We shall denote this as a small downward step, “d”.

The sequence terminates when some $a_n = 1$.

Given any integer, we can list out the sequence of steps.

For instance if $a_1 = 231$, then the sequence $\{a_n\} = \{231, 77, 51, 17, 11, 7, 10, 14, 9, 3, 1\}$ corresponds to the steps “DdDddUUdDD”.

Of course, there are other sequences that begin with that same sequence “DdDddUUdDD...”.

For instance, if $a_1 = 1004064$, then the sequence is DdDddUUdDDdUDUUUdDdUUDDDUdDD.

In fact, 1004064 is the smallest possible $a_1 > 10^6$ that begins with the sequence DdDddUUdDD.

What is the smallest $a_1 > 10^{15}$ that begins with the sequence “UDDDUd-ddDDUDDdDdDddDDUDDdUUdD”?

Problem 278: Linear Combinations of Semiprimes

Link

Tags: Algebra, Combinatorics, Number Theory

Given the values of integers $1 < a_1 < a_2 < \dots < a_n$, consider the linear combination

$q_1 a_1 + q_2 a_2 + \dots + q_n a_n = b$, using only integer values $q_k \geq 0$.

Note that for a given set of a_k , it may be that not all values of b are possible.

For instance, if $a_1 = 5$ and $a_2 = 7$, there are no $q_1 \geq 0$ and $q_2 \geq 0$ such that b could be

1, 2, 3, 4, 6, 8, 9, 11, 13, 16, 18 or 23.

In fact, 23 is the largest impossible value of b for $a_1 = 5$ and $a_2 = 7$. We therefore call $f(5, 7) = 23$. Similarly, it can be shown that $f(6, 10, 15) = 29$ and $f(14, 22, 77) = 195$.

Find $\sum f(pq, pr, qr)$, where p, q and r are prime numbers and $p < q < r < 5000$.

Problem 279: Triangles with Integral Sides and an Integral Angle

Link

Tags: Geometry

How many triangles are there with integral sides, at least one integral angle (measured in degrees), and a perimeter that does not exceed 10^8 ?

Problem 280: Ant and Seeds

[Link](#)

Tags: Combinatorics, Probability

A laborious ant walks randomly on a 5×5 grid. The walk starts from the central square. At each step, the ant moves to an adjacent square at random, without leaving the grid; thus there are 2, 3 or 4 possible moves at each step depending on the ant's position.

At the start of the walk, a seed is placed on each square of the lower row. When the ant isn't carrying a seed and reaches a square of the lower row containing a seed, it will start to carry the seed. The ant will drop the seed on the first empty square of the upper row it eventually reaches.

What's the expected number of steps until all seeds have been dropped in the top row?

Give your answer rounded to 6 decimal places.

Problem 281: Pizza Toppings

[Link](#)

Tags: Geometry

You are given a pizza (perfect circle) that has been cut into $m \cdot n$ equal pieces and you want to have exactly one topping on each slice.

Let $f(m, n)$ denote the number of ways you can have toppings on the pizza with m different toppings ($m \geq 2$), using each topping on exactly n slices ($n \geq 1$). Reflections are considered distinct, rotations are not.

Thus, for instance, $f(2, 1) = 1$, $f(2, 2) = f(3, 1) = 2$ and $f(3, 2) = 16$. $f(3, 2)$ is shown below:

Problem 282: The Ackermann Function

[Link](#)

Tags: Number Theory

For non-negative integers m, n , the Ackermann function $A(m, n)$ is defined as follows:

$$A(m, n) = n + 1$$

if $m=0$ $A(m-1, 1)$ if $m>0$ and $n=0$ $A(m-1, A(m, n-1))$ if $m>0$ and $n>0$

For example $A(1, 0) = 2$, $A(2, 2) = 7$ and $A(3, 4) = 125$.

Find $\sum_{n=0}^6 A(n, n)$ and give your answer mod 14^8 .

Problem 283: Integer Sided Triangles with Integral Area/perimeter Ratio

Link

Tags: Geometry

Consider the triangle with sides 6, 8, and 10. It can be seen that the perimeter and the area are both equal to 24. So the area/perimeter ratio is equal to 1.

Consider also the triangle with sides 13, 14 and 15. The perimeter equals 42 while the area is equal to 84. So for this triangle the area/perimeter ratio is equal to 2.

Find the sum of the perimeters of all integer sided triangles for which the area/perimeter ratios are equal to positive integers not exceeding 1000.

Problem 284: Steady Squares

Link

Tags: Number Theory

The 3-digit number 376 in the decimal numbering system is an example of numbers with the special property that its square ends with the same digits: $376^2 = 141376$. Let's call a number with this property a steady square.

Steady squares can also be observed in other numbering systems. In the base 14 numbering system, the 3-digit number c37 is also a steady square: $c37^2 = aa0c37$, and the sum of its digits is $c+3+7=18$ in the same numbering system. The letters a, b, c and d are used for the 10, 11, 12 and 13 digits respectively, in a manner similar to the hexadecimal numbering system.

For $1 \leq n \leq 9$, the sum of the digits of all the n-digit steady squares in the base 14 numbering system is 2d8 (582 decimal). Steady squares with leading 0's are not allowed.

Find the sum of the digits of all the n -digit steady squares in the base 14 numbering system for

$1 \leq n \leq 10000$ (decimal) and give your answer in the base 14 system using lower case letters where necessary.

Problem 285: Pythagorean Odds

[Link](#)

Tags: Combinatorics, Probability

Albert chooses a positive integer k , then two real numbers a, b are randomly chosen in the interval $[0, 1]$ with uniform distribution.

The square root of the sum $(k \cdot a + 1)^2 + (k \cdot b + 1)^2$ is then computed and rounded to the nearest integer. If the result is equal to k , he scores k points; otherwise he scores nothing.

For example, if $k = 6$, $a = 0.2$ and $b = 0.85$, then $(k \cdot a + 1)^2 + (k \cdot b + 1)^2 = 42.05$.

The square root of 42.05 is $6.484 \dots$ and when rounded to the nearest integer, it becomes 6.

This is equal to k , so he scores 6 points.

It can be shown that if he plays 10 turns with $k = 1, k = 2, \dots, k = 10$, the expected value of his total score, rounded to five decimal places, is 10.20914.

If he plays 10^5 turns with $k = 1, k = 2, k = 3, \dots, k = 10^5$, what is the expected value of his total score, rounded to five decimal places?

Problem 286: Scoring Probabilities

[Link](#)

Tags: Game Theory, Geometry, Probability

Barbara is a mathematician and a basketball player. She has found that the probability of scoring a point when shooting from a distance x is exactly $(1 - x/q)$, where q is a real constant greater than 50.

During each practice run, she takes shots from distances $x = 1, x = 2, \dots, x = 50$ and, according to her records, she has precisely a 2% chance to score a total of exactly 20 points.

Find q and give your answer rounded to 10 decimal places.

Problem 287: Quadtree Encoding (a Simple Compression Algorithm)

Link

Tags: Game Theory, Graph Theory, Sequences

The quadtree encoding allows us to describe a $2^N \times 2^N$ black and white image as a sequence of bits (0 and 1). Those sequences are to be read from left to right like this:

- the first bit deals with the complete $2^N \times 2^N$ region;
- “0” denotes a split:

the current $2^n \times 2^n$ region is divided into 4 sub-regions of dimension $2^{n-1} \times 2^{n-1}$, the next bits contains the description of the top left, top right, bottom left and bottom right sub-regions - in that order;

- “10” indicates that the current region contains only black pixels;
- “11” indicates that the current region contains only white pixels.

Consider the following 4×4 image (colored marks denote places where a split can occur):

Problem 288: An Enormous Factorial

Link

Tags: Game Theory, Number Theory, Probability

For any prime p the number $N(p, q)$ is defined by $N(p, q) = \sum_{n=0}^q T_n \cdot p^n$

with T_n generated by the following random number generator:

$$S_0 = 290797$$

$$S_{n+1} = S_n^2 \bmod 50515093$$

$$T_n = S_n \bmod p$$

Let $\text{Nfac}(p, q)$ be the factorial of $N(p, q)$.

Let $\text{NF}(p, q)$ be the number of factors p in $\text{Nfac}(p, q)$.

You are given that $\text{NF}(3, 10000) \bmod 3^{20} = 624955285$.

Find $\text{NF}(61, 10^7) \bmod 61^{10}$.

Problem 289: Eulerian Cycles

Link

Tags: Combinatorics, Game Theory, Geometry

Let $C(x, y)$ be a circle passing through the points (x, y) , $(x, y + 1)$, $(x + 1, y)$ and $(x + 1, y + 1)$.

For positive integers m and n , let $E(m, n)$ be a configuration which consists of the $m \cdot n$ circles:

$\{C(x, y) : 0 \leq xm, 0 \leq yn, x \text{ and } y \text{ are integers}\}$.

An Eulerian cycle on $E(m, n)$ is a closed path that passes through each arc exactly once.

Many such paths are possible on $E(m, n)$, but we are only interested in those which are not self-crossing: a non-crossing path just touches itself at lattice points, but it never crosses itself.

The image below shows $E(3, 3)$ and an example of an Eulerian non-crossing path.

Problem 290: Digital Signature

Link

Tags: Number Theory

How many integers $0 \leq n < 10^{18}$ have the property that the sum of the digits of n equals the sum of digits of $137n$?

Problem 291: Panaitopol Primes

Link

Tags: Number Theory

A prime number p is called a Panaitopol prime if $p = \frac{x^4 - y^4}{x^3 + y^3}$ for some positive integers x and y .

Find how many Panaitopol primes are less than 5×10^{15} .

Problem 292: Pythagorean Polygons

Link

Tags: Game Theory, Geometry, Graph Theory

We shall define a pythagorean polygon to be a convex polygon with the following properties:

- there are at least three vertices,
- no three vertices are aligned,
- each vertex has integer coordinates,
- each edge has integer length.

For a given integer n , define $P(n)$ as the number of distinct pythagorean polygons for which the perimeter is $\leq n$.

Pythagorean polygons should be considered distinct as long as none is a translation of another.

You are given that $P(4) = 1$, $P(30) = 3655$ and $P(60) = 891045$.

Find $P(120)$.

Problem 293: Pseudo-Fortunate Numbers

Link

Tags: Number Theory

An even positive integer N will be called admissible, if it is a power of 2 or its distinct prime factors are consecutive primes.

The first twelve admissible numbers are 2, 4, 6, 8, 12, 16, 18, 24, 30, 32, 36, 48.

If N is admissible, the smallest integer M such that $N + M$ is prime, will be called the pseudo-Fortunate number for N .

For example, $N = 630$ is admissible since it is even and its distinct prime factors are the consecutive primes 2, 3, 5 and 7.

The next prime number after 631 is 641; hence, the pseudo-Fortunate number for 630 is $M = 11$.

It can also be seen that the pseudo-Fortunate number for 16 is 3.

Find the sum of all distinct pseudo-Fortunate numbers for admissible numbers N less than 10^9 .

Problem 294: Sum of Digits - Experience #23

Link

Tags: Game Theory, Number Theory

For a positive integer k , define $d(k)$ as the sum of the digits of k in its usual decimal representation. Thus $d(42) = 4 + 2 = 6$.

For a positive integer n , define $S(n)$ as the number of positive integers $k10^n$ with the following properties :

- k is divisible by 23 and
- $d(k) = 23$.

You are given that $S(9) = 263626$ and $S(42) = 6377168878570056$.

Find $S(11^{12})$ and give your answer mod 10^9 .

Problem 295: Lenticular Holes

Link

Tags: Combinatorics, Game Theory, Geometry

We call the convex area enclosed by two circles a lenticular hole if:

- The centres of both circles are on lattice points.
- The two circles intersect at two distinct lattice points.
- The interior of the convex area enclosed by both circles does not contain any lattice points.

Consider the circles:

$$C_0: x^2 + y^2 = 25$$

$$C_1: (x + 4)^2 + (y - 4)^2 = 1$$

$$C_2: (x - 12)^2 + (y - 4)^2 = 65$$

The circles C_0 , C_1 and C_2 are drawn in the picture below.

Problem 296: Angular Bisector and Tangent

Link

Tags: Geometry

Given is an integer sided triangle ABC with $BC \leq AC \leq AB$. k is the angular bisector of angle ACB . m is the tangent at C to the circumscribed circle of ABC . n is a line parallel to m through B .

The intersection of n and k is called E .

Problem 297: Zeckendorf Representation

Link

Tags: Sequences

Each new term in the Fibonacci sequence is generated by adding the previous two terms.

Starting with 1 and 2, the first 10 terms will be: 1, 2, 3, 5, 8, 13, 21, 34, 55, 89.

Every positive integer can be uniquely written as a sum of nonconsecutive terms of the Fibonacci sequence. For example, $100 = 3 + 8 + 89$.

Such a sum is called the Zeckendorf representation of the number.

For any integer n , let $z(n)$ be the number of terms in the Zeckendorf representation of n .

Thus, $z(5) = 1$, $z(14) = 2$, $z(100) = 3$ etc.

Also, for $0 \leq n < 10^6$, $\sum z(n) = 7894453$.

Find $\sum z(n)$ for $0 \leq n < 10^{17}$.

Problem 298: Selective Amnesia

Link

Tags: Game Theory, Probability, Sequences

Larry and Robin play a memory game involving a sequence of random numbers between 1 and 10, inclusive, that are called out one at a time. Each player can remember up to 5 previous numbers. When the called number is in a player's memory, that player is awarded a point. If it's not, the player adds the called number to his memory, removing another number if his memory is full.

Both players start with empty memories. Both players always add new missed numbers to their memory but use a different strategy in deciding which number to remove:

Larry's strategy is to remove the number that hasn't been called in the longest time.

Robin's strategy is to remove the number that's been in the memory the longest time.

Example game:

Turn Called number Larry's memory Larry's score Robin's memory Robin's score

```
1 1 1 0 1 0
2 2 1,2 0 1,2 0 3 4 1,2,4 0 1,2,4 0
4 6 1,2,4,6 0 1,2,4,6 0
5 1 1,2,4,6 1 1,2,4,6 1
6 8 1,2,4,6,8 1 1,2,4,6,8 1
7 10 1,4,6,8,10 1 2,4,6,8,10 1
8 2 1,2,6,8,10 1 2,4,6,8,10 2
9 4 1,2,4,8,10 1 2,4,6,8,10 3
10 1 1,2,4,8,10 2 1,4,6,8,10 3
```

Denoting Larry's score by L and Robin's score by R , what is the expected value of $|L-R|$ after 50 turns? Give your answer rounded to eight decimal places using the format `x.xxxxxxxx` .

Problem 299: Three Similar Triangles

[Link](#)

Tags: Geometry

Four points with integer coordinates are selected: $A(a, 0)$, $B(b, 0)$, $C(0, c)$ and $D(0, d)$, with $0ab$ and $0cd$.

Point P , also with integer coordinates, is chosen on the line AC so that the three triangles ABP , CDP and BDP are all similar.

Problem 300: Protein Folding

[Link](#)

Tags: Number Theory

In a very simplified form, we can consider proteins as strings consisting of hydrophobic (H) and polar (P) elements, e.g. HHPPHHHHPHHPH.

For this problem, the orientation of a protein is important; e.g. HPP is considered distinct from PPH. Thus, there are 2^n distinct proteins consisting of n elements.

When one encounters these strings in nature, they are always folded in such a way that the number of H-H contact points is as large as possible, since this is energetically advantageous.

As a result, the H-elements tend to accumulate in the inner part, with the P-elements on the outside.

Natural proteins are folded in three dimensions of course, but we will only consider protein folding in two dimensions.

The figure below shows two possible ways that our example protein could be folded (H-H contact points are shown with red dots).

Problem 301: Nim

Link

Tags: Game Theory

Nim is a game played with heaps of stones, where two players take it in turn to remove any number of stones from any heap until no stones remain.

We'll consider the three-heap normal-play version of Nim, which works as follows:

- At the start of the game there are three heaps of stones.
- On each player's turn, the player may remove any positive number of stones from any single heap.
- The first player unable to move (because no stones remain) loses.

If (n_1, n_2, n_3) indicates a Nim position consisting of heaps of size n_1 , n_2 , and n_3 , then there is a simple function, which you may look up or attempt to deduce for yourself, $X(n_1, n_2, n_3)$ that returns:

- zero if, with perfect strategy, the player about to move will eventually lose; or
- non-zero if, with perfect strategy, the player about to move will eventually win.

For example $X(1, 2, 3) = 0$ because, no matter what the current player does, the opponent can respond with a move that leaves two heaps of equal size, at which point every move by the current player can be mirrored by the opponent until no stones remain; so the current player loses. To illustrate:

- current player moves to $(1, 2, 1)$
- opponent moves to $(1, 0, 1)$

- current player moves to $(0, 0, 1)$
- opponent moves to $(0, 0, 0)$, and so wins.

For how many positive integers $n \leq 2^{30}$ does $X(n, 2n, 3n) = 0$?

Problem 302: Strong Achilles Numbers

Link

Tags: Number Theory

A positive integer n is powerful if p^2 is a divisor of n for every prime factor p in n .

A positive integer n is a perfect power if n can be expressed as a power of another positive integer.

A positive integer n is an Achilles number if n is powerful but not a perfect power. For example, 864 and 1800 are Achilles numbers: $864 = 2^5 \cdot 3^3$ and $1800 = 2^3 \cdot 3^2 \cdot 5^2$.

We shall call a positive integer S a Strong Achilles number if both S and $\phi(S)$ are Achilles numbers.¹

For example, 864 is a Strong Achilles number: $\phi(864) = 288 = 2^5 \cdot 3^2$. However, 1800 isn't a Strong Achilles number because: $\phi(1800) = 480 = 2^5 \cdot 3^1 \cdot 5^1$.

There are 7 Strong Achilles numbers below 10^4 and 656 below 10^8 .

How many Strong Achilles numbers are there below 10^{18} ?

¹ ϕ denotes Euler's totient function.

Problem 303: Multiples with Small Digits

Link

Tags: Number Theory

For a positive integer n , define $f(n)$ as the least positive multiple of n that, written in base 10, uses only digits ≤ 2 .

Thus $f(2) = 2$, $f(3) = 12$, $f(7) = 21$, $f(42) = 210$, $f(89) = 1121222$.

Also, $\sum_{n=1}^{100} \frac{f(n)}{n} = 11363107$.

Find $\sum_{n=1}^{10000} \frac{f(n)}{n}$.

Problem 304: Primonacci

[Link](#)

Tags: Number Theory, Sequences

For any positive integer n the function $\text{next_prime}(n)$ returns the smallest prime p such that pn .

The sequence $a(n)$ is defined by:

$a(1) = \text{next_prime}(10^{14})$ and $a(n) = \text{next_prime}(a(n-1))$ for $n \geq 1$.

The Fibonacci sequence $f(n)$ is defined by: $f(0) = 0$, $f(1) = 1$ and $f(n) = f(n-1) + f(n-2)$ for $n \geq 1$.

The sequence $b(n)$ is defined as $f(a(n))$.

Find $\sum b(n)$ for $1 \leq n \leq 100\,000$. Give your answer mod 1234567891011.

Problem 305: Reflexive Position

[Link](#)

Tags: Number Theory

Let's call S the (infinite) string that is made by concatenating the consecutive positive integers (starting from 1) written down in base 10.

Thus, $S = 1234567891011121314151617181920212223242 \dots$

It's easy to see that any number will show up an infinite number of times in S .

Let's call $f(n)$ the starting position of the n th occurrence of n in S .

For example, $f(1) = 1$, $f(5) = 81$, $f(12) = 271$ and $f(7780) = 111111365$.

Find $\sum f(3^k)$ for $1 \leq k \leq 13$.

Problem 306: Paper-strip Game

[Link](#)

Tags: Game Theory

The following game is a classic example of Combinatorial Game Theory:

Two players start with a strip of n white squares and they take alternate turns.

On each turn, a player picks two contiguous white squares and paints them black.

The first player who cannot make a move loses.

- $n = 1$: No valid moves, so the first player loses automatically.
- $n = 2$: Only one valid move, after which the second player loses.
- $n = 3$: Two valid moves, but both leave a situation where the second player loses.
- $n = 4$: Three valid moves for the first player, who is able to win the game by painting the two middle squares.
- $n = 5$: Four valid moves for the first player (shown below in red), but no matter what the player does, the second player (blue) wins.

Problem 307: Chip Defects

[Link](#)

Tags: Number Theory, Probability

k defects are randomly distributed amongst n integrated-circuit chips produced by a factory (any number of defects may be found on a chip and each defect is independent of the other defects).

Let $p(k, n)$ represent the probability that there is a chip with at least 3 defects.

For instance $p(3, 7) \approx 0.0204081633$.

Find $p(20\,000, 1\,000\,000)$ and give your answer rounded to 10 decimal places in the form 0.abcdefghij.

Problem 308: An Amazing Prime-generating Automaton

[Link](#)

Tags: Game Theory, Number Theory, Sequences

A program written in the programming language Fractran consists of a list of fractions.

The internal state of the Fractran Virtual Machine is a positive integer, which is initially set to a seed value. Each iteration of a Fractran program multiplies the state integer by the first fraction in the list which will leave it an integer.

For example, one of the Fractran programs that John Horton Conway wrote for prime-generation consists of the following 14 fractions:

$$\frac{17}{91}, \frac{78}{85}, \frac{19}{51}, \frac{23}{38}, \frac{29}{33}, \frac{77}{29}, \frac{95}{23}, \frac{77}{19}, \frac{1}{17}, \frac{11}{13}, \frac{13}{11}, \frac{15}{2}, \frac{1}{7}, \frac{55}{1}$$

Starting with the seed integer 2, successive iterations of the program produce the sequence:

15, 825, 725, 1925, 2275, 425, $\dots$, 68, 4, 30, $\dots$, 136, 8, 60, $\dots$, 544, 32, 240, $\dots$

The powers of 2 that appear in this sequence are 22, 23, 25, $\dots$

It can be shown that all the powers of 2 in this sequence have prime exponents and that all the primes appear as exponents of powers of 2, in proper order!

If someone uses the above Fractran program to solve Project Euler Problem 7 (find the 10001st prime), how many iterations would be needed until the program produces 210001st prime ?

Problem 309: Integer Ladders

Link

Tags: Graph Theory

In the classic “Crossing Ladders” problem, we are given the lengths x and y of two ladders resting on the opposite walls of a narrow, level street. We are also given the height h above the street where the two ladders cross and we are asked to find the width of the street (w).

Problem 310: Nim Square

Link

Tags: Game Theory

Alice and Bob play the game Nim Square.

Nim Square is just like ordinary three-heap normal play Nim, but the players may only remove a square number of stones from a heap.

The number of stones in the three heaps is represented by the ordered triple (a, b, c) .

If $0 \leq a \leq b \leq c \leq 29$ then the number of losing positions for the next player is 1160.

Find the number of losing positions for the next player if $0 \leq a \leq b \leq c \leq 100\,000$.

Problem 311: Biclinic Integral Quadrilaterals

Link

Tags: Game Theory

$ABCD$ is a convex, integer sided quadrilateral with $1 \leq ABBCDDAD$.

BD has integer length. O is the midpoint of BD . AO has integer length.

We'll call $ABCD$ a biclinic integral quadrilateral if $AO = CO \leq BO = DO$.

For example, the following quadrilateral is a biclinic integral quadrilateral:

$AB = 19$, $BC = 29$, $CD = 37$, $AD = 43$, $BD = 48$ and $AO = CO = 23$.

Problem 312: Cyclic Paths on Sierpiński Graphs

Link

Tags: Geometry, Graph Theory

- A Sierpiński graph of order-1 (S_1) is an equilateral triangle.
 - S_{n+1} is obtained from S_n by positioning three copies of S_n so that every pair of copies has one common corner.
-

Problem 313: Sliding Game

Link

Tags: Combinatorics, Game Theory, Sequences

In a sliding game a counter may slide horizontally or vertically into an empty space. The objective of the game is to move the red counter from the top left corner of a grid to the bottom right corner; the space always starts in the bottom right corner. For example, the following sequence of pictures show how the game can be completed in five moves on a 2 by 2 grid.

Problem 314: The Mouse on the Moon

Link

Tags: Combinatorics, Game Theory, Geometry, Sequences

The moon has been opened up, and land can be obtained for free, but there is a catch. You have to build a wall around the land that you stake out, and building a wall on the moon is expensive. Every country has been allotted a

500m by 500m square area, but they will possess only that area which they wall in. 251001 posts have been placed in a rectangular grid with 1 meter spacing. The wall must be a closed series of straight lines, each line running from post to post.

The bigger countries of course have built a 2000m wall enclosing the entire 250000m² area. The Duchy of Grand Fenwick, has a tighter budget, and has asked you (their Royal Programmer) to compute what shape would get best maximum enclosed-area/wall-length ratio.

You have done some preliminary calculations on a sheet of paper. For a 2000 meter wall enclosing the 250000m² area the enclosed-area/wall-length ratio is 125.

Although not allowed, but to get an idea if this is anything better: if you place a circle inside the square area touching the four sides the area will be equal to $\pi \times 250^2 m^2$ and the perimeter will be $\pi \times 500m$, so the enclosed-area/wall-length ratio will also be 125.

However, if you cut off from the square four triangles with sides 75m, 75m and $75\sqrt{2}m$ the total area becomes 238750m² and the perimeter becomes $1400 + 300\sqrt{2}m$. So this gives an enclosed-area/wall-length ratio of 130.87, which is significantly better.

Problem 315: Digital Root Clocks

Link

Tags: Number Theory

Problem 316: Numbers in Decimal Expansions

Link

Tags: Number Theory, Probability, Sequences

Let $p = p_1 p_2 p_3 \dots$ be an infinite sequence of random digits, selected from $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ with equal probability.

It can be seen that p corresponds to the real number $0.p_1 p_2 p_3 \dots$

It can also be seen that choosing a random real number from the interval $[0, 1)$ is equivalent to choosing an infinite sequence of random digits selected from $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ with equal probability.

For any positive integer n with d decimal digits, let k be the smallest index such that $p_k, p_{k+1}, \dots, p_{k+d-1}$ are the decimal digits of n , in the same order.

Also, let $g(n)$ be the expected value of k ; it can be proven that $g(n)$ is always finite and, interestingly, always an integer number.

For example, if $n = 535$, then

for $p = 31415926535897 \dots$, we get $k = 9$

for $p = 355287143650049560000490848764084685354 \dots$, we get $k = 36$

etc and we find that $g(535) = 1008$.

Given that $\sum_{n=2}^{999} g\left(\left\lfloor \frac{10^6}{n} \right\rfloor\right) = 27280188$, find $\sum_{n=2}^{999999} g\left(\left\lfloor \frac{10^{16}}{n} \right\rfloor\right)$.

Note: $\lfloor x \rfloor$ represents the floor function.

Problem 317: Firecracker

Link

Tags: Geometry

A firecracker explodes at a height of $100m$ above level ground. It breaks into a large number of very small fragments, which move in every direction; all of them have the same initial velocity of $20m/s$.

We assume that the fragments move without air resistance, in a uniform gravitational field with $g = 9.81m/s^2$.

Find the volume (in m^3) of the region through which the fragments move before reaching the ground. Give your answer rounded to four decimal places.

Problem 318: 2011 Nines

Link

Tags: Game Theory

Consider the real number $\sqrt{2} + \sqrt{3}$.

When we calculate the even powers of $\sqrt{2} + \sqrt{3}$ we get:

$$(\sqrt{2} + \sqrt{3})^2 = 9.898979485566356 \dots$$

$$(\sqrt{2} + \sqrt{3})^4 = 97.98979485566356 \dots$$

$$(\sqrt{2} + \sqrt{3})^6 = 969.998969071069263 \dots$$

$$(\sqrt{2} + \sqrt{3})^8 = 9601.99989585502907 \dots$$

$$(\sqrt{2} + \sqrt{3})^{10} = 95049.999989479221 \dots$$

$$\{(\sqrt{2} + \sqrt{3})^{12}\} = 940897.99999989371855 \dots$$

$$\{(\sqrt{2} + \sqrt{3})^{14}\} = 9313929.99999989263 \dots$$

$$\{(\sqrt{2} + \sqrt{3})^{16}\} = 92198401.9999998915 \dots$$

It looks as if the number of consecutive nines at the beginning of the fractional part of these powers is non-decreasing.

In fact it can be proven that the fractional part of $(\sqrt{2} + \sqrt{3})^{2n}$ approaches 1 for large n .

Consider all real numbers of the form $\sqrt{p} + \sqrt{q}$ with p and q positive integers and $p < q$, such that the fractional part of $(\sqrt{p} + \sqrt{q})^{2n}$ approaches 1 for large n .

Let $C(p, q, n)$ be the number of consecutive nines at the beginning of the fractional part of $(\sqrt{p} + \sqrt{q})^{2n}$.

Let $N(p, q)$ be the minimal value of n such that $C(p, q, n) \geq 2011$.

Find $\sum N(p, q)$ for $p + q \leq 2011$.

Problem 319: Bounded Sequences

Link

Tags: Sequences

Let $x_1, x_2, \dots, x_n$ be a sequence of length n such that:

- $x_1 = 2$
- for all $1 \leq i \leq n$: $x_{i-1} \mid x_i$
- for all i and j with $1 \leq i, j \leq n$: $(x_i)^j \mid (x_j + 1)^i$.

There are only five such sequences of length 2, namely: $\{2, 4\}$, $\{2, 5\}$, $\{2, 6\}$, $\{2, 7\}$ and $\{2, 8\}$.

There are 293 such sequences of length 5; three examples are given below:

$\{2, 5, 11, 25, 55\}$, $\{2, 6, 14, 36, 88\}$, $\{2, 8, 22, 64, 181\}$.

Let $t(n)$ denote the number of such sequences of length n .

You are given that $t(10) = 86195$ and $t(20) = 5227991891$.

Find $t(10^{10})$ and give your answer modulo 10^9 .

Problem 320: Factorials Divisible by a Huge Integer

[Link](#)

Tags: Number Theory

Let $N(i)$ be the smallest integer n such that $n!$ is divisible by $(i!)^{1234567890}$

Let $S(u) = \sum N(i)$ for $10 \leq i \leq u$.

$S(1000) = 614538266565663$.

Find $S(1\,000\,000) \bmod 10^{18}$.

Problem 321: Swapping Counters

[Link](#)

Tags: Number Theory

A horizontal row comprising of $2n + 1$ squares has n red counters placed at one end and n blue counters at the other end, being separated by a single empty square in the centre. For example, when $n = 3$.

Problem 322: Binomial Coefficients Divisible by 10

[Link](#)

Tags: Combinatorics

Let $T(m, n)$ be the number of the binomial coefficients iC_n that are divisible by 10 for $n \leq im$ (i, m and n are positive integers).

You are given that $T(10^9, 10^7 - 10) = 989697000$.

Find $T(10^{18}, 10^{12} - 10)$.

Problem 323: Bitwise-OR Operations on Random Integers

[Link](#)

Tags: Game Theory, Number Theory, Probability, Sequences

Let $y_0, y_1, y_2, \dots$ be a sequence of random unsigned 32-bit integers (i.e. $0 \leq y_i 2^{32}$, every value equally likely).

For the sequence x_i the following recursion is given:

- $x_0 = 0$ and

- $x_i = x_{i-1} \mid y_{i-1}$, for $i \geq 1$. ($\mid$ is the bitwise-OR operator).

It can be seen that eventually there will be an index N such that $x_i = 2^{32} - 1$ (a bit-pattern of all ones) for all $i \geq N$.

Find the expected value of N .

Give your answer rounded to 10 digits after the decimal point.

Problem 324: Building a Tower

[Link](#)

Tags: Number Theory

Let $f(n)$ represent the number of ways one can fill a $3 \times 3 \times n$ tower with blocks of $2 \times 1 \times 1$. You're allowed to rotate the blocks in any way you like; however, rotations, reflections etc of the tower itself are counted as distinct.

For example (with $q = 100000007$): $f(2) = 229$, $f(4) = 117805$, $f(10) \bmod q = 96149360$, $f(10^3) \bmod q = 24806056$, $f(10^6) \bmod q = 30808124$.

Find $f(10^{10000}) \bmod 1000000007$.

Problem 325: Stone Game II

[Link](#)

Tags: Game Theory, Number Theory

A game is played with two piles of stones and two players.

On each player's turn, the player may remove a number of stones from the larger pile.

The number of stones removed must be a positive multiple of the number of stones in the smaller pile.

E.g. Let the ordered pair $(6, 14)$ describe a configuration with 6 stones in the smaller pile and 14 stones in the larger pile, then the first player can remove 6 or 12 stones from the larger pile.

The player taking all the stones from a pile wins the game.

A winning configuration is one where the first player can force a win. For example, $(1, 5)$, $(2, 6)$, and $(3, 12)$ are winning configurations because the first player can immediately remove all stones in the second pile.

A losing configuration is one where the second player can force a win, no matter what the first player does. For example, $(2, 3)$ and $(3, 4)$ are losing configurations: any legal move leaves a winning configuration for the second player.

Define $S(N)$ as the sum of $(x_i + y_i)$ for all losing configurations (x_i, y_i) , $0 \leq x_i, y_i \leq N$.

We can verify that $S(10) = 211$ and $S(10^4) = 230312207313$.

Find $S(10^{16}) \bmod 7^{10}$.

Problem 326: Modulo Summations

Link

Tags: Sequences

Let a_n be a sequence recursively defined by: $a_1 = 1$, $a_n = \left(\sum_{k=1}^{n-1} k \cdot a_k \right) \bmod n$.

So the first 10 elements of a_n are: 1, 1, 0, 3, 0, 3, 5, 4, 1, 9.

Let $f(N, M)$ represent the number of pairs (p, q) such that:

$$1 \leq p \leq q \leq N \quad \text{font-family: inherit; and} \quad \left(\sum_{i=p}^q a_i \right) \bmod M = 0$$

It can be seen that $f(10, 10) = 4$ with the pairs $(3, 3)$, $(5, 5)$, $(7, 9)$ and $(9, 10)$.

You are also given that $f(10^4, 10^3) = 97158$.

Find $f(10^{12}, 10^6)$.

Problem 327: Rooms of Doom

Link

Tags: Sequences

A series of three rooms are connected to each other by automatic doors.

Problem 328: Lowest-cost Search

Link

Tags: Game Theory

We are trying to find a hidden number selected from the set of integers $\{1, 2, \dots, n\}$ by asking questions. Each number (question) we ask, has a cost equal to the number asked and we get one of three possible answers:

- “Your guess is lower than the hidden number”, or
- “Yes, that’s it!”, or
- “Your guess is higher than the hidden number”.

Given the value of n , an optimal strategy minimizes the total cost (i.e. the sum of all the questions asked) for the worst possible case. E.g.

If $n = 3$, the best we can do is obviously to ask the number “2”. The answer will immediately lead us to find the hidden number (at a total cost = 2).

If $n = 8$, we might decide to use a “binary search” type of strategy: Our first question would be “4” and if the hidden number is higher than 4 we will need one or two additional questions.

Let our second question be “6”. If the hidden number is still higher than 6, we will need a third question in order to discriminate between 7 and 8.

Thus, our third question will be “7” and the total cost for this worst-case scenario will be $4 + 6 + 7 = 17$.

We can improve considerably the worst-case cost for $n = 8$, by asking “5” as our first question.

If we are told that the hidden number is higher than 5, our second question will be “7”, then we’ll know for certain what the hidden number is (for a total cost of $5 + 7 = 12$).

If we are told that the hidden number is lower than 5, our second question will be “3” and if the hidden number is lower than 3 our third question will be “1”, giving a total cost of $5 + 3 + 1 = 9$.

Since 129, the worst-case cost for this strategy is 12. That’s better than what we achieved previously with the “binary search” strategy; it is also better than or equal to any other strategy.

So, in fact, we have just described an optimal strategy for $n = 8$.

Let $C(n)$ be the worst-case cost achieved by an optimal strategy for n , as described above.

Thus $C(1) = 0$, $C(2) = 1$, $C(3) = 2$ and $C(8) = 12$.

Similarly, $C(100) = 400$ and $\sum_{n=1}^{100} C(n) = 17575$.

Find $\sum_{n=1}^{200000} C(n)$.

Problem 329: Prime Frog

[Link](#)

Tags: Number Theory, Probability, Sequences

Susan has a prime frog.

Her frog is jumping around over 500 squares numbered 1 to 500. He can only jump one square to the left or to the right, with equal probability, and he cannot jump outside the range $[1; 500]$. (if it lands at either end, it automatically jumps to the only available square on the next move.)

When he is on a square with a prime number on it, he croaks 'P' (PRIME) with probability $2/3$ or 'N' (NOT PRIME) with probability $1/3$ just before jumping to the next square.

When he is on a square with a number on it that is not a prime he croaks 'P' with probability $1/3$ or 'N' with probability $2/3$ just before jumping to the next square.

Given that the frog's starting position is random with the same probability for every square, and given that she listens to his first 15 croaks, what is the probability that she hears the sequence PPPNPPPNPPNPN?

Give your answer as a fraction p/q in reduced form.

Problem 330: Euler's Number

[Link](#)

Tags: Sequences

An infinite sequence of real numbers $a(n)$ is defined for all integers n as follows:

$$a(n) = \begin{cases} 1 & n=0 \\ \sum_{i=1}^{\infty} \frac{a(n-i)}{i!} & n \geq 1 \end{cases}$$

For example,

$$a(0) = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots = e - 1$$

$$a(1) = \frac{e-1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots = 2e - 3$$

$$a(2) = \frac{2e-3}{1!} + \frac{e-1}{2!} + \frac{1}{3!} + \cdots = \frac{7}{2}e - 6$$

with $e = 2.7182818\ldots$ being Euler's constant.

It can be shown that $a(n)$ is of the form $\frac{A(n)e + B(n)}{n!}$ for integers $A(n)$ and $B(n)$.

For example, $a(10) = \frac{328161643e - 652694486}{10!}$.

Find $A(10^9) + B(10^9)$ and give your answer mod 77 777 777.

Problem 331: Cross Flips

Link

Tags: Combinatorics, Game Theory

$N \times N$ disks are placed on a square game board. Each disk has a black side and white side.

At each turn, you may choose a disk and flip all the disks in the same row and the same column as this disk: thus $2 \times N - 1$ disks are flipped. The game ends when all disks show their white side. The following example shows a game on a 5×5 board.

Problem 332: Spherical Triangles

Link

Tags: Geometry

A spherical triangle is a figure formed on the surface of a sphere by three great circular arcs intersecting pairwise in three vertices.

Problem 333: Special Partitions

Link

Tags: Combinatorics, Game Theory, Number Theory

All positive integers can be partitioned in such a way that each and every term of the partition can be expressed as $2^i \times 3^j$, where $i, j \geq 0$.

Let's consider only such partitions where none of the terms can divide any of the other terms.

For example, the partition of $17 = 2 + 6 + 9 = (2^1 \times 3^0 + 2^1 \times 3^1 + 2^0 \times 3^2)$ would not be valid since 2 can divide 6. Neither would the partition $17 = 16 + 1 = (2^4 \times 3^0 + 2^0 \times 3^0)$ since 1 can divide 16. The only valid partition of 17 would be $8 + 9 = (2^3 \times 3^0 + 2^0 \times 3^2)$.

Many integers have more than one valid partition, the first being 11 having the following two partitions.

$$11 = 2 + 9 = (2^1 \times 3^0 + 2^0 \times 3^2)$$

$$11 = 8 + 3 = (2^3 \times 3^0 + 2^0 \times 3^1)$$

Let's define $P(n)$ as the number of valid partitions of n . For example, $P(11) = 2$.

Let's consider only the prime integers q which would have a single valid partition such as $P(17)$.

The sum of the primes q such that $P(q) = 1$ equals 233.

Find the sum of the primes q such that $P(q) = 1$.

Problem 334: Spilling the Beans

[Link](#)

Tags: Game Theory

In Plato's heaven, there exist an infinite number of bowls in a straight line.

Each bowl either contains some or none of a finite number of beans.

A child plays a game, which allows only one kind of move: removing two beans from any bowl, and putting one in each of the two adjacent bowls. The game ends when each bowl contains either one or no beans.

For example, consider two adjacent bowls containing 2 and 3 beans respectively, all other bowls being empty. The following eight moves will finish the game:

Problem 335: Gathering the Beans

[Link](#)

Tags: Geometry

Whenever Peter feels bored, he places some bowls, containing one bean each, in a circle. After this, he takes all the beans out of a certain bowl and drops them one by one in the bowls going clockwise. He repeats this, starting from the bowl he dropped the last bean in, until the initial situation appears again. For example with 5 bowls he acts as follows:

Problem 336: Maximix Arrangements

Link

Tags: Combinatorics

A train is used to transport four carriages in the order: ABCD. However, sometimes when the train arrives to collect the carriages they are not in the correct order.

To rearrange the carriages they are all shunted on to a large rotating turntable. After the carriages are uncoupled at a specific point the train moves off the turntable pulling the carriages still attached with it. The remaining carriages are rotated 180 degrees. All of the carriages are then rejoined and this process is repeated as often as necessary in order to obtain the least number of uses of the turntable.

Some arrangements, such as ADCB, can be solved easily: the carriages are separated between A and D, and after DCB are rotated the correct order has been achieved.

However, Simple Simon, the train driver, is not known for his efficiency, so he always solves the problem by initially getting carriage A in the correct place, then carriage B, and so on.

Using four carriages, the worst possible arrangements for Simon, which we shall call maximix arrangements, are DACB and DBAC; each requiring him five rotations (although, using the most efficient approach, they could be solved using just three rotations). The process he uses for DACB is shown below.

Problem 337: Totient Stairstep Sequences

Link

Tags: Number Theory, Sequences

Let $\{a_1, a_2, \dots, a_n\}$ be an integer sequence of length n such that:

- $a_1 = 6$
- for all $1 \leq i < n$: $\phi(a_i)\phi(a_{i+1})a_i a_{i+1} \mid 1$

Let $S(N)$ be the number of such sequences with $a_n \leq N$.

For example, $S(10) = 4$: $\{6\}$, $\{6, 8\}$, $\{6, 8, 9\}$ and $\{6, 10\}$.

We can verify that $S(100) = 482073668$ and $S(10\,000) \bmod 10^8 = 73808307$.

Find $S(20\,000\,000) \bmod 10^8$.

1 ϕ denotes Euler's totient function.

Problem 338: Cutting Rectangular Grid Paper

Link

Tags: Combinatorics, Geometry

A rectangular sheet of grid paper with integer dimensions $w \times h$ is given. Its grid spacing is 1.

When we cut the sheet along the grid lines into two pieces and rearrange those pieces without overlap, we can make new rectangles with different dimensions.

For example, from a sheet with dimensions 9×4 , we can make rectangles with dimensions 18×2 , 12×3 and 6×6 by cutting and rearranging as below:

Problem 339: Peredur Fab Efracw

Link

Tags: Probability

“And he came towards a valley, through which ran a river; and the borders of the valley were wooded, and on each side of the river were level meadows. And on one side of the river he saw a flock of white sheep, and on the other a flock of black sheep. And whenever one of the white sheep bleated, one of the black sheep would cross over and become white; and when one of the black sheep bleated, one of the white sheep would cross over and become black.” en.wikisource.org

Initially each flock consists of n sheep. Each sheep (regardless of colour) is equally likely to be the next sheep to bleat. After a sheep has bleated and a sheep from the other flock has crossed over, Peredur may remove a number of white sheep in order to maximize the expected final number of black sheep. Let $E(n)$ be the expected final number of black sheep if Peredur uses an optimal strategy.

You are given that $E(5) = 6.871346$ rounded to 6 places behind the decimal point.

Find $E(10\,000)$ and give your answer rounded to 6 places behind the decimal point.

Problem 340: Crazy Function

Link

Tags: Number Theory

For fixed integers a, b, c , define the crazy function $F(n)$ as follows:

$F(n) = n - c$ for all $n \leq b$

$F(n) = F(a + F(a + F(a + F(a + n))))$ for all $n > b$.

Also, define $S(a, b, c) = \sum_{n=0}^b F(n)$.

For example, if $a = 50$, $b = 2000$ and $c = 40$, then $F(0) = 3240$ and $F(2000) = 2040$.

Also, $S(50, 2000, 40) = 5204240$.

Find the last 9 digits of $S(21^7, 7^{21}, 12^7)$.

Problem 341: Golomb's Self-describing Sequence

Link

Tags: Algebra, Sequences

The Golomb's self-describing sequence ($G(n)$) is the only nondecreasing sequence of natural numbers such that n appears exactly $G(n)$ times in the sequence. The values of $G(n)$ for the first few n are

n	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	...			
$G(n)$	1	2	2	3	3	4	4	4	5
5	5	6	6	6	6	...			

Problem 342: The Totient of a Square Is a Cube

Link

Tags: Number Theory

Consider the number 50.

$50^2 = 2500 = 2^2 \times 5^4$, so $\phi(2500) = 2 \times 4 \times 5^3 = 8 \times 5^3 = 2^3 \times 5^3$. 1

So 2500 is a square and $\phi(2500)$ is a cube.

Find the sum of all numbers n , $1 \leq n \leq 10^{10}$ such that $\phi(n^2)$ is a cube.

1 ϕ denotes Euler's totient function.

Problem 343: Fractional Sequences

Link

Tags: Sequences

For any positive integer k , a finite sequence a_i of fractions x_i/y_i is defined by:

$a_1 = 1/k$ and

$a_i = (x_{i-1} + 1)/(y_{i-1} - 1)$ reduced to lowest terms for $i \geq 1$.

When a_i reaches some integer n , the sequence stops. (That is, when $y_i = 1$.)

Define $f(k) = n$.

For example, for $k = 20$:

$1/20 \rightarrow 2/19 \rightarrow 3/18 = 1/6 \rightarrow 2/5 \rightarrow 3/4 \rightarrow 4/3 \rightarrow 5/2 \rightarrow 6/1 = 6$

So $f(20) = 6$.

Also $f(1) = 1$, $f(2) = 2$, $f(3) = 1$ and $\sum f(k^3) = 118937$ for $1 \leq k \leq 100$.

Find $\sum f(k^3)$ for $1 \leq k \leq 2 \times 10^6$.

Problem 344: Silver Dollar Game

Link

Tags: Combinatorics, Game Theory, Probability

One variant of N.G. de Bruijn's silver dollar game can be described as follows:

On a strip of squares a number of coins are placed, at most one coin per square. Only one coin, called the silver dollar, has any value. Two players take turns making moves. At each turn a player must make either a regular or a special move.

A regular move consists of selecting one coin and moving it one or more squares to the left. The coin cannot move out of the strip or jump on or over another coin.

Alternatively, the player can choose to make the special move of pocketing the leftmost coin rather than making a regular move. If no regular moves are possible, the player is forced to pocket the leftmost coin.

The winner is the player who pockets the silver dollar.

Problem 345: Matrix Sum

Link

Tags: Algebra

We define the Matrix Sum of a matrix as the maximum possible sum of matrix elements such that none of the selected elements share the same row or column.

For example, the Matrix Sum of the matrix below equals 3315 ($= 863 + 383 + 343 + 959 + 767$):

7	53	183	439	863
497	383	563	79	973
287	63	343	169	583
627	343	773	959	943
767	473	103	699	303

Find the Matrix Sum of:

7	53	183	439	863	497	383	563	79	973	287	63	343	169	583
627	343	773	959	943	767	473	103	699	303	957	703	583	639	913
447	283	463	29	23	487	463	993	119	883	327	493	423	159	743
217	623	3	399	853	407	103	983	89	463	290	516	212	462	350
960	376	682	962	300	780	486	502	912	800	250	346	172	812	350
870	456	192	162	593	473	915	45	989	873	823	965	425	329	803
973	965	905	919	133	673	665	235	509	613	673	815	165	992	326
322	148	972	962	286	255	941	541	265	323	925	281	601	95	973
445	721	11	525	473	65	511	164	138	672	18	428	154	448	848
414	456	310	312	798	104	566	520	302	248	694	976	430	392	198
184	829	373	181	631	101	969	613	840	740	778	458	284	760	390
821	461	843	513	17	901	711	993	293	157	274	94	192	156	574
34	124	4	878	450	476	712	914	838	669	875	299	823	329	699
815	559	813	459	522	788	168	586	966	232	308	833	251	631	107
813	883	451	509	615	77	281	613	459	205	380	274	302	35	805

Problem 346: Strong Repunits

[Link](#)

Tags: Number Theory

The number 7 is special, because 7 is 111 written in base 2, and 11 written in base 6 (i.e. $7_{10} = 11_6 = 111_2$). In other words, 7 is a repunit in at least two bases b .

We shall call a positive integer with this property a strong repunit. It can be verified that there are 8 strong repunits below 50: $\{1, 7, 13, 15, 21, 31, 40, 43\}$.

Furthermore, the sum of all strong repunits below 1000 equals 15864.

Find the sum of all strong repunits below 10^{12} .

Problem 347: Largest Integer Divisible by Two Primes

[Link](#)

Tags: Number Theory

The largest integer ≤ 100 that is only divisible by both the primes 2 and 3 is 96, as $96 = 32 \times 3 = 2^5 \times 3$. For two distinct primes p and q let $M(p, q, N)$ be the largest positive integer $\leq N$ only divisible by both p and q and $M(p, q, N) = 0$ if such a positive integer does not exist.

E.g. $M(2, 3, 100) = 96$.

$M(3, 5, 100) = 75$ and not 90 because 90 is divisible by 2, 3 and 5.

Also $M(2, 73, 100) = 0$ because there does not exist a positive integer ≤ 100 that is divisible by both 2 and 73.

Let $S(N)$ be the sum of all distinct $M(p, q, N)$. $S(100) = 2262$.

Find $S(10\,000\,000)$.

Problem 348: Sum of a Square and a Cube

[Link](#)

Tags: Number Theory

Many numbers can be expressed as the sum of a square and a cube. Some of them in more than one way.

Consider the palindromic numbers that can be expressed as the sum of a square and a cube, both greater than 1, in exactly 4 different ways.

For example, 5229225 is a palindromic number and it can be expressed in exactly 4 different ways:

$$2285^2 + 20^3$$

$$2223^2 + 66^3$$

$$1810^2 + 125^3$$

$$1197^2 + 156^3$$

Find the sum of the five smallest such palindromic numbers.

Problem 349: Langton's Ant

[Link](#)

Tags: Combinatorics, Game Theory

An ant moves on a regular grid of squares that are coloured either black or white.

The ant is always oriented in one of the cardinal directions (left, right, up or down) and moves from square to adjacent square according to the following rules:

- if it is on a black square, it flips the colour of the square to white, rotates 90 degrees counterclockwise and moves forward one square.
- if it is on a white square, it flips the colour of the square to black, rotates 90 degrees clockwise and moves forward one square.

Starting with a grid that is entirely white, how many squares are black after 10^{18} moves of the ant?

Problem 350: Constraining the Least Greatest and the Greatest Least

[Link](#)

Tags: Number Theory, Sequences

A list of size n is a sequence of n natural numbers. Examples are $(2, 4, 6)$, $(2, 6, 4)$, $(10, 6, 15, 6)$, and (11) .

The greatest common divisor, or gcd, of a list is the largest natural number that divides all entries of the list. Examples: $\gcd(2, 6, 4) = 2$, $\gcd(10, 6, 15, 6) = 1$ and $\gcd(11) = 11$.

The least common multiple, or lcm, of a list is the smallest natural number divisible by each entry of the list. Examples: $\text{lcm}(2, 6, 4) = 12$, $\text{lcm}(10, 6, 15, 6) = 30$ and $\text{lcm}(11) = 11$.

Let $f(G, L, N)$ be the number of lists of size N with $\text{gcd} \geq G$ and $\text{lcm} \leq L$. For example:

$$f(10, 100, 1) = 91.$$

$$f(10, 100, 2) = 327.$$

$$f(10, 100, 3) = 1135.$$

$$f(10, 100, 1000) \bmod 101^4 = 3286053.$$

Find $f(10^6, 10^{12}, 10^{18}) \bmod 101^4$.

Problem 351: Hexagonal Orchards

[Link](#)

Tags: Combinatorics, Game Theory

A hexagonal orchard of order n is a triangular lattice made up of points within a regular hexagon with side n . The following is an example of a hexagonal orchard of order 5:

Problem 352: Blood Tests

[Link](#)

Tags: Game Theory, Probability

Each one of the 25 sheep in a flock must be tested for a rare virus, known to affect 2% of the sheep population. An accurate and extremely sensitive PCR test exists for blood samples, producing a clear positive / negative result, but it is very time-consuming and expensive.

Because of the high cost, the vet-in-charge suggests that instead of performing 25 separate tests, the following procedure can be used instead:

The sheep are split into 5 groups of 5 sheep in each group. For each group, the 5 samples are mixed together and a single test is performed. Then,

- If the result is negative, all the sheep in that group are deemed to be virus-free.
- If the result is positive, 5 additional tests will be performed (a separate test for each animal) to determine the affected individual(s).

Since the probability of infection for any specific animal is only 0.02, the first test (on the pooled samples) for each group will be:

- Negative (and no more tests needed) with probability $0.98^5 = 0.9039207968$.
- Positive (5 additional tests needed) with probability $1 - 0.9039207968 = 0.0960792032$.

Thus, the expected number of tests for each group is $1 + 0.0960792032 \times 5 = 1.480396016$.

Consequently, all 5 groups can be screened using an average of only $1.480396016 \times 5 = \mathbf{7.40198008}$ tests, which represents a huge saving of more than 70%!

Although the scheme we have just described seems to be very efficient, it can still be improved considerably (always assuming that the test is sufficiently sensitive and that there are no adverse effects caused by mixing different samples). E.g.:

- We may start by running a test on a mixture of all the 25 samples. It can be verified that in about 60.35% of the cases this test will be negative, thus no more tests will be needed. Further testing will only be required for the remaining 39.65% of the cases.
- If we know that at least one animal in a group of 5 is infected and the first 4 individual tests come out negative, there is no need to run a test on the fifth animal (we know that it must be infected).
- We can try a different number of groups / different number of animals in each group, adjusting those numbers at each level so that the total expected number of tests will be minimised.

To simplify the very wide range of possibilities, there is one restriction we place when devising the most cost-efficient testing scheme: whenever we start with a mixed sample, all the sheep contributing to that sample must be fully screened (i.e. a verdict of infected / virus-free must be reached for all of them) before we start examining any other animals.

For the current example, it turns out that the most cost-efficient testing scheme (we'll call it the optimal strategy) requires an average of just **4.155452** tests!

Using the optimal strategy, let $T(s, p)$ represent the average number of tests needed to screen a flock of s sheep for a virus having probability p to be present in any individual.

Thus, rounded to six decimal places, $T(25, 0.02) = 4.155452$ and $T(25, 0.10) = 12.702124$.

Find $\sum T(10000, p)$ for $p = 0.01, 0.02, 0.03, \dots 0.50$.

Give your answer rounded to six decimal places.

Problem 353: Risky Moon

Link

Tags: Game Theory, Geometry, Number Theory

A moon could be described by the sphere $C(r)$ with centre $(0, 0, 0)$ and radius r .

There are stations on the moon at the points on the surface of $C(r)$ with integer coordinates. The station at $(0, 0, r)$ is called North Pole station, the station at $(0, 0, -r)$ is called South Pole station.

All stations are connected with each other via the shortest road on the great arc through the stations. A journey between two stations is risky. If d is the length of the road between two stations, $\left(\frac{d}{\pi r}\right)^2$ is a measure for the risk of the journey (let us call it the risk of the road). If the journey includes more than two stations, the risk of the journey is the sum of risks of the used roads.

A direct journey from the North Pole station to the South Pole station has the length πr and risk 1. The journey from the North Pole station to the South Pole station via $(0, r, 0)$ has the same length, but a smaller risk:

$$\left(\frac{\frac{1}{2}\pi r}{\pi r}\right)^2 + \left(\frac{\frac{1}{2}\pi r}{\pi r}\right)^2 = 0.5$$

The minimal risk of a journey from the North Pole station to the South Pole station on $C(r)$ is $M(r)$.

You are given that $M(7) = 0.1784943998$ rounded to 10 digits behind the decimal point.

Find $\sum_{n=1}^{15} M(2^n - 1)$.

Give your answer rounded to 10 digits behind the decimal point in the form a.bcd efghijk.

Problem 354: Distances in a Bee's Honeycomb

Link

Tags: Number Theory

Consider a honey bee's honeycomb where each cell is a perfect regular hexagon with side length 1.

Problem 355: Maximal Coprime Subset

Link

Tags: Combinatorics, Number Theory

Define $\text{Co}(n)$ to be the maximal possible sum of a set of mutually co-prime elements from $\{1, 2, \dots, n\}$. For example $\text{Co}(10)$ is 30 and hits that maximum on the subset $\{1, 5, 7, 8, 9\}$.

You are given that $\text{Co}(30) = 193$ and $\text{Co}(100) = 1356$.

Find $\text{Co}(200000)$.

Problem 356: Largest Roots of Cubic Polynomials

Link

Tags: Algebra, Number Theory

Let a_n be the largest real root of a polynomial $g(x) = x^3 - 2^n \cdot x^2 + n$.

For example, $a_2 = 3.86619826\dots$

Find the last eight digits of $\sum_{i=1}^{30} \lfloor a_i^{987654321} \rfloor$.

Note: $\lfloor a \rfloor$ represents the floor function.

Problem 357: Prime Generating Integers

Link

Tags: Number Theory

Consider the divisors of 30: 1, 2, 3, 5, 6, 10, 15, 30.

It can be seen that for every divisor d of 30, $d + 30/d$ is prime.

Find the sum of all positive integers n not exceeding 100 000 000 such that for every divisor d of n , $d + n/d$ is prime.

Problem 358: Cyclic Numbers

Link

Tags: Number Theory

A cyclic number with n digits has a very interesting property:

When it is multiplied by $1, 2, 3, 4, \dots, n$, all the products have exactly the same digits, in the same order, but rotated in a circular fashion!

The smallest cyclic number is the 6-digit number 142857:

$$142857 \times 1 = 142857$$

$$142857 \times 2 = 285714$$

$$142857 \times 3 = 428571$$

$$142857 \times 4 = 571428$$

$$142857 \times 5 = 714285$$

$$142857 \times 6 = 857142$$

The next cyclic number is 0588235294117647 with 16 digits :

$$0588235294117647 \times 1 = 0588235294117647$$

$$0588235294117647 \times 2 = 1176470588235294$$

$$0588235294117647 \times 3 = 1764705882352941$$

...

$$0588235294117647 \times 16 = 9411764705882352$$

Note that for cyclic numbers, leading zeros are important.

There is only one cyclic number for which, the eleven leftmost digits are 00000000137 and the five rightmost digits are 56789 (i.e., it has the form 00000000137...56789 with an unknown number of digits in the middle). Find the sum of all its digits.

Problem 359: Hilbert's New Hotel

Link

Tags: Game Theory, Number Theory

An infinite number of people (numbered 1, 2, 3, etc.) are lined up to get a room at Hilbert's newest infinite hotel. The hotel contains an infinite number of floors (numbered 1, 2, 3, etc.), and each floor contains an infinite number of rooms (numbered 1, 2, 3, etc.).

Initially the hotel is empty. Hilbert declares a rule on how the n th person is assigned a room: person n gets the first vacant room in the lowest numbered floor satisfying either of the following:

- the floor is empty

- the floor is not empty, and if the latest person taking a room in that floor is person m , then $m + n$ is a perfect square

Person 1 gets room 1 in floor 1 since floor 1 is empty.

Person 2 does not get room 2 in floor 1 since $1 + 2 = 3$ is not a perfect square.

Person 2 instead gets room 1 in floor 2 since floor 2 is empty.

Person 3 gets room 2 in floor 1 since $1 + 3 = 4$ is a perfect square.

Eventually, every person in the line gets a room in the hotel.

Define $P(f, r)$ to be n if person n occupies room r in floor f , and 0 if no person occupies the room. Here are a few examples:

$$P(1, 1) = 1$$

$$P(1, 2) = 3$$

$$P(2, 1) = 2$$

$$P(10, 20) = 440$$

$$P(25, 75) = 4863$$

$$P(99, 100) = 19454$$

Find the sum of all $P(f, r)$ for all positive f and r such that $f \times r = 71328803586048$ and give the last 8 digits as your answer.

Problem 360: Scary Sphere

Link

Tags: Geometry

Given two points (x_1, y_1, z_1) and (x_2, y_2, z_2) in three dimensional space, the Manhattan distance between those points is defined as $|x_1 - x_2| + |y_1 - y_2| + |z_1 - z_2|$.

Let $C(r)$ be a sphere with radius r and center in the origin $O(0, 0, 0)$.

Let $I(r)$ be the set of all points with integer coordinates on the surface of $C(r)$.

Let $S(r)$ be the sum of the Manhattan distances of all elements of $I(r)$ to the origin O .

E.g. $S(45) = 34518$.

Find $S(10^{10})$.

Problem 361: Subsequence of Thue-Morse Sequence

Link

Tags: Number Theory, Sequences

The Thue-Morse sequence $\{T_n\}$ is a binary sequence satisfying:

- $T_0 = 0$
- $T_{2n} = T_n$
- $T_{2n+1} = 1 - T_n$

The first several terms of $\{T_n\}$ are given as follows:

01101001100101101001011001101001...

We define $\{A_n\}$ as the sorted sequence of integers such that the binary expression of each element appears as a subsequence in $\{T_n\}$.

For example, the decimal number 18 is expressed as 10010 in binary. 10010 appears in $\{T_n\}$ (T_8 to T_{12}), so 18 is an element of $\{A_n\}$.

The decimal number 14 is expressed as 1110 in binary. 1110 never appears in $\{T_n\}$, so 14 is not an element of $\{A_n\}$.

The first several terms of $\{A_n\}$ are given as follows:

n 0 1 2 3 4 5 6 7 8 9 10 11 12 ...

A_n 0 1 2 3 4 5 6 9 10 11 12 13 18 ...

We can also verify that $A_{100} = 3251$ and $A_{1000} = 80852364498$.

Find the last 9 digits of $\sum_{k=1}^{18} A_{10^k}$.

Problem 362: Squarefree Factors

Link

Tags: Number Theory

Consider the number 54.

54 can be factored in 7 distinct ways into one or more factors larger than 1:

54, 2×27 , 3×18 , 6×9 , $3 \times 3 \times 6$, $2 \times 3 \times 9$ and $2 \times 3 \times 3 \times 3$.

If we require that the factors are all squarefree only two ways remain: $3 \times 3 \times 6$ and $2 \times 3 \times 3 \times 3$.

Let's call $\text{Fsf}(n)$ the number of ways n can be factored into one or more squarefree factors larger than 1, so $\text{Fsf}(54) = 2$.

Let $S(n)$ be $\sum \text{Fsf}(k)$ for $k = 2$ to n .

$S(100) = 193$.

Find $S(10\,000\,000\,000)$.

Problem 363: Bézier Curves

[Link](#)

Tags: Number Theory

A cubic Bézier curve is defined by four points: P_0, P_1, P_2 , and P_3 .

Problem 364: Comfortable Distance

[Link](#)

Tags: Game Theory

There are N seats in a row. N people come after each other to fill the seats according to the following rules:

- If there is any seat whose adjacent seat(s) are not occupied take such a seat.
- If there is no such seat and there is any seat for which only one adjacent seat is occupied take such a seat.
- Otherwise take one of the remaining available seats.

Let $T(N)$ be the number of possibilities that N seats are occupied by N people with the given rules. The following figure shows $T(4) = 8$.

Problem 365: A Huge Binomial Coefficient

[Link](#)

Tags: Combinatorics, Number Theory

The binomial coefficient $\binom{10^{18}}{10^9}$ is a number with more than 9 billion (9×10^9) digits.

Let $M(n, k, m)$ denote the binomial coefficient $\binom{n}{k}$ modulo m .

Calculate $\sum M(10^{18}, 10^9, p \cdot q \cdot r)$ for $1000pqr5000$ and p, q, r prime.

Problem 366: Stone Game III

Link

Tags: Game Theory

Two players, Anton and Bernhard, are playing the following game.

There is one pile of n stones.

The first player may remove any positive number of stones, but not the whole pile.

Thereafter, each player may remove at most twice the number of stones his opponent took on the previous move.

The player who removes the last stone wins.

E.g. $n = 5$.

If the first player takes anything more than one stone the next player will be able to take all remaining stones.

If the first player takes one stone, leaving four, his opponent will take also one stone, leaving three stones.

The first player cannot take all three because he may take at most $2 \times 1 = 2$ stones. So let's say he takes also one stone, leaving 2. The second player can take the two remaining stones and wins.

So 5 is a losing position for the first player.

For some winning positions there is more than one possible move for the first player.

E.g. when $n = 17$ the first player can remove one or four stones.

Let $M(n)$ be the maximum number of stones the first player can take from a winning position at his first turn and $M(n) = 0$ for any other position.

$\sum M(n)$ for $n \leq 100$ is 728.

Find $\sum M(n)$ for $n \leq 10^{18}$. Give your answer modulo 10^8 .

Problem 367: Bozo Sort

Link

Tags: Combinatorics, Game Theory, Probability, Sequences

Bozo sort, not to be confused with the slightly less efficient bogo sort, consists out of checking if the input sequence is sorted and if not swapping randomly two elements. This is repeated until eventually the sequence is sorted.

If we consider all permutations of the first 4 natural numbers as input the expectation value of the number of swaps, averaged over all $4!$ input sequences is 24.75.

The already sorted sequence takes 0 steps.

In this problem we consider the following variant on bozo sort.

If the sequence is not in order we pick three elements at random and shuffle these three elements randomly.

All $3! = 6$ permutations of those three elements are equally likely.

The already sorted sequence will take 0 steps.

If we consider all permutations of the first 4 natural numbers as input the expectation value of the number of shuffles, averaged over all $4!$ input sequences is 27.5.

Consider as input sequences the permutations of the first 11 natural numbers.

Averaged over all $11!$ input sequences, what is the expected number of shuffles this sorting algorithm will perform?

Give your answer rounded to the nearest integer.

Problem 368: A Kempner-like Series

[Link](#)

Tags: Number Theory, Sequences

The harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$ is well known to be divergent.

If we however omit from this series every term where the denominator has a 9 in it, the series remarkably enough converges to approximately 22.9206766193.

This modified harmonic series is called the Kempner series.

Let us now consider another modified harmonic series by omitting from the harmonic series every term where the denominator has 3 or more equal consecutive digits. One can verify that out of the first 1200 terms of the harmonic series, only 20 terms will be omitted.

These 20 omitted terms are:

$$\frac{1}{111}, \frac{1}{222}, \frac{1}{333}, \frac{1}{444}, \frac{1}{555}, \frac{1}{666}, \frac{1}{777}, \frac{1}{888}, \frac{1}{999}, \frac{1}{1000}, \frac{1}{1110},$$

$$\frac{1}{1111}, \frac{1}{1112}, \frac{1}{1113}, \frac{1}{1114}, \frac{1}{1115}, \frac{1}{1116}, \frac{1}{1117}, \frac{1}{1118}, \frac{1}{1119}.$$

This series converges as well.

Find the value the series converges to.

Give your answer rounded to 10 digits behind the decimal point.

Problem 369: Badugi

[Link](#)

Tags: Combinatorics

In a standard 52 card deck of playing cards, a set of 4 cards is a Badugi if it contains 4 cards with no pairs and no two cards of the same suit.

Let $f(n)$ be the number of ways to choose n cards with a 4 card subset that is a Badugi. For example, there are 2598960 ways to choose five cards from a standard 52 card deck, of which 514800 contain a 4 card subset that is a Badugi, so $f(5) = 514800$.

Find $\sum f(n)$ for $4 \leq n \leq 13$.

Problem 370: Geometric Triangles

[Link](#)

Tags: Geometry

Let us define a geometric triangle as an integer sided triangle with sides $a \leq b \leq c$ so that its sides form a geometric progression, i.e. $b^2 = a \cdot c$

An example of such a geometric triangle is the triangle with sides $a = 144$, $b = 156$ and $c = 169$.

There are 861805 geometric triangles with perimeter $\leq 10^6$.

How many geometric triangles exist with perimeter $\leq 2.5 \cdot 10^{13}$?

Problem 371: Licence Plates

[Link](#)

Tags: Game Theory, Number Theory, Probability

Oregon licence plates consist of three letters followed by a three digit number (each digit can be from $[0..9]$).

While driving to work Seth plays the following game:

Whenever the numbers of two licence plates seen on his trip add to 1000 that's a win.

E.g. MIC-012 and HAN-988 is a win and RYU-500 and SET-500 too (as long as he sees them in the same trip).

Find the expected number of plates he needs to see for a win.

Give your answer rounded to 8 decimal places behind the decimal point.

Note: We assume that each licence plate seen is equally likely to have any three digit number on it.

Problem 372: Pencils of Rays

Link

Tags: Combinatorics

Let $R(M, N)$ be the number of lattice points (x, y) which satisfy $Mx \leq N$, $My \leq N$ and $\left\lfloor \frac{y^2}{x^2} \right\rfloor$ is odd.

We can verify that $R(0, 100) = 3019$ and $R(100, 10000) = 29750422$.

Find $R(2 \cdot 10^6, 10^9)$.

Note: $\lfloor x \rfloor$ represents the floor function.

Problem 373: Circumscribed Circles

Link

Tags: Geometry

Every triangle has a circumscribed circle that goes through the three vertices. Consider all integer sided triangles for which the radius of the circumscribed circle is integral as well.

Let $S(n)$ be the sum of the radii of the circumscribed circles of all such triangles for which the radius does not exceed n .

$S(100) = 4950$ and $S(1200) = 1653605$.

Find $S(10^7)$.

Problem 374: Maximum Integer Partition Product

[Link](#)

Tags: Combinatorics, Number Theory

An integer partition of a number n is a way of writing n as a sum of positive integers.

Partitions that differ only in the order of their summands are considered the same. A partition of n into distinct parts is a partition of n in which every part occurs at most once.

The partitions of 5 into distinct parts are:

5, 4 + 1 and 3 + 2.

Let $f(n)$ be the maximum product of the parts of any such partition of n into distinct parts and let $m(n)$ be the number of elements of any such partition of n with that product.

So $f(5) = 6$ and $m(5) = 2$.

For $n = 10$ the partition with the largest product is $10 = 2 + 3 + 5$, which gives $f(10) = 30$ and $m(10) = 3$.

And their product, $f(10) \cdot m(10) = 30 \cdot 3 = 90$.

It can be verified that

$\sum f(n) \cdot m(n)$ for $1 \leq n \leq 100 = 1683550844462$.

Find $\sum f(n) \cdot m(n)$ for $1 \leq n \leq 10^{14}$.

Give your answer modulo 982451653, the 50 millionth prime.

Problem 375: Minimum of Subsequences

[Link](#)

Tags: Game Theory, Probability, Sequences

Let S_n be an integer sequence produced with the following pseudo-random number generator:

$$\begin{aligned} S_0 &= 290797 \\ S_{n+1} &= S_n^2 \bmod 50515093 \end{aligned}$$

Let $A(i, j)$ be the minimum of the numbers $S_i, S_{i+1}, \dots, S_j$ for $i \leq j$.

Let $M(N) = \sum A(i, j)$ for $1 \leq i \leq j \leq N$.

We can verify that $M(10) = 432256955$ and $M(10\,000) = 3264567774119$.

Find $M(2\,000\,000\,000)$.

Problem 376: Nontransitive Sets of Dice

Link

Tags: Game Theory, Number Theory, Probability

Consider the following set of dice with nonstandard pips:

Die A : 1 4 4 4 4 4

Die B : 2 2 2 5 5 5

Die C : 3 3 3 3 3 6

A game is played by two players picking a die in turn and rolling it. The player who rolls the highest value wins.

If the first player picks die A and the second player picks die B we get

$$P(\text{second player wins}) = 7/121/2.$$

If the first player picks die B and the second player picks die C we get

$$P(\text{second player wins}) = 7/121/2.$$

If the first player picks die C and the second player picks die A we get

$$P(\text{second player wins}) = 25/361/2.$$

So whatever die the first player picks, the second player can pick another die and have a larger than 50% chance of winning.

A set of dice having this property is called a nontransitive set of dice.

We wish to investigate how many sets of nontransitive dice exist. We will assume the following conditions:

- There are three six-sided dice with each side having between 1 and N pips, inclusive.
- Dice with the same set of pips are equal, regardless of which side on the die the pips are located.
- The same pip value may appear on multiple dice; if both players roll the same value neither player wins.
- The sets of dice $\{A, B, C\}$, $\{B, C, A\}$ and $\{C, A, B\}$ are the same set.

For $N = 7$ we find there are 9780 such sets.

How many are there for $N = 30$?

Problem 377: Sum of Digits - Experience #13

Link

Tags: Number Theory

There are 16 positive integers that do not have a zero in their digits and that have a digital sum equal to 5, namely:

5, 14, 23, 32, 41, 113, 122, 131, 212, 221, 311, 1112, 1121, 1211, 2111 and 11111.

Their sum is 17891.

Let $f(n)$ be the sum of all positive integers that do not have a zero in their digits and have a digital sum equal to n .

Find $\sum_{i=1}^{17} f(13^i)$.

Give the last 9 digits as your answer.

Problem 378: Triangle Triples

Link

Tags: Geometry, Number Theory

Let $T(n)$ be the n th triangle number, so $T(n) = \frac{n(n+1)}{2}$.

Let $dT(n)$ be the number of divisors of $T(n)$.

E.g.: $T(7) = 28$ and $dT(7) = 6$.

Let $Tr(n)$ be the number of triples (i, j, k) such that $1 \leq ijk \leq n$ and $dT(i)dT(j)dT(k)$.

$Tr(20) = 14$, $Tr(100) = 5772$, and $Tr(1000) = 11174776$.

Find $Tr(60000000)$.

Give the last 18 digits of your answer.

Problem 379: Least Common Multiple Count

Link

Tags: Number Theory

Let $f(n)$ be the number of couples (x, y) with x and y positive integers, $x \leq y$ and the least common multiple of x and y equal to n .

Let g be the summatory function of f , i.e.: $g(n) = \sum f(i)$ for $1 \leq i \leq n$.

You are given that $g(10^6) = 37429395$.

Find $g(10^{12})$.

Problem 380: Amazing Mazes!

Link

Tags: Combinatorics, Game Theory, Number Theory

An $m \times n$ maze is an $m \times n$ rectangular grid with walls placed between grid cells such that there is exactly one path from the top-left square to any other square. The following are examples of a 9×12 maze and a 15×20 maze:

Let $C(m, n)$ be the number of distinct $m \times n$ mazes. Mazes which can be formed by rotation and reflection from another maze are considered distinct.

It can be verified that $C(1, 1) = 1$, $C(2, 2) = 4$, $C(3, 4) = 2415$, and $C(9, 12) = 2.5720e46$ (in scientific notation rounded to 5 significant digits).

Find $C(100, 500)$ and write your answer in scientific notation rounded to 5 significant digits.

When giving your answer, use a lowercase e to separate mantissa and exponent. E.g. if the answer is 1234567891011 then the answer format would be 1.2346e12.

Problem 381: (prime $- k$) Factorial

Link

Tags: Number Theory

For a prime p let $S(p) = (\sum (p - k)!) \bmod (p)$ for $1 \leq k \leq 5$.

For example, if $p = 7$,

$$(7 - 1)! + (7 - 2)! + (7 - 3)! + (7 - 4)! + (7 - 5)! = 6! + 5! + 4! + 3! + 2! = 720 + 120 + 24 + 6 + 2 = 872.$$

As $872 \bmod (7) = 4$, $S(7) = 4$.

It can be verified that $\sum S(p) = 480$ for $5 \leq p < 100$.

Find $\sum S(p)$ for $5 \leq p < 10^8$.

Problem 382: Generating Polygons

Link

Tags: Combinatorics, Game Theory, Geometry, Number Theory, Sequences

A polygon is a flat shape consisting of straight line segments that are joined to form a closed chain or circuit. A polygon consists of at least three sides and does not self-intersect.

A set S of positive numbers is said to generate a polygon P if:

- no two sides of P are the same length,
- the length of every side of P is in S , and
- S contains no other value.

For example:

The set $\{3, 4, 5\}$ generates a polygon with sides 3, 4, and 5 (a triangle).

The set $\{6, 9, 11, 24\}$ generates a polygon with sides 6, 9, 11, and 24 (a quadrilateral).

The sets $\{1, 2, 3\}$ and $\{2, 3, 4, 9\}$ do not generate any polygon at all.

Consider the sequence s , defined as follows:

- $s_1 = 1, s_2 = 2, s_3 = 3$
- $s_n = s_{n-1} + s_{n-3}$ for $n \geq 3$.

Let U_n be the set $\{s_1, s_2, \dots, s_n\}$. For example, $U_{10} = \{1, 2, 3, 4, 6, 9, 13, 19, 28, 41\}$.

Let $f(n)$ be the number of subsets of U_n which generate at least one polygon.

For example, $f(5) = 7$, $f(10) = 501$ and $f(25) = 18635853$.

Find the last 9 digits of $f(10^{18})$.

Problem 383: Divisibility Comparison Between Factorials

Link

Tags: Number Theory

Let $f_5(n)$ be the largest integer x for which 5^x divides n .

For example, $f_5(625000) = 7$.

Let $T_5(n)$ be the number of integers i which satisfy $f_5((2 \cdot i - 1)!) \geq f_5(i!)$ and $1 \leq i \leq n$.

It can be verified that $T_5(10^3) = 68$ and $T_5(10^9) = 2408210$.

Find $T_5(10^{18})$.

Problem 384: Rudin-Shapiro Sequence

Link

Tags: Sequences

Define the sequence $a(n)$ as the number of adjacent pairs of ones in the binary expansion of n (possibly overlapping).

E.g.: $a(5) = a(101_2) = 0$, $a(6) = a(110_2) = 1$, $a(7) = a(111_2) = 2$.

Define the sequence $b(n) = (-1)^{a(n)}$.

This sequence is called the Rudin-Shapiro sequence.

Also consider the summatory sequence of $b(n)$: $s(n) = \sum_{i=0}^n b(i)$.

The first couple of values of these sequences are:

n 0 1 2 3 4 5 6 7

$a(n)$ 0 0 0 1 0 0 1 2

$b(n)$ 1 1 1 -1 1 1 -1 1

$s(n)$ 1 2 3 2 3 4 3 4

The sequence $s(n)$ has the remarkable property that all elements are positive and every positive integer k occurs exactly k times.

Define $g(t, c)$, with $1 \leq c \leq t$, as the index in $s(n)$ for which t occurs for the c 'th time in $s(n)$.

E.g.: $g(3, 3) = 6$, $g(4, 2) = 7$ and $g(54321, 12345) = 1220847710$.

Let $F(n)$ be the Fibonacci sequence defined by:

$F(0) = F(1) = 1$ and

$F(n) = F(n-1) + F(n-2)$ for $n \geq 1$.

Define $GF(t) = g(F(t), F(t-1))$.

Find $\sum GF(t)$ for $2 \leq t \leq 45$.

Problem 385: Ellipses Inside Triangles

Link

Tags: Geometry

For any triangle T in the plane, it can be shown that there is a unique ellipse with largest area that is completely inside T .

For a given n , consider triangles T such that:

- the vertices of T have integer coordinates with absolute value $\leq n$, and
- the foci of the largest-area ellipse inside T are $(\sqrt{13}, 0)$ and $(-\sqrt{13}, 0)$.

Let $A(n)$ be the sum of the areas of all such triangles.

For example, if $n = 8$, there are two such triangles. Their vertices are $(-4, -3), (-4, 3), (8, 0)$ and $(4, 3), (4, -3), (-8, 0)$, and the area of each triangle is 36. Thus $A(8) = 36 + 36 = 72$.

It can be verified that $A(10) = 252$, $A(100) = 34632$ and $A(1000) = 3529008$.

Find $A(1\,000\,000\,000)$.

¹The foci (plural of focus) of an ellipse are two points A and B such that for every point P on the boundary of the ellipse, $AP + PB$ is constant.

Problem 386: Maximum Length of an Antichain

Link

Tags: Combinatorics, Number Theory

Let n be an integer and $S(n)$ be the set of factors of n .

A subset A of $S(n)$ is called an antichain of $S(n)$ if A contains only one element or if none of the elements of A divides any of the other elements of A .

For example: $S(30) = \{1, 2, 3, 5, 6, 10, 15, 30\}$.

$\{2, 5, 6\}$ is not an antichain of $S(30)$.

$\{2, 3, 5\}$ is an antichain of $S(30)$.

Let $N(n)$ be the maximum length of an antichain of $S(n)$.

Find $\sum N(n)$ for $1 \leq n \leq 10^8$.

Problem 387: Harshad Numbers

Link

Tags: Number Theory

A Harshad or Niven number is a number that is divisible by the sum of its digits.

201 is a Harshad number because it is divisible by 3 (the sum of its digits.)

When we truncate the last digit from 201, we get 20, which is a Harshad number.

When we truncate the last digit from 20, we get 2, which is also a Harshad number.

Let's call a Harshad number that, while recursively truncating the last digit, always results in a Harshad number a right truncatable Harshad number.

Also:

$201/3 = 67$ which is prime.

Let's call a Harshad number that, when divided by the sum of its digits, results in a prime a strong Harshad number.

Now take the number 2011 which is prime.

When we truncate the last digit from it we get 201, a strong Harshad number that is also right truncatable.

Let's call such primes strong, right truncatable Harshad primes.

You are given that the sum of the strong, right truncatable Harshad primes less than 10000 is 90619.

Find the sum of the strong, right truncatable Harshad primes less than 10^{14} .

Problem 388: Distinct Lines

[Link](#)

Tags: Combinatorics, Number Theory

Consider all lattice points (a, b, c) with $0 \leq a, b, c \leq N$.

From the origin $O(0, 0, 0)$ all lines are drawn to the other lattice points.

Let $D(N)$ be the number of distinct such lines.

You are given that $D(1\,000\,000) = 831909254469114121$.

Find $D(10^{10})$. Give as your answer the first nine digits followed by the last nine digits.

Problem 389: Platonic Dice

[Link](#)

Tags: Probability

An unbiased single 4-sided die is thrown and its value, T , is noted. T unbiased 6-sided dice are thrown and their scores are added together. The sum, C , is

noted. C unbiased 8-sided dice are thrown and their scores are added together. The sum, O , is noted. O unbiased 12-sided dice are thrown and their scores are added together. The sum, D , is noted. D unbiased 20-sided dice are thrown and their scores are added together. The sum, I , is noted.

Find the variance of I , and give your answer rounded to 4 decimal places.

Problem 390: Triangles with Non Rational Sides and Integral Area

Link

Tags: Geometry

Consider the triangle with sides $\sqrt{5}$, $\sqrt{65}$ and $\sqrt{68}$. It can be shown that this triangle has area 9.

$S(n)$ is the sum of the areas of all triangles with sides $\sqrt{1+b^2}$, $\sqrt{1+c^2}$ and $\sqrt{b^2+c^2}$ (for positive integers b and c) that have an integral area not exceeding n .

The example triangle has $b = 2$ and $c = 8$.

$S(10^6) = 18018206$.

Find $S(10^{10})$.

Problem 391: Hopping Game

Link

Tags: Combinatorics, Game Theory, Sequences

Let s_k be the number of 1's when writing the numbers from 0 to k in binary.

For example, writing 0 to 5 in binary, we have 0, 1, 10, 11, 100, 101. There are seven 1's, so $s_5 = 7$.

The sequence $S = \{s_k : k \geq 0\}$ starts $\{0, 1, 2, 4, 5, 7, 9, 12, \dots\}$.

A game is played by two players. Before the game starts, a number n is chosen. A counter c starts at 0. At each turn, the player chooses a number from 1 to n (inclusive) and increases c by that number. The resulting value of c must be a member of S . If there are no more valid moves, then the player loses.

For example, with $n = 5$ and starting with $c = 0$:

Player 1 chooses 4, so c becomes $0 + 4 = 4$.

Player 2 chooses 5, so c becomes $4 + 5 = 9$.

Player 1 chooses 3, so c becomes $9 + 3 = 12$.

etc.

Note that c must always belong to S , and each player can increase c by at most n .

Let $M(n)$ be the highest number that the first player could choose at the start to force a win, and $M(n) = 0$ if there is no such move. For example, $M(2) = 2$, $M(7) = 1$, and $M(20) = 4$.

It can be verified that $\sum M(n)^3 = 8150$ for $1 \leq n \leq 20$.

Find $\sum M(n)^3$ for $1 \leq n \leq 1000$.

Problem 392: Enmeshed Unit Circle

Link

Tags: Algebra, Combinatorics, Game Theory, Geometry, Graph Theory, Number Theory

A rectilinear grid is an orthogonal grid where the spacing between the gridlines does not have to be equidistant.

An example of such grid is logarithmic graph paper.

Consider rectilinear grids in the Cartesian coordinate system with the following properties:

- The gridlines are parallel to the axes of the Cartesian coordinate system.
- There are $N + 2$ vertical and $N + 2$ horizontal gridlines. Hence there are $(N + 1) \times (N + 1)$ rectangular cells.
- The equations of the two outer vertical gridlines are $x = -1$ and $x = 1$.
- The equations of the two outer horizontal gridlines are $y = -1$ and $y = 1$.
- The grid cells are colored red if they overlap with the unit circle, black otherwise. For this problem we would like you to find the positions of the remaining N inner horizontal and N inner vertical gridlines so that the area occupied by the red cells is minimized.

E.g. here is a picture of the solution for $N = 10$:

The area occupied by the red cells for $N = 10$ rounded to 10 digits behind the decimal point is 3.3469640797.

Find the positions for $N = 400$.

Give as your answer the area occupied by the red cells rounded to 10 digits behind the decimal point.

Problem 393: Migrating Ants

[Link](#)

Tags: Combinatorics, Graph Theory

An $n \times n$ grid of squares contains n^2 ants, one ant per square.

All ants decide to move simultaneously to an adjacent square (usually 4 possibilities, except for ants on the edge of the grid or at the corners).

We define $f(n)$ to be the number of ways this can happen without any ants ending on the same square and without any two ants crossing the same edge between two squares.

You are given that $f(4) = 88$.

Find $f(10)$.

Problem 394: Eating Pie

[Link](#)

Tags: Game Theory, Probability

Jeff eats a pie in an unusual way.

The pie is circular. He starts with slicing an initial cut in the pie along a radius.

While there is at least a given fraction F of pie left, he performs the following procedure:

- He makes two slices from the pie centre to any point of what is remaining of the pie border, any point on the remaining pie border equally likely. This will divide the remaining pie into three pieces.
- Going counterclockwise from the initial cut, he takes the first two pie pieces and eats them.

When less than a fraction F of pie remains, he does not repeat this procedure. Instead, he eats all of the remaining pie.

For $x \geq 1$, let $E(x)$ be the expected number of times Jeff repeats the procedure above with $F = 1/x$.

It can be verified that $E(1) = 1$, $E(2) \approx 1.2676536759$, and $E(7.5) \approx 2.1215732071$.

Find $E(40)$ rounded to 10 decimal places behind the decimal point.

Problem 395: Pythagorean Tree

[Link](#)

Tags: Game Theory, Geometry, Graph Theory, Probability

The Pythagorean tree is a fractal generated by the following procedure:

Start with a unit square. Then, calling one of the sides its base (in the animation, the bottom side is the base):

- Attach a right triangle to the side opposite the base, with the hypotenuse coinciding with that side and with the sides in a 3-4-5 ratio. Note that the smaller side of the triangle must be on the ‘right’ side with respect to the base (see animation).
- Attach a square to each leg of the right triangle, with one of its sides coinciding with that leg.
- Repeat this procedure for both squares, considering as their bases the sides touching the triangle.

The resulting figure, after an infinite number of iterations, is the Pythagorean tree.

Problem 396: Weak Goodstein Sequence

[Link](#)

Tags: Number Theory, Sequences

For any positive integer n , the n th weak Goodstein sequence $\{g_1, g_2, g_3, \dots\}$ is defined as:

- $g_1 = n$
- for $k \geq 1$, g_k is obtained by writing g_{k-1} in base k , interpreting it as a base $k + 1$ number, and subtracting 1.

The sequence terminates when g_k becomes 0.

For example, the 6th weak Goodstein sequence is $\{6, 11, 17, 25, \dots\}$:

- $g_1 = 6$.
- $g_2 = 11$ since $6 = 110_2$, $110_3 = 12$, and $12 - 1 = 11$.
- $g_3 = 17$ since $11 = 102_3$, $102_4 = 18$, and $18 - 1 = 17$.
- $g_4 = 25$ since $17 = 101_4$, $101_5 = 26$, and $26 - 1 = 25$.

and so on.

It can be shown that every weak Goodstein sequence terminates.

Let $G(n)$ be the number of nonzero elements in the n th weak Goodstein sequence.

It can be verified that $G(2) = 3$, $G(4) = 21$ and $G(6) = 381$.

It can also be verified that $\sum G(n) = 2517$ for $1 \leq n \leq 8$.

Find the last 9 digits of $\sum G(n)$ for $1 \leq n \leq 16$.

Problem 397: Triangle on Parabola

Link

Tags: Geometry

On the parabola $y = x^2/k$, three points $A(a, a^2/k)$, $B(b, b^2/k)$ and $C(c, c^2/k)$ are chosen.

Let $F(K, X)$ be the number of the integer quadruplets (k, a, b, c) such that at least one angle of the triangle ABC is 45-degree, with $1 \leq k \leq K$ and $-X \leq abc \leq X$.

For example, $F(1, 10) = 41$ and $F(10, 100) = 12492$.

Find $F(10^6, 10^9)$.

Problem 398: Cutting Rope

Link

Tags: Combinatorics, Geometry, Number Theory, Probability

Inside a rope of length n , $n - 1$ points are placed with distance 1 from each other and from the endpoints. Among these points, we choose $m - 1$ points at random and cut the rope at these points to create m segments.

Let $E(n, m)$ be the expected length of the second-shortest segment. For example, $E(3, 2) = 2$ and $E(8, 3) = 16/7$. Note that if multiple segments have the same shortest length the length of the second-shortest segment is defined as the same as the shortest length.

Find $E(10^7, 100)$. Give your answer rounded to 5 decimal places behind the decimal point.

Problem 399: Squarefree Fibonacci Numbers

Link

Tags: Number Theory, Sequences

The first 15 Fibonacci numbers are:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610.

It can be seen that 8 and 144 are not squarefree: 8 is divisible by 4 and 144 is divisible by 4 and by 9.

So the first 13 squarefree Fibonacci numbers are:

1, 1, 2, 3, 5, 13, 21, 34, 55, 89, 233, 377 and 610.

The 200th squarefree Fibonacci number is: 971183874599339129547649988289594072811608739584170445.

The last sixteen digits of this number are: 1608739584170445 and in scientific notation this number can be written as $9.7\text{e}53$.

Find the 100 000 000th squarefree Fibonacci number.

Give as your answer its last sixteen digits followed by a comma followed by the number in scientific notation (rounded to one digit after the decimal point).

For the 200th squarefree number the answer would have been: 1608739584170445,9.7e53

Note:

For this problem, assume that for every prime p , the first fibonacci number divisible by p is not divisible by p^2 (this is part of Wall's conjecture). This has been verified for primes $\leq 3 \cdot 10^{15}$, but has not been proven in general.

If it happens that the conjecture is false, then the accepted answer to this problem isn't guaranteed to be the 100 000 000th squarefree Fibonacci number, rather it represents only a lower bound for that number.

Problem 400: Fibonacci Tree Game

Link

Tags: Game Theory, Graph Theory, Number Theory, Sequences

A Fibonacci tree is a binary tree recursively defined as:

- $T(0)$ is the empty tree.
- $T(1)$ is the binary tree with only one node.
- $T(k)$ consists of a root node that has $T(k-1)$ and $T(k-2)$ as children.

On such a tree two players play a take-away game. On each turn a player selects a node and removes that node along with the subtree rooted at that node.

The player who is forced to take the root node of the entire tree loses.

Here are the winning moves of the first player on the first turn for $T(k)$ from $k = 1$ to $k = 6$.

Let $f(k)$ be the number of winning moves of the first player (i.e. the moves for which the second player has no winning strategy) on the first turn of the game when this game is played on $T(k)$.

For example, $f(5) = 1$ and $f(10) = 17$.

Find $f(10000)$. Give the last 18 digits of your answer.

Problem 401: Sum of Squares of Divisors

Link

Tags: Number Theory

The divisors of 6 are 1, 2, 3 and 6.

The sum of the squares of these numbers is $1 + 4 + 9 + 36 = 50$.

Let $\text{sigma}_2(n)$ represent the sum of the squares of the divisors of n . Thus $\text{sigma}_2(6) = 50$.

Let SIGMA_2 represent the summatory function of sigma_2 , that is $\text{SIGMA}_2(n) = \sum \text{sigma}_2(i)$ for $i = 1$ to n .

The first 6 values of SIGMA_2 are: 1, 6, 16, 37, 63 and 113.

Find $\text{SIGMA}_2(10^{15})$ modulo 10^9 .

Problem 402: Integer-valued Polynomials

Link

Tags: Algebra, Number Theory, Sequences

It can be shown that the polynomial $n^4 + 4n^3 + 2n^2 + 5n$ is a multiple of 6 for every integer n . It can also be shown that 6 is the largest integer satisfying this property.

Define $M(a, b, c)$ as the maximum m such that $n^4 + an^3 + bn^2 + cn$ is a multiple of m for all integers n . For example, $M(4, 2, 5) = 6$.

Also, define $S(N)$ as the sum of $M(a, b, c)$ for all $0a, b, c \leq N$.

We can verify that $S(10) = 1972$ and $S(10000) = 2024258331114$.

Let F_k be the Fibonacci sequence:

$F_0 = 0, F_1 = 1$ and

$F_k = F_{k-1} + F_{k-2}$ for $k \geq 2$.

Find the last 9 digits of $\sum S(F_k)$ for $2 \leq k \leq 1234567890123$.

Problem 403: Lattice Points Enclosed by Parabola and Line

Link

Tags: Combinatorics, Game Theory, Geometry

For integers a and b , we define $D(a, b)$ as the domain enclosed by the parabola $y = x^2$ and the line $y = a \cdot x + b$: $D(a, b) = \{(x, y) \mid x^2 \leq y \leq a \cdot x + b\}$.

$L(a, b)$ is defined as the number of lattice points contained in $D(a, b)$.

For example, $L(1, 2) = 8$ and $L(2, -1) = 1$.

We also define $S(N)$ as the sum of $L(a, b)$ for all the pairs (a, b) such that the area of $D(a, b)$ is a rational number and $|a|, |b| \leq N$.

We can verify that $S(5) = 344$ and $S(100) = 26709528$.

Find $S(10^{12})$. Give your answer mod 10^8 .

Problem 404: Crisscross Ellipses

Link

Tags: Algebra, Geometry, Number Theory

E_a is an ellipse with an equation of the form $x^2 + 4y^2 = 4a^2$.

E'_a is the rotated image of E_a by θ degrees counterclockwise around the origin $O(0, 0)$ for $0^\circ \leq \theta < 90^\circ$.

Problem 405: A Rectangular Tiling

Link

Tags: Game Theory, Geometry

We wish to tile a rectangle whose length is twice its width.

Let $T(0)$ be the tiling consisting of a single rectangle.

For $n > 0$, let $T(n)$ be obtained from $T(n - 1)$ by replacing all tiles in the following manner:

Problem 406: Guessing Game

Link

Tags: Game Theory, Sequences

We are trying to find a hidden number selected from the set of integers $\{1, 2, \dots, n\}$ by asking questions. Each number (question) we ask, we get one of three possible answers:

- “Your guess is lower than the hidden number” (and you incur a cost of a),
or
- “Your guess is higher than the hidden number” (and you incur a cost of b),
or
- “Yes, that’s it!” (and the game ends).

Given the value of n , a , and b , an optimal strategy minimizes the total cost for the worst possible case.

For example, if $n = 5$, $a = 2$, and $b = 3$, then we may begin by asking “2” as our first question.

If we are told that 2 is higher than the hidden number (for a cost of $b=3$), then we are sure that “1” is the hidden number (for a total cost of 3).

If we are told that 2 is lower than the hidden number (for a cost of $a=2$), then our next question will be “4”.

If we are told that 4 is higher than the hidden number (for a cost of $b=3$), then we are sure that “3” is the hidden number (for a total cost of $2+3=5$).

If we are told that 4 is lower than the hidden number (for a cost of $a=2$), then we are sure that “5” is the hidden number (for a total cost of $2+2=4$).

Thus, the worst-case cost achieved by this strategy is 5. It can also be shown that this is the lowest worst-case cost that can be achieved. So, in fact, we have just described an optimal strategy for the given values of n , a , and b .

Let $C(n, a, b)$ be the worst-case cost achieved by an optimal strategy for the given values of n , a and b .

Here are a few examples:

$$C(5, 2, 3) = 5$$

$$C(500, \sqrt{2}, \sqrt{3}) = 13.22073197 \dots$$

$$C(20000, 5, 7) = 82$$

$$C(2000000, \sqrt{5}, \sqrt{7}) = 49.63755955 \dots$$

Let F_k be the Fibonacci numbers: $F_k = F_{k-1} + F_{k-2}$ with base cases $F_1 = F_2 = 1$.
 Find $\sum_{k=1}^{30} C(10^{12}, \sqrt{k}, \sqrt{F_k})$, and give your answer rounded to 8 decimal places
 behind the decimal point.

Problem 407: Idempotents

Link

Tags: Number Theory

If we calculate $a^2 \bmod 6$ for $0 \leq a \leq 5$ we get: 0, 1, 4, 3, 4, 1.

The largest value of a such that $a^2 \equiv a \bmod 6$ is 4.

Let's call $M(n)$ the largest value of an such that $a^2 \equiv a \pmod{n}$.

So $M(6) = 4$.

Find $\sum M(n)$ for $1 \leq n \leq 10^7$.

Problem 408: Admissible Paths Through a Grid

Link

Tags: Combinatorics

Let's call a lattice point (x, y) inadmissible if x, y and $x + y$ are all positive perfect squares.

For example, $(9, 16)$ is inadmissible, while $(0, 4)$, $(3, 1)$ and $(9, 4)$ are not.

Consider a path from point (x_1, y_1) to point (x_2, y_2) using only unit steps north or east.

Let's call such a path admissible if none of its intermediate points are inadmissible.

Let $P(n)$ be the number of admissible paths from $(0, 0)$ to (n, n) .

It can be verified that $P(5) = 252$, $P(16) = 596994440$ and $P(1000) \bmod 1\,000\,000\,007 = 341920854$.

Find $P(10\,000\,000) \bmod 1\,000\,000\,007$.

Problem 409: Nim Extreme

Link

Tags: Game Theory

Let n be a positive integer. Consider nim positions where:

- There are n non-empty piles.
- Each pile has size less than 2^n .
- No two piles have the same size.

Let $W(n)$ be the number of winning nim positions satisfying the above conditions (a position is winning if the first player has a winning strategy). For example, $W(1) = 1$, $W(2) = 6$, $W(3) = 168$, $W(5) = 19764360$ and $W(100) \bmod 1\,000\,000\,007 = 384777056$.

Find $W(10\,000\,000) \bmod 1\,000\,000\,007$.

Problem 410: Circle and Tangent Line

Link

Tags: Combinatorics, Geometry

Let C be the circle with radius r , $x^2 + y^2 = r^2$. We choose two points $P(a, b)$ and $Q(-a, c)$ so that the line passing through P and Q is tangent to C .

For example, the quadruplet $(r, a, b, c) = (2, 6, 2, -7)$ satisfies this property.

Let $F(R, X)$ be the number of the integer quadruplets (r, a, b, c) with this property, and with $0r \leq R$ and $0a \leq X$.

We can verify that $F(1, 5) = 10$, $F(2, 10) = 52$ and $F(10, 100) = 3384$.

Find $F(10^8, 10^9) + F(10^9, 10^8)$.

Problem 411: Uphill Paths

Link

Tags: Combinatorics

Let n be a positive integer. Suppose there are stations at the coordinates $(x, y) = (2^i \bmod n, 3^i \bmod n)$ for $0 \leq i \leq 2n$. We will consider stations with the same coordinates as the same station.

We wish to form a path from $(0, 0)$ to (n, n) such that the x and y coordinates never decrease.

Let $S(n)$ be the maximum number of stations such a path can pass through.

For example, if $n = 22$, there are 11 distinct stations, and a valid path can pass through at most 5 stations. Therefore, $S(22) = 5$. The case is illustrated below, with an example of an optimal path:

It can also be verified that $S(123) = 14$ and $S(10000) = 48$.

Find $\sum S(k^5)$ for $1 \leq k \leq 30$.

Problem 412: Gnomon Numbering

[Link](#)

Tags: Combinatorics

For integers m, n ($0 \leq nm$), let $L(m, n)$ be an $m \times m$ grid with the top-right $n \times n$ grid removed.

For example, $L(5, 3)$ looks like this:

We want to number each cell of $L(m, n)$ with consecutive integers $1, 2, 3, \dots$ such that the number in every cell is smaller than the number below it and to the left of it.

For example, here are two valid numberings of $L(5, 3)$:

Let $LC(m, n)$ be the number of valid numberings of $L(m, n)$.

It can be verified that $LC(3, 0) = 42$, $LC(5, 3) = 250250$, $LC(6, 3) = 406029023400$ and $LC(10, 5) \bmod 76543217 = 61251715$.

Find $LC(10000, 5000) \bmod 76543217$.

Problem 413: One-child Numbers

[Link](#)

Tags: Number Theory

We say that a d -digit positive number (no leading zeros) is a one-child number if exactly one of its sub-strings is divisible by d .

For example, 5671 is a 4-digit one-child number. Among all its sub-strings 5, 6, 7, 1, 56, 67, 71, 567, 671 and 5671, only 56 is divisible by 4.

Similarly, 104 is a 3-digit one-child number because only 0 is divisible by 3.

1132451 is a 7-digit one-child number because only 245 is divisible by 7.

Let $F(N)$ be the number of the one-child numbers less than N .

We can verify that $F(10) = 9$, $F(10^3) = 389$ and $F(10^7) = 277674$.
Find $F(10^{19})$.

Problem 414: Kaprekar Constant

Link

Tags: Number Theory

6174 is a remarkable number; if we sort its digits in increasing order and subtract that number from the number you get when you sort the digits in decreasing order, we get $7641 - 1467 = 6174$.

Even more remarkable is that if we start from any 4 digit number and repeat this process of sorting and subtracting, we'll eventually end up with 6174 or immediately with 0 if all digits are equal.

This also works with numbers that have less than 4 digits if we pad the number with leading zeroes until we have 4 digits.

E.g. let's start with the number 0837:

$$8730 - 0378 = 8352$$

$$8532 - 2358 = 6174$$

6174 is called the Kaprekar constant. The process of sorting and subtracting and repeating this until either 0 or the Kaprekar constant is reached is called the Kaprekar routine.

We can consider the Kaprekar routine for other bases and number of digits.

Unfortunately, it is not guaranteed a Kaprekar constant exists in all cases; either the routine can end up in a cycle for some input numbers or the constant the routine arrives at can be different for different input numbers.

However, it can be shown that for 5 digits and a base $b = 6t + 3 \neq 9$, a Kaprekar constant exists.

E.g. base 15: $(10, 4, 14, 9, 5)_{15}$

base 21: $(14, 6, 20, 13, 7)_{21}$

Define C_b to be the Kaprekar constant in base b for 5 digits. Define the function $sb(i)$ to be

- 0 if $i = C_b$ or if i written in base b consists of 5 identical digits
- the number of iterations it takes the Kaprekar routine in base b to arrive at C_b , otherwise

Note that we can define $sb(i)$ for all integers ib^5 . If i written in base b takes less than 5 digits, the number is padded with leading zero digits until we have 5 digits before applying the Kaprekar routine.

Define $S(b)$ as the sum of $sb(i)$ for $0ib^5$.

E.g. $S(15) = 5274369$

$S(111) = 400668930299$

Find the sum of $S(6k + 3)$ for $2 \leq k \leq 300$.

Give the last 18 digits as your answer.

Problem 415: Titanic Sets

Link

Tags: Combinatorics

A set of lattice points S is called a titanic set if there exists a line passing through exactly two points in S .

An example of a titanic set is $S = \{(0, 0), (0, 1), (0, 2), (1, 1), (2, 0), (1, 0)\}$, where the line passing through $(0, 1)$ and $(2, 0)$ does not pass through any other point in S .

On the other hand, the set $\{(0, 0), (1, 1), (2, 2), (4, 4)\}$ is not a titanic set since the line passing through any two points in the set also passes through the other two.

For any positive integer N , let $T(N)$ be the number of titanic sets S whose every point (x, y) satisfies $0 \leq x, y \leq N$. It can be verified that $T(1) = 11$, $T(2) = 494$, $T(4) = 33554178$, $T(111) \bmod 10^8 = 13500401$ and $T(10^5) \bmod 10^8 = 63259062$.

Find $T(10^{11}) \bmod 10^8$.

Problem 416: A Frog's Trip

Link

Tags: Number Theory

A row of n squares contains a frog in the leftmost square. By successive jumps the frog goes to the rightmost square and then back to the leftmost square. On the outward trip he jumps one, two or three squares to the right, and on the homeward trip he jumps to the left in a similar manner. He cannot jump outside the squares. He repeats the round-trip travel m times.

Let $F(m, n)$ be the number of the ways the frog can travel so that at most one square remains unvisited.

For example, $F(1, 3) = 4$, $F(1, 4) = 15$, $F(1, 5) = 46$, $F(2, 3) = 16$ and $F(2, 100) \bmod 10^9 = 429619151$.

Find the last 9 digits of $F(10, 10^{12})$.

Problem 417: Reciprocal Cycles II

[Link](#)

Tags: Number Theory

A unit fraction contains 1 in the numerator. The decimal representation of the unit fractions with denominators 2 to 10 are given:

$$\begin{aligned} 1/2 &= 0.5 \\ 1/3 &= 0.(3) \\ 1/4 &= 0.25 \\ 1/5 &= 0.2 \\ 1/6 &= 0.1(6) \\ 1/7 &= 0.(142857) \\ 1/8 &= 0.125 \\ 1/9 &= 0.(1) \\ 1/10 &= 0.1 \end{aligned}$$

Where $0.1(6)$ means $0.166666\ldots$, and has a 1-digit recurring cycle. It can be seen that $1/7$ has a 6-digit recurring cycle.

Unit fractions whose denominator has no other prime factors than 2 and/or 5 are not considered to have a recurring cycle.

We define the length of the recurring cycle of those unit fractions as 0.

Let $L(n)$ denote the length of the recurring cycle of $1/n$. You are given that $\sum L(n)$ for $3 \leq n \leq 1\,000\,000$ equals 55535191115.

Find $\sum L(n)$ for $3 \leq n \leq 100\,000\,000$.

Problem 418: Factorisation Triples

[Link](#)

Tags: Game Theory, Number Theory

Let n be a positive integer. An integer triple (a, b, c) is called a factorisation triple of n if:

- $1 \leq a \leq b \leq c$
- $a \cdot b \cdot c = n$.

Define $f(n)$ to be $a+b+c$ for the factorisation triple (a, b, c) of n which minimises c/a . One can show that this triple is unique.

For example, $f(165) = 19$, $f(100100) = 142$ and $f(20!) = 4034872$.

Find $f(43!)$.

Problem 419: Look and Say Sequence

Link

Tags: Number Theory, Sequences

The look and say sequence goes 1, 11, 21, 1211, 111221, 312211, 13112221, 1113213211, ...

The sequence starts with 1 and all other members are obtained by describing the previous member in terms of consecutive digits.

It helps to do this out loud:

1 is 'one one' $\rightarrow$ 11

11 is 'two ones' $\rightarrow$ 21

21 is 'one two and one one' $\rightarrow$ 1211

1211 is 'one one, one two and two ones' $\rightarrow$ 111221

111221 is 'three ones, two twos and one one' $\rightarrow$ 312211

...

Define $A(n)$, $B(n)$ and $C(n)$ as the number of ones, twos and threes in the n 'th element of the sequence respectively.

One can verify that $A(40) = 31254$, $B(40) = 20259$ and $C(40) = 11625$.

Find $A(n)$, $B(n)$ and $C(n)$ for $n = 10^{12}$.

Give your answer modulo 2^{30} and separate your values for A , B and C by a comma.

E.g. for $n = 40$ the answer would be 31254,20259,11625

Problem 420: 2×2 Positive Integer Matrix

Link

Tags: Algebra

A positive integer matrix is a matrix whose elements are all positive integers.

Some positive integer matrices can be expressed as a square of a positive integer matrix in two different ways. Here is an example:

$$\begin{pmatrix} 40 & 12 \\ 48 & 40 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 12 & 2 \end{pmatrix}^2 = \begin{pmatrix} 6 & 1 \\ 4 & 6 \end{pmatrix}^2$$

We define $F(N)$ as the number of the 2×2 positive integer matrices which have a trace less than N and which can be expressed as a square of a positive integer matrix in two different ways.

We can verify that $F(50) = 7$ and $F(1000) = 1019$.

Find $F(10^7)$.

Problem 421: Prime Factors of $n^{15} + 1$

Link

Tags: Number Theory

Numbers of the form $n^{15} + 1$ are composite for every integer n .

For positive integers n and m let $s(n, m)$ be defined as the sum of the distinct prime factors of $n^{15} + 1$ not exceeding m .

E.g. $2^{15} + 1 = 3 \times 3 \times 11 \times 331$.

So $s(2, 10) = 3$ and $s(2, 1000) = 3 + 11 + 331 = 345$.

Also $10^{15} + 1 = 7 \times 11 \times 13 \times 211 \times 241 \times 2161 \times 9091$.

So $s(10, 100) = 31$ and $s(10, 1000) = 483$.

Find $\sum s(n, 10^8)$ for $1 \leq n \leq 10^{11}$.

Problem 422: Sequence of Points on a Hyperbola

Link

Tags: Algebra, Sequences

Let H be the hyperbola defined by the equation $12x^2 + 7xy - 12y^2 = 625$.

Next, define X as the point $(7, 1)$. It can be seen that X is in H .

Now we define a sequence of points in H , $\{P_i : i \geq 1\}$, as:

- $P_1 = (13, 61/4)$.
- $P_2 = (-43/6, -4)$.
- For $i \geq 2$, P_i is the unique point in H that is different from P_{i-1} and such that line $P_i P_{i-1}$ is parallel to line $P_{i-2} X$. It can be shown that P_i is well-defined, and that its coordinates are always rational.

Problem 423: Consecutive Die Throws

Link

Tags: Combinatorics, Game Theory, Number Theory

Let n be a positive integer.

A 6-sided die is thrown n times. Let c be the number of pairs of consecutive throws that give the same value.

For example, if $n = 7$ and the values of the die throws are $(1, 1, 5, 6, 6, 6, 3)$, then the following pairs of consecutive throws give the same value:

$(1, 1, 5, 6, 6, 6, 3)$

$(1, 1, 5, 6, 6, 6, 3)$

$(1, 1, 5, 6, 6, 6, 3)$

Therefore, $c = 3$ for $(1, 1, 5, 6, 6, 6, 3)$.

Define $C(n)$ as the number of outcomes of throwing a 6-sided die n times such that c does not exceed $\pi(n)$.¹

For example, $C(3) = 216$, $C(4) = 1290$, $C(11) = 361912500$ and $C(24) = 4727547363281250000$.

Define $S(L)$ as $\sum C(n)$ for $1 \leq n \leq L$.

For example, $S(50) \bmod 1\,000\,000\,007 = 832833871$.

Find $S(50\,000\,000) \bmod 1\,000\,000\,007$.

¹ π denotes the prime-counting function, i.e. $\pi(n)$ is the number of primes $\leq n$.

Problem 424: Kakuro

Link

Tags: Number Theory

Problem 425: Prime Connection

Link

Tags: Number Theory

Two positive numbers A and B are said to be connected (denoted by " $A \leftrightarrow B$ ") if one of these conditions holds:

- (1) A and B have the same length and differ in exactly one digit; for example, $123 \leftrightarrow 173$.
- (2) Adding one digit to the left of A (or B) makes B (or A); for example, $23 \leftrightarrow 223$ and $123 \leftrightarrow 23$.

We call a prime P a 2's relative if there exists a chain of connected primes between 2 and P and no prime in the chain exceeds P .

For example, 127 is a 2's relative. One of the possible chains is shown below:

$$2 \leftrightarrow 3 \leftrightarrow 13 \leftrightarrow 113 \leftrightarrow 103 \leftrightarrow 107 \leftrightarrow 127$$

However, 11 and 103 are not 2's relatives.

Let $F(N)$ be the sum of the primes $\leq N$ which are not 2's relatives.

We can verify that $F(10^3) = 431$ and $F(10^4) = 78728$.

Find $F(10^7)$.

Problem 426: Box-Ball System

Link

Tags: Game Theory, Sequences

Consider an infinite row of boxes. Some of the boxes contain a ball. For example, an initial configuration of 2 consecutive occupied boxes followed by 2 empty boxes, 2 occupied boxes, 1 empty box, and 2 occupied boxes can be denoted by the sequence $(2, 2, 2, 1, 2)$, in which the number of consecutive occupied and empty boxes appear alternately.

A turn consists of moving each ball exactly once according to the following rule: Transfer the leftmost ball which has not been moved to the nearest empty box to its right.

After one turn the sequence $(2, 2, 2, 1, 2)$ becomes $(2, 2, 1, 2, 3)$ as can be seen below; note that we begin the new sequence starting at the first occupied box.

Problem 427: n -sequences

Link

Tags: Sequences

A sequence of integers $S = \{s_i\}$ is called an n -sequence if it has n elements and each element s_i satisfies $1 \leq s_i \leq n$. Thus there are n^n distinct n -sequences in total. For example, the sequence $S = \{1, 5, 5, 10, 7, 7, 7, 2, 3, 7\}$ is a 10-sequence.

For any sequence S , let $L(S)$ be the length of the longest contiguous subsequence of S with the same value. For example, for the given sequence S above, $L(S) = 3$, because of the three consecutive 7's.

Let $f(n) = \sum L(S)$ for all n -sequences S .

For example, $f(3) = 45$, $f(7) = 1403689$ and $f(11) = 481496895121$.

Find $f(7\,500\,000) \bmod 1\,000\,000\,009$.

Problem 428: Necklace of Circles

Link

Tags: Algebra, Geometry

Let a , b and c be positive numbers.

Let W, X, Y, Z be four collinear points where $|WX| = a$, $|XY| = b$, $|YZ| = c$ and $|WZ| = a + b + c$.

Let C_{in} be the circle having the diameter XY .

Let C_{out} be the circle having the diameter WZ .

The triplet (a, b, c) is called a necklace triplet if you can place $k \geq 3$ distinct circles $C_1, C_2, \dots, C_k$ such that:

- C_i has no common interior points with any C_j for $1 \leq i, j \leq k$ and $i \neq j$,
- C_i is tangent to both C_{in} and C_{out} for $1 \leq i \leq k$,
- C_i is tangent to C_{i+1} for $1 \leq i < k$, and
- C_k is tangent to C_1 .

For example, $(5, 5, 5)$ and $(4, 3, 21)$ are necklace triplets, while it can be shown that $(2, 2, 5)$ is not.

Let $T(n)$ be the number of necklace triplets (a, b, c) such that a , b and c are positive integers, and $b \leq n$. For example, $T(1) = 9$, $T(20) = 732$ and $T(3000) = 438106$.

Find $T(1\,000\,000\,000)$.

Problem 429: Sum of Squares of Unitary Divisors

Link

Tags: Number Theory

A unitary divisor d of a number n is a divisor of n that has the property $\gcd(d, n/d) = 1$.

The unitary divisors of $4! = 24$ are 1, 3, 8 and 24.

The sum of their squares is $1^2 + 3^2 + 8^2 + 24^2 = 650$.

Let $S(n)$ represent the sum of the squares of the unitary divisors of n . Thus $S(4!) = 650$.

Find $S(100\,000\,000!)$ modulo $1\,000\,000\,009$.

Problem 430: Range Flips

Link

Tags: Game Theory, Probability

N disks are placed in a row, indexed 1 to N from left to right.

Each disk has a black side and white side. Initially all disks show their white side.

At each turn, two, not necessarily distinct, integers A and B between 1 and N (inclusive) are chosen uniformly at random.

All disks with an index from A to B (inclusive) are flipped.

The following example shows the case $N = 8$. At the first turn $A = 5$ and $B = 2$, and at the second turn $A = 4$ and $B = 6$.

Let $E(N, M)$ be the expected number of disks that show their white side after M turns.

We can verify that $E(3, 1) = 10/9$, $E(3, 2) = 5/3$, $E(10, 4) \approx 5.157$ and $E(100, 10) \approx 51.893$.

Find $E(10^{10}, 4000)$.

Give your answer rounded to 2 decimal places behind the decimal point.

Problem 431: Square Space Silo

[Link](#)

Tags: Combinatorics, Geometry

Fred the farmer arranges to have a new storage silo installed on his farm and having an obsession for all things square he is absolutely devastated when he discovers that it is circular. Quentin, the representative from the company that installed the silo, explains that they only manufacture cylindrical silos, but he points out that it is resting on a square base. Fred is not amused and insists that it is removed from his property.

Quick thinking Quentin explains that when granular materials are delivered from above a conical slope is formed and the natural angle made with the horizontal is called the angle of repose. For example if the angle of repose, $\alpha = 30$ degrees, and grain is delivered at the centre of the silo then a perfect cone will form towards the top of the cylinder. In the case of this silo, which has a diameter of 6m, the amount of space wasted would be approximately 32.648388556m^3 . However, if grain is delivered at a point on the top which has a horizontal distance of x metres from the centre then a cone with a strangely curved and sloping base is formed. He shows Fred a picture.

Problem 432: Totient Sum

[Link](#)

Tags: Number Theory

Let $S(n, m) = \sum \phi(n \times i)$ for $1 \leq i \leq m$. (ϕ is Euler's totient function)

You are given that $S(510510, 10^6) = 45480596821125120$.

Find $S(510510, 10^{11})$.

Give the last 9 digits of your answer.

Problem 433: Steps in Euclid's Algorithm

[Link](#)

Tags: Number Theory

Let $E(x_0, y_0)$ be the number of steps it takes to determine the greatest common divisor of x_0 and y_0 with Euclid's algorithm. More formally: $x_1 = y_0$, $y_1 = x_0 \bmod y_0$, $x_n = y_{n-1}$, $y_n = x_{n-1} \bmod y_{n-1}$

$E(x_0, y_0)$ is the smallest n such that $y_n = 0$.

We have $E(1, 1) = 1$, $E(10, 6) = 3$ and $E(6, 10) = 4$.

Define $S(N)$ as the sum of $E(x, y)$ for $1 \leq x, y \leq N$.

We have $S(1) = 1$, $S(10) = 221$ and $S(100) = 39826$.

Find $S(5 \cdot 10^6)$.

Problem 434: Rigid Graphs

Link

Tags: Combinatorics, Game Theory, Geometry, Graph Theory

Recall that a graph is a collection of vertices and edges connecting the vertices, and that two vertices connected by an edge are called adjacent.

Graphs can be embedded in Euclidean space by associating each vertex with a point in the Euclidean space.

A flexible graph is an embedding of a graph where it is possible to move one or more vertices continuously so that the distance between at least two nonadjacent vertices is altered while the distances between each pair of adjacent vertices is kept constant.

A rigid graph is an embedding of a graph which is not flexible.

Informally, a graph is rigid if by replacing the vertices with fully rotating hinges and the edges with rods that are unbending and inelastic, no parts of the graph can be moved independently from the rest of the graph.

The grid graphs embedded in the Euclidean plane are not rigid, as the following animation demonstrates:

Problem 435: Polynomials of Fibonacci Numbers

Link

Tags: Algebra, Sequences

The Fibonacci numbers $\{f_n, n \geq 0\}$ are defined recursively as $f_n = f_{n-1} + f_{n-2}$ with base cases $f_0 = 0$ and $f_1 = 1$.

Define the polynomials $\{F_n, n \geq 0\}$ as $F_n(x) = \sum_{i=0}^n f_i x^i$.

For example, $F_7(x) = x + x^2 + 2x^3 + 3x^4 + 5x^5 + 8x^6 + 13x^7$, and $F_7(11) = 268\,357\,683$.

Let $n = 10^{15}$. Find the sum $\sum_{x=0}^{100} F_n(x)$ and give your answer modulo $1\,307\,674\,368\,000 (= 15!)$.

Problem 436: Unfair Wager

Link

Tags: Game Theory, Probability

Julie proposes the following wager to her sister Louise.

She suggests they play a game of chance to determine who will wash the dishes.

For this game, they shall use a generator of independent random numbers uniformly distributed between 0 and 1.

The game starts with $S = 0$.

The first player, Louise, adds to S different random numbers from the generator until $S1$ and records her last random number ' x '.

The second player, Julie, continues adding to S different random numbers from the generator until $S2$ and records her last random number ' y '.

The player with the highest number wins and the loser washes the dishes, i.e. if yx the second player wins.

For example, if the first player draws 0.62 and 0.44, the first player turn ends since $0.62 + 0.441$ and $x = 0.44$.

If the second players draws 0.1, 0.27 and 0.91, the second player turn ends since $0.62 + 0.44 + 0.1 + 0.27 + 0.912$ and $y = 0.91$. Since yx , the second player wins.

Louise thinks about it for a second, and objects: "That's not fair".

What is the probability that the second player wins?

Give your answer rounded to 10 places behind the decimal point in the form 0.abcdefghij.

Problem 437: Fibonacci Primitive Roots

Link

Tags: Algebra, Game Theory, Number Theory, Sequences

When we calculate 8^n modulo 11 for $n = 0$ to 9 we get: 1, 8, 9, 6, 4, 10, 3, 2, 5, 7.

As we see all possible values from 1 to 10 occur. So 8 is a primitive root of 11.

But there is more:

If we take a closer look we see:

$$1 + 8 = 9$$

$$8 + 9 = 17 \equiv 6 \pmod{11}$$

$$9 + 6 = 15 \equiv 4 \pmod{11}$$

$$6 + 4 = 10$$

$$4 + 10 = 14 \equiv 3 \pmod{11}$$

$$10 + 3 = 13 \equiv 2 \pmod{11}$$

$$3 + 2 = 5$$

$$2 + 5 = 7$$

$$5 + 7 = 12 \equiv 1 \pmod{11}.$$

So the powers of 8 mod 11 are cyclic with period 10, and $8^n + 8^{n+1} \equiv 8^{n+2} \pmod{11}$.

8 is called a Fibonacci primitive root of 11.

Not every prime has a Fibonacci primitive root.

There are 323 primes less than 10000 with one or more Fibonacci primitive roots and the sum of these primes is 1480491.

Find the sum of the primes less than 100 000 000 with at least one Fibonacci primitive root.

Problem 438: Integer Part of Polynomial Equation's Solutions

Link

Tags: Algebra, Game Theory

For an n -tuple of integers $t = (a_1, \dots, a_n)$, let $(x_1, \dots, x_n)$ be the solutions of the polynomial equation $x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$.

Consider the following two conditions:

- $x_1, \dots, x_n$ are all real.
- If $x_1, \dots, x_n$ are sorted, $\lfloor x_i \rfloor = i$ for $1 \leq i \leq n$. ($\lfloor \cdot \rfloor$: floor function.)

In the case of $n = 4$, there are 12 n -tuples of integers which satisfy both conditions.

We define $S(t)$ as the sum of the absolute values of the integers in t .

For $n = 4$ we can verify that $\sum S(t) = 2087$ for all n -tuples t which satisfy both conditions.

Find $\sum S(t)$ for $n = 7$.

Problem 439: Sum of Sum of Divisors

Link

Tags: Number Theory

Let $d(k)$ be the sum of all divisors of k .

We define the function $S(N) = \sum_{i=1}^N \sum_{j=1}^N d(i \cdot j)$.

For example, $S(3) = d(1) + d(2) + d(3) + d(2) + d(4) + d(6) + d(3) + d(6) + d(9) = 59$.

You are given that $S(10^3) = 563576517282$ and $S(10^5) \bmod 10^9 = 215766508$.

Find $S(10^{11}) \bmod 10^9$.

Problem 440: GCD and Tiling

Link

Tags: Number Theory

We want to tile a board of length n and height 1 completely, with either 1×2 blocks or 1×1 blocks with a single decimal digit on top:

Problem 441: The Inverse Summation of Coprime Couples

Link

Tags: Number Theory

For an integer M , we define $R(M)$ as the sum of $1/(p \cdot q)$ for all the integer pairs p and q which satisfy all of these conditions:

- $1 \leq pq \leq M$
- $p + q \geq M$
- p and q are coprime.

We also define $S(N)$ as the sum of $R(i)$ for $2 \leq i \leq N$.

We can verify that $S(2) = R(2) = 1/2$, $S(10) \approx 6.9147$ and $S(100) \approx 58.2962$.

Find $S(10^7)$. Give your answer rounded to four decimal places.

Problem 442: Eleven-free Integers

Link

Tags: Number Theory

An integer is called eleven-free if its decimal expansion does not contain any substring representing a power of 11 except 1.

For example, 2404 and 13431 are eleven-free, while 911 and 4121331 are not.

Let $E(n)$ be the n th positive eleven-free integer. For example, $E(3) = 3$, $E(200) = 213$ and $E(500\,000) = 531563$.

Find $E(10^{18})$.

Problem 443: GCD Sequence

Link

Tags: Number Theory, Sequences

Let $g(n)$ be a sequence defined as follows:

$$g(4) = 13,$$

$$g(n) = g(n-1) + \gcd(n, g(n-1)) \text{ for } n \geq 4.$$

The first few values are:

$g(4) = 13, g(5) = 14, g(6) = 15, g(7) = 16, g(8) = 17, g(9) = 18, g(10) = 19, \dots$

$g(11) = 20, g(12) = 21, g(13) = 22, g(14) = 23, g(15) = 24, g(16) = 25, g(17) = 26, g(18) = 27, \dots$

Problem 444: The Roundtable Lottery

[Link](#)

Tags: Combinatorics, Game Theory, Number Theory, Probability

A group of p people decide to sit down at a round table and play a lottery-ticket trading game. Each person starts off with a randomly-assigned, unscratched lottery ticket. Each ticket, when scratched, reveals a whole-pound prize ranging anywhere from £1 to £ p , with no two tickets alike. The goal of the game is for all of the players to maximize the winnings of the ticket they hold upon leaving the game.

An arbitrary person is chosen to be the first player. Going around the table, each player has only one of two options:

- The player can choose to scratch the ticket and reveal its worth to everyone at the table.
- If the player's ticket is unscratched, then the player may trade it with a previous player's scratched ticket, and then leaves the game with that ticket. The previous player then scratches the newly-acquired ticket and reveals its worth to everyone at the table.

The game ends once all tickets have been scratched. All players still remaining at the table must leave with their currently-held tickets.

Assume that players will use the optimal strategy for maximizing the expected value of their ticket winnings.

Let $E(p)$ represent the expected number of players left at the table when the game ends in a game consisting of p players.

E.g. $E(111) = 5.2912$ when rounded to 5 significant digits.

Let $S_1(N) = \sum_{p=1}^N E(p)$.

Let $S_k(N) = \sum_{p=1}^N S_{k-1}(p)$ for $k \geq 1$.

Find $S_{20}(10^{14})$ and write the answer in scientific notation rounded to 10 significant digits. Use a lowercase e to separate mantissa and exponent. For example, the answer for $S_3(100)$ would be 5.983679014e5.

Problem 445: Retractions A

[Link](#)

Tags: Number Theory

For every integer $n > 1$, the family of functions $f_{n,a,b}$ is defined by

$f_{n,a,b}(x) \equiv ax + b \pmod{n}$, for a, b, x integer and $0 < a < n, 0 \leq b < n, 0 \leq x < n$.

We will call $f_{n,a,b}$ a retraction if $f_{n,a,b}(f_{n,a,b}(x)) \equiv f_{n,a,b}(x) \pmod{n}$ for every $0 \leq x < n$.

Let $R(n)$ be the number of retractions for n .

You are given that

$$\sum_{k=1}^{99999} R\left(\binom{100000}{k}\right) \equiv 628701600 \pmod{1000000007}.$$

Find $\sum_{k=1}^{999999} R\left(\binom{1000000}{k}\right).$

Give your answer modulo 1000000007.

Problem 446: Retractions B

Link

Tags: Number Theory

For every integer $n > 1$, the family of functions $f_{n,a,b}$ is defined by

$f_{n,a,b}(x) \equiv ax + b \pmod{n}$, for a, b, x integer and $0 < a < n, 0 \leq b < n, 0 \leq x < n$.

We will call $f_{n,a,b}$ a retraction if $f_{n,a,b}(f_{n,a,b}(x)) \equiv f_{n,a,b}(x) \pmod{n}$ for every $0 \leq x < n$.

Let $R(n)$ be the number of retractions for n .

$$F(N) = \sum_{n=1}^N R(n^4 + 4).$$

$$F(1024) = 77532377300600.$$

Find $F(10^7).$

Give your answer modulo 1000000007.

Problem 447: Retractions C

Link

Tags: Number Theory

For every integer $n > 1$, the family of functions $f_{n,a,b}$ is defined by

$$f_{n,a,b}(x) \equiv ax + b \pmod{n}, \text{ for } a, b, x \text{ integer and } 0 < a < n, 0 \leq b < n, 0 \leq x < n.$$

We will call $f_{n,a,b}$ a retraction if $f_{n,a,b}(f_{n,a,b}(x)) \equiv f_{n,a,b}(x) \pmod{n}$ for every $0 \leq x < n$.

Let $R(n)$ be the number of retractions for n .

$$F(N) = \sum_{n=2}^N R(n).$$

$$F(10^7) \equiv 638042271 \pmod{1\,000\,000\,007}.$$

Find $F(10^{14})$.

Give your answer modulo $1\,000\,000\,007$.

Problem 448: Average Least Common Multiple

[Link](#)

Tags: Number Theory

The function $\text{lcm}(a, b)$ denotes the least common multiple of a and b .

Let $A(n)$ be the average of the values of $\text{lcm}(n, i)$ for $1 \leq i \leq n$.

E.g: $A(2) = (2 + 2)/2 = 2$ and $A(10) = (10 + 10 + 30 + 20 + 10 + 30 + 70 + 40 + 90 + 10)/10 = 32$.

Let $S(n) = \sum A(k)$ for $1 \leq k \leq n$.

$$S(100) = 122726.$$

Find $S(99999999019) \pmod{999999017}$.

Problem 449: Chocolate Covered Candy

[Link](#)

Tags: Algebra

Phil the confectioner is making a new batch of chocolate covered candy. Each candy centre is shaped like an ellipsoid of revolution defined by the equation: $b^2x^2 + b^2y^2 + a^2z^2 = a^2b^2$.

Phil wants to know how much chocolate is needed to cover one candy centre with a uniform coat of chocolate one millimeter thick.

If $a = 1$ mm and $b = 1$ mm, the amount of chocolate required is $\frac{28}{3}\pi$ mm³

If $a = 2$ mm and $b = 1$ mm, the amount of chocolate required is approximately 60.35475635 mm³.

Find the amount of chocolate in mm³ required if $a = 3$ mm and $b = 1$ mm. Give your answer as the number rounded to 8 decimal places behind the decimal point.

Problem 450: Hypocycloid and Lattice Points

Link

Tags: Algebra, Geometry

A hypocycloid is the curve drawn by a point on a small circle rolling inside a larger circle. The parametric equations of a hypocycloid centered at the origin, and starting at the right most point is given by:

$$x(t) = (R - r) \cos(t) + r \cos\left(\frac{R - r}{r}t\right)$$

$$y(t) = (R - r) \sin(t) - r \sin\left(\frac{R - r}{r}t\right)$$

Where R is the radius of the large circle and r the radius of the small circle.

Let $C(R, r)$ be the set of distinct points with integer coordinates on the hypocycloid with radius R and r and for which there is a corresponding value of t such that $\sin(t)$ and $\cos(t)$ are rational numbers.

Let $S(R, r) = \sum_{(x,y) \in C(R,r)} |x| + |y|$ be the sum of the absolute values of the x and y coordinates of the points in $C(R, r)$.

Let $T(N) = \sum_{R=3}^N \sum_{r=1}^{\lfloor \frac{R-1}{2} \rfloor} S(R, r)$ be the sum of $S(R, r)$ for R and r positive integers, $R \leq N$ and $2r < R$.

You are given:

$$C(3, 1) = \{(3, 0), (-1, 2), (-1, 0), (-1, -2)\}$$

$$C(2500, 1000) = \{(2500, 0), (772, 2376), (772, -2376), (516, 1792), (516, -1792), (500, 0), (68, 504), (68, -504), (-1356, 1088), (-1356, -1088), (-1500, 1000), (-1500, -1000)\}$$

Note: $(-625, 0)$ is not an element of $C(2500, 1000)$ because $\sin(t)$ is not a rational number for the corresponding values of t .

$$S(3, 1) = (|3| + |0|) + (|-1| + |2|) + (|-1| + |0|) + (|-1| + |-2|) = 10$$

$$T(3) = 10; T(10) = 524; T(100) = 580442; T(10^3) = 583108600.$$

Find $T(10^6)$.

Problem 451: Modular Inverses

Link

Tags: Number Theory

Consider the number 15.

There are eight positive numbers less than 15 which are coprime to 15: 1, 2, 4, 7, 8, 11, 13, 14.

The modular inverses of these numbers modulo 15 are: 1, 8, 4, 13, 2, 11, 7, 14 because

$$1 \cdot 1 \bmod 15 = 1$$

$$2 \cdot 8 = 16 \bmod 15 = 1$$

$$4 \cdot 4 = 16 \bmod 15 = 1$$

$$7 \cdot 13 = 91 \bmod 15 = 1$$

$$11 \cdot 11 = 121 \bmod 15 = 1$$

$$14 \cdot 14 = 196 \bmod 15 = 1$$

Let $I(n)$ be the largest positive number m smaller than $n - 1$ such that the modular inverse of m modulo n equals m itself.

So $I(15) = 11$.

Also $I(100) = 51$ and $I(7) = 1$.

Find $\sum I(n)$ for $3 \leq n \leq 2 \times 10^7$.

Problem 452: Long Products

Link

Tags: Number Theory

Define $F(m, n)$ as the number of n -tuples of positive integers for which the product of the elements doesn't exceed m .

$$F(10, 10) = 571.$$

$$F(10^6, 10^6) \bmod 1\,234\,567\,891 = 252903833.$$

Find $F(10^9, 10^9) \bmod 1\,234\,567\,891$.

Problem 453: Lattice Quadrilaterals

Link

Tags: Combinatorics, Geometry

A simple quadrilateral is a polygon that has four distinct vertices, has no straight angles and does not self-intersect.

Let $Q(m, n)$ be the number of simple quadrilaterals whose vertices are lattice points with coordinates (x, y) satisfying $0 \leq x \leq m$ and $0 \leq y \leq n$.

For example, $Q(2, 2) = 94$ as can be seen below:

It can also be verified that $Q(3, 7) = 39590$, $Q(12, 3) = 309000$ and $Q(123, 45) = 70542215894646$.

Find $Q(12345, 6789) \bmod 135707531$.

Problem 454: Diophantine Reciprocals III

Link

Tags: Algebra, Game Theory

In the following equation x , y , and n are positive integers.

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

For a limit L we define $F(L)$ as the number of solutions which satisfy $xy \leq L$.

We can verify that $F(15) = 4$ and $F(1000) = 1069$.

Find $F(10^{12})$.

Problem 455: Powers with Trailing Digits

Link

Tags: Number Theory

Let $f(n)$ be the largest positive integer x less than 10^9 such that the last 9 digits of n^x form the number x (including leading zeros), or zero if no such integer exists.

For example:

- $f(4) = 411728896$ ($4^{411728896} = \dots 490411728896$)
- $f(10) = 0$
- $f(157) = 743757$ ($157^{743757} = \dots 567000743757$)
- $\sum_{2 \leq n \leq 10^3} f(n) = 442530011399$

Find $\sum_{2 \leq n \leq 10^6} f(n)$.

Problem 456: Triangles Containing the Origin II

Link

Tags: Geometry

Define: $x_n = (1248^n \bmod 32323) - 16161$ $y_n = (8421^n \bmod 30103) - 15051$

$P_n = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

For example, $P_8 = \{(-14913, -6630), (-10161, 5625), (5226, 11896), (8340, -10778), (15852, -5203), (-15165, 1129), (1129, -15165), (-5203, 15852)\}$

Let $C(n)$ be the number of triangles whose vertices are in P_n which contain the origin in the interior.

Examples:

$C(8) = 20$

$C(600) = 8950634$

$C(40\,000) = 2666610948988$

Find $C(2\,000\,000)$.

Problem 457: A Polynomial Modulo the Square of a Prime

Link

Tags: Number Theory

Let $f(n) = n^2 - 3n - 1$.

Let p be a prime.

Let $R(p)$ be the smallest positive integer n such that $f(n) \bmod p^2 = 0$ if such an integer n exists, otherwise $R(p) = 0$.

Let $SR(L)$ be $\sum R(p)$ for all primes not exceeding L .

Find $SR(10^7)$.

Problem 458: Permutations of Project

[Link](#)

Tags: Combinatorics, Number Theory

Consider the alphabet A made out of the letters of the word “project”: $A = \{c, e, j, o, p, r, t\}$.

Let $T(n)$ be the number of strings of length n consisting of letters from A that do not have a substring that is one of the 5040 permutations of “project”.

$$T(7) = 7^7 - 7! = 818503.$$

Find $T(10^{12})$. Give the last 9 digits of your answer.

Problem 459: Flipping Game

[Link](#)

Tags: Combinatorics, Game Theory, Geometry

The flipping game is a two player game played on an N by N square board.

Each square contains a disk with one side white and one side black.

The game starts with all disks showing their white side.

A turn consists of flipping all disks in a rectangle with the following properties:

- the upper right corner of the rectangle contains a white disk
- the rectangle width is a perfect square $(1, 4, 9, 16, \dots)$
- the rectangle height is a triangular number $(1, 3, 6, 10, \dots)$

Players alternate turns. A player wins by turning the grid all black.

Let $W(N)$ be the number of winning moves for the first player on an N by N board with all disks white, assuming perfect play.

$$W(1) = 1, W(2) = 0, W(5) = 8 \text{ and } W(10^2) = 31395.$$

For $N = 5$, the first player's eight winning first moves are:

Find $W(10^6)$.

Problem 460: An Ant on the Move

[Link](#)

Tags: Combinatorics

On the Euclidean plane, an ant travels from point $A(0, 1)$ to point $B(d, 1)$ for an integer d .

In each step, the ant at point (x_0, y_0) chooses one of the lattice points (x_1, y_1) which satisfy $x_1 \geq 0$ and $y_1 \geq 1$ and goes straight to (x_1, y_1) at a constant velocity v . The value of v depends on y_0 and y_1 as follows:

- If $y_0 = y_1$, the value of v equals y_0 .
- If $y_0 \neq y_1$, the value of v equals $(y_1 - y_0)/(\ln(y_1) - \ln(y_0))$.

The left image is one of the possible paths for $d = 4$. First the ant goes from $A(0, 1)$ to $P_1(1, 3)$ at velocity $(3 - 1)/(\ln(3) - \ln(1)) \approx 1.8205$. Then the required time is $\sqrt{5}/1.8205 \approx 1.2283$.

From $P_1(1, 3)$ to $P_2(3, 3)$ the ant travels at velocity 3 so the required time is $2/3 \approx 0.6667$. From $P_2(3, 3)$ to $B(4, 1)$ the ant travels at velocity $(1 - 3)/(\ln(1) - \ln(3)) \approx 1.8205$ so the required time is $\sqrt{5}/1.8205 \approx 1.2283$.

Thus the total required time is $1.2283 + 0.6667 + 1.2283 = 3.1233$.

The right image is another path. The total required time is calculated as $0.98026 + 1 + 0.98026 = 2.96052$. It can be shown that this is the quickest path for $d = 4$.

Let $F(d)$ be the total required time if the ant chooses the quickest path. For example, $F(4) \approx 2.960516287$.

We can verify that $F(10) \approx 4.668187834$ and $F(100) \approx 9.217221972$.

Find $F(10000)$. Give your answer rounded to nine decimal places.

Problem 461: Almost Pi

Link

Tags: Game Theory

Let $f_n(k) = e^{k/n} - 1$, for all non-negative integers k .

Remarkably, $f_{200}(6) + f_{200}(75) + f_{200}(89) + f_{200}(226) = \underline{3.141592644529} \dots \approx \pi$.

In fact, it is the best approximation of π of the form $f_n(a) + f_n(b) + f_n(c) + f_n(d)$ for $n = 200$.

Let $g(n) = a^2 + b^2 + c^2 + d^2$ for a, b, c, d that minimize the error: $|f_n(a) + f_n(b) + f_n(c) + f_n(d) - \pi|$

(where $|x|$ denotes the absolute value of x).

You are given $g(200) = 6^2 + 75^2 + 89^2 + 226^2 = 64658$.

Find $g(10000)$.

Problem 462: Permutation of 3-smooth Numbers

Link

Tags: Combinatorics, Number Theory

A 3-smooth number is an integer which has no prime factor larger than 3. For an integer N , we define $S(N)$ as the set of 3-smooth numbers less than or equal to N . For example, $S(20) = \{1, 2, 3, 4, 6, 8, 9, 12, 16, 18\}$.

We define $F(N)$ as the number of permutations of $S(N)$ in which each element comes after all of its proper divisors.

This is one of the possible permutations for $N = 20$.

- 1, 2, 4, 3, 9, 8, 16, 6, 18, 12.

This is not a valid permutation because 12 comes before its divisor 6.

- 1, 2, 4, 3, 9, 8, **12**, 16, **6**, 18.

We can verify that $F(6) = 5$, $F(8) = 9$, $F(20) = 450$ and $F(1000) \approx 8.8521816557\text{e}21$.

Find $F(10^{18})$. Give as your answer its scientific notation rounded to ten digits after the decimal point.

When giving your answer, use a lowercase e to separate mantissa and exponent. E.g. if the answer is 112 233 445 566 778 899 then the answer format would be 1.1223344557e17.

Problem 463: A Weird Recurrence Relation

Link

Tags: Number Theory

The function f is defined for all positive integers as follows:

- $f(1) = 1$
- $f(3) = 3$
- $f(2n) = f(n)$
- $f(4n + 1) = 2f(2n + 1) - f(n)$
- $f(4n + 3) = 3f(2n + 1) - 2f(n)$

The function $S(n)$ is defined as $\sum_{i=1}^n f(i)$.

$S(8) = 22$ and $S(100) = 3604$.

Find $S(3^{37})$. Give the last 9 digits of your answer.

Problem 464: Möbius Function and Intervals

Link

Tags: Number Theory

The Möbius function, denoted $\mu(n)$, is defined as:

- $\mu(n) = (-1)^{\omega(n)}$ if n is squarefree (where $\omega(n)$ is the number of distinct prime factors of n)
- $\mu(n) = 0$ if n is not squarefree.

Let $P(a, b)$ be the number of integers n in the interval $[a, b]$ such that $\mu(n) = 1$.

Let $N(a, b)$ be the number of integers n in the interval $[a, b]$ such that $\mu(n) = -1$.

For example, $P(2, 10) = 2$ and $N(2, 10) = 4$.

Let $C(n)$ be the number of integer pairs (a, b) such that:

- $1 \leq a \leq b \leq n$,
- $99 \cdot N(a, b) \leq 100 \cdot P(a, b)$, and
- $99 \cdot P(a, b) \leq 100 \cdot N(a, b)$.

For example, $C(10) = 13$, $C(500) = 16676$ and $C(10\,000) = 20155319$.

Find $C(20\,000\,000)$.

Problem 465: Polar Polygons

Link

Tags: Algebra, Game Theory, Geometry, Graph Theory

The kernel of a polygon is defined by the set of points from which the entire polygon's boundary is visible. We define a polar polygon as a polygon for which the origin is strictly contained inside its kernel.

For this problem, a polygon can have collinear consecutive vertices. However, a polygon still cannot have self-intersection and cannot have zero area.

For example, only the first of the following is a polar polygon (the kernels of the second, third, and fourth do not strictly contain the origin, and the fifth does not have a kernel at all):

Notice that the first polygon has three consecutive collinear vertices.

Let $P(n)$ be the number of polar polygons such that the vertices (x, y) have integer coordinates whose absolute values are not greater than n .

Note that polygons should be counted as different if they have different set of edges, even if they enclose the same area. For example, the polygon with vertices $[(0, 0), (0, 3), (1, 1), (3, 0)]$ is distinct from the polygon with vertices $[(0, 0), (0, 3), (1, 1), (3, 0), (1, 0)]$.

For example, $P(1) = 131$, $P(2) = 1648531$, $P(3) = 1099461296175$ and $P(343) \bmod 1\,000\,000\,007 = 937293740$.

Find $P(7^{13}) \bmod 1\,000\,000\,007$.

Problem 466: Distinct Terms in a Multiplication Table

Link

Tags: Number Theory

Let $P(m, n)$ be the number of distinct terms in an $m \times n$ multiplication table.

For example, a 3×4 multiplication table looks like this:

$\times$ 12341 12342 24683 36912

There are 8 distinct terms $\{1, 2, 3, 4, 6, 8, 9, 12\}$, therefore $P(3, 4) = 8$.

You are given that:

$P(64, 64) = 1263$,

$P(12, 345) = 1998$, and

$P(32, 10^{15}) = 13826382602124302$.

Find $P(64, 10^{16})$.

Problem 467: Superinteger

Link

Tags: Algebra, Number Theory, Sequences

An integer s is called a superinteger of another integer n if the digits of n form a subsequence of the digits of s .

For example, 2718281828 is a superinteger of 18828, while 314159 is not a superinteger of 151.

Let $p(n)$ be the n th prime number, and let $c(n)$ be the n th composite number. For example, $p(1) = 2$, $p(10) = 29$, $c(1) = 4$ and $c(10) = 18$.

$$\{p(i) : i \geq 1\} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \dots\}$$

$$\{c(i) : i \geq 1\} = \{4, 6, 8, 9, 10, 12, 14, 15, 16, 18, \dots\}$$

Let P^D be the sequence of the digital roots of $\{p(i)\}$ (C^D is defined similarly for $\{c(i)\}$):

$$P^D = \{2, 3, 5, 7, 2, 4, 8, 1, 5, 2, \dots\}$$

$$C^D = \{4, 6, 8, 9, 1, 3, 5, 6, 7, 9, \dots\}$$

Let P_n be the integer formed by concatenating the first n elements of P^D (C_n is defined similarly for C^D).

$$P_{10} = 2357248152$$

$$C_{10} = 4689135679$$

Let $f(n)$ be the smallest positive integer that is a common superinteger of P_n and C_n . For example, $f(10) = 2357246891352679$, and $f(100) \bmod 1\,000\,000\,007 = 771661825$.

Find $f(10\,000) \bmod 1\,000\,000\,007$.

Problem 468: Smooth Divisors of Binomial Coefficients

Link

Tags: Combinatorics, Number Theory

An integer is called B -smooth if none of its prime factors is greater than B .

Let $S_B(n)$ be the largest B -smooth divisor of n .

Examples:

$$S_1(10) = 1$$

$$S_4(2100) = 12$$

$$S_{17}(2496144) = 5712$$

Define $F(n) = \sum_{B=1}^n \sum_{r=0}^n S_B\left(\binom{n}{r}\right)$. Here, $\binom{n}{r}$ denotes the binomial coefficient.

Examples:

$$F(11) = 3132$$

$$F(1111) \bmod 1\,000\,000\,993 = 706036312$$

$$F(111\,111) \bmod 1\,000\,000\,993 = 22156169$$

Find $F(11\,111\,111) \bmod 1\,000\,000\,993$.

Problem 469: Empty Chairs

Link

Tags: Combinatorics, Probability

In a room N chairs are placed around a round table.

Knights enter the room one by one and choose at random an available empty chair.

To have enough elbow room the knights always leave at least one empty chair between each other.

When there aren't any suitable chairs left, the fraction C of empty chairs is determined.

We also define $E(N)$ as the expected value of C .

We can verify that $E(4) = 1/2$ and $E(6) = 5/9$.

Find $E(10^{18})$. Give your answer rounded to fourteen decimal places in the form 0.abcdefghijklmn.

Problem 470: Super Ramvok

Link

Tags: Game Theory, Probability

Consider a single game of Ramvok:

Let t represent the maximum number of turns the game lasts. If $t = 0$, then the game ends immediately. Otherwise, on each turn i , the player rolls a die. After rolling, if $i < t$ the player can either stop the game and receive a prize equal to the value of the current roll, or discard the roll and try again next turn. If $i = t$, then the roll cannot be discarded and the prize must be accepted. Before the game begins, t is chosen by the player, who must then pay an up-front cost ct for some constant c . For $c = 0$, t can be chosen to be infinite (with an up-front cost of 0). Let $R(d, c)$ be the expected profit (i.e. net gain) that the player receives from a single game of optimally-played Ramvok, given a fair d -sided die and cost constant c . For example, $R(4, 0.2) = 2.65$. Assume that the player has sufficient funds for paying any/all up-front costs.

Now consider a game of Super Ramvok:

In Super Ramvok, the game of Ramvok is played repeatedly, but with a slight modification. After each game, the die is altered. The alteration process is as follows: The die is rolled once, and if the resulting face has its pips visible, then

that face is altered to be blank instead. If the face is already blank, then it is changed back to its original value. After the alteration is made, another game of Ramvok can begin (and during such a game, at each turn, the die is rolled until a face with a value on it appears). The player knows which faces are blank and which are not at all times. The game of Super Ramvok ends once all faces of the die are blank.

Let $S(d, c)$ be the expected profit that the player receives from an optimally-played game of Super Ramvok, given a fair d -sided die to start (with all sides visible), and cost constant c . For example, $S(6, 1) = 208.3$.

Let $F(n) = \sum_{4 \leq d \leq n} \sum_{0 \leq c \leq n} S(d, c)$.

Calculate $F(20)$, rounded to the nearest integer.

Problem 471: Triangle Inscribed in Ellipse

[Link](#)

Tags: Algebra, Geometry, Number Theory

The triangle $\triangle ABC$ is inscribed in an ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $0 < a, b$ integers.

Let $r(a, b)$ be the radius of the incircle of $\triangle ABC$ when the incircle has center $(2b, 0)$ and A has coordinates $\left(\frac{a}{2}, \frac{\sqrt{3}}{2}b\right)$.

For example, $r(3, 1) = \frac{1}{2}$, $r(6, 2) = 1$, $r(12, 3) = 2$.

Let $G(n) = \sum_{a=3}^n \sum_{b=1}^{\lfloor \frac{a-1}{2} \rfloor} r(a, b)$

You are given $G(10) = 20.59722222$, $G(100) = 19223.60980$ (rounded to 10 significant digits).

Find $G(10^{11})$.

Give your answer in scientific notation rounded to 10 significant digits. Use a lowercase e to separate mantissa and exponent.

For $G(10)$ the answer would have been 2.059722222e1.

Problem 472: Comfortable Distance II

[Link](#)

Tags: Combinatorics, Game Theory, Number Theory

There are N seats in a row. N people come one after another to fill the seats according to the following rules:

- No person sits beside another.
- The first person chooses any seat.
- Each subsequent person chooses the seat furthest from anyone else already seated, as long as it does not violate rule 1. If there is more than one choice satisfying this condition, then the person chooses the leftmost choice.

Note that due to rule 1, some seats will surely be left unoccupied, and the maximum number of people that can be seated is less than N (for $N > 1$).

Here are the possible seating arrangements for $N = 15$:

We see that if the first person chooses correctly, the 15 seats can seat up to 7 people.

We can also see that the first person has 9 choices to maximize the number of people that may be seated.

Let $f(N)$ be the number of choices the first person has to maximize the number of occupants for N seats in a row. Thus, $f(1) = 1$, $f(15) = 9$, $f(20) = 6$, and $f(500) = 16$.

Also, $\sum_{N=1}^{20} f(N) = 83$ for $1 \leq N \leq 20$ and $\sum_{N=1}^{500} f(N) = 13343$ for $1 \leq N \leq 500$.

Find $\sum_{N=1}^{10^{12}} f(N)$ for $1 \leq N \leq 10^{12}$. Give the last 8 digits of your answer.

Problem 473: Phigital Number Base

Link

Tags: Number Theory

Let φ be the golden ratio: $\varphi = \frac{1+\sqrt{5}}{2}$.

Remarkably it is possible to write every positive integer as a sum of powers of φ even if we require that every power of φ is used at most once in this sum.

Even then this representation is not unique.

We can make it unique by requiring that no powers with consecutive exponents are used and that the representation is finite.

E.g: $2 = \varphi + \varphi^{-2}$ and $3 = \varphi^2 + \varphi^{-2}$

To represent this sum of powers of φ we use a string of 0's and 1's with a point to indicate where the negative exponents start.

We call this the representation in the phigital numberbase.

So $1 = 1_\varphi$, $2 = 10.01_\varphi$, $3 = 100.01_\varphi$ and $14 = 100100.001001_\varphi$.

The strings representing 1, 2 and 14 in the phigital number base are palindromic, while the string representing 3 is not. (the phigital point is not the middle character).

The sum of the positive integers not exceeding 1000 whose phigital representation is palindromic is 4345.

Find the sum of the positive integers not exceeding 10^{10} whose phigital representation is palindromic.

Problem 474: Last Digits of Divisors

[Link](#)

Tags: Number Theory

For a positive integer n and digits d , we define $F(n, d)$ as the number of the divisors of n whose last digits equal d .

For example, $F(84, 4) = 3$. Among the divisors of 84 (1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84), three of them (4, 14, 84) have the last digit 4.

We can also verify that $F(12!, 12) = 11$ and $F(50!, 123) = 17888$.

Find $F(10^6!, 65432)$ modulo $(10^{16} + 61)$.

Problem 475: Music Festival

[Link](#)

Tags: Number Theory

$12n$ musicians participate at a music festival. On the first day, they form $3n$ quartets and practice all day.

It is a disaster. At the end of the day, all musicians decide they will never again agree to play with any member of their quartet.

On the second day, they form $4n$ trios, with every musician avoiding any previous quartet partners.

Let $f(12n)$ be the number of ways to organize the trios amongst the $12n$ musicians.

You are given $f(12) = 576$ and $f(24) \bmod 1\,000\,000\,007 = 509089824$.

Find $f(600) \bmod 1\,000\,000\,007$.

Problem 476: Circle Packing II

Link

Tags: Geometry, Graph Theory

Let $R(a, b, c)$ be the maximum area covered by three non-overlapping circles inside a triangle with edge lengths a , b and c .

Let $S(n)$ be the average value of $R(a, b, c)$ over all integer triplets (a, b, c) such that $1 \leq a \leq b \leq ca + b \leq n$.

You are given $S(2) = R(1, 1, 1) \approx 0.31998$, $S(5) \approx 1.25899$.

Find $S(1803)$ rounded to 5 decimal places behind the decimal point.

Problem 477: Number Sequence Game

Link

Tags: Game Theory, Sequences

The number sequence game starts with a sequence S of N numbers written on a line.

Two players alternate turns. The players on their respective turns must select and remove either the first or the last number remaining in the sequence.

A player's own score is (determined by) the sum of all the numbers that player has taken. Each player attempts to maximize their own sum.

If $N = 4$ and $S = \{1, 2, 10, 3\}$, then each player maximizes their own score as follows: - Player 1: removes the first number (1)

- Player 2: removes the last number from the remaining sequence (3)
- Player 1: removes the last number from the remaining sequence (10)
- Player 2: removes the remaining number (2)

Player 1 score is $1 + 10 = 11$.

Let $F(N)$ be the score of player 1 if both players follow the optimal strategy for the sequence $S = \{s_1, s_2, \dots, s_N\}$ defined as:

- $s_1 = 0$
- $s_{i+1} = (s_i^2 + 45)$ modulo 1 000 000 007

The sequence begins with $S = \{0, 45, 2070, 4284945, 753524550, 478107844, 894218625, \dots\}$.

You are given $F(2) = 45$, $F(4) = 4284990$, $F(100) = 26365463243$, $F(10^4) = 2495838522951$.

Find $F(10^8)$.

Problem 478: Mixtures

Link

Tags: Combinatorics, Game Theory, Number Theory

Let us consider mixtures of three substances: A, B and C. A mixture can be described by a ratio of the amounts of A, B, and C in it, i.e., $(a : b : c)$. For example, a mixture described by the ratio $(2 : 3 : 5)$ contains 20% A, 30% B and 50% C.

For the purposes of this problem, we cannot separate the individual components from a mixture. However, we can combine different amounts of different mixtures to form mixtures with new ratios.

For example, say we have three mixtures with ratios $(3 : 0 : 2)$, $(3 : 6 : 11)$ and $(3 : 3 : 4)$. By mixing 10 units of the first, 20 units of the second and 30 units of the third, we get a new mixture with ratio $(6 : 5 : 9)$, since:

$$(10 \cdot \frac{3}{5} + 20 \cdot \frac{3}{20} + 30 \cdot \frac{3}{10} : 10 \cdot \frac{0}{5} + 20 \cdot \frac{6}{20} + 30 \cdot \frac{3}{10} : 10 \cdot \frac{2}{5} + 20 \cdot \frac{11}{20} + 30 \cdot \frac{4}{10}) = (18 : 15 : 27) = (6 : 5 : 9)$$

However, with the same three mixtures, it is impossible to form the ratio $(3 : 2 : 1)$, since the amount of B is always less than the amount of C.

Let n be a positive integer. Suppose that for every triple of integers (a, b, c) with $0 \leq a, b, c \leq n$ and $\gcd(a, b, c) = 1$, we have a mixture with ratio $(a : b : c)$. Let $M(n)$ be the set of all such mixtures.

For example, $M(2)$ contains the 19 mixtures with the following ratios:

$$\begin{aligned} &\{(0 : 0 : 1), (0 : 1 : 0), (0 : 1 : 1), (0 : 1 : 2), (0 : 2 : 1), \\ &\quad (1 : 0 : 0), (1 : 0 : 1), (1 : 0 : 2), (1 : 1 : 0), (1 : 1 : 1), \\ &\quad (1 : 1 : 2), (1 : 2 : 0), (1 : 2 : 1), (1 : 2 : 2), (2 : 0 : 1), \\ &\quad (2 : 1 : 0), (2 : 1 : 1), (2 : 1 : 2), (2 : 2 : 1)\}. \end{aligned}$$

Let $E(n)$ be the number of subsets of $M(n)$ which can produce the mixture with ratio $(1 : 1 : 1)$, i.e., the mixture with equal parts A, B and C.

We can verify that $E(1) = 103$, $E(2) = 520447$, $E(10) \bmod 11^8 = 82608406$ and $E(500) \bmod 11^8 = 13801403$.

Find $E(10\,000\,000) \bmod 11^8$.

Problem 479: Roots on the Rise

[Link](#)

Tags: Algebra

Let a_k , b_k , and c_k represent the three solutions (real or complex numbers) to the equation $\frac{1}{x} = (\frac{k}{x})^2(k + x^2) - kx$.

For instance, for $k = 5$, we see that $\{a_5, b_5, c_5\}$ is approximately $\{5.727244, -0.363622 + 2.057397i, -0.363622 - 2.057397i\}$.

$$\text{Let } S(n) = \sum_{p=1}^n \sum_{k=1}^n (a_k + b_k)^p (b_k + c_k)^p (c_k + a_k)^p.$$

Interestingly, $S(n)$ is always an integer. For example, $S(4) = 51160$.

Find $S(10^6)$ modulo 1 000 000 007.

Problem 480: The Last Question

[Link](#)

Tags: Number Theory

Consider all the words which can be formed by selecting letters, in any order, from the phrase:

thereisasyetinsufficientdataforameaningfulanswer

Problem 481: Chef Showdown

[Link](#)

Tags: Combinatorics, Game Theory, Probability, Sequences

A group of chefs (numbered #1, #2, etc) participate in a turn-based strategic cooking competition. On each chef's turn, he/she cooks up a dish to the best of his/her ability and gives it to a separate panel of judges for taste-testing. Let $S(k)$ represent chef # k 's skill level (which is publicly known). More specifically, $S(k)$ is the probability that chef # k 's dish will be assessed favorably by the judges (on any/all turns). If the dish receives a favorable rating, then the chef must choose one other chef to be eliminated from the competition. The last chef remaining in the competition is the winner.

The game always begins with chef #1, with the turn order iterating sequentially over the rest of the chefs still in play. Then the cycle repeats from the lowest-numbered chef. All chefs aim to optimize their chances of winning within the rules as stated, assuming that the other chefs behave in the same manner. In the

event that a chef has more than one equally-optimal elimination choice, assume that the chosen chef is always the one with the next-closest turn.

Define $W_n(k)$ as the probability that chef $\#k$ wins in a competition with n chefs. If we have $S(1) = 0.25$, $S(2) = 0.5$, and $S(3) = 1$, then $W_3(1) = 0.29375$.

Going forward, we assign $S(k) = F_k/F_{n+1}$ over all $1 \leq k \leq n$, where F_k is a Fibonacci number: $F_k = F_{k-1} + F_{k-2}$ with base cases $F_1 = F_2 = 1$. Then, for example, when considering a competition with $n = 7$ chefs, we have $W_7(1) = 0.08965042$, $W_7(2) = 0.20775702$, $W_7(3) = 0.15291406$, $W_7(4) = 0.14554098$, $W_7(5) = 0.15905291$, $W_7(6) = 0.10261412$, and $W_7(7) = 0.14247050$, rounded to 8 decimal places each.

Let $E(n)$ represent the expected number of dishes cooked in a competition with n chefs. For instance, $E(7) = 42.28176050$.

Find $E(14)$ rounded to 8 decimal places.

Problem 482: The Incenter of a Triangle

Link

Tags: Geometry

ABC is an integer sided triangle with incenter I and perimeter p .

The segments IA , IB and IC have integral length as well.

Let $L = p + |IA| + |IB| + |IC|$.

Let $S(P) = \sum L$ for all such triangles where $p \leq P$. For example, $S(10^3) = 3619$.

Find $S(10^7)$.

Problem 483: Repeated Permutation

Link

Tags: Combinatorics, Number Theory

We define a permutation as an operation that rearranges the order of the elements $\{1, 2, 3, \dots, n\}$. There are $n!$ such permutations, one of which leaves the elements in their initial order. For $n = 3$ we have $3! = 6$ permutations:

- P_1 = keep the initial order
- P_2 = exchange the 1st and 2nd elements
- P_3 = exchange the 1st and 3rd elements
- P_4 = exchange the 2nd and 3rd elements

- P_5 = rotate the elements to the right
- P_6 = rotate the elements to the left

If we select one of these permutations, and we re-apply the same permutation repeatedly, we eventually restore the initial order. For a permutation P_i , let $f(P_i)$ be the number of steps required to restore the initial order by applying the permutation P_i repeatedly. For $n = 3$, we obtain:

- $f(P_1) = 1 : (1, 2, 3) \rightarrow (1, 2, 3)$
- $f(P_2) = 2 : (1, 2, 3) \rightarrow (2, 1, 3) \rightarrow (1, 2, 3)$
- $f(P_3) = 2 : (1, 2, 3) \rightarrow (3, 2, 1) \rightarrow (1, 2, 3)$
- $f(P_4) = 2 : (1, 2, 3) \rightarrow (1, 3, 2) \rightarrow (1, 2, 3)$
- $f(P_5) = 3 : (1, 2, 3) \rightarrow (3, 1, 2) \rightarrow (2, 3, 1) \rightarrow (1, 2, 3)$
- $f(P_6) = 3 : (1, 2, 3) \rightarrow (2, 3, 1) \rightarrow (3, 1, 2) \rightarrow (1, 2, 3)$

Let $g(n)$ be the average value of $f^2(P_i)$ over all permutations P_i of length n .
 $g(3) = (1^2 + 2^2 + 2^2 + 2^2 + 3^2 + 3^2)/3! = 31/6 \approx 5.166666667$
 $g(5) = 2081/120 \approx 1.734166667$
 $g(20) = 12422728886023769167301/2432902008176640000 \approx 5.106136147$

Find $g(350)$ and write the answer in scientific notation rounded to 10 significant digits, using a lowercase e to separate mantissa and exponent, as in the examples above.

Problem 484: Arithmetic Derivative

Link

Tags: Number Theory

The arithmetic derivative is defined by

- $p' = 1$ for any prime p
- $(ab)' = a'b + ab'$ for all integers a, b (Leibniz rule)

For example, $20' = 24$.

Find $\sum \mathbf{gcd}(k, k')$ for $1 \leq k \leq 5 \times 10^{15}$.

Note: $\mathbf{gcd}(x, y)$ denotes the greatest common divisor of x and y .

Problem 485: Maximum Number of Divisors

Link

Tags: Number Theory

Let $d(n)$ be the number of divisors of n .

Let $M(n, k)$ be the maximum value of $d(j)$ for $n \leq j \leq n + k - 1$.

Let $S(u, k)$ be the sum of $M(n, k)$ for $1 \leq n \leq u - k + 1$.

You are given that $S(1000, 10) = 17176$.

Find $S(100\,000\,000, 100\,000)$.

Problem 486: Palindrome-containing Strings

Link

Tags: Number Theory

Let $F_5(n)$ be the number of strings s such that:

- s consists only of '0's and '1's,
- s has length at most n , and
- s contains a palindromic substring of length at least 5.

For example, $F_5(4) = 0$, $F_5(5) = 8$, $F_5(6) = 42$ and $F_5(11) = 3844$.

Let $D(L)$ be the number of integers n such that $5 \leq n \leq L$ and $F_5(n)$ is divisible by 87654321.

For example, $D(10^7) = 0$ and $D(5 \cdot 10^9) = 51$.

Find $D(10^{18})$.

Problem 487: Sums of Power Sums

Link

Tags: Number Theory

Let $f_k(n)$ be the sum of the k th powers of the first n positive integers.

For example, $f_2(10) = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2 = 385$.

Let $S_k(n)$ be the sum of $f_k(i)$ for $1 \leq i \leq n$. For example, $S_4(100) = 35375333830$.

What is $\sum (S_{10000}(10^{12}) \bmod p)$ over all primes p between $2 \cdot 10^9$ and $2 \cdot 10^9 + 2000$?

Problem 488: Unbalanced Nim

Link

Tags: Combinatorics, Game Theory, Number Theory

Alice and Bob have enjoyed playing Nim every day. However, they finally got bored of playing ordinary three-heap Nim.

So, they added an extra rule:

- Must not make two heaps of the same size.

The triple (a, b, c) indicates the size of three heaps.

Under this extra rule, $(2, 4, 5)$ is one of the losing positions for the next player.

To illustrate:

- Alice moves to $(2, 4, 3)$
- Bob moves to $(0, 4, 3)$
- Alice moves to $(0, 2, 3)$
- Bob moves to $(0, 2, 1)$

Unlike ordinary three-heap Nim, $(0, 1, 2)$ and its permutations are the end states of this game.

For an integer N , we define $F(N)$ as the sum of $a + b + c$ for all the losing positions for the next player, with $0abcN$.

For example, $F(8) = 42$, because there are 4 losing positions for the next player, $(1, 3, 5)$, $(1, 4, 6)$, $(2, 3, 6)$ and $(2, 4, 5)$.

We can also verify that $F(128) = 496062$.

Find the last 9 digits of $F(10^{18})$.

Problem 489: Common Factors Between Two Sequences

Link

Tags: Number Theory

Let $G(a, b)$ be the smallest non-negative integer n for which $\gcd(n^3 + b, (n+a)^3 + b)$ is maximized.

For example, $G(1, 1) = 5$ because $\gcd(n^3 + 1, (n+1)^3 + 1)$ reaches its maximum value of 7 for $n = 5$, and is smaller for $0 \leq n < 5$.

Let $H(m, n) = \sum G(a, b)$ for $1 \leq a \leq m, 1 \leq b \leq n$.

You are given $H(5, 5) = 128878$ and $H(10, 10) = 32936544$.

Find $H(18, 1900)$.

Problem 490: Jumping Frog

Link

Tags: Number Theory

There are n stones in a pond, numbered 1 to n . Consecutive stones are spaced one unit apart.

A frog sits on stone 1. He wishes to visit each stone exactly once, stopping on stone n . However, he can only jump from one stone to another if they are at most 3 units apart. In other words, from stone i , he can reach a stone j if $1 \leq j \leq n$ and j is in the set $\{i - 3, i - 2, i - 1, i + 1, i + 2, i + 3\}$.

Let $f(n)$ be the number of ways he can do this. For example, $f(6) = 14$, as shown below:

$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$

$1 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 4 \rightarrow 6$

$1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 5 \rightarrow 6$

$1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 6$

$1 \rightarrow 2 \rightarrow 5 \rightarrow 3 \rightarrow 4 \rightarrow 6$

$1 \rightarrow 2 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 6$

$1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 6$

$1 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 4 \rightarrow 6$

$1 \rightarrow 3 \rightarrow 4 \rightarrow 2 \rightarrow 5 \rightarrow 6$

$1 \rightarrow 3 \rightarrow 5 \rightarrow 2 \rightarrow 4 \rightarrow 6$

$1 \rightarrow 4 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 6$

$1 \rightarrow 4 \rightarrow 2 \rightarrow 5 \rightarrow 3 \rightarrow 6$

$1 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 6$

$1 \rightarrow 4 \rightarrow 5 \rightarrow 2 \rightarrow 3 \rightarrow 6$

Other examples are $f(10) = 254$ and $f(40) = 1439682432976$.

Let $S(L) = \sum f(n)^3$ for $1 \leq n \leq L$.

Examples:

$$S(10) = 18230635$$

$$S(20) = 104207881192114219$$

$$S(1\,000) \bmod 10^9 = 225031475$$

$$S(1\,000\,000) \bmod 10^9 = 363486179$$

Find $S(10^{14}) \bmod 10^9$.

Problem 491: Double Pandigital Number Divisible by 11

[Link](#)

Tags: Number Theory

We call a positive integer double pandigital if it uses all the digits 0 to 9 exactly twice (with no leading zero). For example, 40561817703823564929 is one such number.

How many double pandigital numbers are divisible by 11?

Problem 492: Exploding Sequence

[Link](#)

Tags: Number Theory, Sequences

Define the sequence $a_1, a_2, a_3, \dots$ as:

- $a_1 = 1$
- $a_{n+1} = 6a_n^2 + 10a_n + 3$ for $n \geq 1$.

Examples:

$$a_3 = 2359$$

$$a_6 = 269221280981320216750489044576319$$

$$a_6 \bmod 1\,000\,000\,007 = 203064689$$

$$a_{100} \bmod 1\,000\,000\,007 = 456482974$$

Define $B(x, y, n)$ as $\sum (a_n \bmod p)$ for every prime p such that $x \leq p \leq x + y$.

Examples:

$$B(10^9, 10^3, 10^3) = 23674718882$$

$$B(10^9, 10^3, 10^{15}) = 20731563854$$

Find $B(10^9, 10^7, 10^{15})$.

Problem 493: Under the Rainbow

Link

Tags: Number Theory, Probability

70 coloured balls are placed in an urn, 10 for each of the seven rainbow colours.

What is the expected number of distinct colours in 20 randomly picked balls?

Give your answer with nine digits after the decimal point (a.bcd efghij).

Problem 494: Collatz Prefix Families

Link

Tags: Sequences

The Collatz sequence is defined as: $a_{i+1} = \begin{cases} \frac{a_i}{2} & \text{if } a_i \text{ is even} \\ 3a_i + 1 & \text{if } a_i \text{ is odd} \end{cases}$.

The Collatz conjecture states that starting from any positive integer, the sequence eventually reaches the cycle 1, 4, 2, 1, ...

We shall define the sequence prefix $p(n)$ for the Collatz sequence starting with $a_1 = n$ as the sub-sequence of all numbers not a power of 2 ($2^0 = 1$ is considered a power of 2 for this problem). For example: $p(13) = \{13, 40, 20, 10, 5\}$ $p(8) = \{\}$

Any number invalidating the conjecture would have an infinite length sequence prefix.

Let S_m be the set of all sequence prefixes of length m . Two sequences $\{a_1, a_2, \dots, a_m\}$ and $\{b_1, b_2, \dots, b_m\}$ in S_m are said to belong to the same prefix family if $a_i a_j$ if and only if $b_i b_j$ for all $1 \leq i, j \leq m$.

For example, in S_4 , $\{6, 3, 10, 5\}$ is in the same family as $\{454, 227, 682, 341\}$, but not $\{113, 340, 170, 85\}$.

Let $f(m)$ be the number of distinct prefix families in S_m .

You are given $f(5) = 5$, $f(10) = 55$, $f(20) = 6771$.

Find $f(90)$.

Problem 495: Writing n as the Product of k Distinct Positive Integers

Link

Tags: Combinatorics

Let $W(n, k)$ be the number of ways in which n can be written as the product of k distinct positive integers.

For example, $W(144, 4) = 7$. There are 7 ways in which 144 can be written as a product of 4 distinct positive integers:

- $144 = 1 \times 2 \times 4 \times 18$
- $144 = 1 \times 2 \times 8 \times 9$
- $144 = 1 \times 2 \times 3 \times 24$
- $144 = 1 \times 2 \times 6 \times 12$
- $144 = 1 \times 3 \times 4 \times 12$
- $144 = 1 \times 3 \times 6 \times 8$
- $144 = 2 \times 3 \times 4 \times 6$

Note that permutations of the integers themselves are not considered distinct.

Furthermore, $W(100!, 10)$ modulo $1\,000\,000\,007 = 287549200$.

Find $W(10000!, 30)$ modulo $1\,000\,000\,007$.

Problem 496: Incenter and Circumcenter of Triangle

Link

Tags: Geometry

Given an integer sided triangle ABC :

Let I be the incenter of ABC .

Let D be the intersection between the line AI and the circumcircle of ABC ($A \neq D$).

We define $F(L)$ as the sum of BC for the triangles ABC that satisfy $AC = DI$ and $BC \leq L$.

For example, $F(15) = 45$ because the triangles ABC with $(BC, AC, AB) = (6, 4, 5), (12, 8, 10), (12, 9, 7), (15, 9, 16)$ satisfy the conditions.

Find $F(10^9)$.

Problem 497: Drunken Tower of Hanoi

[Link](#)

Tags: Game Theory, Number Theory, Probability

Bob is very familiar with the famous mathematical puzzle/game, “Tower of Hanoi,” which consists of three upright rods and disks of different sizes that can slide onto any of the rods. The game begins with a stack of n disks placed on the leftmost rod in descending order by size. The objective of the game is to move all of the disks from the leftmost rod to the rightmost rod, given the following restrictions:

- Only one disk can be moved at a time.
- A valid move consists of taking the top disk from one stack and placing it onto another stack (or an empty rod).
- No disk can be placed on top of a smaller disk.

Moving on to a variant of this game, consider a long room k units (square tiles) wide, labeled from 1 to k in ascending order. Three rods are placed at squares a , b , and c , and a stack of n disks is placed on the rod at square a .

Bob begins the game standing at square b . His objective is to play the Tower of Hanoi game by moving all of the disks to the rod at square c . However, Bob can only pick up or set down a disk if he is on the same square as the rod/stack in question.

Unfortunately, Bob is also drunk. On a given move, Bob will either stumble one square to the left or one square to the right with equal probability, unless Bob is at either end of the room, in which case he can only move in one direction. Despite Bob’s inebriated state, he is still capable of following the rules of the game itself, as well as choosing when to pick up or put down a disk.

The following animation depicts a side-view of a sample game for $n = 3$, $k = 7$, $a = 2$, $b = 4$, and $c = 6$:

Let $E(n, k, a, b, c)$ be the expected number of squares that Bob travels during a single optimally-played game. A game is played optimally if the number of disk-pickups is minimized.

Interestingly enough, the result is always an integer. For example, $E(2, 5, 1, 3, 5) = 60$ and $E(3, 20, 4, 9, 17) = 2358$.

Find the last nine digits of $\sum_{1 \leq n \leq 10000} E(n, 10^n, 3^n, 6^n, 9^n)$.

Problem 498: Remainder of Polynomial Division

[Link](#)

Tags: Algebra

For positive integers n and m , we define two polynomials $F_n(x) = x^n$ and $G_m(x) = (x - 1)^m$.

We also define a polynomial $R_{n,m}(x)$ as the remainder of the division of $F_n(x)$ by $G_m(x)$.

For example, $R_{6,3}(x) = 15x^2 - 24x + 10$.

Let $C(n, m, d)$ be the absolute value of the coefficient of the d -th degree term of $R_{n,m}(x)$.

We can verify that $C(6, 3, 1) = 24$ and $C(100, 10, 4) = 227197811615775$.

Find $C(10^{13}, 10^{12}, 10^4) \bmod 999999937$.

Problem 499: St. Petersburg Lottery

Link

Tags: Game Theory, Probability, Sequences

A gambler decides to participate in a special lottery. In this lottery the gambler plays a series of one or more games.

Each game costs m pounds to play and starts with an initial pot of 1 pound. The gambler flips an unbiased coin. Every time a head appears, the pot is doubled and the gambler continues. When a tail appears, the game ends and the gambler collects the current value of the pot. The gambler is certain to win at least 1 pound, the starting value of the pot, at the cost of m pounds, the initial fee.

The game ends if the gambler's fortune falls below m pounds. Let $p_m(s)$ denote the probability that the gambler will never run out of money in this lottery given an initial fortune s and the cost per game m .

For example $p_2(2) \approx 0.2522$, $p_2(5) \approx 0.6873$ and $p_6(10\,000) \approx 0.9952$ (note: $p_m(s) = 0$ for sm).

Find $p_{15}(10^9)$ and give your answer rounded to 7 decimal places behind the decimal point in the form 0.abcdefg.

Problem 500: Problem 500!!!

Link

Tags: Number Theory

The number of divisors of 120 is 16.

In fact 120 is the smallest number having 16 divisors.

Find the smallest number with 2^{500500} divisors.

Give your answer modulo 500500507.

Problem 501: Eight Divisors

[Link](#)

Tags: Number Theory

The eight divisors of 24 are 1, 2, 3, 4, 6, 8, 12 and 24. The ten numbers not exceeding 100 having exactly eight divisors are 24, 30, 40, 42, 54, 56, 66, 70, 78 and 88. Let $f(n)$ be the count of numbers not exceeding n with exactly eight divisors.

You are given $f(100) = 10$, $f(1000) = 180$ and $f(10^6) = 224427$.

Find $f(10^{12})$.

Problem 502: Counting Castles

[Link](#)

Tags: Combinatorics, Game Theory, Geometry, Graph Theory

We define a block to be a rectangle with a height of 1 and an integer-valued length. Let a castle be a configuration of stacked blocks.

Given a game grid that is w units wide and h units tall, a castle is generated according to the following rules:

- Blocks can be placed on top of other blocks as long as nothing sticks out past the edges or hangs out over open space.
- All blocks are aligned/snapped to the grid.
- Any two neighboring blocks on the same row have at least one unit of space between them.
- The bottom row is occupied by a block of length w .
- The maximum achieved height of the entire castle is exactly h .
- The castle is made from an even number of blocks.

The following is a sample castle for $w = 8$ and $h = 5$:

Let $F(w, h)$ represent the number of valid castles, given grid parameters w and h .

For example, $F(4, 2) = 10$, $F(13, 10) = 3729050610636$, $F(10, 13) = 37959702514$, and $F(100, 100) \bmod 1\,000\,000\,007 = 841913936$.

Find $(F(10^{12}, 100) + F(10000, 10000) + F(100, 10^{12})) \bmod 1\,000\,000\,007$.

Problem 503: Compromise or Persist

Link

Tags: Game Theory, Probability

Alice is playing a game with n cards numbered 1 to n .

A game consists of iterations of the following steps.

- (1) Alice picks one of the cards at random.
- (2) Alice cannot see the number on it. Instead, Bob, one of her friends, sees the number and tells Alice how many previously-seen numbers are bigger than the number which he is seeing.
- (3) Alice can end or continue the game. If she decides to end, the number becomes her score. If she decides to continue, the card is removed from the game and she returns to (1). If there is no card left, she is forced to end the game.

Let $F(n)$ be Alice's expected score if she takes the optimized strategy to minimize her score.

For example, $F(3) = 5/3$. At the first iteration, she should continue the game. At the second iteration, she should end the game if Bob says that one previously-seen number is bigger than the number which he is seeing, otherwise she should continue the game.

We can also verify that $F(4) = 15/8$ and $F(10) \approx 2.5579365079$.

Find $F(10^6)$. Give your answer rounded to 10 decimal places behind the decimal point.

Problem 504: Square on the Inside

Link

Tags: Combinatorics

Let $ABCD$ be a quadrilateral whose vertices are lattice points lying on the coordinate axes as follows:

$A(a, 0)$, $B(0, b)$, $C(-c, 0)$, $D(0, -d)$, where $1 \leq a, b, c, d \leq m$ and a, b, c, d, m are integers.

It can be shown that for $m = 4$ there are exactly 256 valid ways to construct $ABCD$. Of these 256 quadrilaterals, 42 of them strictly contain a square number of lattice points.

How many quadrilaterals $ABCD$ strictly contain a square number of lattice points for $m = 100$?

Problem 505: Bidirectional Recurrence

Link

Tags: Number Theory

Let:

$$\begin{aligned} x(0) &= 0 \\ x(1) &= 1 \\ x(2k) &= (3x(k) + 2x(\lfloor \frac{k}{2} \rfloor)) \bmod 2^{60} \text{ for } k \geq 1, \text{ where } \lfloor \cdot \rfloor \text{ is the floor function} \\ x(2k+1) &= (2x(k) + 3x(\lfloor \frac{k}{2} \rfloor)) \bmod 2^{60} \text{ for } k \geq 1 \\ y_n(k) &= \begin{cases} x(k) & \text{if } k \geq n \\ 2^{60} - 1 - \max(y_n(2k), y_n(2k+1)) & \text{if } k < n \end{cases} \\ A(n) &= y_n(1) \end{aligned}$$

You are given:

$$\begin{aligned} x(2) &= 3 \\ x(3) &= 2 \\ x(4) &= 11 \\ y_4(4) &= 11 \\ y_4(3) &= 2^{60} - 9 \\ y_4(2) &= 2^{60} - 12 \\ y_4(1) &= A(4) = 8 \\ A(10) &= 2^{60} - 34 \\ A(10^3) &= 101881 \end{aligned}$$

Find $A(10^{12})$.

Problem 506: Clock Sequence

Link

Tags: Number Theory, Sequences

Consider the infinite repeating sequence of digits:

1234321234321234321...

Amazingly, you can break this sequence of digits into a sequence of integers such that the sum of the digits in the n -th value is n .

The sequence goes as follows:

1, 2, 3, 4, 32, 123, 43, 2123, 432, 1234, 32123, ...

Let v_n be the n -th value in this sequence. For example, $v_2 = 2$, $v_5 = 32$ and $v_{11} = 32123$.

Let $S(n)$ be $v_1 + v_2 + \dots + v_n$. For example, $S(11) = 36120$, and $S(1000) \bmod 123454321 = 18232686$.

Find $S(10^{14}) \bmod 123454321$.

Problem 507: Shortest Lattice Vector

Link

Tags: Game Theory

Let t_n be the tribonacci numbers defined as:

$$t_0 = t_1 = 0;$$

$$t_2 = 1;$$

$$t_n = t_{n-1} + t_{n-2} + t_{n-3} \text{ for } n \geq 3$$

and let $r_n = t_n \bmod 10^7$.

For each pair of Vectors $V_n = (v_1, v_2, v_3)$ and $W_n = (w_1, w_2, w_3)$ with $v_1 = r_{12n-11} - r_{12n-10}$, $v_2 = r_{12n-9} + r_{12n-8}$, $v_3 = r_{12n-7} \cdot r_{12n-6}$ and $w_1 = r_{12n-5} - r_{12n-4}$, $w_2 = r_{12n-3} + r_{12n-2}$, $w_3 = r_{12n-1} \cdot r_{12n}$

we define $S(n)$ as the minimal value of the manhattan length of the vector $D = k \cdot V_n + l \cdot W_n$ measured as $|k \cdot v_1 + l \cdot w_1| + |k \cdot v_2 + l \cdot w_2| + |k \cdot v_3 + l \cdot w_3|$ for any integers k and l with $(k, l) \neq (0, 0)$.

The first vector pair is $(-1, 3, 28)$, $(-11, 125, 40826)$.

You are given that $S(1) = 32$ and $\sum_{n=1}^{10} S(n) = 130762273722$.

Find $\sum_{n=1}^{20000000} S(n)$.

Problem 508: Integers in Base $i - 1$

Link

Tags: Number Theory, Sequences

Consider the Gaussian integer $i - 1$. A base $i - 1$ representation of a Gaussian integer $a + bi$ is a finite sequence of digits $d_{n-1}d_{n-2}\cdots d_1d_0$ such that:

- $a + bi = d_{n-1}(i - 1)^{n-1} + d_{n-2}(i - 1)^{n-2} + \cdots + d_1(i - 1) + d_0$
- Each d_k is in $\{0, 1\}$
- There are no leading zeroes, i.e. $d_{n-1} \neq 0$, unless $a + bi$ is itself 0

Here are base $i - 1$ representations of a few Gaussian integers:

$$11 + 24i \rightarrow 111010110001101$$

$$24 - 11i \rightarrow 110010110011$$

$$8 + 0i \rightarrow 111000000$$

$$-5 + 0i \rightarrow 11001101$$

$$0 + 0i \rightarrow 0$$

Remarkably, every Gaussian integer has a unique base $i - 1$ representation!

Define $f(a + bi)$ as the number of 1s in the unique base $i - 1$ representation of $a + bi$. For example, $f(11 + 24i) = 9$ and $f(24 - 11i) = 7$.

Define $B(L)$ as the sum of $f(a + bi)$ for all integers a, b such that $|a| \leq L$ and $|b| \leq L$. For example, $B(500) = 10795060$.

Find $B(10^{15}) \bmod 1\,000\,000\,007$.

Problem 509: Divisor Nim

Link

Tags: Game Theory, Number Theory

Anton and Bertrand love to play three pile Nim.

However, after a lot of games of Nim they got bored and changed the rules somewhat.

They may only take a number of stones from a pile that is a proper divisor of the number of stones present in the pile. E.g. if a pile at a certain moment contains 24 stones they may take only 1, 2, 3, 4, 6, 8 or 12 stones from that pile.

So if a pile contains one stone they can't take the last stone from it as 1 isn't a proper divisor of 1.

The first player that can't make a valid move loses the game.

Of course both Anton and Bertrand play optimally.

The triple (a, b, c) indicates the number of stones in the three piles.

Let $S(n)$ be the number of winning positions for the next player for $1 \leq a, b, c \leq n$.
 $S(10) = 692$ and $S(100) = 735494$.

Find $S(123456787654321)$ modulo 1234567890.

Problem 510: Tangent Circles

Link

Tags: Geometry

Circles A and B are tangent to each other and to line L at three distinct points.

Circle C is inside the space between A , B and L , and tangent to all three.

Let r_A , r_B and r_C be the radii of A , B and C respectively.

Problem 511: Sequences with Nice Divisibility Properties

Link

Tags: Number Theory, Sequences

Let $Seq(n, k)$ be the number of positive-integer sequences $\{a_i\}_{1 \leq i \leq n}$ of length n such that:

- n is divisible by a_i for $1 \leq i \leq n$, and
 - $n + a_1 + a_2 + \cdots + a_n$ is divisible by k .

Examples:

$Seq(3, 4) = 4$, and the 4 sequences are:

$\{1, 1, 3\}$

$\{1, 3, 1\}$

$\{3, 1, 1\}$

$\{3, 3, 3\}$

$Seq(4, 11) = 8$, and the 8 sequences are:

$\{1, 1, 1, 4\}$

$\{1, 1, 4, 1\}$

$\{1, 4, 1, 1\}$

$\{4, 1, 1, 1\}$

$\{2, 2, 2, 1\}$

$\{2, 2, 1, 2\}$

$\{2, 1, 2, 2\}$

$\{1, 2, 2, 2\}$

The last nine digits of $Seq(1111, 24)$ are 840643584.

Find the last nine digits of $Seq(1234567898765, 4321)$.

Problem 512: Sums of Totients of Powers

Link

Tags: Number Theory

Let $\varphi(n)$ be Euler's totient function.

Let $f(n) = (\sum_{i=1}^n \varphi(n^i)) \bmod (n+1)$.

Let $g(n) = \sum_{i=1}^n f(i)$.

$g(100) = 2007$.

Find $g(5 \times 10^8)$.

Problem 513: Integral Median

Link

Tags: Geometry

ABC is an integral sided triangle with sides $a \leq b \leq c$.

m_C is the median connecting C and the midpoint of AB .

$F(n)$ is the number of such triangles with $c \leq n$ for which m_C has integral length as well.

$F(10) = 3$ and $F(50) = 165$.

Find $F(100000)$.

Problem 514: Geoboard Shapes

Link

Tags: Combinatorics, Probability

A geoboard (of order N) is a square board with equally-spaced pins protruding from the surface, representing an integer point lattice for coordinates $0 \leq x, y \leq N$.

John begins with a pinless geoboard. Each position on the board is a hole that can be filled with a pin. John decides to generate a random integer between 1 and $N + 1$ (inclusive) for each hole in the geoboard. If the random integer is equal to 1 for a given hole, then a pin is placed in that hole.

After John is finished generating numbers for all $(N + 1)^2$ holes and placing any/all corresponding pins, he wraps a tight rubberband around the entire group of pins protruding from the board. Let S represent the shape that is formed. S can also be defined as the smallest convex shape that contains all the pins.

Problem 515: Dissonant Numbers

[Link](#)

Tags: Number Theory

Let $d(p, n, 0)$ be the multiplicative inverse of n modulo prime p , defined as $n \times d(p, n, 0) = 1 \pmod p$.

Let $d(p, n, k) = \sum_{i=1}^n d(p, i, k - 1)$ for $k \geq 1$.

Let $D(a, b, k) = \sum (d(p, p - 1, k) \pmod p)$ for all primes $a \leq pa + b$.

You are given:

- $D(101, 1, 10) = 45$
- $D(10^3, 10^2, 10^2) = 8334$
- $D(10^6, 10^3, 10^3) = 38162302$

Find $D(10^9, 10^5, 10^5)$.

Problem 516: 5-smooth Totients

[Link](#)

Tags: Number Theory

5-smooth numbers are numbers whose largest prime factor doesn't exceed 5.

5-smooth numbers are also called Hamming numbers.

Let $S(L)$ be the sum of the numbers n not exceeding L such that Euler's totient function $\phi(n)$ is a Hamming number.

$S(100) = 3728$.

Find $S(10^{12})$. Give your answer modulo 2^{32} .

Problem 517: A Real Recursion

Link

Tags: Number Theory, Sequences

For every real number a is given the sequence g_a by:

$$g_a(x) = 1 \text{ for } x < a$$

$$g_a(x) = g_a(x-1) + g_a(x-a) \text{ for } x \geq a$$

$$G(n) = g_{\sqrt{n}}(n)$$

$$G(90) = 7564511.$$

Find $\sum G(p)$ for p prime and $10000000 < p < 10010000$

Give your answer modulo 1000000007.

Problem 518: Prime Triples and Geometric Sequences

Link

Tags: Game Theory, Number Theory, Sequences

Let $S(n) = \sum a + b + c$ over all triples (a, b, c) such that:

- a, b and c are prime numbers.
- $abc \mid n$.
- $a+1, b+1$, and $c+1$ form a geometric sequence.

For example, $S(100) = 1035$ with the following triples:

$(2, 5, 11), (2, 11, 47), (5, 11, 23), (5, 17, 53), (7, 11, 17), (7, 23, 71), (11, 23, 47),$
 $(17, 23, 31), (17, 41, 97), (31, 47, 71), (71, 83, 97)$

Find $S(10^8)$.

Problem 519: Tricoloured Coin Fountains

Link

Tags: Combinatorics, Probability

An arrangement of coins in one or more rows with the bottom row being a block without gaps and every coin in a higher row touching exactly two coins in the

row below is called a fountain of coins. Let $f(n)$ be the number of possible fountains with n coins. For 4 coins there are three possible arrangements:

Problem 520: Simbers

Link

Tags: Number Theory

We define a simber to be a positive integer in which any odd digit, if present, occurs an odd number of times, and any even digit, if present, occurs an even number of times.

For example, 141221242 is a 9-digit simber because it has three 1's, four 2's and two 4's.

Let $Q(n)$ be the count of all simbers with at most n digits.

You are given $Q(7) = 287975$ and $Q(100) \bmod 1\,000\,000\,123 = 123864868$.

Find $(\sum_{1 \leq u \leq 39} Q(2^u)) \bmod 1\,000\,000\,123$.

Problem 521: Smallest Prime Factor

Link

Tags: Number Theory

Let $\text{smpf}(n)$ be the smallest prime factor of n .

$\text{smpf}(91) = 7$ because $91 = 7 \times 13$ and $\text{smpf}(45) = 3$ because $45 = 3 \times 3 \times 5$.

Let $S(n)$ be the sum of $\text{smpf}(i)$ for $2 \leq i \leq n$.

E.g. $S(100) = 1257$.

Find $S(10^{12}) \bmod 10^9$.

Problem 522: Hilbert's Blackout

Link

Tags: Combinatorics, Game Theory, Probability

Despite the popularity of Hilbert's infinite hotel, Hilbert decided to try managing extremely large finite hotels, instead.

To cut costs, Hilbert wished to power the new hotel with his own special generator. Each floor would send power to the floor above it, with the top floor sending

power back down to the bottom floor. That way, Hilbert could have the generator placed on any given floor (as he likes having the option) and have electricity flow freely throughout the entire hotel.

Unfortunately, the contractors misinterpreted the schematics when they built the hotel. They informed Hilbert that each floor sends power to another floor at random, instead. This may compromise Hilbert's freedom to have the generator placed anywhere, since blackouts could occur on certain floors.

For example, consider a sample flow diagram for a three-story hotel:

If the generator were placed on the first floor, then every floor would receive power. But if it were placed on the second or third floors instead, then there would be a blackout on the first floor. Note that while a given floor can receive power from many other floors at once, it can only send power to one other floor.

To resolve the blackout concerns, Hilbert decided to have a minimal number of floors rewired. To rewire a floor is to change the floor it sends power to. In the sample diagram above, all possible blackouts can be avoided by rewiring the second floor to send power to the first floor instead of the third floor.

Let $F(n)$ be the sum of the minimum number of floor rewirings needed over all possible power-flow arrangements in a hotel of n floors. For example, $F(3) = 6$, $F(8) = 16276736$, and $F(100) \bmod 135707531 = 84326147$.

Find $F(12344321) \bmod 135707531$.

Problem 523: First Sort I

Link

Tags: Combinatorics, Game Theory, Number Theory, Probability

Consider the following algorithm for sorting a list:

- 1. Starting from the beginning of the list, check each pair of adjacent elements in turn.
- 2. If the elements are out of order:
 - a. Move the smallest element of the pair at the beginning of the list.
 - b. Restart the process from step 1.
- 3. If all pairs are in order, stop.

For example, the list $\{4\ 1\ 3\ 2\}$ is sorted as follows:

- 4132 (4 and 1 are out of order so move 1 to the front of the list)
- 1432 (4 and 3 are out of order so move 3 to the front of the list)
- 3142 (3 and 1 are out of order so move 1 to the front of the list)

- 1 3 4 2 (4 and 2 are out of order so move 2 to the front of the list)
- 2 1 3 4 (2 and 1 are out of order so move 1 to the front of the list)
- 1 2 3 4 (The list is now sorted)

Let $F(L)$ be the number of times step 2a is executed to sort list L . For example, $F(\{4\ 1\ 3\ 2\}) = 5$.

Let $E(n)$ be the expected value of $F(P)$ over all permutations P of the integers $\{1, 2, \dots, n\}$.

You are given $E(4) = 3.25$ and $E(10) = 115.725$.

Find $E(30)$. Give your answer rounded to two digits after the decimal point.

Problem 524: First Sort II

Link

Tags: Combinatorics, Game Theory, Graph Theory

Consider the following algorithm for sorting a list:

1. Starting from the beginning of the list, check each pair of adjacent elements in turn.
2. If the elements are out of order:
 - a. Move the smallest element of the pair at the beginning of the list.
 - b. Restart the process from step 1.
3. If all pairs are in order, stop.

For example, the list $\{4\ 1\ 3\ 2\}$ is sorted as follows:

- 4 1 3 2 (4 and 1 are out of order so move 1 to the front of the list)
- 1 4 3 2 (4 and 3 are out of order so move 3 to the front of the list)
- 3 1 4 2 (3 and 1 are out of order so move 1 to the front of the list)
- 1 3 4 2 (4 and 2 are out of order so move 2 to the front of the list)
- 2 1 3 4 (2 and 1 are out of order so move 1 to the front of the list)
- 1 2 3 4 (The list is now sorted)

Let $F(L)$ be the number of times step 2a is executed to sort list L . For example, $F(\{4\ 1\ 3\ 2\}) = 5$.

We can list all permutations P of the integers $\{1, 2, \dots, n\}$ in lexicographical order, and assign to each permutation an index $I_n(P)$ from 1 to $n!$ corresponding to its position in the list.

Let $Q(n, k) = \min(I_n(P))$ for $F(P) = k$, the index of the first permutation requiring exactly k steps to sort with First Sort. If there is no permutation for which $F(P) = k$, then $Q(n, k)$ is undefined.

For $n = 4$ we have:

PI4(P)F(P) {1, 2, 3, 4}10Q(4, 0) = 1 {1, 2, 4, 3}24Q(4, 4) = 2 {1, 3, 2, 4}32Q(4, 2) = 3 {1, 3, 4, 2}42 {1, 4, 2, 3}56Q(4, 6) = 5 {1, 4, 3, 2}64 {2, 1, 3, 4}71Q(4, 1) = 7 {2, 1, 4, 3}85Q(4, 5) = 8 {2, 3, 1, 4}91 {2, 3, 4, 1}101 {2, 4, 1, 3}115 {2, 4, 3, 1}123Q(4, 3) = 12 {3, 1, 2, 4}133 {3, 1, 4, 2}143 {3, 2, 1, 4}152 {3, 2, 4, 1}162 {3, 4, 1, 2}173 {3, 4, 2, 1}182 {4, 1, 2, 3}197Q(4, 7) = 19 {4, 1, 3, 2}205 {4, 2, 1, 3}216 {4, 2, 3, 1}224 {4, 3, 1, 2}234 {4, 3, 2, 1}243

Let $R(k) = \min(Q(n, k))$ over all n for which $Q(n, k)$ is defined.

Find $R(12^{12})$.

Problem 525: Rolling Ellipse

Link

Tags: Algebra, Geometry

An ellipse $E(a, b)$ is given at its initial position by equation:

$$\frac{x^2}{a^2} + \frac{(y-b)^2}{b^2} = 1$$

The ellipse rolls without slipping along the x axis for one complete turn. Interestingly, the length of the curve generated by a focus is independent from the size of the minor axis:

$$F(a, b) = 2\pi \max(a, b)$$

Problem 526: Largest Prime Factors of Consecutive Numbers

Link

Tags: Number Theory

Let $f(n)$ be the largest prime factor of n .

Let $g(n) = f(n) + f(n+1) + f(n+2) + f(n+3) + f(n+4) + f(n+5) + f(n+6) + f(n+7) + f(n+8)$, the sum of the largest prime factor of each of nine consecutive numbers starting with n .

Let $h(n)$ be the maximum value of $g(k)$ for $2 \leq k \leq n$.

You are given:

- $f(100) = 5$
- $f(101) = 101$
- $g(100) = 409$
- $h(100) = 417$
- $h(10^9) = 4896292593$

Find $h(10^{16})$.

Problem 527: Randomized Binary Search

[Link](#)

Tags: Game Theory, Probability

A secret integer t is selected at random within the range $1 \leq t \leq n$.

The goal is to guess the value of t by making repeated guesses, via integer g . After a guess is made, there are three possible outcomes, in which it will be revealed that either gt , $g = t$, or gt . Then the process can repeat as necessary.

Normally, the number of guesses required on average can be minimized with a binary search: Given a lower bound L and upper bound H (initialized to $L = 1$ and $H = n$), let $g = \lfloor (L + H)/2 \rfloor$ where $\lfloor \cdot \rfloor$ is the integer floor function. If $g = t$, the process ends. Otherwise, if gt , set $L = g + 1$, but if gt instead, set $H = g - 1$. After setting the new bounds, the search process repeats, and ultimately ends once t is found. Even if t can be deduced without searching, assume that a search will be required anyway to confirm the value.

Your friend Bob believes that the standard binary search is not that much better than his randomized variant: Instead of setting $g = \lfloor (L + H)/2 \rfloor$, simply let g be a random integer between L and H , inclusive. The rest of the algorithm is the same as the standard binary search. This new search routine will be referred to as a random binary search.

Given that $1 \leq t \leq n$ for random t , let $B(n)$ be the expected number of guesses needed to find t using the standard binary search, and let $R(n)$ be the expected number of guesses needed to find t using the random binary search. For example, $B(6) = 2.33333333$ and $R(6) = 2.71666667$ when rounded to 8 decimal places.

Find $R(10^{10}) - B(10^{10})$ rounded to 8 decimal places.

Problem 528: Constrained Sums

[Link](#)

Tags: Number Theory

Let $S(n, k, b)$ represent the number of valid solutions to $x_1 + x_2 + \cdots + x_k \leq n$, where $0 \leq x_m \leq b^m$ for all $1 \leq m \leq k$.

For example, $S(14, 3, 2) = 135$, $S(200, 5, 3) = 12949440$, and $S(1000, 10, 5) \bmod 1\,000\,000\,007 = 624839075$.

Find $(\sum_{10 \leq k \leq 15} S(10^k, k, k)) \bmod 1\,000\,000\,007$.

Problem 529: 10-substrings

Link

Tags: Number Theory

A 10-substring of a number is a substring of its digits that sum to 10. For example, the 10-substrings of the number 3523014 are:

- 3523014
- 3523014
- 3523014
- 3523014

A number is called 10-substring-friendly if every one of its digits belongs to a 10-substring. For example, 3523014 is 10-substring-friendly, but 28546 is not.

Let $T(n)$ be the number of 10-substring-friendly numbers from 1 to 10^n (inclusive).

For example $T(2) = 9$ and $T(5) = 3492$.

Find $T(10^{18}) \bmod 1\,000\,000\,007$.

Problem 530: GCD of Divisors

Link

Tags: Number Theory

Every divisor d of a number n has a complementary divisor n/d .

Let $f(n)$ be the sum of the greatest common divisor of d and n/d over all positive divisors d of n , that is $f(n) = \sum_{d|n} \gcd(d, \frac{n}{d})$.

Let F be the summatory function of f , that is $F(k) = \sum_{n=1}^k f(n)$.

You are given that $F(10) = 32$ and $F(1000) = 12776$.

Find $F(10^{15})$.

Problem 531: Chinese Leftovers

[Link](#)

Tags: Number Theory

Let $g(a, n, b, m)$ be the smallest non-negative solution x to the system:

$$x = a \bmod n$$

$$x = b \bmod m$$

if such a solution exists, otherwise 0.

E.g. $g(2, 4, 4, 6) = 10$, but $g(3, 4, 4, 6) = 0$.

Let $\phi(n)$ be Euler's totient function.

Let $f(n, m) = g(\phi(n), n, \phi(m), m)$

Find $\sum f(n, m)$ for $1000000 \leq nm \leq 1005000$.

Problem 532: Nanobots on Geodesics

[Link](#)

Tags: Combinatorics, Geometry

Bob is a manufacturer of nanobots and wants to impress his customers by giving them a ball coloured by his new nanobots as a present.

His nanobots can be programmed to select and locate exactly one other bot precisely and, after activation, move towards this bot along the shortest possible path and draw a coloured line onto the surface while moving. Placed on a plane, the bots will start to move towards their selected bots in a straight line. In contrast, being placed on a ball, they will start to move along a geodesic as the shortest possible path. However, in both cases, whenever their target moves they will adjust their direction instantaneously to the new shortest possible path. All bots will move at the same speed after their simultaneous activation until each bot reaches its goal.

Now Bob places n bots on the ball (with radius 1) equidistantly on a small circle with radius 0.999 and programs each of them to move toward the next nanobot sitting counterclockwise on that small circle. After activation, the bots move in a sort of spiral until they finally meet at one point on the ball.

Using three bots, Bob finds that every bot will draw a line of length 2.84, resulting in a total length of 8.52 for all three bots, each time rounded to two decimal places. The coloured ball looks like this:

Problem 533: Minimum Values of the Carmichael Function

Link

Tags: Number Theory

The Carmichael function $\lambda(n)$ is defined as the smallest positive integer m such that $a^m = 1$ modulo n for all integers a coprime with n .

For example $\lambda(8) = 2$ and $\lambda(240) = 4$.

Define $L(n)$ as the smallest positive integer m such that $\lambda(k) \geq n$ for all $k \geq m$.

For example, $L(6) = 241$ and $L(100) = 20\,174\,525\,281$.

Find $L(20\,000\,000)$. Give the last 9 digits of your answer.

Problem 534: Weak Queens

Link

Tags: Game Theory, Number Theory

The classical eight queens puzzle is the well known problem of placing eight chess queens on an 8×8 chessboard so that no two queens threaten each other. Allowing configurations to reappear in rotated or mirrored form, a total of 92 distinct configurations can be found for eight queens. The general case asks for the number of distinct ways of placing n queens on an $n \times n$ board, e.g. you can find 2 distinct configurations for $n = 4$.

Let's define a weak queen on an $n \times n$ board to be a piece which can move any number of squares if moved horizontally, but a maximum of $n - 1 - w$ squares if moved vertically or diagonally, $0 \leq w \leq n$ being the "weakness factor". For example, a weak queen on an $n \times n$ board with a weakness factor of $w = 1$ located in the bottom row will not be able to threaten any square in the top row as the weak queen would need to move $n - 1$ squares vertically or diagonally to get there, but may only move $n - 2$ squares in these directions. In contrast, the weak queen is not handicapped horizontally, thus threatening every square in its own row, independently from its current position in that row.

Let $Q(n, w)$ be the number of ways n weak queens with weakness factor w can be placed on an $n \times n$ board so that no two queens threaten each other. It can be shown, for example, that $Q(4, 0) = 2$, $Q(4, 2) = 16$ and $Q(4, 3) = 256$.

Let $S(n) = \sum_{w=0}^{n-1} Q(n, w)$.

You are given that $S(4) = 276$ and $S(5) = 3347$.

Find $S(14)$.

Problem 535: Fractal Sequence

Link

Tags: Game Theory, Geometry, Number Theory, Sequences

Consider the infinite integer sequence S starting with:

$S = 1, 1, 2, 1, 3, 2, 4, 1, 5, 3, 6, 2, 7, 8, 4, 9, 1, 10, 11, 5, \dots$

Circle the first occurrence of each integer.

$S = \text{circle}1, 1, \text{circle}2, 1, \text{circle}3, 2, \text{circle}4, 1, \text{circle}5, 3, \text{circle}6, 2, \text{circle}7, \text{circle}8, 4, \text{circle}9, 1, \text{circle}10, \text{circle}11,$

The sequence is characterized by the following properties:

- The circled numbers are consecutive integers starting with 1.
- Immediately preceding each non-circled numbers a_i , there are exactly $\lfloor \sqrt{a_i} \rfloor$ adjacent circled numbers, where $\lfloor \cdot \rfloor$ is the floor function.
- If we remove all circled numbers, the remaining numbers form a sequence identical to S , so S is a fractal sequence.

Let $T(n)$ be the sum of the first n elements of the sequence.

You are given $T(1) = 1$, $T(20) = 86$, $T(10^3) = 364089$ and $T(10^9) = 498676527978348241$.

Find $T(10^{18})$. Give the last 9 digits of your answer.

Problem 536: Modulo Power Identity

Link

Tags: Game Theory

Let $S(n)$ be the sum of all positive integers m not exceeding n having the following property:

$a^{m+4} \equiv a \pmod{m}$ for all integers a .

The values of $m \leq 100$ that satisfy this property are 1, 2, 3, 5 and 21, thus $S(100) = 1 + 2 + 3 + 5 + 21 = 32$.

You are given $S(10^6) = 22868117$.

Find $S(10^{12})$.

Problem 537: Counting Tuples

Link

Tags: Combinatorics, Number Theory

Let $\pi(x)$ be the prime counting function, i.e. the number of prime numbers less than or equal to x .

For example, $\pi(1) = 0$, $\pi(2) = 1$, $\pi(100) = 25$.

Let $T(n, k)$ be the number of k -tuples $(x_1, \dots, x_k)$ which satisfy:

1. every x_i is a positive integer;

$$2. \sum_{i=1}^k \pi(x_i) = n$$

For example $T(3, 3) = 19$.

The 19 tuples are $(1, 1, 5)$, $(1, 5, 1)$, $(5, 1, 1)$, $(1, 1, 6)$, $(1, 6, 1)$, $(6, 1, 1)$, $(1, 2, 3)$, $(1, 3, 2)$, $(2, 1, 3)$, $(2, 3, 1)$, $(3, 1, 2)$, $(3, 2, 1)$, $(1, 2, 4)$, $(1, 4, 2)$, $(2, 1, 4)$, $(2, 4, 1)$, $(4, 1, 2)$, $(4, 2, 1)$, $(2, 2, 2)$.

You are given $T(10, 10) = 869\,985$ and $T(10^3, 10^3) \equiv 578\,270\,566 \pmod{1\,004\,535\,809}$.

Find $T(20\,000, 20\,000) \pmod{1\,004\,535\,809}$.

Problem 538: Maximum Quadrilaterals

Link

Tags: Combinatorics, Geometry, Sequences

Consider a positive integer sequence $S = (s_1, s_2, \dots, s_n)$.

Let $f(S)$ be the perimeter of the maximum-area quadrilateral whose side lengths are 4 elements (s_i, s_j, s_k, s_l) of S (all i, j, k, l distinct). If there are many quadrilaterals with the same maximum area, then choose the one with the largest perimeter.

For example, if $S = (8, 9, 14, 9, 27)$, then we can take the elements $(9, 14, 9, 27)$ and form an isosceles trapezium with parallel side lengths 14 and 27 and both leg lengths 9. The area of this quadrilateral is $127.611470879 \dots$. It can be shown

that this is the largest area for any quadrilateral that can be formed using side lengths from S . Therefore, $f(S) = 9 + 14 + 9 + 27 = 59$.

Let $u_n = 2^{B(3n)} + 3^{B(2n)} + B(n+1)$, where $B(k)$ is the number of 1 bits of k in base 2.

For example, $B(6) = 2$, $B(10) = 2$ and $B(15) = 4$, and $u_5 = 2^4 + 3^2 + 2 = 27$.

Also, let U_n be the sequence $(u_1, u_2, \dots, u_n)$.

For example, $U_{10} = (8, 9, 14, 9, 27, 16, 36, 9, 27, 28)$.

It can be shown that $f(U_5) = 59$, $f(U_{10}) = 118$, $f(U_{150}) = 3223$.

It can also be shown that $\sum f(U_n) = 234761$ for $4 \leq n \leq 150$.

Find $\sum f(U_n)$ for $4 \leq n \leq 3\,000\,000$.

Problem 539: Odd Elimination

Link

Tags: Number Theory

Start from an ordered list of all integers from 1 to n . Going from left to right, remove the first number and every other number afterward until the end of the list. Repeat the procedure from right to left, removing the right most number and every other number from the numbers left. Continue removing every other numbers, alternating left to right and right to left, until a single number remains.

Starting with $n = 9$, we have:

1 2 3 4 5 6 7 8 9

2 4 6 8

2 6

6

Let $P(n)$ be the last number left starting with a list of length n .

Let $S(n) = \sum_{k=1}^n P(k)$.

You are given $P(1) = 1$, $P(9) = 6$, $P(1000) = 510$, $S(1000) = 268271$.

Find $S(10^{18}) \bmod 987654321$.

Problem 540: Counting Primitive Pythagorean Triples

Link

Tags: Geometry, Number Theory

A Pythagorean triple consists of three positive integers a, b and c satisfying $a^2 + b^2 = c^2$.

The triple is called primitive if a, b and c are relatively prime.

Let $P(n)$ be the number of primitive Pythagorean triples with $abc \leq n$.

For example $P(20) = 3$, since there are three triples: $(3, 4, 5)$, $(5, 12, 13)$ and $(8, 15, 17)$.

You are given that $P(10^6) = 159139$.

Find $P(3141592653589793)$.

Problem 541: Divisibility of Harmonic Number Denominators

Link

Tags: Number Theory

The n th harmonic number H_n is defined as the sum of the multiplicative inverses of the first n positive integers, and can be written as a reduced fraction a_n/b_n .

$$H_n = \sum_{k=1}^n \frac{1}{k} = \frac{a_n}{b_n}, \text{ with } \gcd(a_n, b_n) = 1.$$

Let $M(p)$ be the largest value of n such that b_n is not divisible by p .

For example, $M(3) = 68$ because $H_{68} = \frac{a_{68}}{b_{68}} = \frac{14094018321907827923954201611}{2933773379069966367528193600}$, $b_{68} = 2933773379069966367528193600$ is not divisible by 3, but all larger harmonic numbers have denominators divisible by 3.

You are given $M(7) = 719102$.

Find $M(137)$.

Problem 542: Geometric Progression with Maximum Sum

Link

Tags: Game Theory

Let $S(k)$ be the sum of three or more distinct positive integers having the following properties:

- No value exceeds k .
- The values form a geometric progression.
- The sum is maximal.

$$S(4) = 4 + 2 + 1 = 7$$

$$S(10) = 9 + 6 + 4 = 19$$

$$S(12) = 12 + 6 + 3 = 21$$

$$S(1000) = 1000 + 900 + 810 + 729 = 3439$$

$$\text{Let } T(n) = \sum_{k=4}^n (-1)^k S(k).$$

$$T(1000) = 2268$$

$$\text{Find } T(10^{17}).$$

Problem 543: Prime-Sum Numbers

Link

Tags: Number Theory, Sequences

Define function $P(n, k) = 1$ if n can be written as the sum of k prime numbers (with repetitions allowed), and $P(n, k) = 0$ otherwise.

For example, $P(10, 2) = 1$ because 10 can be written as either $3 + 7$ or $5 + 5$, but $P(11, 2) = 0$ because no two primes can sum to 11.

Let $S(n)$ be the sum of all $P(i, k)$ over $1 \leq i, k \leq n$.

For example, $S(10) = 20$, $S(100) = 2402$, and $S(1000) = 248838$.

Let $F(k)$ be the k th Fibonacci number (with $F(0) = 0$ and $F(1) = 1$).

Find the sum of all $S(F(k))$ over $3 \leq k \leq 44$.

Problem 544: Chromatic Conundrum

Link

Tags: Combinatorics

Let $F(r, c, n)$ be the number of ways to colour a rectangular grid with r rows and c columns using at most n colours such that no two adjacent cells share the same colour. Cells that are diagonal to each other are not considered adjacent.

For example, $F(2, 2, 3) = 18$, $F(2, 2, 20) = 130340$, and $F(3, 4, 6) = 102923670$.

Let $S(r, c, n) = \sum_{k=1}^n F(r, c, k)$.

For example, $S(4, 4, 15) \bmod 10^9 + 7 = 325951319$.

Find $S(9, 10, 1112131415) \bmod 10^9 + 7$.

Problem 545: Faulhaber's Formulas

Link

Tags: Algebra

The sum of the k th powers of the first n positive integers can be expressed as a polynomial of degree $k + 1$ with rational coefficients, the Faulhaber's Formulas:

$$1^k + 2^k + \dots + n^k = \sum_{i=1}^n i^k = \sum_{i=1}^{k+1} a_i n^i = a_1 n + a_2 n^2 + \dots + a_k n^k + a_{k+1} n^{k+1},$$

where a_i 's are rational coefficients that can be written as reduced fractions p_i/q_i (if $a_i = 0$, we shall consider $q_i = 1$).

For example, $1^4 + 2^4 + \dots + n^4 = -\frac{1}{30}n + \frac{1}{3}n^3 + \frac{1}{2}n^4 + \frac{1}{5}n^5$.

Define $D(k)$ as the value of q_1 for the sum of k th powers (i.e. the denominator of the reduced fraction a_1).

Define $F(m)$ as the m th value of $k \geq 1$ for which $D(k) = 20010$.

You are given $D(4) = 30$ (since $a_1 = -1/30$), $D(308) = 20010$, $F(1) = 308$, $F(10) = 96404$.

Find $F(10^5)$.

Problem 546: The Floor's Revenge

Link

Tags: Number Theory

Define $f_k(n) = \sum_{i=0}^n f_k(\lfloor \frac{i}{k} \rfloor)$ where $f_k(0) = 1$ and $\lfloor x \rfloor$ denotes the floor function.

For example, $f_5(10) = 18$, $f_7(100) = 1003$, and $f_2(10^3) = 264830889564$.

Find $(\sum_{k=2}^{10} f_k(10^{14})) \bmod (10^9 + 7)$.

Problem 547: Distance of Random Points Within Hollow Square Laminae

Link

Tags: Game Theory, Geometry, Number Theory, Probability

Assuming that two points are chosen randomly (with uniform distribution) within a rectangle, it is possible to determine the expected value of the distance between these two points.

For example, the expected distance between two random points in a unit square is about 0.521405, while the expected distance between two random points in a rectangle with side lengths 2 and 3 is about 1.317067.

Now we define a hollow square lamina of size n to be an integer sized square with side length $n \geq 3$ consisting of n^2 unit squares from which a rectangle consisting of $x \times y$ unit squares ($1 \leq x, y \leq n - 2$) within the original square has been removed.

For $n = 3$ there exists only one hollow square lamina:

For $n = 4$ you can find 9 distinct hollow square laminae, allowing shapes to reappear in rotated or mirrored form:

Let $S(n)$ be the sum of the expected distance between two points chosen randomly within each of the possible hollow square laminae of size n . The two points have to lie within the area left after removing the inner rectangle, i.e. the gray-colored areas in the illustrations above.

For example, $S(3) = 1.6514$ and $S(4) = 19.6564$, rounded to four digits after the decimal point.

Find $S(40)$ rounded to four digits after the decimal point.

Problem 548: Gozinta Chains

Link

Tags: Sequences

A gozinta chain for n is a sequence $\{1, a, b, \dots, n\}$ where each element properly divides the next.

There are eight gozinta chains for 12:

$\{1, 12\}$, $\{1, 2, 12\}$, $\{1, 2, 4, 12\}$, $\{1, 2, 6, 12\}$, $\{1, 3, 12\}$, $\{1, 3, 6, 12\}$, $\{1, 4, 12\}$ and $\{1, 6, 12\}$.

Let $g(n)$ be the number of gozinta chains for n , so $g(12) = 8$.

$g(48) = 48$ and $g(120) = 132$.

Find the sum of the numbers n not exceeding 10^{16} for which $g(n) = n$.

Problem 549: Divisibility of Factorials

Link

Tags: Number Theory

The smallest number m such that 10 divides $m!$ is $m = 5$.

The smallest number m such that 25 divides $m!$ is $m = 10$.

Let $s(n)$ be the smallest number m such that n divides $m!$.

So $s(10) = 5$ and $s(25) = 10$.

Let $S(n)$ be $\sum s(i)$ for $2 \leq i \leq n$.

$S(100) = 2012$.

Find $S(10^8)$.

Problem 550: Divisor Game

Link

Tags: Combinatorics, Game Theory, Number Theory

Two players are playing a game, alternating turns. There are k piles of stones. On each turn, a player has to choose a pile and replace it with two piles of stones under the following two conditions:

- Both new piles must have a number of stones more than one and less than the number of stones of the original pile.
- The number of stones of each of the new piles must be a divisor of the number of stones of the original pile.

The first player unable to make a valid move loses.

Let $f(n, k)$ be the number of winning positions for the first player, assuming perfect play, when the game is played with k piles each having between 2 and n stones (inclusively). $f(10, 5) = 40085$.

Find $f(10^7, 10^{12})$. Give your answer modulo 987654321.

Problem 551: Sum of Digits Sequence

Link

Tags: Number Theory, Sequences

Let $a_0, a_1, \dots$ be an integer sequence defined by:

- $a_0 = 1$;
- for $n \geq 1$, a_n is the sum of the digits of all preceding terms.

The sequence starts with $1, 1, 2, 4, 8, 16, 23, 28, 38, 49, \dots$

You are given $a_{10^6} = 31054319$.

Find $a_{10^{15}}$.

Problem 552: Chinese Leftovers II

[Link](#)

Tags: Algebra, Number Theory, Sequences

Let A_n be the smallest positive integer satisfying $A_n \bmod p_i = i$ for all $1 \leq i \leq n$, where p_i is the i -th prime.

For example $A_2 = 5$, since this is the smallest positive solution of the system of equations

- $A_2 \bmod 2 = 1$
- $A_2 \bmod 3 = 2$

The system of equations for A_3 adds another constraint. That is, A_3 is the smallest positive solution of

- $A_3 \bmod 2 = 1$
- $A_3 \bmod 3 = 2$
- $A_3 \bmod 5 = 3$

and hence $A_3 = 23$. Similarly, one gets $A_4 = 53$ and $A_5 = 1523$.

Let $S(n)$ be the sum of all primes up to n that divide at least one element in the sequence A .

For example, $S(50) = 69 = 5 + 23 + 41$, since 5 divides A_2 , 23 divides A_3 and 41 divides $A_{10} = 5765999453$. No other prime number up to 50 divides an element in A .

Find $S(300000)$.

Problem 553: Power Sets of Power Sets

[Link](#)

Tags: Combinatorics, Game Theory, Graph Theory

Let $P(n)$ be the set of the first n positive integers $\{1, 2, \dots, n\}$.

Let $Q(n)$ be the set of all the non-empty subsets of $P(n)$.

Let $R(n)$ be the set of all the non-empty subsets of $Q(n)$.

An element $X \in R(n)$ is a non-empty subset of $Q(n)$, so it is itself a set.

From X we can construct a graph as follows:

- Each element $Y \in X$ corresponds to a vertex and labeled with Y ;
- Two vertices Y_1 and Y_2 are connected if $Y_1 \cap Y_2 \neq \emptyset$.

For example, $X = \{\{1\}, \{1, 2, 3\}, \{3\}, \{5, 6\}, \{6, 7\}\}$ results in the following graph:

Problem 554: Centaurs on a Chess Board

Link

Tags: Number Theory

On a chess board, a centaur moves like a king or a knight. The diagram below shows the valid moves of a centaur (represented by an inverted king) on an 8×8 board.

Problem 555: McCarthy 91 Function

Link

Tags: Number Theory

The McCarthy 91 function is defined as follows:

$$M_{91}(n) = \begin{cases} n - 10 & \text{if } n > 100 \\ M_{91}(M_{91}(n + 11)) & \text{if } 0 \leq n \leq 100 \end{cases}$$

We can generalize this definition by abstracting away the constants into new variables:

$$M_{m,k,s}(n) = \begin{cases} n - s & \text{if } n > m \\ M_{m,k,s}(M_{m,k,s}(n + k)) & \text{if } 0 \leq n \leq m \end{cases}$$

This way, we have $M_{91} = M_{100,11,10}$.

Let $F_{m,k,s}$ be the set of fixed points of $M_{m,k,s}$. That is,

$$F_{m,k,s} = \{n \in \mathbb{N} \mid M_{m,k,s}(n) = n\}$$

For example, the only fixed point of M_{91} is $n = 91$. In other words, $F_{100,11,10} = \{91\}$.

Now, define $SF(m, k, s)$ as the sum of the elements in $F_{m,k,s}$ and let $S(p, m) = \sum_{1 \leq s < k \leq p} SF(m, k, s)$.

For example, $S(10, 10) = 225$ and $S(1000, 1000) = 208724467$.

Find $S(10^6, 10^6)$.

Problem 556: Squarefree Gaussian Integers

Link

Tags: Combinatorics, Number Theory

A Gaussian integer is a number $z = a + bi$ where a, b are integers and $i^2 = -1$.

Gaussian integers are a subset of the complex numbers, and the integers are the subset of Gaussian integers for which $b = 0$.

A Gaussian integer unit is one for which $a^2 + b^2 = 1$, i.e. one of $1, i, -1, -i$.

Let's define a proper Gaussian integer as one for which $a \neq 0$ and $b \geq 0$.

A Gaussian integer $z_1 = a_1 + b_1i$ is said to be divisible by $z_2 = a_2 + b_2i$ if $z_3 = a_3 + b_3i = z_1/z_2$ is a Gaussian integer.

$$\frac{z_1}{z_2} = \frac{a_1 + b_1i}{a_2 + b_2i} = \frac{(a_1 + b_1i)(a_2 - b_2i)}{(a_2 + b_2i)(a_2 - b_2i)} = \frac{a_1a_2 + b_1b_2}{a_2^2 + b_2^2} + \frac{a_2b_1 - a_1b_2}{a_2^2 + b_2^2}i = a_3 + b_3i$$

So, z_1 is divisible by z_2 if $\frac{a_1a_2 + b_1b_2}{a_2^2 + b_2^2}$ and $\frac{a_2b_1 - a_1b_2}{a_2^2 + b_2^2}$ are integers.

For example, 2 is divisible by $1 + i$ because $2/(1 + i) = 1 - i$ is a Gaussian integer.

A Gaussian prime is a Gaussian integer that is divisible only by a unit, itself or itself times a unit.

For example, $1 + 2i$ is a Gaussian prime, because it is only divisible by $1, i, -1, -i, 1 + 2i, i(1 + 2i) = i - 2, -(1 + 2i) = -1 - 2i$ and $-i(1 + 2i) = 2 - i$.

2 is not a Gaussian prime as it is divisible by $1 + i$.

A Gaussian integer can be uniquely factored as the product of a unit and proper Gaussian primes.

For example $2 = -i(1 + i)^2$ and $1 + 3i = (1 + i)(2 + i)$.

A Gaussian integer is said to be squarefree if its prime factorization does not contain repeated proper Gaussian primes.

So 2 is not squarefree over the Gaussian integers, but $1 + 3i$ is.

Units and Gaussian primes are squarefree by definition.

Let $f(n)$ be the count of proper squarefree Gaussian integers with $a^2 + b^2 \leq n$.

For example $f(10) = 7$ because $1, 1 + i, 1 + 2i, 1 + 3i = (1 + i)(2 + i), 2 + i, 3$ and $3 + i = -i(1 + i)(1 + 2i)$ are squarefree, while $2 = -i(1 + i)^2$ and $2 + 2i = -i(1 + i)^3$ are not.

You are given $f(10^2) = 54$, $f(10^4) = 5218$ and $f(10^8) = 52126906$.

Find $f(10^{14})$.

Problem 557: Cutting Triangles

Link

Tags: Combinatorics, Geometry, Graph Theory

A triangle is cut into four pieces by two straight lines, each starting at one vertex and ending on the opposite edge. This results in forming three smaller triangular pieces, and one quadrilateral. If the original triangle has an integral area, it is often possible to choose cuts such that all of the four pieces also have integral area. For example, the diagram below shows a triangle of area 55 that has been cut in this way.

Problem 558: Irrational Base

Link

Tags: Algebra

Let r be the real root of the equation $x^3 = x^2 + 1$.

Every positive integer can be written as the sum of distinct increasing powers of r .

If we require the number of terms to be finite and the difference between any two exponents to be three or more, then the representation is unique.

For example, $3 = r^{-10} + r^{-5} + r^{-1} + r^2$ and $10 = r^{-10} + r^{-7} + r^6$.

Interestingly, the relation holds for the complex roots of the equation.

Let $w(n)$ be the number of terms in this unique representation of n . Thus $w(3) = 4$ and $w(10) = 3$.

More formally, for all positive integers n , we have:

$$n = \sum_{k=-\infty}^{\infty} b_k r^k$$

under the conditions that:

b_k is 0 or 1 for all k ;

$b_k + b_{k+1} + b_{k+2} \leq 1$ for all k ;

$$w(n) = \sum_{k=-\infty}^{\infty} b_k \text{ is finite.}$$

$$\text{Let } S(m) = \sum_{j=1}^m w(j^2).$$

You are given $S(10) = 61$ and $S(1000) = 19403$.

Find $S(5\,000\,000)$.

Problem 559: Permuted Matrices

Link

Tags: Algebra, Combinatorics, Game Theory, Number Theory

An ascent of a column j in a matrix occurs if the value of column j is smaller than the value of column $j + 1$ in all rows.

Let $P(k, r, n)$ be the number of $r \times n$ matrices with the following properties:

- The rows are permutations of $\{1, 2, 3, \dots, n\}$.
- Numbering the first column as 1, a column ascent occurs at column jn if and only if j is not a multiple of k .

For example, $P(1, 2, 3) = 19$, $P(2, 4, 6) = 65508751$ and $P(7, 5, 30) \bmod 1000000123 = 161858102$.

$$\text{Let } Q(n) = \sum_{k=1}^n P(k, n, n).$$

For example, $Q(5) = 21879393751$ and $Q(50) \bmod 1000000123 = 819573537$.

Find $Q(50000) \bmod 1000000123$.

Problem 560: Coprime Nim

Link

Tags: Game Theory, Number Theory

Coprime Nim is just like ordinary normal play Nim, but the players may only remove a number of stones from a pile that is coprime with the current size of the pile. Two players remove stones in turn. The player who removes the last stone wins.

Let $L(n, k)$ be the number of losing starting positions for the first player, assuming perfect play, when the game is played with k piles, each having between 1 and $n - 1$ stones inclusively.

For example, $L(5, 2) = 6$ since the losing initial positions are $(1, 1)$, $(2, 2)$, $(2, 4)$, $(3, 3)$, $(4, 2)$ and $(4, 4)$.

You are also given $L(10, 5) = 9964$, $L(10, 10) = 472400303$, $L(10^3, 10^3) \bmod 1\,000\,000\,007 = 954021836$.

Find $L(10^7, 10^7) \bmod 1\,000\,000\,007$.

Problem 561: Divisor Pairs

Link

Tags: Game Theory, Number Theory

Let $S(n)$ be the number of pairs (a, b) of distinct divisors of n such that a divides b .

For $n = 6$ we get the following pairs: $(1, 2)$, $(1, 3)$, $(1, 6)$, $(2, 6)$ and $(3, 6)$. So $S(6) = 5$.

Let $p_m\#$ be the product of the first m prime numbers, so $p_2\# = 2 * 3 = 6$.

Let $E(m, n)$ be the highest integer k such that 2^k divides $S((p_m\#)^n)$.

$E(2, 1) = 0$ since 2^0 is the highest power of 2 that divides $S(6)=5$.

Let $Q(n) = \sum_{i=1}^n E(904961, i)$

$Q(8) = 2714886$.

Evaluate $Q(10^{12})$.

Problem 562: Maximal Perimeter

Link

Tags: Combinatorics, Geometry, Graph Theory

Construct triangle ABC such that:

- Vertices A , B and C are lattice points inside or on the circle of radius r centered at the origin;

- the triangle contains no other lattice point inside or on its edges;
- the perimeter is maximum.

Let R be the circumradius of triangle ABC and $T(r) = R/r$.

For $r = 5$, one possible triangle has vertices $(-4, -3)$, $(4, 2)$ and $(1, 0)$ with perimeter $\sqrt{13} + \sqrt{34} + \sqrt{89}$ and circumradius $R = \sqrt{\frac{19669}{2}}$, so $T(5) = \sqrt{\frac{19669}{50}}$.

You are given $T(10) \approx 97.26729$ and $T(100) \approx 9157.64707$.

Find $T(10^7)$. Give your answer rounded to the nearest integer.

Problem 563: Robot Welders

Link

Tags: Game Theory, Geometry, Graph Theory

A company specialises in producing large rectangular metal sheets, starting from unit square metal plates. The welding is performed by a range of robots of increasing size. Unfortunately, the programming options of these robots are rather limited. Each one can only process up to 25 identical rectangles of metal, which they can weld along either edge to produce a larger rectangle. The only programmable variables are the number of rectangles to be processed (up to and including 25), and whether to weld the long or short edge.

For example, the first robot could be programmed to weld together 11 raw unit square plates to make a 11×1 strip. The next could take 10 of these 11×1 strips, and weld them either to make a longer 110×1 strip, or a 11×10 rectangle. Many, but not all, possible dimensions of metal sheets can be constructed in this way.

One regular customer has a particularly unusual order: The finished product should have an exact area, and the long side must not be more than 10% larger than the short side. If these requirements can be met in more than one way, in terms of the exact dimensions of the two sides, then the customer will demand that all variants be produced. For example, if the order calls for a metal sheet of area 889200, then there are three final dimensions that can be produced: 900×988 , 912×975 and 936×950 . The target area of 889200 is the smallest area which can be manufactured in three different variants, within the limitations of the robot welders.

Let $M(n)$ be the minimal area that can be manufactured in exactly n variants with the longer edge not greater than 10% bigger than the shorter edge. Hence $M(3) = 889200$.

Find $\sum_{n=2}^{100} M(n)$.

Problem 564: Maximal Polygons

Link

Tags: Geometry, Probability, Sequences

A line segment of length $2n - 3$ is randomly split into n segments of integer length ($n \geq 3$). In the sequence given by this split, the segments are then used as consecutive sides of a convex n -polygon, formed in such a way that its area is maximal. All of the $\binom{2n-4}{n-1}$ possibilities for splitting up the initial line segment occur with the same probability.

Let $E(n)$ be the expected value of the area that is obtained by this procedure.

For example, for $n = 3$ the only possible split of the line segment of length 3 results in three line segments with length 1, that form an equilateral triangle with an area of $\frac{1}{4}\sqrt{3}$. Therefore $E(3) = 0.433013$, rounded to 6 decimal places.

For $n = 4$ you can find 4 different possible splits, each of which is composed of three line segments with length 1 and one line segment with length 2. All of these splits lead to the same maximal quadrilateral with an area of $\frac{3}{4}\sqrt{3}$, thus $E(4) = 1.299038$, rounded to 6 decimal places.

$$\text{Let } S(k) = \sum_{n=3}^k E(n).$$

For example, $S(3) = 0.433013$, $S(4) = 1.732051$, $S(5) = 4.604767$ and $S(10) = 66.955511$, rounded to 6 decimal places each.

Find $S(50)$, rounded to 6 decimal places.

Problem 565: Divisibility of Sum of Divisors

Link

Tags: Number Theory

Let $\sigma(n)$ be the sum of the divisors of n .

E.g. the divisors of 4 are 1, 2 and 4, so $\sigma(4) = 7$.

The numbers n not exceeding 20 such that 7 divides $\sigma(n)$ are: 4, 12, 13 and 20, the sum of these numbers being 49.

Let $S(n, d)$ be the sum of the numbers i not exceeding n such that d divides $\sigma(i)$.

So $S(20, 7) = 49$.

You are given: $S(10^6, 2017) = 150850429$ and $S(10^9, 2017) = 249652238344557$.

Find $S(10^{11}, 2017)$.

Problem 566: Cake Icing Puzzle

Link

Tags: Game Theory, Geometry

Adam plays the following game with his birthday cake.

He cuts a piece forming a circular sector of 60 degrees and flips the piece upside down, with the icing on the bottom.

He then rotates the cake by 60 degrees counterclockwise, cuts an adjacent 60 degree piece and flips it upside down.

He keeps repeating this, until after a total of twelve steps, all the icing is back on top.

Amazingly, this works for any piece size, even if the cutting angle is an irrational number: all the icing will be back on top after a finite number of steps.

Now, Adam tries something different: he alternates cutting pieces of size $x = \frac{360}{9}$ degrees, $y = \frac{360}{10}$ degrees and $z = \frac{360}{\sqrt{11}}$ degrees. The first piece he cuts has size x and he flips it. The second has size y and he flips it. The third has size z and he flips it. He repeats this with pieces of size x , y and z in that order until all the icing is back on top, and discovers he needs 60 flips altogether.

Problem 567: Reciprocal Games I

Link

Tags: Game Theory, Probability

Tom has built a random generator that is connected to a row of n light bulbs. Whenever the random generator is activated each of the n lights is turned on with the probability of $\frac{1}{2}$, independently of its former state or the state of the other light bulbs.

While discussing with his friend Jerry how to use his generator, they invent two different games, they call the reciprocal games:

Both games consist of n turns. Each turn is started by choosing a number k randomly between (and including) 1 and n , with equal probability of $\frac{1}{n}$ for each number, while the possible win for that turn is the reciprocal of k , that is $\frac{1}{k}$.

In game A, Tom activates his random generator once in each turn. If the number of lights turned on is the same as the previously chosen number k , Jerry wins and gets $\frac{1}{k}$, otherwise he will receive nothing for that turn. Jerry's expected

win after playing the total game A consisting of n turns is called $J_A(n)$. For example $J_A(6) = 0.39505208$, rounded to 8 decimal places.

For each turn in game B, after k has been randomly selected, Tom keeps reactivating his random generator until exactly k lights are turned on. After that Jerry takes over and reactivates the random generator until he, too, has generated a pattern with exactly k lights turned on. If this pattern is identical to Tom's last pattern, Jerry wins and gets $\frac{1}{k}$, otherwise he will receive nothing. Jerry's expected win after the total game B consisting of n turns is called $J_B(n)$. For example $J_B(6) = 0.43333333$, rounded to 8 decimal places.

Let $S(m) = \sum_{n=1}^m (J_A(n) + J_B(n))$. For example $S(6) = 7.58932292$, rounded to 8 decimal places.

Find $S(123456789)$, rounded to 8 decimal places.

Problem 568: Reciprocal Games II

Link

Tags: Game Theory, Number Theory, Probability

Tom has built a random generator that is connected to a row of n light bulbs. Whenever the random generator is activated each of the n lights is turned on with the probability of $\frac{1}{2}$, independently of its former state or the state of the other light bulbs.

While discussing with his friend Jerry how to use his generator, they invent two different games, they call the reciprocal games:

Both games consist of n turns. Each turn is started by choosing a number k randomly between (and including) 1 and n , with equal probability of $\frac{1}{n}$ for each number, while the possible win for that turn is the reciprocal of k , that is $\frac{1}{k}$.

In game A, Tom activates his random generator once in each turn. If the number of lights turned on is the same as the previously chosen number k , Jerry wins and gets $\frac{1}{k}$, otherwise he will receive nothing for that turn. Jerry's expected win after playing the total game A consisting of n turns is called $J_A(n)$. For example $J_A(6) = 0.39505208$, rounded to 8 decimal places.

For each turn in game B, after k has been randomly selected, Tom keeps reactivating his random generator until exactly k lights are turned on. After that Jerry takes over and reactivates the random generator until he, too, has generated a pattern with exactly k lights turned on. If this pattern is identical to Tom's last pattern, Jerry wins and gets $\frac{1}{k}$, otherwise he will receive nothing. Jerry's expected win after the total game B consisting of n turns is called $J_B(n)$. For example $J_B(6) = 0.43333333$, rounded to 8 decimal places.

Let $D(n) = J_B(n)J_A(n)$. For example, $D(6) = 0.03828125$.

Find the 7 most significant digits of $D(123456789)$ after removing all leading zeros.

(If, for example, we had asked for the 7 most significant digits of $D(6)$, the answer would have been 3828125.)

Problem 569: Prime Mountain Range

[Link](#)

Tags: Number Theory

A mountain range consists of a line of mountains with slopes of exactly 45° , and heights governed by the prime numbers, p_n . The up-slope of the k th mountain is of height p_{2k-1} , and the downslope is p_{2k} . The first few foot-hills of this range are illustrated below.

Problem 570: Snowflakes

[Link](#)

Tags: Geometry

A snowflake of order n is formed by overlaying an equilateral triangle (rotated by 180 degrees) onto each equilateral triangle of the same size in a snowflake of order $n - 1$. A snowflake of order 1 is a single equilateral triangle.

Problem 571: Super Pandigital Numbers

[Link](#)

Tags: Number Theory

A positive number is pandigital in base b if it contains all digits from 0 to $b - 1$ at least once when written in base b .

An n -super-pandigital number is a number that is simultaneously pandigital in all bases from 2 to n inclusively.

For example $978 = 1111010010_2 = 1100020_3 = 33102_4 = 12403_5$ is the smallest 5-super-pandigital number.

Similarly, 1093265784 is the smallest 10-super-pandigital number.

The sum of the 10 smallest 10-super-pandigital numbers is 20319792309.

What is the sum of the 10 smallest 12-super-pandigital numbers?

Problem 572: Idempotent Matrices

Link

Tags: Algebra

A matrix M is called idempotent if $M^2 = M$.

Let M be a three by three matrix : $M = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$.

Let $C(n)$ be the number of idempotent three by three matrices M with integer elements such that

$0 \leq a, b, c, d, e, f, g, h, i \leq n$.

$C(1) = 164$ and $C(2) = 848$.

Find $C(200)$.

Problem 573: Unfair Race

Link

Tags: Game Theory, Number Theory, Probability

n runners in very different training states want to compete in a race. Each one of them is given a different starting number k ($1 \leq k \leq n$) according to the runner's (constant) individual racing speed being $v_k = \frac{k}{n}$.

In order to give the slower runners a chance to win the race, n different starting positions are chosen randomly (with uniform distribution) and independently from each other within the racing track of length 1. After this, the starting position nearest to the goal is assigned to runner 1, the next nearest starting position to runner 2 and so on, until finally the starting position furthest away from the goal is assigned to runner n . The winner of the race is the runner who reaches the goal first.

Interestingly, the expected running time for the winner is $\frac{1}{2}$, independently of the number of runners. Moreover, while it can be shown that all runners will have the same expected running time of $\frac{n}{n+1}$, the race is still unfair, since the winning chances may differ significantly for different starting numbers:

Let $P_{n,k}$ be the probability for runner k to win a race with n runners and $E_n = \sum_{k=1}^n k P_{n,k}$ be the expected starting number of the winner in that race.

It can be shown that, for example, $P_{3,1} = \frac{4}{9}$, $P_{3,2} = \frac{2}{9}$, $P_{3,3} = \frac{1}{3}$ and $E_3 = \frac{17}{9}$ for a race with 3 runners.

You are given that $E_4 = 2.21875$, $E_5 = 2.5104$ and $E_{10} = 3.66021568$.

Find $E_{1000000}$ rounded to 4 digits after the decimal point.

Problem 574: Verifying Primes

[Link](#)

Tags: Game Theory, Number Theory

Let q be a prime and $A \geq B > 0$ be two integers with the following properties:

- A and B have no prime factor in common, that is $\gcd(A, B) = 1$.
- The product AB is divisible by every prime less than q .

It can be shown that, given these conditions, any sum $A + B < q^2$ and any difference $1 < A - B < q^2$ has to be a prime number. Thus you can verify that a number p is prime by showing that either $p = A + B < q^2$ or $p = A - B < q^2$ for some A, B, q fulfilling the conditions listed above.

Let $V(p)$ be the smallest possible value of A in any sum $p = A + B$ and any difference $p = A - B$, that verifies p being prime. Examples:

$V(2) = 1$, since $2 = 1 + 1 < 2^2$.

$V(37) = 22$, since $37 = 22 + 15 = 2 \cdot 11 + 3 \cdot 5 < 7^2$ is the associated sum with the smallest possible A .

$V(151) = 165$ since $151 = 165 - 14 = 3 \cdot 5 \cdot 11 - 2 \cdot 7 < 13^2$ is the associated difference with the smallest possible A .

Let $S(n)$ be the sum of $V(p)$ for all primes $p < n$. For example, $S(10) = 10$ and $S(200) = 7177$.

Find $S(3800)$.

Problem 575: Wandering Robots

[Link](#)

Tags: Algebra, Combinatorics, Game Theory, Graph Theory

It was quite an ordinary day when a mysterious alien vessel appeared as if from nowhere. After waiting several hours and receiving no response it is decided to send a team to investigate, of which you are included. Upon entering the

vessel you are met by a friendly holographic figure, Katharina, who explains the purpose of the vessel, Eulertopia.

She claims that Eulertopia is almost older than time itself. Its mission was to take advantage of a combination of incredible computational power and vast periods of time to discover the answer to life, the universe, and everything. Hence the resident cleaning robot, Leonhard, along with his housekeeping responsibilities, was built with a powerful computational matrix to ponder the meaning of life as he wanders through a massive 1000 by 1000 square grid of rooms. She goes on to explain that the rooms are numbered sequentially from left to right, row by row. So, for example, if Leonhard was wandering around a 5 by 5 grid then the rooms would be numbered in the following way.

Problem 576: Irrational Jumps

Link

Tags: Geometry, Number Theory

A bouncing point moves counterclockwise along a circle with circumference 1 with jumps of constant length $l1$, until it hits a gap of length $g1$, that is placed in a distance d counterclockwise from the starting point. The gap does not include the starting point, that is $g + d1$.

Let $S(l, g, d)$ be the sum of the length of all jumps, until the point falls into the gap. It can be shown that $S(l, g, d)$ is finite for any irrational jump size l , regardless of the values of g and d .

Examples:

$$S(\sqrt{\frac{1}{2}}, 0.06, 0.7) = 0.7071 \dots, \quad S(\sqrt{\frac{1}{2}}, 0.06, 0.3543) = 1.4142 \dots \quad \text{and} \\ S(\sqrt{\frac{1}{2}}, 0.06, 0.2427) = 16.2634 \dots$$

Let $M(n, g)$ be the maximum of $\sum S(\sqrt{\frac{1}{p}}, g, d)$ for all primes $p \leq n$ and any valid value of d .

Examples:

$$M(3, 0.06) = 29.5425 \dots, \quad \text{since } S(\sqrt{\frac{1}{2}}, 0.06, 0.2427) + S(\sqrt{\frac{1}{3}}, 0.06, 0.2427) = 29.5425 \dots \text{ is the maximal reachable sum for } g = 0.06.$$

$$M(10, 0.01) = 266.9010 \dots$$

Find $M(100, 0.00002)$, rounded to 4 decimal places.

Problem 577: Counting Hexagons

Link

Tags: Combinatorics, Geometry

An equilateral triangle with integer side length $n \geq 3$ is divided into n^2 equilateral triangles with side length 1 as shown in the diagram below.

The vertices of these triangles constitute a triangular lattice with $\frac{(n+1)(n+2)}{2}$ lattice points.

Let $H(n)$ be the number of all regular hexagons that can be found by connecting 6 of these points.

Problem 578: Integers with Decreasing Prime Powers

Link

Tags: Number Theory

Any positive integer can be written as a product of prime powers: $p_1^{a_1} \times p_2^{a_2} \times \cdots \times p_k^{a_k}$,

where p_i are distinct prime integers, $a_i \geq 0$ and $p_i p_j$ if $i < j$.

A decreasing prime power positive integer is one for which $a_i \geq a_j$ if $i < j$.

For example, 1, 2, 15 = 3×5 , 360 = $2^3 \times 3^2 \times 5$ and 1000 = $2^3 \times 5^3$ are decreasing prime power integers.

Let $C(n)$ be the count of decreasing prime power positive integers not exceeding n .

$C(100) = 94$ since all positive integers not exceeding 100 have decreasing prime powers except 18, 50, 54, 75, 90 and 98.

You are given $C(10^6) = 922052$.

Find $C(10^{13})$.

Problem 579: Lattice Points in Lattice Cubes

Link

Tags: Combinatorics

A lattice cube is a cube in which all vertices have integer coordinates. Let $C(n)$ be the number of different lattice cubes in which the coordinates of all vertices range between (and including) 0 and n . Two cubes are hereby considered different if any of their vertices have different coordinates.

For example, $C(1) = 1$, $C(2) = 9$, $C(4) = 100$, $C(5) = 229$, $C(10) = 4469$ and $C(50) = 8154671$.

Different cubes may contain different numbers of lattice points.

For example, the cube with the vertices

$(0, 0, 0)$, $(3, 0, 0)$, $(0, 3, 0)$, $(0, 0, 3)$, $(0, 3, 3)$, $(3, 0, 3)$, $(3, 3, 0)$, $(3, 3, 3)$ contains 64 lattice points (56 lattice points on the surface including the 8 vertices and 8 points within the cube).

In contrast, the cube with the vertices

$(0, 2, 2)$, $(1, 4, 4)$, $(2, 0, 3)$, $(2, 3, 0)$, $(3, 2, 5)$, $(3, 5, 2)$, $(4, 1, 1)$, $(5, 3, 3)$ contains only 40 lattice points (20 points on the surface and 20 points within the cube), although both cubes have the same side length 3.

Let $S(n)$ be the sum of the lattice points contained in the different lattice cubes in which the coordinates of all vertices range between (and including) 0 and n .

For example, $S(1) = 8$, $S(2) = 91$, $S(4) = 1878$, $S(5) = 5832$, $S(10) = 387003$ and $S(50) = 29948928129$.

Find $S(5000) \bmod 10^9$.

Problem 580: Squarefree Hilbert Numbers

[Link](#)

Tags: Number Theory

A Hilbert number is any positive integer of the form $4k + 1$ for integer $k \geq 0$. We shall define a squarefree Hilbert number as a Hilbert number which is not divisible by the square of any Hilbert number other than one. For example, 117 is a squarefree Hilbert number, equaling 9×13 . However 6237 is a Hilbert number that is not squarefree in this sense, as it is divisible by 9^2 . The number 3969 is also not squarefree, as it is divisible by both 9^2 and 21^2 .

There are 2327192 squarefree Hilbert numbers below 10^7 .

How many squarefree Hilbert numbers are there below 10^{16} ?

Problem 581: 47-smooth Triangular Numbers

[Link](#)

Tags: Number Theory, Probability, Sequences

A number is p -smooth if it has no prime factors larger than p .

Let T be the sequence of triangular numbers, i.e. $T(n) = n(n+1)/2$.
Find the sum of all indices n such that $T(n)$ is 47-smooth.

Problem 582: Nearly Isosceles 120 Degree Triangles

Link

Tags: Geometry

Let a, b and c be the sides of an integer sided triangle with one angle of 120 degrees, $a \leq b \leq c$ and $b - a \leq 100$.

Let $T(n)$ be the number of such triangles with $c \leq n$.

$T(1000) = 235$ and $T(10^8) = 1245$.

Find $T(10^{100})$.

Problem 583: Heron Envelopes

Link

Tags: Geometry

A standard envelope shape is a convex figure consisting of an isosceles triangle (the flap) placed on top of a rectangle. An example of an envelope with integral sides is shown below. Note that to form a sensible envelope, the perpendicular height of the flap (BCD) must be smaller than the height of the rectangle ($ABDE$).

Problem 584: Birthday Problem Revisited

Link

Tags: Probability

A long long time ago in a galaxy far far away, the Wimwians, inhabitants of planet WimWi, discovered an unmanned drone that had landed on their planet. On examining the drone, they uncovered a device that sought the answer for the so called "Birthday Problem". The description of the problem was as follows:

If people on your planet were to enter a very large room one by one, what will be the expected number of people in the room when you first find 3 people with Birthdays within 1 day from each other.

The description further instructed them to enter the answer into the device and send the drone into space again. Startled by this turn of events, the Wimwians

consulted their best mathematicians. Each year on Wimwi has 10 days and the mathematicians assumed equally likely birthdays and ignored leap years (leap years in Wimwi have 11 days), and found 5.78688636 to be the required answer. As such, the Wimwians entered this answer and sent the drone back into space.

After traveling light years away, the drone then landed on planet Joka. The same events ensued except this time, the numbers in the device had changed due to some unknown technical issues. The description read:

If people on your planet were to enter a very large room one by one, what will be the expected number of people in the room when you first find 3 people with Birthdays within 7 days from each other.

With a 100-day year on the planet, the Jokars (inhabitants of Joka) found the answer to be 8.48967364 (rounded to 8 decimal places because the device allowed only 8 places after the decimal point) assuming equally likely birthdays. They too entered the answer into the device and launched the drone into space again.

This time the drone landed on planet Earth. As before the numbers in the problem description had changed. It read:

If people on your planet were to enter a very large room one by one, what will be the expected number of people in the room when you first find 4 people with Birthdays within 7 days from each other.

What would be the answer (rounded to eight places after the decimal point) the people of Earth have to enter into the device for a year with 365 days? Ignore leap years. Also assume that all birthdays are equally likely and independent of each other.

Problem 585: Nested Square Roots

Link

Tags: Algebra, Combinatorics

Consider the term $\sqrt{x + \sqrt{y} + \sqrt{z}}$ that is representing a nested square root. x , y and z are positive integers and y and z are not allowed to be perfect squares, so the number below the outer square root is irrational. Still it can be shown that for some combinations of x , y and z the given term can be simplified into a sum and/or difference of simple square roots of integers, actually denesting the square roots in the initial expression.

Here are some examples of this denesting:

$$\sqrt{3 + \sqrt{2} + \sqrt{2}} = \sqrt{2} + \sqrt{1} = \sqrt{2} + 1$$

$$\sqrt{8 + \sqrt{15} + \sqrt{15}} = \sqrt{5} + \sqrt{3}$$

$$\sqrt{20 + \sqrt{96} + \sqrt{12}} = \sqrt{9} + \sqrt{6} + \sqrt{3} - \sqrt{2} = 3 + \sqrt{6} + \sqrt{3} - \sqrt{2}$$

$$\sqrt{28 + \sqrt{160} + \sqrt{108}} = \sqrt{15} + \sqrt{6} + \sqrt{5} - \sqrt{2}$$

As you can see the integers used in the denested expression may also be perfect squares resulting in further simplification.

Let $F(n)$ be the number of different terms $\sqrt{x + \sqrt{y} + \sqrt{z}}$, that can be denested into the sum and/or difference of a finite number of square roots, given the additional condition that $0 < x \leq n$. That is,

$$\sqrt{x + \sqrt{y} + \sqrt{z}} = \sum_{i=1}^k s_i \sqrt{a_i}$$

with k, x, y, z and all a_i being positive integers, all $s_i = \pm 1$ and $x \leq n$. Furthermore y and z are not allowed to be perfect squares.

Nested roots with the same value are not considered different, for example $\sqrt{7 + \sqrt{3} + \sqrt{27}}$, $\sqrt{7 + \sqrt{12} + \sqrt{12}}$ and $\sqrt{7 + \sqrt{27} + \sqrt{3}}$, that can all three be denested into $2 + \sqrt{3}$, would only be counted once.

You are given that $F(10) = 17$, $F(15) = 46$, $F(20) = 86$, $F(30) = 213$ and $F(100) = 2918$ and $F(5000) = 11134074$.

Find $F(5000000)$.

Problem 586: Binary Quadratic Form

Link

Tags: Number Theory

The number 209 can be expressed as $a^2 + 3ab + b^2$ in two distinct ways:

$$\text{\$} \quad 209 = 8^2 + 3 \cdot 8 \cdot 5 + 5^2$$

$$\text{\$} \quad 209 = 13^2 + 3 \cdot 13 \cdot 1 + 1^2$$

Let $f(n, r)$ be the number of integers k not exceeding n that can be expressed as $k = a^2 + 3ab + b^2$, with a, b integers, in exactly r different ways.

You are given that $f(10^5, 4) = 237$ and $f(10^8, 6) = 59517$.

Find $f(10^{15}, 40)$.

Problem 587: Concave Triangle

Link

Tags: Geometry

A square is drawn around a circle as shown in the diagram below on the left.

We shall call the blue shaded region the L-section.

A line is drawn from the bottom left of the square to the top right as shown in the diagram on the right.

We shall call the orange shaded region a concave triangle.

Problem 588: Quintinomial Coefficients

[Link](#)

Tags: Combinatorics

The coefficients in the expansion of $(x+1)^k$ are called binomial coefficients.

Analogously the coefficients in the expansion of $(x^4 + x^3 + x^2 + x + 1)^k$ are called quintinomial coefficients. (quintus= Latin for fifth).

Consider the expansion of $(x^4 + x^3 + x^2 + x + 1)^3$:

$$x^{12} + 3x^{11} + 6x^{10} + 10x^9 + 15x^8 + 18x^7 + 19x^6 + 18x^5 + 15x^4 + 10x^3 + 6x^2 + 3x + 1$$

As we can see 7 out of the 13 quintinomial coefficients for $k=3$ are odd.

Let $Q(k)$ be the number of odd coefficients in the expansion of $(x^4 + x^3 + x^2 + x + 1)^k$.

So $Q(3) = 7$.

You are given $Q(10) = 17$ and $Q(100) = 35$.

Find $\sum_{k=1}^{18} Q(10^k)$.

Problem 589: Poohsticks Marathon

[Link](#)

Tags: Game Theory, Probability

Christopher Robin and Pooh Bear love the game of Poohsticks so much that they invented a new version which allows them to play for longer before one of them wins and they have to go home for tea. The game starts as normal with both dropping a stick simultaneously on the upstream side of a bridge. But rather than the game ending when one of the sticks emerges on the downstream side, instead they fish their sticks out of the water, and drop them back in again on the upstream side. The game only ends when one of the sticks emerges from under the bridge ahead of the other one having also ‘lapped’ the other stick -

that is, having made one additional journey under the bridge compared to the other stick.

On a particular day when playing this game, the time taken for a stick to travel under the bridge varies between a minimum of 30 seconds, and a maximum of 60 seconds. The time taken to fish a stick out of the water and drop it back in again on the other side is 5 seconds. The current under the bridge has the unusual property that the sticks' journey time is always an integral number of seconds, and it is equally likely to emerge at any of the possible times between 30 and 60 seconds (inclusive). It turns out that under these circumstances, the expected time for playing a single game is 1036.15 seconds (rounded to 2 decimal places). This time is measured from the point of dropping the sticks for the first time, to the point where the winning stick emerges from under the bridge having lapped the other.

The stream flows at different rates each day, but maintains the property that the journey time in seconds is equally distributed amongst the integers from a minimum, n , to a maximum, m , inclusive. Let the expected time of play in seconds be $E(m, n)$. Hence $E(60, 30) = 1036.15\dots$

Let $S(k) = \sum_{m=2}^k \sum_{n=1}^{m-1} E(m, n)$.

For example $S(5) = 7722.82$ rounded to 2 decimal places.

Find $S(100)$ and give your answer rounded to 2 decimal places.

Problem 590: Sets with a Given Least Common Multiple

Link

Tags: Game Theory, Number Theory

Let $H(n)$ denote the number of sets of positive integers such that the least common multiple of the integers in the set equals n .

E.g.:

The integers in the following ten sets all have a least common multiple of 6:

$\{2, 3\}$, $\{1, 2, 3\}$, $\{6\}$, $\{1, 6\}$, $\{2, 6\}$, $\{1, 2, 6\}$, $\{3, 6\}$, $\{1, 3, 6\}$, $\{2, 3, 6\}$ and $\{1, 2, 3, 6\}$.

Thus $H(6) = 10$.

Let $L(n)$ denote the least common multiple of the numbers 1 through n .

E.g. $L(6)$ is the least common multiple of the numbers 1, 2, 3, 4, 5, 6 and $L(6)$ equals 60.

Let $HL(n)$ denote $H(L(n))$.

You are given $HL(4) = H(12) = 44$.

Find $HL(50000)$. Give your answer modulo 10^9 .

Problem 591: Best Approximations by Quadratic Integers

[Link](#)

Tags: Algebra, Game Theory

Given a non-square integer d , any real x can be approximated arbitrarily close by quadratic integers $a + b\sqrt{d}$, where a, b are integers. For example, the following inequalities approximate π with precision 10^{-13} :

$$4375636191520\sqrt{2} - 6188084046055 < \pi < 721133315582\sqrt{2} - 1019836515172$$

We call $BQA_d(x, n)$ the quadratic integer closest to x with the absolute values of a, b not exceeding n . We also define the integral part of a quadratic integer as $I_d(a + b\sqrt{d}) = a$.

You are given that:

- $BQA_2(\pi, 10) = 6 - 2\sqrt{2}$
- $BQA_5(\pi, 100) = 26\sqrt{5} - 55$
- $BQA_7(\pi, 10^6) = 560323 - 211781\sqrt{7}$
- $I_2(BQA_2(\pi, 10^{13})) = -6188084046055$

Find the sum of $|I_d(BQA_d(\pi, 10^{13}))|$ for all non-square positive integers less than 100.

Problem 592: Factorial Trailing Digits 2

[Link](#)

Tags: Number Theory, Sequences

For any N , let $f(N)$ be the last twelve hexadecimal digits before the trailing zeroes in $N!$.

For example, the hexadecimal representation of $20!$ is $21C3677C82B40000$,

so $f(20)$ is the digit sequence $21C3677C82B4$.

Find $f(20!)$. Give your answer as twelve hexadecimal digits, using uppercase for the digits A to F.

Problem 593: Fleeting Medians

[Link](#)

Tags: Number Theory, Sequences

We define two sequences $S = \{S(1), S(2), \dots, S(n)\}$ and $S_2 = \{S_2(1), S_2(2), \dots, S_2(n)\}$:

$S(k) = (p_k)^k \bmod 10007$ where p_k is the k th prime number.

$S_2(k) = S(k) + S(\lfloor \frac{k}{10000} \rfloor + 1)$ where $\lfloor \cdot \rfloor$ denotes the floor function.

Then let $M(i, j)$ be the median of elements $S_2(i)$ through $S_2(j)$, inclusive. For example, $M(1, 10) = 2021.5$ and $M(10^2, 10^3) = 4715.0$.

Let $F(n, k) = \sum_{i=1}^{n-k+1} M(i, i+k-1)$. For example, $F(100, 10) = 463628.5$ and $F(10^5, 10^4) = 675348207.5$.

Find $F(10^7, 10^5)$. If the sum is not an integer, use .5 to denote a half. Otherwise, use .0 instead.

Problem 594: Rhombus Tilings

[Link](#)

Tags: Geometry, Graph Theory

For a polygon P , let $t(P)$ be the number of ways in which P can be tiled using rhombi and squares with edge length 1. Distinct rotations and reflections are counted as separate tilings.

For example, if O is a regular octagon with edge length 1, then $t(O) = 8$. As it happens, all these 8 tilings are rotations of one another:

Problem 595: Incremental Random Sort

[Link](#)

Tags: Combinatorics, Game Theory, Probability, Sequences

A deck of cards numbered from 1 to n is shuffled randomly such that each permutation is equally likely.

The cards are to be sorted into ascending order using the following technique:

- Look at the initial sequence of cards. If it is already sorted, then there is no need for further action. Otherwise, if any subsequences of cards happen to be in the correct place relative to one another (ascending with no gaps), then those subsequences are fixed by attaching the cards together. For

example, with 7 cards initially in the order 4123756, the cards labelled 1, 2 and 3 would be attached together, as would 5 and 6.

- The cards are ‘shuffled’ by being thrown into the air, but note that any correctly sequenced cards remain attached, so their orders are maintained. The cards (or bundles of attached cards) are then picked up randomly. You should assume that this randomisation is unbiased, despite the fact that some cards are single, and others are grouped together.
- Repeat steps 1 and 2 until the cards are sorted.

Let $S(n)$ be the expected number of shuffles needed to sort the cards. Since the order is checked before the first shuffle, $S(1) = 0$. You are given that $S(2) = 1$, and $S(5) = 4213/871$.

Find $S(52)$, and give your answer rounded to 8 decimal places.

Problem 596: Number of Lattice Points in a Hyperball

[Link](#)

Tags: Combinatorics

Let $T(r)$ be the number of integer quadruplets x, y, z, t such that $x^2 + y^2 + z^2 + t^2 \leq r^2$. In other words, $T(r)$ is the number of lattice points in the four-dimensional hyperball of radius r .

You are given that $T(2) = 89$, $T(5) = 3121$, $T(100) = 493490641$ and $T(10^4) = 49348022079085897$.

Find $T(10^8) \bmod 1000000007$.

Problem 597: Torpids

[Link](#)

Tags: Combinatorics, Game Theory, Geometry, Number Theory, Probability

The Torpids are rowing races held annually in Oxford, following some curious rules:

- A division consists of n boats (typically 13), placed in order based on past performance.
- All boats within a division start at 40 metre intervals along the river, in order with the highest-placed boat starting furthest upstream.
- The boats all start rowing simultaneously, upstream, trying to catch the boat in front while avoiding being caught by boats behind.

- Each boat continues rowing until either it reaches the finish line or it catches up with (“bumps”) a boat in front.
- The finish line is a distance L metres (the course length, in reality about 1800 metres) upstream from the starting position of the lowest-placed boat. (Because of the staggered starting positions, higher-placed boats row a slightly shorter course than lower-placed boats.)
- When a “bump” occurs, the “bumping” boat takes no further part in the race. The “bumped” boat must continue, however, and may even be “bumped” again by boats that started two or more places behind it.
- After the race, boats are assigned new places within the division, based on the bumps that occurred. Specifically, for any boat A that started in a lower place than B , then A will be placed higher than B in the new order if and only if one of the following occurred:
 - A bumped B directly
 - A bumped another boat that went on to bump B
 - A bumped another boat, that bumped yet another boat, that bumped B
 - etc

NOTE: For the purposes of this problem you may disregard the boats’ lengths, and assume that a bump occurs precisely when the two boats draw level. (In reality, a bump is awarded as soon as physical contact is made, which usually occurs when there is much less than a full boat length’s overlap.)

Suppose that, in a particular race, each boat B_j rows at a steady speed $v_j = -\log X_j$ metres per second, where the X_j are chosen randomly (with uniform distribution) between 0 and 1, independently from one another. These speeds are relative to the riverbank: you may disregard the flow of the river.

Let $p(n, L)$ be the probability that the new order is an even permutation of the starting order, when there are n boats in the division and L is the course length.

For example, with $n = 3$ and $L = 160$, labelling the boats as A, B, C in starting order with C highest, the different possible outcomes of the race are as follows:

Bumps occurring New order Permutation Probability

none A, B, C even $4/15$

B bumps C A, C, B odd $8/45$

A bumps B B, A, C odd $1/3$

B bumps C , then A bumps C C, A, B even $4/27$

A bumps B , then B bumps C C, B, A odd $2/27$

Therefore, $p(3, 160) = 4/15 + 4/27 = 56/135$.

You are also given that $p(4, 400) = 0.5107843137$, rounded to 10 digits after the decimal point.

Find $p(13, 1800)$ rounded to 10 digits after the decimal point.

Problem 598: Split Divisibilities

Link

Tags: Number Theory

Consider the number 48.

There are five pairs of integers a and b ($a \leq b$) such that $a \times b = 48$: $(1, 48)$, $(2, 24)$, $(3, 16)$, $(4, 12)$ and $(6, 8)$.

It can be seen that both 6 and 8 have 4 divisors.

So of those five pairs one consists of two integers with the same number of divisors.

In general:

Let $C(n)$ be the number of pairs of positive integers $a \times b = n$, ($a \leq b$) such that a and b have the same number of divisors; so $C(48) = 1$.

You are given $C(10!) = 3$: $(1680, 2160)$, $(1800, 2016)$ and $(1890, 1920)$.

Find $C(100!)$.

Problem 599: Distinct Colourings of a Rubik's Cube

Link

Tags: Combinatorics

The well-known Rubik's Cube puzzle has many fascinating mathematical properties. The $2 \times 2 \times 2$ variant has 8 cubelets with a total of 24 visible faces, each with a coloured sticker. Successively turning faces will rearrange the cubelets, although not all arrangements of cubelets are reachable without dismantling the puzzle.

Suppose that we wish to apply new stickers to a $2 \times 2 \times 2$ Rubik's cube in a non-standard colouring. Specifically, we have n different colours available (with an unlimited supply of stickers of each colour), and we place one sticker on each of the 24 faces in any arrangement that we please. We are not required to use all the colours, and if desired the same colour may appear in more than one face of a single cubelet.

We say that two such colourings c_1, c_2 are essentially distinct if a cube coloured according to c_1 cannot be made to match a cube coloured according to c_2 by performing mechanically possible Rubik's Cube moves.

For example, with two colours available, there are 183 essentially distinct colourings.

How many essentially distinct colourings are there with 10 different colours available?

Problem 600: Integer Sided Equiangular Hexagons

Link

Tags: Number Theory

Let $H(n)$ be the number of distinct integer sided equiangular convex hexagons with perimeter not exceeding n .

Hexagons are distinct if and only if they are not congruent.

You are given $H(6) = 1$, $H(12) = 10$, $H(100) = 31248$.

Find $H(55106)$.

Equiangular hexagons with perimeter not exceeding 12

Problem 601: Divisibility Streaks

Link

Tags: Number Theory

For every positive number n we define the function $streak(n) = k$ as the smallest positive integer k such that $n + k$ is not divisible by $k + 1$.

E.g:

13 is divisible by 1

14 is divisible by 2

15 is divisible by 3

16 is divisible by 4

17 is NOT divisible by 5

So $streak(13) = 4$.

Similarly:

120 is divisible by 1

121 is NOT divisible by 2

So $\text{streak}(120) = 1$.

Define $P(s, N)$ to be the number of integers n , $1 \leq n \leq N$, for which $\text{streak}(n) = s$.

So $P(3, 14) = 1$ and $P(6, 10^6) = 14286$.

Find the sum, as i ranges from 1 to 31, of $P(i, 4^i)$.

Problem 602: Product of Head Counts

[Link](#)

Tags: Algebra, Probability, Sequences

Alice enlists the help of some friends to generate a random number, using a single unfair coin. She and her friends sit around a table and, starting with Alice, they take it in turns to toss the coin. Everyone keeps a count of how many heads they obtain individually. The process ends as soon as Alice obtains a Head. At this point, Alice multiplies all her friends' Head counts together to obtain her random number.

As an illustration, suppose Alice is assisted by Bob, Charlie, and Dawn, who are seated round the table in that order, and that they obtain the sequence of Head/Tail outcomes THHH—TTTT—THHT—H beginning and ending with Alice. Then Bob and Charlie each obtain 2 heads, and Dawn obtains 1 head. Alice's random number is therefore $2 \times 2 \times 1 = 4$.

Define $e(n, p)$ to be the expected value of Alice's random number, where n is the number of friends helping (excluding Alice herself), and p is the probability of the coin coming up Tails.

It turns out that, for any fixed n , $e(n, p)$ is always a polynomial in p . For example, $e(3, p) = p^3 + 4p^2 + p$.

Define $c(n, k)$ to be the coefficient of p^k in the polynomial $e(n, p)$. So $c(3, 1) = 1$, $c(3, 2) = 4$, and $c(3, 3) = 1$.

You are given that $c(100, 40) \equiv 986699437 \pmod{10^9 + 7}$.

Find $c(10000000, 4000000) \pmod{10^9 + 7}$.

Problem 603: Substring Sums of Prime Concatenations

[Link](#)

Tags: Number Theory

Let $S(n)$ be the sum of all contiguous integer-substrings that can be formed from the integer n . The substrings need not be distinct.

For example, $S(2024) = 2 + 0 + 2 + 4 + 20 + 02 + 24 + 202 + 024 + 2024 = 2304$.

Let $P(n)$ be the integer formed by concatenating the first n primes together. For example, $P(7) = 2357111317$.

Let $C(n, k)$ be the integer formed by concatenating k copies of $P(n)$ together. For example, $C(7, 3) = 235711131723571113172357111317$.

Evaluate $S(C(10^6, 10^{12})) \bmod (10^9 + 7)$.

Problem 604: Convex Path in Square

[Link](#)

Tags: Combinatorics, Graph Theory

Let $F(N)$ be the maximum number of lattice points in an axis-aligned $N \times N$ square that the graph of a single strictly convex increasing function can pass through.

You are given that $F(1) = 2$, $F(3) = 3$, $F(9) = 6$, $F(11) = 7$, $F(100) = 30$ and $F(50000) = 1898$.

Below is the graph of a function reaching the maximum 3 for $N = 3$:

Problem 605: Pairwise Coin-Tossing Game

[Link](#)

Tags: Game Theory, Number Theory, Probability

Consider an n -player game played in consecutive pairs: Round 1 takes place between players 1 and 2, round 2 takes place between players 2 and 3, and so on and so forth, all the way up to round n , which takes place between players n and 1. Then round $n + 1$ takes place between players 1 and 2 as the entire cycle starts again.

In other words, during round r , player $((r - 1) \bmod n) + 1$ faces off against player $(r \bmod n) + 1$.

During each round, a fair coin is tossed to decide which of the two players wins that round. If any given player wins both rounds r and $r + 1$, then that player wins the entire game.

Let $P_n(k)$ be the probability that player k wins in an n -player game, in the form of a reduced fraction. For example, $P_3(1) = 12/49$ and $P_6(2) = 368/1323$.

Let $M_n(k)$ be the product of the reduced numerator and denominator of $P_n(k)$.
For example, $M_3(1) = 588$ and $M_6(2) = 486864$.

Find the last 8 digits of $M_{10^8+7}(10^4 + 7)$.

Problem 606: Gozinta Chains II

Link

Tags: Number Theory, Sequences

A gozinta chain for n is a sequence $\{1, a, b, \dots, n\}$ where each element properly divides the next.

For example, there are eight distinct gozinta chains for 12:

$\{1, 12\}$, $\{1, 2, 12\}$, $\{1, 2, 4, 12\}$, $\{1, 2, 6, 12\}$, $\{1, 3, 12\}$, $\{1, 3, 6, 12\}$, $\{1, 4, 12\}$ and $\{1, 6, 12\}$.

Let $S(n)$ be the sum of all numbers, k , not exceeding n , which have 252 distinct gozinta chains.

You are given $S(10^6) = 8462952$ and $S(10^{12}) = 623291998881978$.

Find $S(10^{36})$, giving the last nine digits of your answer.

Problem 607: Marsh Crossing

Link

Tags: Combinatorics

Frodo and Sam need to travel 100 leagues due East from point A to point B. On normal terrain, they can cover 10 leagues per day, and so the journey would take 10 days. However, their path is crossed by a long marsh which runs exactly South-West to North-East, and walking through the marsh will slow them down. The marsh is 50 leagues wide at all points, and the mid-point of AB is located in the middle of the marsh. A map of the region is shown in the diagram below:

Problem 608: Divisor Sums

Link

Tags: Number Theory

Let $D(m, n) = \sum_{d|m} \sum_{k=1}^n \sigma_0(kd)$ where d runs through all divisors of m and $\sigma_0(n)$ is the number of divisors of n .
 You are given $D(3!, 10^2) = 3398$ and $D(4!, 10^6) = 268882292$.
 Find $D(200!, 10^{12}) \bmod (10^9 + 7)$.

Problem 609: π Sequences

Link

Tags: Combinatorics, Number Theory, Sequences

For every $n \geq 1$ the prime-counting function $\pi(n)$ is equal to the number of primes not exceeding n .

E.g. $\pi(6) = 3$ and $\pi(100) = 25$.

We say that a sequence of integers $u = (u_0, \dots, u_m)$ is a π sequence if

- $u_n \geq 1$ for every n
- $u_{n+1} = \pi(u_n)$
- u has two or more elements

For $u_0 = 10$ there are three distinct π sequences: $(10, 4)$, $(10, 4, 2)$ and $(10, 4, 2, 1)$.

Let $c(u)$ be the number of elements of u that are not prime.

Let $p(n, k)$ be the number of π sequences u for which $u_0 \leq n$ and $c(u) = k$.

Let $P(n)$ be the product of all $p(n, k)$ that are larger than 0.

You are given: $P(10) = 3 \times 8 \times 9 \times 3 = 648$ and $P(100) = 31038676032$.

Find $P(10^8)$. Give your answer modulo 1000000007.

Problem 610: Roman Numerals II

Link

Tags: Combinatorics, Game Theory, Probability, Sequences

A random generator produces a sequence of symbols drawn from the set $\{I, V, X, L, C, D, M, \#\}$. Each item in the sequence is determined by selecting one of these symbols at random, independently of the other items in the sequence. At each step, the seven letters are equally likely to be selected, with probability 14% each, but the $\#$ symbol only has a 2% chance of selection.

We write down the sequence of letters from left to right as they are generated, and we stop at the first occurrence of the # symbol (without writing it). However, we stipulate that what we have written down must always (when non-empty) be a valid Roman numeral representation in minimal form. If appending the next letter would contravene this then we simply skip it and try again with the next symbol generated.

Please take careful note of About... Roman Numerals for the definitive rules for this problem on what constitutes a “valid Roman numeral representation” and “minimal form”. For example, the (only) sequence that represents 49 is XLIX. The subtractive combination IL is invalid because of rule (ii), while XXXIX is valid but not minimal. The rules do not place any restriction on the number of occurrences of M, so all positive integers have a valid representation. These are the same rules as were used in Problem 89, and members are invited to solve that problem first.

Find the expected value of the number represented by what we have written down when we stop. (If nothing is written down then count that as zero.) Give your answer rounded to 8 places after the decimal point.

Problem 611: Hallway of Square Steps

[Link](#)

Tags: Game Theory, Sequences

Peter moves in a hallway with $N + 1$ doors consecutively numbered from 0 through N . All doors are initially closed. Peter starts in front of door 0, and repeatedly performs the following steps:

- First, he walks a positive square number of doors away from his position.
- Then he walks another, larger square number of doors away from his new position.
- He toggles the door he faces (opens it if closed, closes it if open).
- And finally returns to door 0.

We call an action any sequence of those steps. Peter never performs the exact same action twice, and makes sure to perform all possible actions that don't bring him past the last door.

Let $F(N)$ be the number of doors that are open after Peter has performed all possible actions. You are given that $F(5) = 1$, $F(100) = 27$, $F(1000) = 233$ and $F(10^6) = 112168$.

Find $F(10^{12})$.

Problem 612: Friend Numbers

[Link](#)

Tags: Number Theory

Let's call two numbers friend numbers if their representation in base 10 has at least one common digit. E.g. 1123 and 3981 are friend numbers.

Let $f(n)$ be the number of pairs (p, q) with $1 \leq p, q \leq n$ such that p and q are friend numbers.

$$f(100) = 1539.$$

Find $f(10^{18}) \bmod 1000267129$.

Problem 613: Pythagorean Ant

[Link](#)

Tags: Game Theory, Geometry, Number Theory, Probability

Dave is doing his homework on the balcony and, preparing a presentation about Pythagorean triangles, has just cut out a triangle with side lengths 30cm, 40cm and 50cm from some cardboard, when a gust of wind blows the triangle down into the garden.

Another gust blows a small ant straight onto this triangle. The poor ant is completely disoriented and starts to crawl straight ahead in random direction in order to get back into the grass.

Assuming that all possible positions of the ant within the triangle and all possible directions of moving on are equiprobable, what is the probability that the ant leaves the triangle along its longest side?

Give your answer rounded to 10 digits after the decimal point.

Problem 614: Special Partitions 2

[Link](#)

Tags: Combinatorics

An integer partition of a number n is a way of writing n as a sum of positive integers. Partitions that differ only by the order of their summands are considered the same.

We call an integer partition special if 1) all its summands are distinct, and 2) all its even summands are also divisible by 4. For example, the special partitions of

10 are:

$$10 = 1 + 4 + 5 = 3 + 7 = 1 + 9$$

The number 10 admits many more integer partitions (a total of 42), but only those three are special.

Let be $P(n)$ the number of special integer partitions of n . You are given that $P(1) = 1$, $P(2) = 0$, $P(3) = 1$, $P(6) = 1$, $P(10) = 3$, $P(100) = 37076$ and $P(1000) = 3699177285485660336$.

Find $\sum_{i=1}^{10^7} P(i)$. Give the result modulo $10^9 + 7$.

Problem 615: The Millionth Number with at Least One Million Prime Factors

Link

Tags: Number Theory

Consider the natural numbers having at least 5 prime factors, which don't have to be distinct. Sorting these numbers by size gives a list which starts with:

- $32 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$
- $48 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3$
- $64 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$
- $72 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3$
- $80 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5$
- $96 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3$
- $\dots$

So, for example, the fifth number with at least 5 prime factors is 80.

Find the millionth number with at least one million prime factors. Give your answer modulo 123454321.

Problem 616: Creative Numbers

Link

Tags: Game Theory, Number Theory

Alice plays the following game, she starts with a list of integers L and on each step she can either:

- remove two elements a and b from L and add a^b to L
- or conversely remove an element c from L that can be written as a^b , with a and b being two integers such that $a, b > 1$, and add both a and b to L

For example starting from the list $L = \{8\}$, Alice can remove 8 and add 2 and 3 resulting in $L = \{2, 3\}$ in a first step. Then she can obtain $L = \{9\}$ in a second step.

Note that the same integer is allowed to appear multiple times in the list.

An integer $n > 1$ is said to be creative if for any integer m Alice can obtain a list that contains m starting from $L = \{n\}$.

Find the sum of all creative integers less than or equal to 10^{12} .

Problem 617: Mirror Power Sequence

Link

Tags: Sequences

For two integers n, e , we define an (n, e) -MPS (Mirror Power Sequence) to be an infinite sequence of integers $(a_i)_{i \geq 0}$ such that for all $i \geq 0$, $a_{i+1} = \min(a_i^e, n - a_i^e)$ and $a_i \neq 1$. Examples of such sequences are the two $(18, 2)$ -MPS sequences made of alternating 2 and 4.

Note that even though such a sequence is uniquely determined by n, e and a_0 , for most values such a sequence does not exist. For example, no (n, e) -MPS exists for $n \leq 6$.

Define $C(n)$ to be the number of (n, e) -MPS for some e , and $D(N) = \sum_{n=2}^N C(n)$.

You are given that $D(10) = 2$, $D(100) = 21$, $D(1000) = 69$, $D(10^6) = 1303$ and $D(10^{12}) = 1014800$.

Find $D(10^{18})$.

Problem 618: Numbers with a Given Prime Factor Sum

Link

Tags: Number Theory, Sequences

Consider the numbers 15, 16 and 18:

$$15 = 3 \times 5 \text{ and } 3 + 5 = 8.$$

$$16 = 2 \times 2 \times 2 \times 2 \text{ and } 2 + 2 + 2 + 2 = 8.$$

$18 = 2 \times 3 \times 3$ and $2 + 3 + 3 = 8$.

15, 16 and 18 are the only numbers that have 8 as sum of the prime factors (counted with multiplicity).

We define $S(k)$ to be the sum of all numbers n where the sum of the prime factors (with multiplicity) of n is k .

Hence $S(8) = 15 + 16 + 18 = 49$.

Other examples: $S(1) = 0$, $S(2) = 2$, $S(3) = 3$, $S(5) = 5 + 6 = 11$.

The Fibonacci sequence is $F_1 = 1$, $F_2 = 1$, $F_3 = 2$, $F_4 = 3$, $F_5 = 5$, $\dots$

Find the last nine digits of $\sum_{k=2}^{24} S(F_k)$.

Problem 619: Square Subsets

[Link](#)

Tags: Combinatorics

For a set of positive integers $\{a, a+1, a+2, \dots, b\}$, let $C(a, b)$ be the number of non-empty subsets in which the product of all elements is a perfect square.

For example $C(5, 10) = 3$, since the products of all elements of $\{5, 8, 10\}$, $\{5, 8, 9, 10\}$ and $\{9\}$ are perfect squares, and no other subsets of $\{5, 6, 7, 8, 9, 10\}$ have this property.

You are given that $C(40, 55) = 15$, and $C(1000, 1234) \bmod 1000000007 = 975523611$.

Find $C(1000000, 1234567) \bmod 1000000007$.

Problem 620: Planetary Gears

[Link](#)

Tags: Combinatorics, Game Theory, Geometry

A circle C of circumference c centimetres has a smaller circle S of circumference s centimetres lying off-centre within it. Four other distinct circles, which we call “planets”, with circumferences p, p, q, q centimetres respectively ($p < q$), are inscribed within C but outside S , with each planet touching both C and S tangentially. The planets are permitted to overlap one another, but the boundaries of S and C must be at least 1cm apart at their closest point.

Now suppose that these circles are actually gears with perfectly meshing teeth at a pitch of 1cm. C is an internal gear with teeth on the inside. We require

that c, s, p, q are all integers (as they are the numbers of teeth), and we further stipulate that any gear must have at least 5 teeth.

Note that “perfectly meshing” means that as the gears rotate, the ratio between their angular velocities remains constant, and the teeth of one gear perfectly align with the grooves of the other gear and vice versa. Only for certain gear sizes and positions will it be possible for S and C each to mesh perfectly with all the planets. Arrangements where not all gears mesh perfectly are not valid.

Define $g(c, s, p, q)$ to be the number of such gear arrangements for given values of c, s, p, q : it turns out that this is finite as only certain discrete arrangements are possible satisfying the above conditions. For example, $g(16, 5, 5, 6) = 9$.

Here is one such arrangement:

Problem 621: Expressing an Integer as the Sum of Triangular Numbers

[Link](#)

Tags: Combinatorics

Gauss famously proved that every positive integer can be expressed as the sum of three triangular numbers (including 0 as the lowest triangular number). In fact most numbers can be expressed as a sum of three triangular numbers in several ways.

Let $G(n)$ be the number of ways of expressing n as the sum of three triangular numbers, regarding different arrangements of the terms of the sum as distinct.

For example, $G(9) = 7$, as 9 can be expressed as: $3 + 3 + 3$, $0 + 3 + 6$, $0 + 6 + 3$, $3 + 0 + 6$, $3 + 6 + 0$, $6 + 0 + 3$, $6 + 3 + 0$.

You are given $G(1000) = 78$ and $G(10^6) = 2106$.

Find $G(17526 \times 10^9)$.

Problem 622: Riffle Shuffles

[Link](#)

Tags: Game Theory

A riffle shuffle is executed as follows: a deck of cards is split into two equal halves, with the top half taken in the left hand and the bottom half taken in the right hand. Next, the cards are interleaved exactly, with the top card in the right half inserted just after the top card in the left half, the 2nd card in the

right half just after the 2nd card in the left half, etc. (Note that this process preserves the location of the top and bottom card of the deck)

Let $s(n)$ be the minimum number of consecutive riffle shuffles needed to restore a deck of size n to its original configuration, where n is a positive even number.

Amazingly, a standard deck of 52 cards will first return to its original configuration after only 8 perfect shuffles, so $s(52) = 8$. It can be verified that a deck of 86 cards will also return to its original configuration after exactly 8 shuffles, and the sum of all values of n that satisfy $s(n) = 8$ is 412.

Find the sum of all values of n that satisfy $s(n) = 60$.

Problem 623: Lambda Count

Link

Tags: Game Theory

The lambda-calculus is a universal model of computation at the core of functional programming languages. It is based on lambda-terms, a minimal programming language featuring only function definitions, function calls and variables. Lambda-terms are built according to the following rules:

- Any variable x (single letter, from some infinite alphabet) is a lambda-term.
- If M and N are lambda-terms, then (MN) is a lambda-term, called the application of M to N .
- If x is a variable and M is a term, then $(\lambda x.M)$ is a lambda-term, called an abstraction. An abstraction defines an anonymous function, taking x as parameter and sending back M .

A lambda-term T is said to be closed if for all variables x , all occurrences of x within T are contained within some abstraction $(\lambda x.M)$ in T . The smallest such abstraction is said to bind the occurrence of the variable x . In other words, a lambda-term is closed if all its variables are bound to parameters of enclosing functions definitions. For example, the term $(\lambda x.x)$ is closed, while the term $(\lambda x.(xy))$ is not because y is not bound.

Also, we can rename variables as long as no binding abstraction changes. This means that $(\lambda x.x)$ and $(\lambda y.y)$ should be considered equivalent since we merely renamed a parameter. Two terms equivalent modulo such renaming are called α -equivalent. Note that $(\lambda x.(\lambda y.(xy)))$ and $(\lambda x.(\lambda x.(xx)))$ are not α -equivalent, since the abstraction binding the first variable was the outer one and becomes the inner one. However, $(\lambda x.(\lambda y.(xy)))$ and $(\lambda y.(\lambda x.(yx)))$ are α -equivalent.

The following table regroups the lambda-terms that can be written with at most 15 symbols, symbols being parenthesis, λ , dot and variables.

$(\lambda x.x)$	$(\lambda x.(xx))$	$(\lambda x.(\lambda y.x))$	$(\lambda x.(\lambda y.y))$
$(\lambda x.(x(xx)))$	$(\lambda x.((xx)x))$	$(\lambda x.(\lambda y.(xx)))$	$(\lambda x.(\lambda y.(xy)))$
$(\lambda x.(\lambda y.(yx)))$	$(\lambda x.(\lambda y.(yy)))$	$(\lambda x.(x(\lambda y.x)))$	$(\lambda x.(x(\lambda y.y)))$
$(\lambda x.((\lambda y.x)x))$	$(\lambda x.((\lambda y.y)x))$	$((\lambda x.x)(\lambda x.x))$	$(\lambda x.(x(x(xx))))$
$(\lambda x.(x((xx)x)))$	$(\lambda x.((xx)(xx)))$	$(\lambda x.((x(xx))x))$	$(\lambda x.(((xx)x)x))$

Let be $\Lambda(n)$ the number of distinct closed lambda-terms that can be written using at most n symbols, where terms that are α -equivalent to one another should be counted only once. You are given that $\Lambda(6) = 1$, $\Lambda(9) = 2$, $\Lambda(15) = 20$ and $\Lambda(35) = 3166438$.

Find $\Lambda(2000)$. Give the answer modulo 1 000 000 007.

Problem 624: Two Heads Are Better Than One

Link

Tags: Number Theory, Probability

An unbiased coin is tossed repeatedly until two consecutive heads are obtained. Suppose these occur on the $(M - 1)$ th and M th toss.

Let $P(n)$ be the probability that M is divisible by n . For example, the outcomes HH, HTHH, and THTTHH all count towards $P(2)$, but THH and HTTHH do not.

You are given that $P(2) = \frac{3}{5}$ and $P(3) = \frac{9}{31}$. Indeed, it can be shown that $P(n)$ is always a rational number.

For a prime p and a fully reduced fraction $\frac{a}{b}$, define $Q(\frac{a}{b}, p)$ to be the smallest positive q for which $a \equiv bq \pmod{p}$.

For example $Q(P(2), 109) = Q(\frac{3}{5}, 109) = 66$, because $5 \cdot 66 = 330 \equiv 3 \pmod{109}$ and 66 is the smallest positive such number.

Similarly $Q(P(3), 109) = 46$.

Find $Q(P(10^{18}), 1\,000\,000\,009)$.

Problem 625: Gcd Sum

Link

Tags: Number Theory

$$G(N) = \sum_{j=1}^N \sum_{i=1}^j \gcd(i, j).$$

You are given: $G(10) = 122$.

Find $G(10^{11})$. Give your answer modulo 998244353.

Problem 626: Counting Binary Matrices

Link

Tags: Algebra, Game Theory, Sequences

A binary matrix is a matrix consisting entirely of 0s and 1s. Consider the following transformations that can be performed on a binary matrix:

- Swap any two rows
- Swap any two columns
- Flip all elements in a single row (1s become 0s, 0s become 1s)
- Flip all elements in a single column

Two binary matrices A and B will be considered equivalent if there is a sequence of such transformations that when applied to A yields B . For example, the following two matrices are equivalent:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

via the sequence of two transformations “Flip all elements in column 3” followed by “Swap rows 1 and 2”.

Define $c(n)$ to be the maximum number of $n \times n$ binary matrices that can be found such that no two are equivalent. For example, $c(3) = 3$. You are also given that $c(5) = 39$ and $c(8) = 656108$.

Find $c(20)$, and give your answer modulo 1 001 001 011.

Problem 627: Counting Products

Link

Tags: Number Theory

Consider the set S of all possible products of n positive integers not exceeding m , that is

$$S = \{x_1 x_2 \cdots x_n \mid 1 \leq x_1, x_2, \dots, x_n \leq m\}.$$

Let $F(m, n)$ be the number of the distinct elements of the set S .

For example, $F(9, 2) = 36$ and $F(30, 2) = 308$.

Find $F(30, 10001) \bmod 1\,000\,000\,007$.

Problem 628: Open Chess Positions

Link

Tags: Combinatorics, Game Theory

A position in chess is an (orientated) arrangement of chess pieces placed on a chessboard of given size. In the following, we consider all positions in which n pawns are placed on a $n \times n$

board in such a way, that there is a single pawn in every row and every column.

We call such a position an open position, if a rook, starting at the (empty) lower left corner and using only moves towards the right or upwards, can reach the upper right corner without moving onto any field occupied by a pawn.

Let $f(n)$ be the number of open positions for a $n \times n$ chessboard.

For example, $f(3) = 2$, illustrated by the two open positions for a 3×3 chessboard below.

You are also given $f(5) = 70$.

Find $f(10^8)$ modulo $1\,008\,691\,207$.

Problem 629: Scatterstone Nim

Link

Tags: Game Theory

Alice and Bob are playing a modified game of Nim called Scatterstone Nim, with Alice going first, alternating turns with Bob. The game begins with an arbitrary set of stone piles with a total number of stones equal to n .

During a player's turn, he/she must pick a pile having at least 2 stones and perform a split operation, dividing the pile into an arbitrary set of p non-empty, arbitrarily-sized piles where $2 \leq p \leq k$ for some fixed constant k . For example, a pile of size 4 can be split into $\{1, 3\}$ or $\{2, 2\}$, or $\{1, 1, 2\}$ if $k = 3$ and in addition $\{1, 1, 1, 1\}$ if $k = 4$.

If no valid move is possible on a given turn, then the other player wins the game.

A winning position is defined as a set of stone piles where a player can ultimately ensure victory no matter what the other player does.

Let $f(n, k)$ be the number of winning positions for Alice on her first turn, given parameters n and k . For example, $f(5, 2) = 3$ with winning positions $\{1, 1, 1, 2\}, \{1, 4\}, \{2, 3\}$. In contrast, $f(5, 3) = 5$ with winning positions $\{1, 1, 1, 2\}, \{1, 1, 3\}, \{1, 4\}, \{2, 3\}, \{5\}$.

Let $g(n)$ be the sum of $f(n, k)$ over all $2 \leq k \leq n$. For example, $g(7) = 66$ and $g(10) = 291$.

Find $g(200) \bmod (10^9 + 7)$.

Problem 630: Crossed Lines

[Link](#)

Tags: Number Theory

Given a set, L , of unique lines, let $M(L)$ be the number of lines in the set and let $S(L)$ be the sum over every line of the number of times that line is crossed by another line in the set. For example, two sets of three lines are shown below:

Problem 631: Constrained Permutations

[Link](#)

Tags: Combinatorics, Sequences

Let $(p_1 p_2 \dots p_k)$ denote the permutation of the set $1, \dots, k$ that maps $p_i \mapsto i$. Define the length of the permutation to be k ; note that the empty permutation $()$ has length zero.

Define an occurrence of a permutation $p = (p_1 p_2 \dots p_k)$ in a permutation $P = (P_1 P_2 \dots P_n)$ to be a sequence $1 \leq t_1 t_2 \dots t_k \leq n$ such that $p_i p_j$ if and only if $P_{t_i} P_{t_j}$ for all $i, j \in \{1, \dots, k\}$.

For example, (1243) occurs twice in the permutation (314625) : once as the 1st, 3rd, 4th and 6th elements $(3 \ 46 \ 5)$, and once as the 2nd, 3rd, 4th and 6th elements $(\ 146 \ 5)$.

Let $f(n, m)$ be the number of permutations P of length at most n such that there is no occurrence of the permutation 1243 in P and there are at most m occurrences of the permutation 21 in P .

For example, $f(2, 0) = 3$, with the permutations $()$, (1) , $(1, 2)$ but not $(2, 1)$.

You are also given that $f(4, 5) = 32$ and $f(10, 25) = 294\,400$.

Find $f(10^{18}, 40)$ modulo $1\,000\,000\,007$.

Problem 632: Square Prime Factors

Link

Tags: Number Theory

For an integer n , we define the square prime factors of n to be the primes whose square divides n . For example, the square prime factors of $1500 = 2^2 \times 3 \times 5^3$ are 2 and 5.

Let $C_k(N)$ be the number of integers between 1 and N inclusive with exactly k square prime factors. You are given some values of $C_k(N)$ in the table below.

\$\$

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
$N = 10$	7	3	0	0	0	0
$N = 10^2$	61	36	3	0	0	0
$N = 10^3$	608	343	48	1	0	0
$N = 10^4$	6083	3363	533	21	0	0
$N = 10^5$	60794	33562	5345	297	2	0
$N = 10^6$	607926	335438	53358	3218	60	0
$N = 10^7$	6079291	3353956	533140	32777	834	2
$N = 10^8$	60792694	33539196	5329747	329028	9257	78

\$\$

Find the product of all non-zero $C_k(10^{16})$. Give the result reduced modulo 1 000 000 007.

Problem 633: Square Prime Factors II

Link

Tags: Game Theory, Number Theory

For an integer n , we define the square prime factors of n to be the primes whose square divides n . For example, the square prime factors of $1500 = 2^2 \times 3 \times 5^3$ are 2 and 5.

Let $C_k(N)$ be the number of integers between 1 and N inclusive with exactly k square prime factors. It can be shown that with growing N the ratio $\frac{C_k(N)}{N}$ gets arbitrarily close to a constant c_k^∞ , as suggested by the table below.

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$
$C_k(10)$	7	3	0	0	0
$C_k(10^2)$	61	36	3	0	0
$C_k(10^3)$	608	343	48	1	0
$C_k(10^4)$	6083	3363	533	21	0
$C_k(10^5)$	60794	33562	5345	297	2
$C_k(10^6)$	607926	335438	53358	3218	60
$C_k(10^7)$	6079291	3353956	533140	32777	834
$C_k(10^8)$	60792694	33539196	5329747	329028	9257
$C_k(10^9)$	607927124	335389706	53294365	3291791	95821
c_k^∞	$\frac{6}{\pi^2}$	3.3539×10^{-1}	5.3293×10^{-2}	3.2921×10^{-3}	9.7046×10^{-5}

Find c_7^∞ . Give the result in scientific notation rounded to 5 significant digits, using a e to separate mantissa and exponent. E.g. if the answer were 0.000123456789, then the answer format would be 1.2346e−4.

Problem 634: Numbers of the Form a^2b^3

Link

Tags: Number Theory

Define $F(n)$ to be the number of integers xn that can be written in the form $x = a^2b^3$, where a and b are integers not necessarily different and both greater than 1.

For example, $32 = 2^2 \times 2^3$ and $72 = 3^2 \times 2^3$ are the only two integers less than 100 that can be written in this form. Hence, $F(100) = 2$.

Further you are given $F(2 \times 10^4) = 130$ and $F(3 \times 10^6) = 2014$.

Find $F(9 \times 10^{18})$.

Problem 635: Subset Sums

Link

Tags: Combinatorics, Number Theory

Let $A_q(n)$ be the number of subsets, B , of the set $\{1, 2, \dots, q \cdot n\}$ that satisfy two conditions:

- 1) B has exactly n elements;
- 2) the sum of the elements of B is divisible by n .

E.g. $A_2(5) = 52$ and $A_3(5) = 603$.

Let $S_q(L)$ be $\sum A_q(p)$ where the sum is taken over all primes $p \leq L$.

E.g. $S_2(10) = 554$, $S_2(100) \bmod 1\,000\,000\,009 = 100433628$ and $S_3(100) \bmod 1\,000\,000\,009 = 855618282$.

Find $S_2(10^8) + S_3(10^8)$. Give your answer modulo $1\,000\,000\,009$.

Problem 636: Restricted Factorisations

[Link](#)

Tags: Combinatorics

Consider writing a natural number as product of powers of natural numbers with given exponents, additionally requiring different base numbers for each power.

For example, 256 can be written as a product of a square and a fourth power in three ways such that the base numbers are different.

That is, $256 = 1^2 \times 4^4 = 4^2 \times 2^4 = 16^2 \times 1^4$

Though 4^2 and 2^4 are both equal, we are concerned only about the base numbers in this problem. Note that permutations are not considered distinct, for example $16^2 \times 1^4$ and $1^4 \times 16^2$ are considered to be the same.

Similarly, $10!$ can be written as a product of one natural number, two squares and three cubes in two ways ($10! = 42 \times 5^2 \times 4^2 \times 3^3 \times 2^3 \times 1^3 = 21 \times 5^2 \times 2^2 \times 4^3 \times 3^3 \times 1^3$) whereas $20!$ can be given the same representation in 41680 ways.

Let $F(n)$ denote the number of ways in which n can be written as a product of one natural number, two squares, three cubes and four fourth powers.

You are given that $F(25!) = 4933$, $F(100!) \bmod 1\,000\,000\,007 = 693\,952\,493$, and $F(1\,000!) \bmod 1\,000\,000\,007 = 6\,364\,496$.

Find $F(1\,000\,000!) \bmod 1\,000\,000\,007$.

Problem 637: Flexible Digit Sum

[Link](#)

Tags: Number Theory

Given any positive integer n , we can construct a new integer by inserting plus signs between some of the digits of the base B representation of n , and then carrying out the additions.

For example, from $n = 123_{10}$ (n in base 10) we can construct the four base 10 integers 123_{10} , $1 + 23 = 24_{10}$, $12 + 3 = 15_{10}$ and $1 + 2 + 3 = 6_{10}$.

Let $f(n, B)$ be the smallest number of steps needed to arrive at a single-digit number in base B . For example, $f(7, 10) = 0$ and $f(123, 10) = 1$.

Let $g(n, B_1, B_2)$ be the sum of the positive integers i not exceeding n such that $f(i, B_1) = f(i, B_2)$.

You are given $g(100, 10, 3) = 3302$.

Find $g(10^7, 10, 3)$.

Problem 638: Weighted Lattice Paths

Link

Tags: Combinatorics, Game Theory, Geometry

Let $P_{a,b}$ denote a path in a $a \times b$ lattice grid with following properties:

- The path begins at $(0, 0)$ and ends at (a, b) .
- The path consists only of unit moves upwards or to the right; that is the coordinates are increasing with every move.

Denote $A(P_{a,b})$ to be the area under the path. For the example of a $P_{4,3}$ path given below, the area equals 6.

Problem 639: Summing a Multiplicative Function

Link

Tags: Number Theory

A multiplicative function $f(x)$ is a function over positive integers satisfying $f(1) = 1$ and $f(ab) = f(a)f(b)$ for any two coprime positive integers a and b .

For integer k let $f_k(n)$ be a multiplicative function additionally satisfying $f_k(p^e) = p^k$ for any prime p and any integer $e > 0$.

For example, $f_1(2) = 2$, $f_1(4) = 2$, $f_1(18) = 6$ and $f_2(18) = 36$.

Let $S_k(n) = \sum_{i=1}^n f_k(i)$. For example, $S_1(10) = 41$, $S_1(100) = 3512$,

$S_2(100) = 208090$, $S_1(10000) = 35252550$ and $\sum_{k=1}^3 S_k(10^8) \equiv 338787512 \pmod{1\,000\,000\,007}$.

Find $\sum_{k=1}^{50} S_k(10^{12}) \bmod 1\,000\,000\,007$.

Problem 640: Shut the Box

[Link](#)

Tags: Combinatorics, Game Theory, Probability

Bob plays a single-player game of chance using two standard 6-sided dice and twelve cards numbered 1 to 12. When the game starts, all cards are placed face up on a table.

Each turn, Bob rolls both dice, getting numbers x and y respectively, each in the range $1, \dots, 6$. He must choose amongst three options: turn over card x , card y , or card $x + y$. (If the chosen card is already face down, it is turned to face up, and vice versa.)

If Bob manages to have all twelve cards face down at the same time, he wins.

Alice plays a similar game, except that instead of dice she uses two fair coins, counting heads as 2 and tails as 1, and that she uses four cards instead of twelve. Alice finds that, with the optimal strategy for her game, the expected number of turns taken until she wins is approximately 5.673651.

Assuming that Bob plays with an optimal strategy, what is the expected number of turns taken until he wins? Give your answer rounded to 6 places after the decimal point.

Problem 641: A Long Row of Dice

[Link](#)

Tags: Game Theory, Probability

Consider a row of n dice all showing 1.

First turn every second die, $(2, 4, 6, \dots)$, so that the number showing is increased by 1. Then turn every third die. The sixth die will now show a 3. Then turn every fourth die and so on until every n th die (only the last die) is turned. If the die to be turned is showing a 6 then it is changed to show a 1.

Let $f(n)$ be the number of dice that are showing a 1 when the process finishes. You are given $f(100) = 2$ and $f(10^8) = 69$.

Find $f(10^{36})$.

Problem 642: Sum of Largest Prime Factors

Link

Tags: Number Theory

Let $f(n)$ be the largest prime factor of n and $F(n) = \sum_{i=2}^n f(i)$.

For example $F(10) = 32$, $F(100) = 1915$ and $F(10000) = 10118280$.

Find $F(201820182018)$. Give your answer modulus 10^9 .

Problem 643: 2-Friendly

Link

Tags: Number Theory

Two positive integers a and b are 2-friendly when $\gcd(a, b) = 2^t, t \geq 0$. For example, 24 and 40 are 2-friendly because $\gcd(24, 40) = 8 = 2^3$ while 24 and 36 are not because $\gcd(24, 36) = 12 = 2^2 \cdot 3$ is not a power of 2.

Let $f(n)$ be the number of pairs, (p, q) , of positive integers with $1 \leq pq \leq n$ such that p and q are 2-friendly. You are given $f(10^2) = 1031$ and $f(10^6) = 321418433$ modulo 1 000 000 007.

Find $f(10^{11})$ modulo 1 000 000 007.

Problem 644: Squares on the Line

Link

Tags: Game Theory

Sam and Tom are trying a game of (partially) covering a given line segment of length L by taking turns in placing unit squares onto the line segment.

As illustrated below, the squares may be positioned in two different ways, either “straight” by placing the midpoints of two opposite sides on the line segment, or “diagonal” by placing two opposite corners on the line segment. Newly placed squares may touch other squares, but are not allowed to overlap any other square laid down before.

The player who is able to place the last unit square onto the line segment wins.

Problem 645: Every Day Is a Holiday

Link

Tags: Game Theory, Number Theory, Probability

On planet J, a year lasts for D days. Holidays are defined by the two following rules.

- At the beginning of the reign of the current Emperor, his birthday is declared a holiday from that year onwards.
- If both the day before and after a day d are holidays, then d also becomes a holiday.

Initially there are no holidays. Let $E(D)$ be the expected number of Emperors to reign before all the days of the year are holidays, assuming that their birthdays are independent and uniformly distributed throughout the D days of the year.

You are given $E(2) = 1$, $E(5) = 31/6$, $E(365) \approx 1174.3501$.

Find $E(10000)$. Give your answer rounded to 4 digits after the decimal point.

Problem 646: Bounded Divisors

Link

Tags: Number Theory

Let n be a natural number and $p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ its prime factorisation.

Define the Liouville function $\lambda(n)$ as $\lambda(n) = (-1)^{\sum_{i=1}^k \alpha_i}$.

(i.e. -1 if the sum of the exponents α_i is odd and 1 if the sum of the exponents is even.)

Let $S(n, L, H)$ be the sum $\lambda(d) \cdot d$ over all divisors d of n for which $L \leq d \leq H$.

You are given:

- $S(10!, 100, 1000) = 1457$
- $S(15!, 10^3, 10^5) = -107974$
- $S(30!, 10^8, 10^{12}) = 9766732243224$.

Find $S(70!, 10^{20}, 10^{60})$ and give your answer modulo 1 000 000 007.

Problem 647: Linear Transformations of Polygonal Numbers

Link

Tags: Geometry, Sequences

It is possible to find positive integers A and B such that given any triangular number, T_n , then $AT_n + B$ is always a triangular number. We define $F_3(N)$ to be the sum of $(A + B)$ over all such possible pairs (A, B) with $\max(A, B) \leq N$. For example $F_3(100) = 184$.

Polygonal numbers are generalisations of triangular numbers. Polygonal numbers with parameter k we call k -gonal numbers. The formula for the n th k -gonal number is $\frac{1}{2}n(n(k-2) + 4 - k)$ where $n \geq 1$. For example when $k = 3$ we get $\frac{1}{2}n(n+1)$ the formula for triangular numbers.

The statement above is true for pentagonal, heptagonal and in fact any k -gonal number with k odd. For example when $k = 5$ we get the pentagonal numbers and we can find positive integers A and B such that given any pentagonal number, P_n , then $AP_n + B$ is always a pentagonal number. We define $F_5(N)$ to be the sum of $(A + B)$ over all such possible pairs (A, B) with $\max(A, B) \leq N$.

Similarly we define $F_k(N)$ for odd k . You are given $\sum_k F_k(10^3) = 14993$ where the sum is over all odd $k = 3, 5, 7, \dots$

Find $\sum_k F_k(10^{12})$ where the sum is over all odd $k = 3, 5, 7, \dots$

Problem 648: Skipping Squares

Link

Tags: Probability, Sequences

For some fixed $\rho \in [0, 1]$, we begin a sum s at 0 and repeatedly apply a process: With probability ρ , we add 1 to s , otherwise we add 2 to s .

The process ends when either s is a perfect square or s exceeds 10^{18} , whichever occurs first. For example, if s goes through 0, 2, 3, 5, 7, 9, the process ends at $s = 9$, and two squares 1 and 4 were skipped over.

Let $f(\rho)$ be the expected number of perfect squares skipped over when the process finishes.

It can be shown that the power series for $f(\rho)$ is $\sum_{k=0}^{\infty} a_k \rho^k$ for a suitable (unique) choice of coefficients a_k . Some of the first few coefficients are $a_0 = 1$, $a_1 = 0$, $a_5 = -18$, $a_{10} = 45176$.

Let $F(n) = \sum_{k=0}^n a_k$. You are given that $F(10) = 53964$ and $F(50) \equiv 842418857 \pmod{10^9}$.

Find $F(1000)$, and give your answer modulo 10^9 .

Problem 649: Low-Prime Chessboard Nim

Link

Tags: Combinatorics, Game Theory, Graph Theory, Number Theory, Probability

Alice and Bob are taking turns playing a game consisting of c different coins on a chessboard of size n by n .

The game may start with any arrangement of c coins in squares on the board. It is possible at any time for more than one coin to occupy the same square on the board at the same time. The coins are distinguishable, so swapping two coins gives a different arrangement if (and only if) they are on different squares.

On a given turn, the player must choose a coin and move it either left or up 2, 3, 5, or 7 spaces in a single direction. The only restriction is that the coin cannot move off the edge of the board.

The game ends when a player is unable to make a valid move, thereby granting the other player the victory.

Assuming that Alice goes first and that both players are playing optimally, let $M(n, c)$ be the number of possible starting arrangements for which Alice can ensure her victory, given a board of size n by n with c distinct coins.

For example, $M(3, 1) = 4$, $M(3, 2) = 40$, and $M(9, 3) = 450304$.

What are the last 9 digits of $M(10\,000\,019, 100)$?

Problem 650: Divisors of Binomial Product

Link

Tags: Combinatorics, Number Theory

Let $B(n) = \prod_{k=0}^n \binom{n}{k}$, a product of binomial coefficients.

For example, $B(5) = \binom{5}{0} \times \binom{5}{1} \times \binom{5}{2} \times \binom{5}{3} \times \binom{5}{4} \times \binom{5}{5} = 1 \times 5 \times 10 \times 10 \times 5 \times 1 = 2500$.

Let $D(n) = \sum_{d|B(n)} d$, the sum of the divisors of $B(n)$.

For example, the divisors of $B(5)$ are 1, 2, 4, 5, 10, 20, 25, 50, 100, 125, 250, 500, 625, 1250 and 2500,

so $D(5) = 1 + 2 + 4 + 5 + 10 + 20 + 25 + 50 + 100 + 125 + 250 + 500 + 625 + 1250 + 2500 = 5467$.

Let $S(n) = \sum_{k=1}^n D(k)$.

You are given $S(5) = 5736$, $S(10) = 141740594713218418$ and $S(100) \bmod 1\,000\,000\,007 = 332792866$.

Find $S(20\,000) \bmod 1\,000\,000\,007$.

Problem 651: Patterned Cylinders

Link

Tags: Combinatorics, Graph Theory, Number Theory, Sequences

An infinitely long cylinder has its curved surface fully covered with different coloured but otherwise identical rectangular stickers, without overlapping. The stickers are aligned with the cylinder, so two of their edges are parallel with the cylinder's axis, with four stickers meeting at each corner.

Let $a > 0$ and suppose that the colouring is periodic along the cylinder, with the pattern repeating every a stickers. (The period is allowed to be any divisor of a .) Let b be the number of stickers that fit round the circumference of the cylinder.

Let $f(m, a, b)$ be the number of different such periodic patterns that use exactly m distinct colours of stickers. Translations along the axis, reflections in any plane, rotations in any axis, (or combinations of such operations) applied to a pattern are to be counted as the same as the original pattern.

You are given that $f(2, 2, 3) = 11$, $f(3, 2, 3) = 56$, and $f(2, 3, 4) = 156$. Furthermore, $f(8, 13, 21) \equiv 49718354 \pmod{1\,000\,000\,007}$, and $f(13, 144, 233) \equiv 907081451 \pmod{1\,000\,000\,007}$.

Find $\sum_{i=4}^{40} f(i, F_{i-1}, F_i) \bmod 1\,000\,000\,007$, where F_i are the Fibonacci numbers starting at $F_0 = 0$, $F_1 = 1$.

Problem 652: Distinct Values of a Proto-logarithmic Function

Link

Tags: Number Theory

Consider the values of $\log_2(8)$, $\log_4(64)$ and $\log_3(27)$. All three are equal to 3.

Generally, the function $f(m, n) = \log_m(n)$ over integers $m, n \geq 2$ has the property that

$f(m_1, n_1) = f(m_2, n_2)$ if

- $m_1 = a^e, n_1 = a^f, m_2 = b^e, n_2 = b^f$ for some integers a, b, e, f or
- $m_1 = a^e, n_1 = b^e, m_2 = a^f, n_2 = b^f$ for some integers a, b, e, f

We call a function $g(m, n)$ over integers $m, n \geq 2$ proto-logarithmic if

- $g(m_1, n_1) = g(m_2, n_2)$ if any integers a, b, e, f fulfilling 1. or 2. can be found
- and $g(m_1, n_1) \neq g(m_2, n_2)$ if no integers a, b, e, f fulfilling 1. or 2. can be found.

Let $D(N)$ be the number of distinct values that any proto-logarithmic function $g(m, n)$ attains over $2 \leq m, n \leq N$.

For example, $D(5) = 13$, $D(10) = 69$, $D(100) = 9607$ and $D(10000) = 99959605$.

Find $D(10^{18})$, and give the last 9 digits as answer.

Note: According to the four exponentials conjecture the function $\log_m(n)$ is proto-logarithmic. While this conjecture is yet unproven in general, $\log_m(n)$ can be used to calculate $D(N)$ for small values of N .

Problem 653: Frictionless Tube

Link

Tags: Combinatorics, Game Theory, Geometry, Probability, Sequences

Consider a horizontal frictionless tube with length L millimetres, and a diameter of 20 millimetres. The east end of the tube is open, while the west end is sealed. The tube contains N marbles of diameter 20 millimetres at designated starting locations, each one initially moving either westward or eastward with common speed v .

Since there are marbles moving in opposite directions, there are bound to be some collisions. We assume that the collisions are perfectly elastic, so both marbles involved instantly change direction and continue with speed v away from the collision site. Similarly, if the west-most marble collides with the sealed end of the tube, it instantly changes direction and continues eastward at speed v . On the other hand, once a marble reaches the unsealed east end, it exits the tube and has no further interaction with the remaining marbles.

To obtain the starting positions and initial directions, we use the pseudo-random sequence r_j defined by:

$$r_1 = 6\,563\,116$$

$$r_{j+1} = r_j^2 \bmod 32\,745\,673$$

The west-most marble is initially positioned with a gap of $(r_1 \bmod 1000) + 1$ millimetres between it and the sealed end of the tube, measured from the west-most point of the surface of the marble. Then, for $2 \leq j \leq N$, counting from the west, the gap between the $(j - 1)$ th and j th marbles, as measured from their closest points, is given by $(r_j \bmod 1000) + 1$ millimetres. Furthermore, the j th marble is initially moving eastward if $r_j \leq 10\,000\,000$, and westward if $r_j > 10\,000\,000$.

For example, with $N = 3$, the sequence specifies gaps of 117, 432, and 173 millimetres. The marbles' centres are therefore 127, 579, and 772 millimetres from the sealed west end of the tube. The west-most marble initially moves eastward, while the other two initially move westward.

Under this setup, and with a five metre tube ($L = 5000$), it turns out that the middle (second) marble travels 5519 millimetres before its centre reaches the east-most end of the tube.

Let $d(L, N, j)$ be the distance in millimetres that the j th marble travels before its centre reaches the eastern end of the tube. So $d(5000, 3, 2) = 5519$. You are also given that $d(10\,000, 11, 6) = 11\,780$ and $d(100\,000, 101, 51) = 114\,101$.

Find $d(1\,000\,000\,000, 1\,000\,001, 500\,001)$.

Problem 654: Neighbourly Constraints

Link

Tags: Game Theory

Let $T(n, m)$ be the number of m -tuples of positive integers such that the sum of any two neighbouring elements of the tuple is $\leq n$.

For example, $T(3, 4) = 8$, via the following eight 4-tuples:

$(1, 1, 1, 1)$

$(1, 1, 1, 2)$

$(1, 1, 2, 1)$

$(1, 2, 1, 1)$

$(1, 2, 1, 2)$

$(2, 1, 1, 1)$

$(2, 1, 1, 2)$

$(2, 1, 2, 1)$

You are also given that $T(5, 5) = 246$, $T(10, 10^2) \equiv 862820094 \pmod{1\,000\,000\,007}$ and $T(10^2, 10) \equiv 782136797 \pmod{1\,000\,000\,007}$.

Find $T(5000, 10^{12}) \bmod 1\,000\,000\,007$.

Problem 655: Divisible Palindromes

[Link](#)

Tags: Number Theory

The numbers 545, 5 995 and 15 151 are the three smallest palindromes divisible by 109. There are nine palindromes less than 100 000 which are divisible by 109.

How many palindromes less than 10^{32} are divisible by 10 000 019 ?

Problem 656: Palindromic Sequences

[Link](#)

Tags: Number Theory, Sequences

Given an irrational number α , let $S_\alpha(n)$ be the sequence $S_\alpha(n) = \lfloor \alpha \cdot n \rfloor - \lfloor \alpha \cdot (n-1) \rfloor$ for $n \geq 1$.

($\lfloor \dots \rfloor$ is the floor-function.)

It can be proven that for any irrational α there exist infinitely many values of n such that the subsequence $\{S_\alpha(1), S_\alpha(2) \dots S_\alpha(n)\}$ is palindromic.

The first 20 values of n that give a palindromic subsequence for $\alpha = \sqrt{31}$ are: 1, 3, 5, 7, 44, 81, 118, 273, 3158, 9201, 15244, 21287, 133765, 246243, 358721, 829920, 9600319, 27971037, 46341755, 64712473.

Let $H_g(\alpha)$ be the sum of the first g values of n for which the corresponding subsequence is palindromic.

So $H_{20}(\sqrt{31}) = 150243655$.

Let $T = \{2, 3, 5, 6, 7, 8, 10, \dots, 1000\}$ be the set of positive integers, not exceeding 1000, excluding perfect squares.

Calculate the sum of $H_{100}(\sqrt{\beta})$ for $\beta \in T$. Give the last 15 digits of your answer.

Problem 657: Incomplete Words

[Link](#)

Tags: Sequences

In the context of formal languages, any finite sequence of letters of a given alphabet Σ is called a word over Σ . We call a word incomplete if it does not contain every letter of Σ .

For example, using the alphabet $\Sigma = \{a, b, c\}$, ‘ ab ’, ‘ $abab$ ’ and ‘’ (the empty word) are incomplete words over Σ , while ‘ $abac$ ’ is a complete word over Σ .

Given an alphabet Σ of α letters, we define $I(\alpha, n)$ to be the number of incomplete words over Σ with a length not exceeding n .

For example, $I(3, 0) = 1$, $I(3, 2) = 13$ and $I(3, 4) = 79$.

Find $I(10^7, 10^{12})$. Give your answer modulo 1 000 000 007.

Problem 658: Incomplete Words II

[Link](#)

Tags: Sequences

In the context of formal languages, any finite sequence of letters of a given alphabet Σ is called a word over Σ . We call a word incomplete if it does not contain every letter of Σ .

For example, using the alphabet $\Sigma = \{a, b, c\}$, ‘ ab ’, ‘ $abab$ ’ and ‘’ (the empty word) are incomplete words over Σ , while ‘ $abac$ ’ is a complete word over Σ .

Given an alphabet Σ of α letters, we define $I(\alpha, n)$ to be the number of incomplete words over Σ with a length not exceeding n .

For example, $I(3, 0) = 1$, $I(3, 2) = 13$ and $I(3, 4) = 79$.

Let
$$S(k, n) = \sum_{\alpha=1}^k I(\alpha, n).$$

For example, $S(4, 4) = 406$, $S(8, 8) = 27902680$ and $S(10, 100) \equiv 983602076 \pmod{1\,000\,000\,007}$.

Find $S(10^7, 10^{12})$. Give your answer modulo 1 000 000 007.

Problem 659: Largest Prime

[Link](#)

Tags: Number Theory, Sequences

Consider the sequence $n^2 + 3$ with $n \geq 1$.

If we write down the first terms of this sequence we get:

4, 7, 12, 19, 28, 39, 52, 67, 84, 103, 124, 147, 172, 199, 228, 259, 292, 327, 364, . . .

We see that the terms for $n = 6$ and $n = 7$ (39 and 52) are both divisible by 13.

In fact 13 is the largest prime dividing any two successive terms of this sequence.

Let $P(k)$ be the largest prime that divides any two successive terms of the sequence $n^2 + k^2$.

Find the last 18 digits of $\sum_{k=1}^{10\,000\,000} P(k)$.

Problem 660: Pandigital Triangles

[Link](#)

Tags: Geometry, Number Theory

We call an integer sided triangle n -pandigital if it contains one angle of 120 degrees and, when the sides of the triangle are written in base n , together they use all n digits of that base exactly once.

For example, the triangle (217, 248, 403) is 9-pandigital because it contains one angle of 120 degrees and the sides written in base 9 are $261_9, 305_9, 487_9$ using each of the 9 digits of that base once.

Find the sum of the largest sides of all n -pandigital triangles with $9 \leq n \leq 18$.

Problem 661: A Long Chess Match

[Link](#)

Tags: Game Theory, Number Theory, Probability

Two friends A and B are great fans of Chess. They both enjoy playing the game, but after each game the player who lost the game would like to continue (to get back at the other player) and the player who won would prefer to stop (to finish on a high).

So they come up with a plan. After every game, they would toss a (biased) coin with probability p of Heads (and hence probability $1 - p$ of Tails). If they get Tails, they will continue with the next game. Otherwise they end the match. Also, after every game the players make a note of who is leading in the match.

Let p_A denote the probability of A winning a game and p_B the probability of B winning a game. Accordingly $1 - p_A - p_B$ is the probability that a game ends in a draw. Let $\mathbb{E}_A(p_A, p_B, p)$ denote the expected number of times A was leading in the match.

For example, $\mathbb{E}_A(0.25, 0.25, 0.5) \approx 0.585786$ and $\mathbb{E}_A(0.47, 0.48, 0.001) \approx 377.471736$, both rounded to six places after the decimal point.

$$\text{Let } H(n) = \sum_{k=3}^n \mathbb{E}_A \left(\frac{1}{\sqrt{k+3}}, \frac{1}{\sqrt{k+3}} + \frac{1}{k^2}, \frac{1}{k^3} \right)$$

For example $H(3) \approx 6.8345$, rounded to 4 digits after the decimal point.

Find $H(50)$, rounded to 4 digits after the decimal point.

Problem 662: Fibonacci Paths

Link

Tags: Combinatorics, Geometry, Sequences

Alice walks on a lattice grid. She can step from one lattice point $A(a, b)$ to another $B(a+x, b+y)$ providing distance $AB = \sqrt{x^2 + y^2}$ is a Fibonacci number $\{1, 2, 3, 5, 8, 13, \dots\}$ and $x \geq 0, y \geq 0$.

In the lattice grid below Alice can step from the blue point to any of the red points.

Let $F(W, H)$ be the number of paths Alice can take from $(0, 0)$ to (W, H) .

You are given $F(3, 4) = 278$ and $F(10, 10) = 215846462$.

Find $F(10\,000, 10\,000) \bmod 1\,000\,000\,007$.

Problem 663: Sums of Subarrays

Link

Tags: Number Theory

Let t_k be the tribonacci numbers defined as:

$$t_0 = t_1 = 0;$$

$$t_2 = 1;$$

$$t_k = t_{k-1} + t_{k-2} + t_{k-3} \quad \text{for } k \geq 3.$$

For a given integer n , let A_n be an array of length n (indexed from 0 to $n-1$), that is initially filled with zeros.

The array is changed iteratively by replacing $A_n[(t_{2i-2} \bmod n)]$ with $A_n[(t_{2i-2} \bmod n)] + 2(t_{2i-1} \bmod n) - n + 1$ in each step i .

After each step i , define $M_n(i)$ to be $\max\{\sum_{j=p}^q A_n[j] : 0 \leq p \leq qn\}$, the maximal sum of any contiguous subarray of A_n .

The first 6 steps for $n = 5$ are illustrated below:

Initial state: $A_5 = \{0, 0, 0, 0, 0\}$

Step 1: $\Rightarrow A_5 = \{-4, 0, 0, 0, 0\}, M_5(1) = 0$

Step 2: $\Rightarrow A_5 = \{-4, -2, 0, 0, 0\}, M_5(2) = 0$

Step 3: $\Rightarrow A_5 = \{-4, -2, 4, 0, 0\}, M_5(3) = 4$

Step 4: $\Rightarrow A_5 = \{-4, -2, 6, 0, 0\}, M_5(4) = 6$

Step 5: $\Rightarrow A_5 = \{-4, -2, 6, 0, 4\}, M_5(5) = 10$

Step 6: $\Rightarrow A_5 = \{-4, 2, 6, 0, 4\}, M_5(6) = 12$

Let $S(n, l) = \sum_{i=1}^l M_n(i)$. Thus $S(5, 6) = 32$.

You are given $S(5, 100) = 2416$, $S(14, 100) = 3881$ and $S(107, 1000) = 1618572$.

Find $S(10\,000\,003, 10\,200\,000) - S(10\,000\,003, 10\,000\,000)$.

Problem 664: An Infinite Game

Link

Tags: Combinatorics, Game Theory

Peter is playing a solitaire game on an infinite checkerboard, each square of which can hold an unlimited number of tokens.

Each move of the game consists of the following steps:

- Choose one token T to move. This may be any token on the board, as long as not all of its four adjacent squares are empty.
- Select and discard one token D from a square adjacent to that of T .
- Move T to any one of its four adjacent squares (even if that square is already occupied).

Problem 665: Proportionate Nim

Link

Tags: Combinatorics, Game Theory

Two players play a game with two piles of stones, alternating turns.

On each turn, the corresponding player chooses a positive integer n and does one of the following:

- removes n stones from one pile;
- removes n stones from both piles; or
- removes n stones from one pile and $2n$ stones from the other pile.

The player who removes the last stone wins.

We denote by (n, m) the position in which the piles have n and m stones remaining. Note that (n, m) is considered to be the same position as (m, n) .

Then, for example, if the position is $(2, 6)$, the next player may reach the following positions:

$(0, 2), (0, 4), (0, 5), (0, 6), (1, 2), (1, 4), (1, 5), (1, 6), (2, 2), (2, 3), (2, 4), (2, 5)$

A position is a losing position if the player to move next cannot force a win. For example, $(1, 3), (2, 6), (4, 5)$ are the first few losing positions.

Let $f(M)$ be the sum of $n + m$ for all losing positions (n, m) with $n \leq m$ and $n + m \leq M$. For example, $f(10) = 21$, by considering the losing positions $(1, 3), (2, 6), (4, 5)$.

You are given that $f(100) = 1164$ and $f(1000) = 117002$.

Find $f(10^7)$.

Problem 666: Polymorphic Bacteria

Link

Tags: Combinatorics, Game Theory, Probability, Sequences

Members of a species of bacteria occur in two different types: α and β . Individual bacteria are capable of multiplying and mutating between the types according to the following rules:

- Every minute, each individual will simultaneously undergo some kind of transformation.
- Each individual A of type α will, independently, do one of the following (at random with equal probability):
 - clone itself, resulting in a new bacterium of type α (alongside A who remains)
 - split into 3 new bacteria of type β (replacing A)

- Each individual B of type β will, independently, do one of the following (at random with equal probability):
- spawn a new bacterium of type α (alongside B who remains)
- die

If a population starts with a single bacterium of type α , then it can be shown that there is a 0.07243802 probability that the population will eventually die out, and a 0.92756198 probability that the population will last forever. These probabilities are given rounded to 8 decimal places.

Now consider another species of bacteria, $S_{k,m}$ (where k and m are positive integers), which occurs in k different types α_i for $0 \leq i < k$. The rules governing this species' lifecycle involve the sequence r_n defined by:

- $r_0 = 306$
- $r_{n+1} = r_n^2 \bmod 10\,007$

Every minute, for each i , each bacterium A of type α_i will independently choose an integer j uniformly at random in the range $0 \leq j < m$. What it then does depends on $q = r_{im+j} \bmod 5$:

- If $q = 0$, A dies.
- If $q = 1$, A clones itself, resulting in a new bacterium of type α_i (alongside A who remains).
- If $q = 2$, A mutates, changing into type $\alpha_{(2i) \bmod k}$.
- If $q = 3$, A splits into 3 new bacteria of type $\alpha_{(i^2+1) \bmod k}$ (replacing A).
- If $q = 4$, A spawns a new bacterium of type $\alpha_{(i+1) \bmod k}$ (alongside A who remains).

In fact, our original species was none other than $S_{2,2}$, with $\alpha = \alpha_0$ and $\beta = \alpha_1$.

Let $P_{k,m}$ be the probability that a population of species $S_{k,m}$, starting with a single bacterium of type α_0 , will eventually die out. So $P_{2,2} = 0.07243802$. You are also given that $P_{4,3} = 0.18554021$ and $P_{10,5} = 0.53466253$, all rounded to 8 decimal places.

Find $P_{500,10}$, and give your answer rounded to 8 decimal places.

Problem 667: Moving Pentagon

Link

Tags: Combinatorics, Geometry, Graph Theory, Number Theory

After buying a Gerver Sofa from the Moving Sofa Company, Jack wants to buy a matching cocktail table from the same company. Most important for him is

that the table can be pushed through his L-shaped corridor into the living room without having to be lifted from its table legs.

Unfortunately, the simple square model offered to him is too small for him, so he asks for a bigger model.

He is offered the new pentagonal model illustrated below:

Note, while the shape and size can be ordered individually, due to the production process, all edges of the pentagonal table have to have the same length.

Given optimal form and size, what is the biggest pentagonal cocktail table (in terms of area) that Jack can buy that still fits through his unit wide L-shaped corridor?

Give your answer rounded to 10 digits after the decimal point (if Jack had chosen the square model instead the answer would have been 1.0000000000).

Problem 668: Square Root Smooth Numbers

[Link](#)

Tags: Number Theory

A positive integer is called square root smooth if all of its prime factors are strictly less than its square root.

Including the number 1, there are 29 square root smooth numbers not exceeding 100.

How many square root smooth numbers are there not exceeding 10 000 000 000?

Problem 669: The King's Banquet

[Link](#)

Tags: Combinatorics, Game Theory, Sequences

The Knights of the Order of Fibonacci are preparing a grand feast for their king. There are n knights, and each knight is assigned a distinct number from 1 to n .

When the knights sit down at the roundtable for their feast, they follow a peculiar seating rule: two knights can only sit next to each other if their respective numbers sum to a Fibonacci number.

When the n knights all try to sit down around a circular table with n chairs, they are unable to find a suitable seating arrangement for any $n > 2$ despite their best efforts. Just when they are about to give up, they remember that the king will sit on his throne at the table as well.

Suppose there are $n = 7$ knights and 7 chairs at the roundtable, in addition to the king's throne. After some trial and error, they come up with the following seating arrangement (K represents the king):

Problem 670: Colouring a Strip

[Link](#)

Tags: Combinatorics, Game Theory, Geometry

A certain type of tile comes in three different sizes - 1×1 , 1×2 , and 1×3 - and in four different colours: blue, green, red and yellow. There is an unlimited number of tiles available in each combination of size and colour.

These are used to tile a $2 \times n$ rectangle, where n is a positive integer, subject to the following conditions:

- The rectangle must be fully covered by non-overlapping tiles.
- It is not permitted for four tiles to have their corners meeting at a single point.
- Adjacent tiles must be of different colours.

For example, the following is an acceptable tiling of a 2×12 rectangle:

Problem 671: Colouring a Loop

[Link](#)

Tags: Combinatorics, Game Theory

A certain type of flexible tile comes in three different sizes - 1×1 , 1×2 , and 1×3 - and in k different colours. There is an unlimited number of tiles available in each combination of size and colour.

These are used to tile a closed loop of width 2 and length (circumference) n , where n is a positive integer, subject to the following conditions:

- The loop must be fully covered by non-overlapping tiles.
- It is not permitted for four tiles to have their corners meeting at a single point.
- Adjacent tiles must be of different colours.

For example, the following is an acceptable tiling of a 2×23 loop with $k = 4$ (blue, green, red and yellow):

Problem 672: One More One

Link

Tags: Game Theory

Consider the following process that can be applied recursively to any positive integer n :

- if $n = 1$ do nothing and the process stops,
- if n is divisible by 7 divide it by 7,
- otherwise add 1.

Define $g(n)$ to be the number of 1's that must be added before the process ends. For example:

$125 \xrightarrow{+1} 126 \xrightarrow{\div 7} 18 \xrightarrow{+1} 19 \xrightarrow{+1} 20 \xrightarrow{+1} 21 \xrightarrow{\div 7} 3 \xrightarrow{+1} 4 \xrightarrow{+1} 5 \xrightarrow{+1} 6 \xrightarrow{+1} 7 \xrightarrow{\div 7} 1$.

Eight 1's are added so $g(125) = 8$. Similarly $g(1000) = 9$ and $g(10000) = 21$.

Define $S(N) = \sum_{n=1}^N g(n)$ and $H(K) = S\left(\frac{7^K - 1}{11}\right)$. You are given $H(10) = 690409338$.

Find $H(10^9)$ modulo 1 117 117 717.

Problem 673: Beds and Desks

Link

Tags: Combinatorics, Game Theory, Probability

At Euler University, each of the n students (numbered from 1 to n) occupies a bed in the dormitory and uses a desk in the classroom.

Some of the beds are in private rooms which a student occupies alone, while the others are in double rooms occupied by two students as roommates. Similarly, each desk is either a single desk for the sole use of one student, or a twin desk at which two students sit together as desk partners.

We represent the bed and desk sharing arrangements each by a list of pairs of student numbers. For example, with $n = 4$, if $(2, 3)$ represents the bed pairing and $(1, 3)(2, 4)$ the desk pairing, then students 2 and 3 are roommates while 1 and 4 have single rooms, and students 1 and 3 are desk partners, as are students 2 and 4.

The new chancellor of the university decides to change the organisation of beds and desks: a permutation σ of the numbers $1, 2, \dots, n$ will be chosen, and each student k will be given both the bed and the desk formerly occupied by student number $\sigma(k)$.

The students agree to this change, under the conditions that:

- Any two students currently sharing a room will still be roommates.
- Any two students currently sharing a desk will still be desk partners.

In the example above, there are only two ways to satisfy these conditions: either take no action (σ is the identity permutation), or reverse the order of the students.

With $n = 6$, for the bed pairing $(1, 2)(3, 4)(5, 6)$ and the desk pairing $(3, 6)(4, 5)$, there are 8 permutations which satisfy the conditions. One example is the mapping $(1, 2, 3, 4, 5, 6) \mapsto (1, 2, 5, 6, 3, 4)$.

With $n = 36$, if we have bed pairing:

$(2, 13)(4, 30)(5, 27)(6, 16)(10, 18)(12, 35)(14, 19)(15, 20)(17, 26)(21, 32)(22, 33)(24, 34)(25, 28)$

and desk pairing

$(1, 35)(2, 22)(3, 36)(4, 28)(5, 25)(7, 18)(9, 23)(13, 19)(14, 33)(15, 34)(20, 24)(26, 29)(27, 30)$

then among the $36!$ possible permutations (including the identity permutation), 663552 of them satisfy the conditions stipulated by the students.

The downloadable text files `beds.txt` and `desks.txt` contain pairings for $n = 500$. Each pairing is written on its own line, with the student numbers of the two roommates (or desk partners) separated with a comma. For example, the desk pairing in the $n = 4$ example above would be represented in this file format as:

1,3 2,4

With these pairings, find the number of permutations that satisfy the students' conditions. Give your answer modulo 999 999 937.

Problem 674: Solving $\mathcal{I}$ -equations

Link

Tags: Algebra, Game Theory, Number Theory

We define the $\mathcal{I}$ operator as the function

$$\mathcal{I}(x, y) = (1 + x + y)^2 + y - x$$

and $\mathcal{I}$ -expressions as arithmetic expressions built only from variable names and applications of $\mathcal{I}$. A variable name may consist of one or more letters. For example, the three expressions x , $\mathcal{I}(x, y)$, and $\mathcal{I}(\mathcal{I}(x, ab), x)$ are all $\mathcal{I}$ -expressions.

For two $\mathcal{I}$ -expressions e_1 and e_2 such that the equation $e_1 = e_2$ has a solution in non-negative integers, we define the least simultaneous value of e_1 and e_2 to be the minimum value taken by e_1 and e_2 on such a solution. If the equation $e_1 = e_2$

has no solution in non-negative integers, we define the least simultaneous value of e_1 and e_2 to be 0. For example, consider the following three $\mathcal{I}$ -expressions:

$$\begin{aligned} A &= \mathcal{I}(x, \mathcal{I}(z, t)) \\ B &= \mathcal{I}(\mathcal{I}(y, z), y) \\ C &= \mathcal{I}(\mathcal{I}(x, z), y) \end{aligned}$$

The least simultaneous value of A and B is 23, attained for $x = 3, y = 1, z = t = 0$. On the other hand, $A = C$ has no solutions in non-negative integers, so the least simultaneous value of A and C is 0. The total sum of least simultaneous pairs made of $\mathcal{I}$ -expressions from $\{A, B, C\}$ is 26.

Find the sum of least simultaneous values of all $\mathcal{I}$ -expressions pairs made of distinct expressions from file I-expressions.txt (pairs (e_1, e_2) and (e_2, e_1) are considered to be identical). Give the last nine digits of the result as the answer.

Problem 675: $2^{\omega(n)}$

[Link](#)

Tags: Number Theory

Let $\omega(n)$ denote the number of distinct prime divisors of a positive integer n .

So $\omega(1) = 0$ and $\omega(360) = \omega(2^3 \times 3^2 \times 5) = 3$.

Let $S(n)$ be $\sum_{d|n} \omega(d)$.

E.g. $S(6) = 2^{\omega(1)} + 2^{\omega(2)} + 2^{\omega(3)} + 2^{\omega(6)} = 2^0 + 2^1 + 2^1 + 2^2 = 9$.

Let $F(n) = \sum_{i=2}^n S(i!)$. $F(10) = 4821$.

Find $F(10\,000\,000)$. Give your answer modulo 1 000 000 087.

Problem 676: Matching Digit Sums

[Link](#)

Tags: Number Theory

Let $d(i, b)$ be the digit sum of the number i in base b . For example $d(9, 2) = 2$, since $9 = 1001_2$. When using different bases, the respective digit sums most of the time deviate from each other, for example $d(9, 4) = 3 \neq d(9, 2)$.

However, for some numbers i there will be a match, like $d(17, 4) = d(17, 2) = 2$. Let $M(n, b_1, b_2)$ be the sum of all natural numbers $i \leq n$ for which $d(i, b_1) = d(i, b_2)$. For example, $M(10, 8, 2) = 18$, $M(100, 8, 2) = 292$ and $M(10^6, 8, 2) = 19173952$.

Find $\sum_{k=3}^6 \sum_{l=1}^{k-2} M(10^{16}, 2^k, 2^l)$, giving the last 16 digits as the answer.

Problem 677: Coloured Graphs

Link

Tags: Game Theory, Graph Theory, Number Theory

Let $g(n)$ be the number of undirected graphs with n nodes satisfying the following properties:

- The graph is connected and has no cycles or multiple edges.
- Each node is either red, blue, or yellow.
- A red node may have no more than 4 edges connected to it.
- A blue or yellow node may have no more than 3 edges connected to it.
- An edge may not directly connect a yellow node to a yellow node.

For example, $g(2) = 5$, $g(3) = 15$, and $g(4) = 57$.

You are also given that $g(10) = 710249$ and $g(100) \equiv 919747298 \pmod{1\,000\,000\,007}$.

Find $g(10\,000) \pmod{1\,000\,000\,007}$.

Problem 678: Fermat-like Equations

Link

Tags: Geometry

If a triple of positive integers (a, b, c) satisfies $a^2 + b^2 = c^2$, it is called a Pythagorean triple. No triple (a, b, c) satisfies $a^e + b^e = c^e$ when $e \geq 3$ (Fermat's Last Theorem). However, if the exponents of the left-hand side and right-hand side differ, this is not true. For example, $3^3 + 6^3 = 3^5$.

Let a, b, c, e, f be all positive integers, $0 < a, b, c$, $e \geq 2$, $f \geq 3$ and $c^f \leq N$. Let $F(N)$ be the number of (a, b, c, e, f) such that $a^e + b^e = c^f$. You are given $F(10^3) = 7$, $F(10^5) = 53$ and $F(10^7) = 287$.

Find $F(10^{18})$.

Problem 679: Freefarea

Link

Tags: Geometry

Let S be the set consisting of the four letters $\{\text{'A'}, \text{'E'}, \text{'F'}, \text{'R'}\}$.

For $n \geq 0$, let $S^*(n)$ denote the set of words of length n consisting of letters belonging to S .

We designate the words **FREE**, **FARE**, **AREA**, **REEF** as keywords.

Let $f(n)$ be the number of words in $S^*(n)$ that contains all four keywords exactly once.

This first happens for $n = 9$, and indeed there is a unique 9 lettered word that contain each of the keywords once: **FREEFAREA**

So, $f(9) = 1$.

You are also given that $f(15) = 72863$.

Find $f(30)$.

Problem 680: Yarra Gnisrever

Link

Tags: Game Theory, Probability, Sequences

Let N and K be two positive integers.

F_n is the n -th Fibonacci number: $F_1 = F_2 = 1$, $F_n = F_{n-1} + F_{n-2}$ for all $n \geq 3$.

Let $s_n = F_{2n-1} \bmod N$ and let $t_n = F_{2n} \bmod N$.

Start with an array of integers $A = (A[0], \dots, A[N-1])$ where initially every $A[i]$ is equal to i . Now perform K successive operations on A , where the j -th operation consists of reversing the order of those elements in A with indices between s_j and t_j (both ends inclusive).

Define $R(N, K)$ to be $\sum_{i=0}^{N-1} i \times A[i]$ after K operations.

For example, $R(5, 4) = 27$, as can be seen from the following procedure:

Initial position: $(0, 1, 2, 3, 4)$

Step 1 - Reverse $A[1]$ to $A[1]$: $(0, 1, 2, 3, 4)$

Step 2 - Reverse $A[2]$ to $A[3]$: $(0, 1, 3, 2, 4)$

Step 3 - Reverse $A[0]$ to $A[3]$: $(2, 3, 1, 0, 4)$

Step 4 - Reverse $A[3]$ to $A[1]$: $(2, 0, 1, 3, 4)$

$$R(5, 4) = 0 \times 2 + 1 \times 0 + 2 \times 1 + 3 \times 3 + 4 \times 4 = 27$$

$$\text{Also, } R(10^2, 10^2) = 246597 \text{ and } R(10^4, 10^4) = 249275481640.$$

Find $R(10^{18}, 10^6)$ giving your answer modulo 10^9 .

Problem 681: Maximal Area

Link

Tags: Geometry, Graph Theory

Given positive integers $a \leq b \leq c \leq d$, it may be possible to form quadrilaterals with edge lengths a, b, c, d (in any order). When this is the case, let $M(a, b, c, d)$ denote the maximal area of such a quadrilateral. For example, $M(2, 2, 3, 3) = 6$, attained e.g. by a 2×3 rectangle.

Let $SP(n)$ be the sum of $a + b + c + d$ over all choices $a \leq b \leq c \leq d$ for which $M(a, b, c, d)$ is a positive integer not exceeding n .

$$SP(10) = 186 \text{ and } SP(100) = 23238.$$

Find $SP(1\,000\,000)$.

Problem 682: 5-Smooth Pairs

Link

Tags: Number Theory

5-smooth numbers are numbers whose largest prime factor doesn't exceed 5.

5-smooth numbers are also called Hamming numbers.

Let $\Omega(a)$ be the count of prime factors of a (counted with multiplicity).

Let $s(a)$ be the sum of the prime factors of a (with multiplicity).

$$\text{For example, } \Omega(300) = 5 \text{ and } s(300) = 2 + 2 + 3 + 5 + 5 = 17.$$

Let $f(n)$ be the number of pairs, (p, q) , of Hamming numbers such that $\Omega(p) = \Omega(q)$ and $s(p) + s(q) = n$.

You are given $f(10) = 4$ (the pairs are $(4, 9), (5, 5), (6, 6), (9, 4)$) and $f(10^2) = 3629$.

Find $f(10^7) \bmod 1\,000\,000\,007$.

Problem 683: The Chase II

[Link](#)

Tags: Game Theory, Number Theory, Probability

Consider the following variant of “The Chase” game. This game is played between n players sitting around a circular table using two dice. It consists of $n - 1$ rounds, and at the end of each round one player is eliminated and has to pay a certain amount of money into a pot. The last player remaining is the winner and receives the entire contents of the pot.

At the beginning of a round, each die is given to a randomly selected player. A round then consists of a number of turns.

During each turn, each of the two players with a die rolls it. If a player rolls a 1 or a 2, the die is passed to the neighbour on the left; if the player rolls a 5 or a 6, the die is passed to the neighbour on the right; otherwise, the player keeps the die for the next turn.

The round ends when one player has both dice at the beginning of a turn. At which point that player is immediately eliminated and has to pay s^2 where s is the number of completed turns in this round. Note that if both dice happen to be handed to the same player at the beginning of a round, then no turns are completed, so the player is eliminated without having to pay any money into the pot.

Let $G(n)$ be the expected amount that the winner will receive. For example $G(5)$ is approximately 96.544, and $G(50)$ is 2.82491788e6 in scientific notation rounded to 9 significant digits.

Find $G(500)$, giving your answer in scientific notation rounded to 9 significant digits.

Problem 684: Inverse Digit Sum

[Link](#)

Tags: Number Theory, Sequences

Define $s(n)$ to be the smallest number that has a digit sum of n . For example $s(10) = 19$.

Let $S(k) = \sum_{n=1}^k s(n)$. You are given $S(20) = 1074$.

Further let f_i be the Fibonacci sequence defined by $f_0 = 0, f_1 = 1$ and $f_i = f_{i-2} + f_{i-1}$ for all $i \geq 2$.

Find $\sum_{i=2}^{90} S(f_i)$. Give your answer modulo 1 000 000 007.

Problem 685: Inverse Digit Sum II

Link

Tags: Number Theory

Writing down the numbers which have a digit sum of 10 in ascending order, we get: 19, 28, 37, 46, 55, 64, 73, 82, 91, 109, 118, ...

Let $f(n, m)$ be the m^{th} occurrence of the digit sum n . For example, $f(10, 1) = 19$, $f(10, 10) = 109$ and $f(10, 100) = 1423$.

Let $S(k) = \sum_{n=1}^k f(n^3, n^4)$. For example $S(3) = 7128$ and $S(10) \equiv 32287064 \pmod{1\,000\,000\,007}$.

Find $S(10\,000)$ modulo 1 000 000 007.

Problem 686: Powers of Two

Link

Tags: Number Theory

$2^7 = 128$ is the first power of two whose leading digits are “12”.

The next power of two whose leading digits are “12” is 2^{80} .

Define $p(L, n)$ to be the n th-smallest value of j such that the base 10 representation of 2^j begins with the digits of L .

So $p(12, 1) = 7$ and $p(12, 2) = 80$.

You are also given that $p(123, 45) = 12710$.

Find $p(123, 678910)$.

Problem 687: Shuffling Cards

Link

Tags: Number Theory, Probability

A standard deck of 52 playing cards, which consists of thirteen ranks (Ace, Two, ..., Ten, King, Queen and Jack) each in four suits (Clubs, Diamonds, Hearts

and Spades), is randomly shuffled. Let us call a rank perfect if no two cards of that same rank appear next to each other after the shuffle.

It can be seen that the expected number of ranks that are perfect after a random shuffle equals $\frac{4324}{425} \approx 10.1741176471$.

Find the probability that the number of perfect ranks is prime. Give your answer rounded to 10 decimal places.

Problem 688: Piles of Plates

Link

Tags: Number Theory

We stack n plates into k non-empty piles where each pile is a different size. Define $f(n, k)$ to be the maximum number of plates possible in the smallest pile. For example when $n = 10$ and $k = 3$ the piles 2, 3, 5 is the best that can be done and so $f(10, 3) = 2$. It is impossible to divide 10 into 5 non-empty differently-sized piles and hence $f(10, 5) = 0$.

Define $F(n)$ to be the sum of $f(n, k)$ for all possible pile sizes $k \geq 1$. For example $F(100) = 275$.

Further define $S(N) = \sum_{n=1}^N F(n)$. You are given $S(100) = 12656$.

Find $S(10^{16})$ giving your answer modulo 1 000 000 007.

Problem 689: Binary Series

Link

Tags: Number Theory, Probability

For $0 \leq x < 1$, define $d_i(x)$ to be the i th digit after the binary point of the binary representation of x .

For example $d_2(0.25) = 1$, $d_i(0.25) = 0$ for $i \neq 2$.

Let $f(x) = \sum_{i=1}^{\infty} \frac{d_i(x)}{i^2}$.

Let $p(a)$ be probability that $f(x) < a$, given that x is uniformly distributed between 0 and 1.

Find $p(0.5)$. Give your answer rounded to 8 digits after the decimal point.

Problem 690: Tom and Jerry

Link

Tags: Game Theory, Graph Theory

Tom (the cat) and Jerry (the mouse) are playing on a simple graph G .

Every vertex of G is a mousehole, and every edge of G is a tunnel connecting two mouseholes.

Originally, Jerry is hiding in one of the mouseholes.

Every morning, Tom can check one (and only one) of the mouseholes. If Jerry happens to be hiding there then Tom catches Jerry and the game is over.

Every evening, if the game continues, Jerry moves to a mousehole which is adjacent (i.e. connected by a tunnel, if there is one available) to his current hiding place. The next morning Tom checks again and the game continues like this.

Let us call a graph G a Tom graph, if our super-smart Tom, who knows the configuration of the graph but does not know the location of Jerry, can guarantee to catch Jerry in finitely many days. For example consider all graphs on 3 nodes:

Problem 691: Long Substring with Many Repetitions

Link

Tags: Sequences

Given a character string s , we define $L(k, s)$ to be the length of the longest substring of s which appears at least k times in s , or 0 if such a substring does not exist. For example, $L(3, \text{"bbabcbcabcbacba"}) = 4$ because of the three occurrences of the substring "abca", and $L(2, \text{"bbabcbcabcbacba"}) = 7$ because of the repeated substring "abcbacba". Note that the occurrences can overlap.

Let a_n , b_n and c_n be the 0/1 sequences defined by:

- $a_0 = 0$
- $a_{2n} = a_n$
- $a_{2n+1} = 1 - a_n$
- $b_n = \lfloor \frac{n+1}{\varphi} \rfloor - \lfloor \frac{n}{\varphi} \rfloor$ (where φ is the golden ratio)
- $c_n = a_n + b_n - 2a_nb_n$

and S_n the character string $c_0 \dots c_{n-1}$. You are given that $L(2, S_{10}) = 5$, $L(3, S_{10}) = 2$, $L(2, S_{100}) = 14$, $L(4, S_{100}) = 6$, $L(2, S_{1000}) = 86$, $L(3, S_{1000}) = 45$, $L(5, S_{1000}) = 31$, and that the sum of non-zero $L(k, S_{1000})$ for $k \geq 1$ is 2460.

Find the sum of non-zero $L(k, S_{5000000})$ for $k \geq 1$.

Problem 692: Siegbert and Jo

Link

Tags: Combinatorics, Game Theory

Siegbert and Jo take turns playing a game with a heap of N pebbles:

1. Siegbert is the first to take some pebbles. He can take as many pebbles as he wants. (Between 1 and N inclusive.)
2. In each of the following turns the current player must take at least one pebble and at most twice the amount of pebbles taken by the previous player.
3. The player who takes the last pebble wins.

Although Siegbert can always win by taking all the pebbles on his first turn, to make the game more interesting he chooses to take the smallest number of pebbles that guarantees he will still win (assuming both Siegbert and Jo play optimally for the rest of the game).

Let $H(N)$ be that minimal amount for a heap of N pebbles.

$H(1) = 1$, $H(4) = 1$, $H(17) = 1$, $H(8) = 8$ and $H(18) = 5$.

Let $G(n)$ be $\sum_{k=1}^n H(k)$.

$G(13) = 43$.

Find $G(23416728348467685)$.

Problem 693: Finite Sequence Generator

Link

Tags: Game Theory, Sequences

Two positive integers x and y ($x > y$) can generate a sequence in the following manner:

- $a_x = y$ is the first term,
- $a_{z+1} = a_z^2 \bmod z$ for $z = x, x+1, x+2, \dots$ and
- the generation stops when a term becomes 0 or 1.

The number of terms in this sequence is denoted $l(x, y)$.

For example, with $x = 5$ and $y = 3$, we get $a_5 = 3$, $a_6 = 3^2 \bmod 5 = 4$, $a_7 = 4^2 \bmod 6 = 4$, etc. Giving the sequence of 29 terms:

\$ 3,4,4,2,4,7,9,4,4,3,9,6,4,16,4,16,16,4,16,3,9,6,10,19,25,16,16,8,0 \$

Hence $l(5, 3) = 29$.

$g(x)$ is defined to be the maximum value of $l(x, y)$ for yx . For example, $g(5) = 29$.

Further, define $f(n)$ to be the maximum value of $g(x)$ for $x \leq n$. For example, $f(100) = 145$ and $f(10\,000) = 8824$.

Find $f(3\,000\,000)$.

Problem 694: Cube-full Divisors

[Link](#)

Tags: Number Theory

A positive integer n is considered cube-full, if for every prime p that divides n , so does p^3 . Note that 1 is considered cube-full.

Let $s(n)$ be the function that counts the number of cube-full divisors of n . For example, 1, 8 and 16 are the three cube-full divisors of 16. Therefore, $s(16) = 3$.

Let $S(n)$ represent the summatory function of $s(n)$, that is $S(n) = \sum_{i=1}^n s(i)$.

You are given $S(16) = 19$, $S(100) = 126$ and $S(10\,000) = 13344$.

Find $S(10^{18})$.

Problem 695: Random Rectangles

[Link](#)

Tags: Geometry, Probability

Three points, P_1 , P_2 and P_3 , are randomly selected within a unit square. Consider the three rectangles with sides parallel to the sides of the unit square and a diagonal that is one of the three line segments $\overline{P_1P_2}$, $\overline{P_1P_3}$ or $\overline{P_2P_3}$ (see picture below).

Problem 696: Mahjong

Link

Tags: Combinatorics, Game Theory

The game of Mahjong is played with tiles belonging to s suits. Each tile also has a number in the range $1 \dots n$, and for each suit/number combination there are exactly four indistinguishable tiles with that suit and number. (The real Mahjong game also contains other bonus tiles, but those will not feature in this problem.)

A winning hand is a collection of $3t + 2$ Tiles (where t is a fixed integer) that can be arranged as t Triples and one Pair, where:

- A Triple is either a Chow or a Pung
- A Chow is three tiles of the same suit and consecutive numbers
- A Pung is three identical tiles (same suit and same number)
- A Pair is two identical tiles (same suit and same number)

For example, here is a winning hand with $n = 9$, $s = 3$, $t = 4$, consisting in this case of two Chows, two Pungs, and one Pair:

Problem 697: Randomly Decaying Sequence

Link

Tags: Combinatorics, Game Theory, Probability, Sequences

Given a fixed real number c , define a random sequence $(X_n)_{n \geq 0}$ by the following random process:

- $X_0 = c$ (with probability 1).
- For $n > 0$, $X_n = U_n X_{n-1}$ where U_n is a real number chosen at random between zero and one, uniformly, and independently of all previous choices $(U_m)_{m < n}$.

If we desire there to be precisely a 25% probability that $X_{100} < 1$, then this can be arranged by fixing c such that $\log_{10} c \approx 46.27$.

Suppose now that c is set to a different value, so that there is precisely a 25% probability that $X_{10\,000\,000} < 1$.

Find $\log_{10} c$ and give your answer rounded to two places after the decimal point.

Problem 698: 123 Numbers

Link

Tags: Number Theory

We define 123-numbers as follows:

- 1 is the smallest 123-number.
- When written in base 10 the only digits that can be present are “1”, “2” and “3” and if present the number of times they each occur is also a 123-number.

So 2 is a 123-number, since it consists of one digit “2” and 1 is a 123-number. Therefore, 33 is a 123-number as well since it consists of two digits “3” and 2 is a 123-number.

On the other hand, 1111 is not a 123-number, since it contains 4 digits “1” and 4 is not a 123-number.

In ascending order, the first 123-numbers are:

1, 2, 3, 11, 12, 13, 21, 22, 23, 31, 32, 33, 111, 112, 113, 121, 122, 123, 131, ...

Let $F(n)$ be the n -th 123-number. For example $F(4) = 11$, $F(10) = 31$, $F(40) = 1112$, $F(1000) = 1223321$ and $F(6000) = 2333333333323$.

Find $F(111\,111\,111\,111\,222\,333)$. Give your answer modulo 123 123 123.

Problem 699: Trifle Numbers

Link

Tags: Number Theory

Let $\sigma(n)$ be the sum of all the divisors of the positive integer n , for example:

$$\sigma(10) = 1 + 2 + 5 + 10 = 18.$$

Define $T(N)$ to be the sum of all numbers $n \leq N$ such that when the fraction $\frac{\sigma(n)}{n}$ is written in its lowest form $\frac{a}{b}$, the denominator is a power of 3 i.e. $b = 3^k$, $k > 0$.

You are given $T(100) = 270$ and $T(10^6) = 26089287$.

Find $T(10^{14})$.

Problem 700: Eulercoin

Link

Tags: Probability, Sequences

Leonhard Euler was born on 15 April 1707.

Consider the sequence $1504170715041707n \bmod 4503599627370517$.

An element of this sequence is defined to be an Eulercoin if it is strictly smaller than all previously found Eulercoins.

For example, the first term is 1504170715041707 which is the first Eulercoin. The second term is 3008341430083414 which is greater than 1504170715041707 so is not an Eulercoin. However, the third term is 8912517754604 which is small enough to be a new Eulercoin.

The sum of the first 2 Eulercoins is therefore 1513083232796311.

Find the sum of all Eulercoins.

Problem 701: Random Connected Area

Link

Tags: Geometry, Graph Theory, Probability

Consider a rectangle made up of $W \times H$ square cells each with area 1. Each cell is independently coloured black with probability 0.5 otherwise white. Black cells sharing an edge are assumed to be connected. Consider the maximum area of connected cells.

Define $E(W, H)$ to be the expected value of this maximum area. For example, $E(2, 2) = 1.875$, as illustrated below.

Problem 702: Jumping Flea

Link

Tags: Geometry

A regular hexagon table of side length N is divided into equilateral triangles of side length 1. The picture below is an illustration of the case $N = 3$.

Problem 703: Circular Logic II

Link

Tags: Sequences

Given an integer n , $n \geq 3$, let $B = \{\text{false}, \text{true}\}$ and let B^n be the set of sequences of n values from B . The function f from B^n to B^n is defined by $f(b_1 \dots b_n) = c_1 \dots c_n$ where:

- $c_i = b_{i+1}$ for $1 \leq i < n$.
- $c_n = b_1$ AND $(b_2 \text{ XOR } b_3)$, where AND and XOR are the logical AND and exclusive OR operations.

Let $S(n)$ be the number of functions T from B^n to B such that for all x in B^n , $T(x)$ AND $T(f(x)) = \text{false}$. You are given that $S(3) = 35$ and $S(4) = 2118$.

Find $S(20)$. Give your answer modulo 1 001 001 011.

Problem 704: Factors of Two in Binomial Coefficients

Link

Tags: Number Theory

Define $g(n, m)$ to be the largest integer k such that 2^k divides $\binom{n}{m}$. For example, $\binom{12}{5} = 792 = 2^3 \cdot 3^2 \cdot 11$, hence $g(12, 5) = 3$. Then define $F(n) = \max\{g(n, m) : 0 \leq m \leq n\}$. $F(10) = 3$ and $F(100) = 6$.

Let $S(N) = \sum_{n=1}^N F(n)$. You are given that $S(100) = 389$ and $S(10^7) = 203222840$.

Find $S(10^{16})$.

Problem 705: Total Inversion Count of Divided Sequences

Link

Tags: Number Theory, Sequences

The inversion count of a sequence of digits is the smallest number of adjacent pairs that must be swapped to sort the sequence.

For example, 34214 has inversion count of 5: $34214 \rightarrow 32414 \rightarrow 23414 \rightarrow 23144 \rightarrow 21344 \rightarrow 12344$.

If each digit of a sequence is replaced by one of its divisors a divided sequence is obtained.

For example, the sequence 332 has 8 divided sequences: $\{332, 331, 312, 311, 132, 131, 112, 111\}$.

Define $G(N)$ to be the concatenation of all primes less than N , ignoring any zero digit.

For example, $G(20) = 235711131719$.

Define $F(N)$ to be the sum of the inversion count for all possible divided sequences from the master sequence $G(N)$.

You are given $F(20) = 3312$ and $F(50) = 338079744$.

Find $F(10^8)$. Give your answer modulo 1 000 000 007.

Problem 706: 3-Like Numbers

[Link](#)

Tags: Number Theory

For a positive integer n , define $f(n)$ to be the number of non-empty substrings of n that are divisible by 3. For example, the string “2573” has 10 non-empty substrings, three of which represent numbers that are divisible by 3, namely 57, 573 and 3. So $f(2573) = 3$.

If $f(n)$ is divisible by 3 then we say that n is 3-like.

Define $F(d)$ to be how many d digit numbers are 3-like. For example, $F(2) = 30$ and $F(6) = 290898$.

Find $F(10^5)$. Give your answer modulo 1 000 000 007.

Problem 707: Lights Out

[Link](#)

Tags: Combinatorics, Graph Theory

Consider a $w \times h$ grid. A cell is either ON or OFF. When a cell is selected, that cell and all cells connected to that cell by an edge are toggled on-off, off-on. See the diagram for the 3 cases of selecting a corner cell, an edge cell or central cell in a grid that has all cells on (white).

Problem 708: Twos Are All You Need

[Link](#)

Tags: Number Theory

A positive integer, n , is factorised into prime factors. We define $f(n)$ to be the product when each prime factor is replaced with 2. In addition we define $f(1) = 1$.

For example, $90 = 2 \times 3 \times 3 \times 5$, then replacing the primes, $2 \times 2 \times 2 \times 2 = 16$, hence $f(90) = 16$.

Let $S(N) = \sum_{n=1}^N f(n)$. You are given $S(10^8) = 9613563919$.

Find $S(10^{14})$.

Problem 709: Even Stevens

[Link](#)

Tags: Number Theory

Every day for the past n days Even Stevens brings home his groceries in a plastic bag. He stores these plastic bags in a cupboard. He either puts the plastic bag into the cupboard with the rest, or else he takes an even number of the existing bags (which may either be empty or previously filled with other bags themselves) and places these into the current bag.

After 4 days there are 5 possible packings and if the bags are numbered 1 (oldest), 2, 3, 4, they are:

- Four empty bags,
- 1 and 2 inside 3, 4 empty,
- 1 and 3 inside 4, 2 empty,
- 1 and 2 inside 4, 3 empty,
- 2 and 3 inside 4, 1 empty.

Note that 1, 2, 3 inside 4 is invalid because every bag must contain an even number of bags.

Define $f(n)$ to be the number of possible packings of n bags. Hence $f(4) = 5$. You are also given $f(8) = 1\,385$.

Find $f(24\,680)$ giving your answer modulo $1\,020\,202\,009$.

Problem 710: One Million Members

[Link](#)

Tags: Number Theory

On Sunday 5 April 2020 the Project Euler membership first exceeded one million members. We would like to present this problem to celebrate that milestone. Thank you to everyone for being a part of Project Euler.

The number 6 can be written as a palindromic sum in exactly eight different ways:

$(1, 1, 1, 1, 1, 1), (1, 1, 2, 1, 1), (1, 2, 2, 1), (1, 4, 1), (2, 1, 1, 2), (2, 2, 2), (3, 3), (6)$

We shall define a twopal to be a palindromic tuple having at least one element with a value of 2. It should also be noted that elements are not restricted to single digits. For example, $(3, 2, 13, 6, 13, 2, 3)$ is a valid twopal.

If we let $t(n)$ be the number of twopals whose elements sum to n , then it can be seen that $t(6) = 4$:

$(1, 1, 2, 1, 1), (1, 2, 2, 1), (2, 1, 1, 2), (2, 2, 2)$

Similarly, $t(20) = 824$.

In searching for the answer to the ultimate question of life, the universe, and everything, it can be verified that $t(42) = 1999923$, which happens to be the first value of $t(n)$ that exceeds one million.

However, your challenge to the “ultimatest” question of life, the universe, and everything is to find the least value of n_{42} such that $t(n)$ is divisible by one million.

Problem 711: Binary Blackboard

Link

Tags: Game Theory

Oscar and Eric play the following game. First, they agree on a positive integer n , and they begin by writing its binary representation on a blackboard. They then take turns, with Oscar going first, to write a number on the blackboard in binary representation, such that the sum of all written numbers does not exceed $2n$.

The game ends when there are no valid moves left. Oscar wins if the number of 1s on the blackboard is odd, and Eric wins if it is even.

Let $S(N)$ be the sum of all $n \leq 2^N$ for which Eric can guarantee winning, assuming optimal play.

For example, the first few values of n for which Eric can guarantee winning are 1, 3, 4, 7, 15, 16. Hence $S(4) = 46$.

You are also given that $S(12) = 54532$ and $S(1234) \equiv 690421393 \pmod{1\,000\,000\,007}$.

Find $S(12345678)$. Give your answer modulo $1\,000\,000\,007$.

Problem 712: Exponent Difference

[Link](#)

Tags: Number Theory

For any integer $n > 0$ and prime number p , define $\nu_p(n)$ as the greatest integer r such that p^r divides n .

Define

$$D(n, m) = \sum_{p \text{ prime}} |\nu_p(n) - \nu_p(m)|.$$

For example, $D(14, 24) = 4$.

Furthermore, define

$$S(N) = \sum_{1 \leq n, m \leq N} D(n, m).$$

You are given $S(10) = 210$ and $S(10^2) = 37018$.

Find $S(10^{12})$. Give your answer modulo 1 000 000 007.

Problem 713: Turán's Water Heating System

[Link](#)

Tags: Combinatorics, Sequences

Turan has the electrical water heating system outside his house in a shed. The electrical system uses two fuses in series, one in the house and one in the shed. (Nowadays old fashioned fuses are often replaced with reusable mini circuit breakers, but Turan's system still uses old fashioned fuses.) For the heating system to work both fuses must work.

Turan has N fuses. He knows that m of them are working and the rest are blown. However, he doesn't know which ones are blown. So he tries different combinations until the heating system turns on.

We denote by $T(N, m)$ the smallest number of tries required to ensure the heating system turns on.

$T(3, 2) = 3$ and $T(8, 4) = 7$.

Let $L(N)$ be the sum of all $T(N, m)$ for $2 \leq m \leq N$.

$L(10^3) = 3281346$.

Find $L(10^7)$.

Problem 714: Duodigits

Link

Tags: Number Theory

We call a natural number a duodigit if its decimal representation uses no more than two different digits. For example, 12, 110 and 33333 are duodigits, while 102 is not.

It can be shown that every natural number has duodigit multiples. Let $d(n)$ be the smallest (positive) multiple of the number n that happens to be a duodigit. For example, $d(12) = 12$, $d(102) = 1122$, $d(103) = 515$, $d(290) = 11011010$ and $d(317) = 211122$.

Let $D(k) = \sum_{n=1}^k d(n)$. You are given $D(110) = 11\,047$, $D(150) = 53\,312$ and $D(500) = 29\,570\,988$.

Find $D(50\,000)$. Give your answer in scientific notation rounded to 13 significant digits (12 after the decimal point). If, for example, we had asked for $D(500)$ instead, the answer format would have been 2.957098800000e7.

Problem 715: Sextuplet Norms

Link

Tags: Number Theory

Let $f(n)$ be the number of 6-tuples $(x_1, x_2, x_3, x_4, x_5, x_6)$ such that:

- All x_i are integers with $0 \leq x_i < n$
- $\gcd(x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2, n^2) = 1$

Let $G(n) = \sum_{k=1}^n \frac{f(k)}{k^2 \varphi(k)}$

where $\varphi(n)$ is Euler's totient function.

For example, $G(10) = 3053$ and $G(10^5) \equiv 157612967 \pmod{1\,000\,000\,007}$.

Find $G(10^{12}) \pmod{1\,000\,000\,007}$.

Problem 716: Grid Graphs

Link

Tags: Combinatorics, Game Theory, Graph Theory

Consider a directed graph made from an orthogonal lattice of $H \times W$ nodes. The edges are the horizontal and vertical connections between adjacent nodes. W vertical directed lines are drawn and all the edges on these lines inherit that direction. Similarly, H horizontal directed lines are drawn and all the edges on these lines inherit that direction.

Two nodes, A and B in a directed graph, are strongly connected if there is both a path, along the directed edges, from A to B as well as from B to A . Note that every node is strongly connected to itself.

A strongly connected component in a directed graph is a non-empty set M of nodes satisfying the following two properties:

- All nodes in M are strongly connected to each other.
- M is maximal, in the sense that no node in M is strongly connected to any node outside of M .

There are $2^H \times 2^W$ ways of drawing the directed lines. Each way gives a directed graph $\mathcal{G}$. We define $S(\mathcal{G})$ to be the number of strongly connected components in $\mathcal{G}$.

The illustration below shows a directed graph with $H = 3$ and $W = 4$ that consists of four different strongly connected components (indicated by the different colours).

Problem 717: Summation of a Modular Formula

[Link](#)

Tags: Number Theory

For an odd prime p , define $f(p) = \left\lfloor \frac{2^{(2^p)}}{p} \right\rfloor \bmod 2^p$

For example, when $p = 3$, $\lfloor 2^8/3 \rfloor = 85 \equiv 5 \pmod{8}$ and so $f(3) = 5$.

Further define $g(p) = f(p) \bmod p$. You are given $g(31) = 17$.

Now define $G(N)$ to be the summation of $g(p)$ for all odd primes less than N .

You are given $G(100) = 474$ and $G(10^4) = 2819236$.

Find $G(10^7)$.

Problem 718: Unreachable Numbers

[Link](#)

Tags: Algebra

Consider the equation $17^p a + 19^p b + 23^p c = n$ where a, b, c and p are positive integers, i.e. $a, b, c, p > 0$.

For a given p there are some values of $n > 0$ for which the equation cannot be solved. We call these unreachable values.

Define $G(p)$ to be the sum of all unreachable values of n for the given value of p . For example $G(1) = 8253$ and $G(2) = 60258000$.

Find $G(6)$. Give your answer modulo 1 000 000 007.

Problem 719: Number Splitting

[Link](#)

Tags: Number Theory

We define an S -number to be a natural number, n , that is a perfect square and its square root can be obtained by splitting the decimal representation of n into 2 or more numbers then adding the numbers.

For example, 81 is an S -number because $\sqrt{81} = 8 + 1$.

6724 is an S -number: $\sqrt{6724} = 6 + 72 + 4$.

8281 is an S -number: $\sqrt{8281} = 8 + 2 + 81 = 82 + 8 + 1$.

9801 is an S -number: $\sqrt{9801} = 98 + 0 + 1$.

Further we define $T(N)$ to be the sum of all S numbers $n \leq N$. You are given $T(10^4) = 41333$.

Find $T(10^{12})$.

Problem 720: Unpredictable Permutations

[Link](#)

Tags: Combinatorics, Graph Theory

Consider all permutations of $\{1, 2, \dots, N\}$, listed in lexicographic order. For example, for $N = 4$, the list starts as follows:

$(1, 2, 3, 4) \ (1, 2, 4, 3) \ (1, 3, 2, 4) \ (1, 3, 4, 2) \ (1, 4, 2, 3) \ (1, 4, 3, 2) \ (2, 1, 3, 4) \ :$

Problem 721: High Powers of Irrational Numbers

Link

Tags: Number Theory

Given is the function $f(a, n) = \lfloor (\lceil \sqrt{a} \rceil + \sqrt{a})^n \rfloor$.

$\lfloor \cdot \rfloor$ denotes the floor function and $\lceil \cdot \rceil$ denotes the ceiling function.

$f(5, 2) = 27$ and $f(5, 5) = 3935$.

$$G(n) = \sum_{a=1}^n f(a, a^2).$$

$\$G(1000) \bmod 999,999,937 = 163861845.\$$

Find $G(5\,000\,000)$. Give your answer modulo $999\,999\,937$.

Problem 722: Slowly Converging Series

Link

Tags: Number Theory, Sequences

For a non-negative integer k , define

$$E_k(q) = \sum_{n=1}^{\infty} \sigma_k(n) q^n$$

where $\sigma_k(n) = \sum_{d|n} d^k$ is the sum of the k -th powers of the positive divisors of n .

It can be shown that, for every k , the series $E_k(q)$ converges for any $0 < q < 1$.

For example,

$$E_1(1 - \frac{1}{2^4}) = 3.872155809243\text{e}2$$

$$E_3(1 - \frac{1}{2^8}) = 2.767385314772\text{e}10$$

$$E_7(1 - \frac{1}{2^{15}}) = 6.725803486744\text{e}39$$

All the above values are given in scientific notation rounded to twelve digits after the decimal point.

Find the value of $E_{15}(1 - \frac{1}{2^{25}})$.

Give the answer in scientific notation rounded to twelve digits after the decimal point.

Problem 723: Pythagorean Quadrilaterals

[Link](#)

Tags: Algebra, Combinatorics, Geometry

A pythagorean triangle with catheti a and b and hypotenuse c is characterized by the well-known equation $a^2 + b^2 = c^2$. However, this can also be formulated differently:

When inscribed into a circle with radius r , a triangle with sides a , b and c is pythagorean, if and only if $a^2 + b^2 + c^2 = 8r^2$.

Analogously, we call a quadrilateral $ABCD$ with sides a , b , c and d , inscribed in a circle with radius r , a pythagorean quadrilateral, if $a^2 + b^2 + c^2 + d^2 = 8r^2$.

We further call a pythagorean quadrilateral a pythagorean lattice grid quadrilateral, if all four vertices are lattice grid points with the same distance r from the origin O (which then happens to be the centre of the circumcircle).

Let $f(r)$ be the number of different pythagorean lattice grid quadrilaterals for which the radius of the circumcircle is r . For example $f(1) = 1$, $f(\sqrt{2}) = 1$, $f(\sqrt{5}) = 38$ and $f(5) = 167$.

Two of the pythagorean lattice grid quadrilaterals with $r = \sqrt{5}$ are illustrated below:

Problem 724: Drone Delivery

[Link](#)

Tags: Geometry, Probability

A depot uses n drones to disperse packages containing essential supplies along a long straight road.

Initially all drones are stationary, loaded with a supply package.

Every second, the depot selects a drone at random and sends it this instruction:

- If you are stationary, start moving at one centimetre per second along the road.
- If you are moving, increase your speed by one centimetre per second along the road without changing direction.

The road is wide enough that drones can overtake one another without risk of collision.

Eventually, there will only be one drone left at the depot waiting to receive its first instruction. As soon as that drone has flown one centimetre along the road, all drones drop their packages and return to the depot.

Let $E(n)$ be the expected distance in centimetres from the depot that the supply packages land.

For example, $E(2) = \frac{7}{2}$, $E(5) = \frac{12019}{720}$, and $E(100) \approx 1427.193470$.

Find $E(10^8)$. Give your answer rounded to the nearest integer.

Problem 725: Digit Sum Numbers

[Link](#)

Tags: Number Theory

A number where one digit is the sum of the other digits is called a digit sum number or DS-number for short. For example, 352, 3003 and 32812 are DS-numbers.

We define $S(n)$ to be the sum of all DS-numbers of n digits or less.

You are given $S(3) = 63270$ and $S(7) = 85499991450$.

Find $S(2020)$. Give your answer modulo 10^{16} .

Problem 726: Falling Bottles

[Link](#)

Tags: Game Theory

Consider a stack of bottles of wine. There are n layers in the stack with the top layer containing only one bottle and the bottom layer containing n bottles. For $n = 4$ the stack looks like the picture below.

Problem 727: Triangle of Circular Arcs

[Link](#)

Tags: Geometry

Let r_a , r_b and r_c be the radii of three circles that are mutually and externally tangent to each other. The three circles then form a triangle of circular arcs between their tangency points as shown for the three blue circles in the picture below.

Problem 728: Circle of Coins

[Link](#)

Tags: Combinatorics, Game Theory, Geometry, Probability, Sequences

Consider n coins arranged in a circle where each coin shows heads or tails. A move consists of turning over k consecutive coins: tail-head or head-tail. Using a sequence of these moves the objective is to get all the coins showing heads.

Consider the example, shown below, where $n = 8$ and $k = 3$ and the initial state is one coin showing tails (black). The example shows a solution for this state.

Problem 729: Range of Periodic Sequence

[Link](#)

Tags: Game Theory, Sequences

Consider the sequence of real numbers a_n defined by the starting value a_0 and the recurrence $a_{n+1} = a_n - \frac{1}{a_n}$ for any $n \geq 0$.

For some starting values a_0 the sequence will be periodic. For example, $a_0 = \sqrt{\frac{1}{2}}$ yields the sequence: $\sqrt{\frac{1}{2}}, -\sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}}, \dots$

We are interested in the range of such a periodic sequence which is the difference between the maximum and minimum of the sequence. For example, the range of the sequence above would be $\sqrt{\frac{1}{2}} - (-\sqrt{\frac{1}{2}}) = \sqrt{2}$.

Let $S(P)$ be the sum of the ranges of all such periodic sequences with a period not exceeding P .

For example, $S(2) = 2\sqrt{2} \approx 2.8284$, being the sum of the ranges of the two sequences starting with $a_0 = \sqrt{\frac{1}{2}}$ and $a_0 = -\sqrt{\frac{1}{2}}$.

You are given $S(3) \approx 14.6461$ and $S(5) \approx 124.1056$.

Find $S(25)$, rounded to 4 decimal places.

Problem 730: Shifted Pythagorean Triples

[Link](#)

Tags: Geometry, Number Theory

For a non-negative integer k , the triple (p, q, r) of positive integers is called a k -shifted Pythagorean triple if

$$p^2 + q^2 + k = r^2$$

(p, q, r) is said to be primitive if $\gcd(p, q, r) = 1$.

Let $P_k(n)$ be the number of primitive k -shifted Pythagorean triples such that $1 \leq p \leq q \leq r$ and $p + q + r \leq n$. For example, $P_0(10^4) = 703$ and $P_{20}(10^4) = 1979$.

Define

$$S(m, n) = \sum_{k=0}^m P_k(n).$$

You are given that $S(10, 10^4) = 10956$.

Find $S(10^2, 10^8)$.

Problem 731: A Stoneham Number

Link

Tags: Number Theory

$$A = \sum_{i=1}^{\infty} \frac{1}{3^i 10^{3^i}}$$

Define $A(n)$ to be the 10 decimal digits from the n th digit onward. For example, $A(100) = 4938271604$ and $A(10^8) = 2584642393$.

Find $A(10^{16})$.

Problem 732: Standing on the Shoulders of Trolls

Link

Tags: Geometry

N trolls are in a hole that is D_N cm deep. The n -th troll is characterized by:

- the distance from his feet to his shoulders in cm, h_n
- the length of his arms in cm, l_n
- his IQ (Irascibility Quotient), q_n .

Trolls can pile up on top of each other, with each troll standing on the shoulders of the one below him. A troll can climb out of the hole and escape if his hands can reach to the surface. Once a troll escapes he cannot participate any further in the escaping effort.

The trolls execute an optimal strategy for maximizing the total IQ of the escaping trolls, defined as $Q(N)$.

Let

$$r_n = [(5^n \bmod (10^9 + 7)) \bmod 101] + 50$$

$$h_n = r_{3n}$$

$$l_n = r_{3n+1}$$

$$q_n = r_{3n+2}$$

$$D_N = \frac{1}{\sqrt{2}} \sum_{n=0}^{N-1} h_n.$$

For example, the first troll ($n = 0$) is 51cm tall to his shoulders, has 55cm long arms, and has an IQ of 75.

You are given that $Q(5) = 401$ and $Q(15) = 941$.

Find $Q(1000)$.

Problem 733: Ascending Subsequences

Link

Tags: Sequences

Let a_i be the sequence defined by $a_i = 153^i \bmod 10\,000\,019$ for $i \geq 1$.

The first terms of a_i are: 153, 23409, 3581577, 7980255, 976697, 9434375, ...

Consider the subsequences consisting of 4 terms in ascending order. For the part of the sequence shown above, these are:

153, 23409, 3581577, 7980255

153, 23409, 3581577, 9434375

153, 23409, 7980255, 9434375

153, 23409, 976697, 9434375

153, 3581577, 7980255, 9434375 and

23409, 3581577, 7980255, 9434375.

Define $S(n)$ to be the sum of the terms for all such subsequences within the first n terms of a_i . Thus $S(6) = 94513710$.

You are given that $S(100) = 4465488724217$.

Find $S(10^6)$ modulo 1 000 000 007.

Problem 734: A Bit of Prime

Link

Tags: Number Theory

The logical-OR of two bits is 0 if both bits are 0, otherwise it is 1.

The bitwise-OR of two positive integers performs a logical-OR operation on each pair of corresponding bits in the binary expansion of its inputs.

For example, the bitwise-OR of 10 and 6 is 14 because $10 = 1010_2$, $6 = 0110_2$ and $14 = 1110_2$.

Let $T(n, k)$ be the number of k -tuples $(x_1, x_2, \dots, x_k)$ such that

- every x_i is a prime $\leq n$
- the bitwise-OR of the tuple is a prime $\leq n$

For example, $T(5, 2) = 5$. The five 2-tuples are $(2, 2)$, $(2, 3)$, $(3, 2)$, $(3, 3)$ and $(5, 5)$.

You are given $T(100, 3) = 3355$ and $T(1000, 10) \equiv 2071632 \pmod{1\,000\,000\,007}$.

Find $T(10^6, 999983)$. Give your answer modulo 1 000 000 007.

Problem 735: Divisors of $2n^2$

Link

Tags: Number Theory

Let $f(n)$ be the number of divisors of $2n^2$ that are no greater than n . For example, $f(15) = 8$ because there are 8 such divisors: 1, 2, 3, 5, 6, 9, 10, 15. Note that 18 is also a divisor of 2×15^2 but it is not counted because it is greater than 15.

Let $F(N) = \sum_{n=1}^N f(n)$. You are given $F(15) = 63$, and $F(1000) = 15066$.

Find $F(10^{12})$.

Problem 736: Paths to Equality

Link

Tags: Combinatorics, Sequences

Define two functions on lattice points:

$$r(x, y) = (x + 1, 2y) \quad s(x, y) = (2x, y + 1)$$

A path to equality of length n for a pair (a, b) is a sequence $((a_1, b_1), (a_2, b_2), \dots, (a_n, b_n))$, where:

- $(a_1, b_1) = (a, b)$
- $(a_k, b_k) = r(a_{k-1}, b_{k-1})$ or $(a_k, b_k) = s(a_{k-1}, b_{k-1})$ for $k > 1$
- $a_k \neq b_k$ for $k < n$
- $a_n = b_n$

$a_n = b_n$ is called the final value.

For example,

$$(45, 90) \xrightarrow{r} (46, 180) \xrightarrow{s} (92, 181) \xrightarrow{s} (184, 182) \xrightarrow{s} (368, 183) \xrightarrow{s} (736, 184) \xrightarrow{r} (737, 368) \xrightarrow{s} (1474, 369) \xrightarrow{r} (1475, 738) \xrightarrow{r} (1476, 1476)$$

This is a path to equality for $(45, 90)$ and is of length 10 with final value 1476. There is no path to equality of $(45, 90)$ with smaller length.

Find the unique path to equality for $(45, 90)$ with smallest odd length. Enter the final value as your answer.

Problem 737: Coin Loops

Link

Tags: Game Theory, Probability

A game is played with many identical, round coins on a flat table.

Consider a line perpendicular to the table.

The first coin is placed on the table touching the line.

Then, one by one, the coins are placed horizontally on top of the previous coin and touching the line.

The complete stack of coins must be balanced after every placement.

The diagram below [not to scale] shows a possible placement of 8 coins where point P represents the line.

Problem 738: Counting Ordered Factorisations

Link

Tags: Number Theory

Define $d(n, k)$ to be the number of ways to write n as a product of k ordered integers

$$n = x_1 \times x_2 \times x_3 \times \dots \times x_k \quad 1 \leq x_1 \leq x_2 \leq \dots \leq x_k$$

Further define $D(N, K)$ to be the sum of $d(n, k)$ for $1 \leq n \leq N$ and $1 \leq k \leq K$.

You are given that $D(10, 10) = 153$ and $D(100, 100) = 35384$.

Find $D(10^{10}, 10^{10})$ giving your answer modulo 1 000 000 007.

Problem 739: Summation of Summations

Link

Tags: Sequences

Take a sequence of length n . Discard the first term then make a sequence of the partial summations. Continue to do this over and over until we are left with a single term. We define this to be $f(n)$.

Consider the example where we start with a sequence of length 8:

\$

1	1	1	1	1	1	1	1
	1	2	3	4	5	6	7
		2	5	9	14	20	27
			5	14	28	48	75
				14	42	90	165
					42	132	297
						132	429
							429

\$

Then the final number is 429, so $f(8) = 429$.

For this problem we start with the sequence 1, 3, 4, 7, 11, 18, 29, 47, ...

This is the Lucas sequence where two terms are added to get the next term.

Applying the same process as above we get $f(8) = 2663$.

You are also given $f(20) = 742296999$ modulo 1 000 000 007

Find $f(10^8)$. Give your answer modulo 1 000 000 007.

Problem 740: Secret Santa

[Link](#)

Tags: Probability

Secret Santa is a process that allows n people to give each other presents, so that each person gives a single present and receives a single present. At the beginning each of the n people write their name on a slip of paper and put the slip into a hat. Each person takes a random slip from the hat. If the slip has their name they draw another random slip from the hat and then put the slip with their name back into the hat. At the end everyone buys a Christmas present for the person whose name is on the slip they are holding. This process will fail if the last person draws their own name.

In this variation each of the n people gives and receives two presents. At the beginning each of the n people writes their name on two slips of paper and puts the slips into a hat (there will be $2n$ slips of paper in the hat). As before each person takes from the hat a random slip that does not contain their own name. Then the same person repeats this process thus ending up with two slips, neither of which contains that person's own name. Then the next person draws two slips in the same way, and so on. The process will fail if the last person gets at least one slip with their own name.

Define $q(n)$ to be the probability of this happening. You are given $q(3) = 0.3611111111$ and $q(5) = 0.2476095994$ both rounded to 10 decimal places.

Find $q(100)$ rounded to 10 decimal places.

Problem 741: Binary Grid Colouring

[Link](#)

Tags: Combinatorics

Let $f(n)$ be the number of ways an $n \times n$ square grid can be coloured, each cell either black or white, such that each row and each column contains exactly two black cells.

For example, $f(4) = 90$, $f(7) = 3110940$ and $f(8) = 187530840$.

Let $g(n)$ be the number of colourings in $f(n)$ that are unique up to rotations and reflections.

You are given $g(4) = 20$, $g(7) = 390816$ and $g(8) = 23462347$ giving $g(7) + g(8) = 23853163$.

Find $g(7^7) + g(8^8)$. Give your answer modulo 1 000 000 007.

Problem 742: Minimum Area of a Convex Grid Polygon

Link

Tags: Combinatorics, Geometry

A symmetrical convex grid polygon is a polygon such that:

- All its vertices have integer coordinates.
- All its internal angles are strictly smaller than 180° .
- It has both horizontal and vertical symmetry.

For example, the left polygon is a convex grid polygon which has neither horizontal nor vertical symmetry, while the right one is a valid symmetrical convex grid polygon with six vertices:

Problem 743: Window into a Matrix

Link

Tags: Algebra, Game Theory

A window into a matrix is a contiguous sub matrix.

Consider a $2 \times n$ matrix where every entry is either 0 or 1.

Let $A(k, n)$ be the total number of these matrices such that the sum of the entries in every $2 \times k$ window is k .

You are given that $A(3, 9) = 560$ and $A(4, 20) = 1060870$.

Find $A(10^8, 10^{16})$. Give your answer modulo 1 000 000 007.

Problem 744: What? Where? When?

Link

Tags: Game Theory, Probability

“What? Where? When?” is a TV game show in which a team of experts attempt to answer questions. The following is a simplified version of the game.

It begins with $2n + 1$ envelopes. $2n$ of them contain a question and one contains a RED card.

In each round one of the remaining envelopes is randomly chosen. If the envelope contains the RED card the game ends. If the envelope contains a question

the expert gives their answer. If their answer is correct they earn one point, otherwise the viewers earn one point. The game ends normally when either the expert obtains n points or the viewers obtain n points.

Assuming that the expert provides the correct answer with a fixed probability p , let $f(n, p)$ be the probability that the game ends normally (i.e. RED card never turns up).

You are given (rounded to 10 decimal places) that

$$f(6, \frac{1}{2}) = 0.2851562500,$$

$$f(10, \frac{3}{7}) = 0.2330040743,$$

$$f(10^4, 0.3) = 0.2857499982.$$

Find $f(10^{11}, 0.4999)$. Give your answer rounded to 10 places behind the decimal point.

Problem 745: Sum of Squares II

[Link](#)

Tags: Number Theory

For a positive integer, n , define $g(n)$ to be the maximum perfect square that divides n .

For example, $g(18) = 9$, $g(19) = 1$.

Also define

$$S(N) = \sum_{n=1}^N g(n)$$

For example, $S(10) = 24$ and $S(100) = 767$.

Find $S(10^{14})$. Give your answer modulo 1 000 000 007.

Problem 746: A Messy Dinner

[Link](#)

Tags: Number Theory

n families, each with four members, a father, a mother, a son and a daughter, were invited to a restaurant. They were all seated at a large circular table with $4n$ seats such that men and women alternate.

Let $M(n)$ be the number of ways the families can be seated such that none of the families were seated together. A family is considered to be seated together only when all the members of a family sit next to each other.

For example, $M(1) = 0$, $M(2) = 896$, $M(3) = 890880$ and $M(10) \equiv 170717180 \pmod{1\,000\,000\,007}$.

Let $S(n) = \sum_{k=2}^n M(k)$.

For example, $S(10) \equiv 399291975 \pmod{1\,000\,000\,007}$.

Find $S(2021)$. Give your answer modulo $1\,000\,000\,007$.

Problem 747: Triangular Pizza

Link

Tags: Combinatorics, Geometry

Mamma Triangolo baked a triangular pizza. She wants to cut the pizza into n pieces. She first chooses a point P in the interior (not boundary) of the triangle pizza, and then performs n cuts, which all start from P and extend straight to the boundary of the pizza so that the n pieces are all triangles and all have the same area.

Let $\psi(n)$ be the number of different ways for Mamma Triangolo to cut the pizza, subject to the constraints.

For example, $\psi(3) = 7$.

Problem 748: Upside Down Diophantine Equation

Link

Tags: Algebra, Geometry, Number Theory

Upside Down is a modification of the famous Pythagorean equation:

$$\frac{1}{x^2} + \frac{1}{y^2} = \frac{13}{z^2}.$$

A solution (x, y, z) to this equation with x, y and z positive integers is a primitive solution if $\gcd(x, y, z) = 1$.

Let $S(N)$ be the sum of $x + y + z$ over primitive Upside Down solutions such that $1 \leq x, y, z \leq N$ and $x \leq y$.

For $N = 100$ the primitive solutions are $(2, 3, 6)$ and $(5, 90, 18)$, thus $S(10^2) = 124$.

It can be checked that $S(10^3) = 1470$ and $S(10^5) = 2340084$.

Find $S(10^{16})$ and give the last 9 digits as your answer.

Problem 749: Near Power Sums

[Link](#)

Tags: Number Theory

A positive integer, n , is a near power sum if there exists a positive integer, k , such that the sum of the k th powers of the digits in its decimal representation is equal to either $n + 1$ or $n - 1$. For example 35 is a near power sum number because $3^2 + 5^2 = 34$.

Define $S(d)$ to be the sum of all near power sum numbers of d digits or less. Then $S(2) = 110$ and $S(6) = 2562701$.

Find $S(16)$.

Problem 750: Optimal Card Stacking

[Link](#)

Tags: Game Theory, Geometry, Sequences

Card Stacking is a game on a computer starting with an array of N cards labelled $1, 2, \dots, N$. A stack of cards can be moved by dragging horizontally with the mouse to another stack but only when the resulting stack is in sequence. The goal of the game is to combine the cards into a single stack using minimal total drag distance.

Problem 751: Concatenation Coincidence

[Link](#)

Tags: Game Theory, Sequences

A non-decreasing sequence of integers a_n can be generated from any positive real value θ by the following procedure:

$$\begin{aligned} b_1 &= \theta \\ b_n &= \lfloor b_{n-1} \rfloor (b_{n-1} - \lfloor b_{n-1} \rfloor + 1) \quad \forall n \geq 2 \\ a_n &= \lfloor b_n \rfloor \end{aligned}$$

Where $\lfloor \cdot \rfloor$ is the floor function.

For example, $\theta = 2.956938891377988\dots$ generates the Fibonacci sequence: 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

The concatenation of a sequence of positive integers a_n is a real value denoted τ constructed by concatenating the elements of the sequence after the decimal point, starting at a_1 : $a_1.a_2a_3a_4\dots$

For example, the Fibonacci sequence constructed from $\theta = 2.956938891377988\dots$ yields the concatenation $\tau = 2.3581321345589\dots$ Clearly, $\tau \neq \theta$ for this value of θ .

Find the only value of θ for which the generated sequence starts at $a_1 = 2$ and the concatenation of the generated sequence equals the original value: $\tau = \theta$. Give your answer rounded to 24 places after the decimal point.

Problem 752: Powers of $1 + \sqrt{7}$

Link

Tags: Number Theory

When $(1 + \sqrt{7})$ is raised to an integral power, n , we always get a number of the form $(a + b\sqrt{7})$.

We write $(1 + \sqrt{7})^n = \alpha(n) + \beta(n)\sqrt{7}$.

For a given number x we define $g(x)$ to be the smallest positive integer n such that:

$$\begin{aligned}\alpha(n) &\equiv 1 \pmod{x} & \text{and} \\ \beta(n) &\equiv 0 \pmod{x}\end{aligned}$$

and $g(x) = 0$ if there is no such value of n . For example, $g(3) = 0$, $g(5) = 12$.

Further define

$$G(N) = \sum_{x=2}^N g(x)$$

You are given $G(10^2) = 28891$ and $G(10^3) = 13131583$.

Find $G(10^6)$.

Problem 753: Fermat Equation

Link

Tags: Algebra, Number Theory

Fermat's Last Theorem states that no three positive integers a, b, c satisfy the equation

$$a^n + b^n = c^n$$

for any integer value of n greater than 2.

For this problem we are only considering the case $n = 3$. For certain values of p , it is possible to solve the congruence equation:

$$a^3 + b^3 \equiv c^3 \pmod{p}$$

For a prime p , we define $F(p)$ as the number of integer solutions to this equation for $1 \leq a, b, c < p$.

You are given $F(5) = 12$ and $F(7) = 0$.

Find the sum of $F(p)$ over all primes p less than 6 000 000.

Problem 754: Product of Gauss Factorials

Link

Tags: Number Theory

The Gauss Factorial of a number n is defined as the product of all positive numbers $\leq n$ that are relatively prime to n . For example $g(10) = 1 \times 3 \times 7 \times 9 = 189$.

Also we define

$$G(n) = \prod_{i=1}^n g(i)$$

You are given $G(10) = 23044331520000$.

Find $G(10^8)$. Give your answer modulo 1 000 000 007.

Problem 755: Not Zeckendorf

Link

Tags: Sequences

Consider the Fibonacci sequence $\{1, 2, 3, 5, 8, 13, 21, \dots\}$.

We let $f(n)$ be the number of ways of representing an integer $n \geq 0$ as the sum of different Fibonacci numbers.

For example, $16 = 3 + 13 = 1 + 2 + 13 = 3 + 5 + 8 = 1 + 2 + 5 + 8$ and hence $f(16) = 4$. By convention $f(0) = 1$.

Further we define

$$S(n) = \sum_{k=0}^n f(k).$$

You are given $S(100) = 415$ and $S(10^4) = 312807$.

Find $S(10^{13})$.

Problem 756: Approximating a Sum

Link

Tags: Combinatorics, Number Theory, Probability

Consider a function $f(k)$ defined for all positive integers $k > 0$. Let S be the sum of the first n values of f . That is,

$$S = f(1) + f(2) + f(3) + \cdots + f(n) = \sum_{k=1}^n f(k).$$

In this problem, we employ randomness to approximate this sum. That is, we choose a random, uniformly distributed, m -tuple of positive integers $(X_1, X_2, X_3, \dots, X_m)$ such that $0 = X_0 X_1 X_2 \cdots X_m \leq n$ and calculate a modified sum S^* as follows.

$$S^* = \sum_{i=1}^m f(X_i)(X_i - X_{i-1})$$

We now define the error of this approximation to be $\Delta = S - S^*$.

Let $\mathbb{E}(\Delta|f(k), n, m)$ be the expected value of the error given the function $f(k)$, the number of terms n in the sum and the length of random sample m .

For example, $\mathbb{E}(\Delta|k, 100, 50) = 2525/1326 \approx 1.904223$ and $\mathbb{E}(\Delta|\varphi(k), 10^4, 10^2) \approx 5842.849907$, where $\varphi(k)$ is Euler's totient function.

Find $\mathbb{E}(\Delta|\varphi(k), 12345678, 12345)$ rounded to six places after the decimal point.

Problem 757: Stealthy Numbers

Link

Tags: Number Theory

A positive integer N is stealthy, if there exist positive integers a, b, c, d such that $ab = cd = N$ and $a + b = c + d + 1$.

For example, $36 = 4 \times 9 = 6 \times 6$ is stealthy.

You are also given that there are 2851 stealthy numbers not exceeding 10^6 .

How many stealthy numbers are there that don't exceed 10^{14} ?

Problem 758: Buckets of Water

Link

Tags: Number Theory

There are 3 buckets labelled S (small) of 3 litres, M (medium) of 5 litres and L (large) of 8 litres.

Initially S and M are full of water and L is empty. By pouring water between the buckets exactly one litre of water can be measured.

Since there is no other way to measure, once a pouring starts it cannot stop until either the source bucket is empty or the destination bucket is full.

At least four pourings are needed to get one litre:

$$(3, 5, 0) \xrightarrow{M \rightarrow L} (3, 0, 5) \xrightarrow{S \rightarrow M} (0, 3, 5) \xrightarrow{L \rightarrow S} (3, 3, 2) \xrightarrow{S \rightarrow M} (1, 5, 2)$$

Problem 759: A Squared Recurrence Relation

Link

Tags: Number Theory

The function f is defined for all positive integers as follows:

$$\begin{aligned} f(1) &= 1 \\ f(2n) &= 2f(n) \\ f(2n+1) &= 2n+1 + 2f(n) + \frac{1}{n}f(n) \end{aligned}$$

It can be proven that $f(n)$ is integer for all values of n .

The function $S(n)$ is defined as $S(n) = \sum_{i=1}^n f(i)^2$.

For example, $S(10) = 1530$ and $S(10^2) = 4798445$.

Find $S(10^{16})$. Give your answer modulo 1 000 000 007.

Problem 760: Sum over Bitwise Operators

Link

Tags: Graph Theory

Define

$$g(m, n) = (m \oplus n) + (m \vee n) + (m \wedge n)$$

where $\oplus, \vee, \wedge$ are the bitwise XOR, OR and AND operator respectively.

Also set

$$G(N) = \sum_{n=0}^N \sum_{k=0}^n g(k, n-k)$$

For example, $G(10) = 754$ and $G(10^2) = 583766$.

Find $G(10^{18})$. Give your answer modulo 1 000 000 007.

Problem 761: Runner and Swimmer

Link

Tags: Game Theory, Geometry, Graph Theory, Number Theory

Two friends, a runner and a swimmer, are playing a sporting game: The swimmer is swimming within a circular pool while the runner moves along the pool edge. While the runner tries to catch the swimmer at the very moment that the swimmer leaves the pool, the swimmer tries to reach the edge before the runner arrives there. They start the game with the swimmer located in the middle of the pool, while the runner is located anywhere at the edge of the pool.

We assume that the swimmer can move with any velocity up to 1 in any direction and the runner can move with any velocity up to v in either direction around the edge of the pool. Moreover we assume that both players can react immediately to any change of movement of their opponent.

Assuming optimal strategy of both players, it can be shown that the swimmer can always win by escaping the pool at some point at the edge before the runner gets there, if v is less than the critical speed $V_{\text{Circle}} \approx 4.60333885$ and can never win if $v \geq V_{\text{Circle}}$.

Now the two players play the game in a perfectly square pool. Again the swimmer starts in the middle of the pool, while the runner starts at the midpoint of one of the edges of the pool. It can be shown that the critical maximal speed of the runner below which the swimmer can always escape and above which the runner can always catch the swimmer when trying to leave the pool is $V_{\text{Square}} \approx 5.78859314$.

At last, both players decide to play the game in a pool in the form of regular hexagon. Giving the same conditions as above, with the swimmer starting in the middle of the pool and the runner at the midpoint of one of the edges of the pool, find the critical maximal speed V_{Hexagon} of the runner, below which the swimmer can always escape and above which the runner can always catch the swimmer. Give your answer rounded to 8 digits after the decimal point.

Problem 762: Amoebas in a 2D Grid

[Link](#)

Tags: Combinatorics, Game Theory

Consider a two dimensional grid of squares. The grid has 4 rows but infinitely many columns.

An amoeba in square (x, y) can divide itself into two amoebas to occupy the squares $(x + 1, y)$ and $(x + 1, (y + 1) \bmod 4)$, provided these squares are empty.

The following diagrams show two cases of an amoeba placed in square A of each grid. When it divides, it is replaced with two amoebas, one at each of the squares marked with B:

Problem 763: Amoebas in a 3D Grid

[Link](#)

Tags: Combinatorics, Number Theory

Consider a three dimensional grid of cubes. An amoeba in cube (x, y, z) can divide itself into three amoebas to occupy the cubes $(x + 1, y, z)$, $(x, y + 1, z)$ and $(x, y, z + 1)$, provided these cubes are empty.

Originally there is only one amoeba in the cube $(0, 0, 0)$. After N divisions there will be $2N + 1$ amoebas arranged in the grid. An arrangement may be reached in several different ways but it is only counted once. Let $D(N)$ be the number of different possible arrangements after N divisions.

For example, $D(2) = 3$, $D(10) = 44499$, $D(20) = 9204559704$ and the last nine digits of $D(100)$ are 780166455.

Find $D(10\,000)$, enter the last nine digits as your answer.

Problem 764: Asymmetric Diophantine Equation

[Link](#)

Tags: Algebra, Game Theory, Number Theory

Consider the following Diophantine equation:

$$16x^2 + y^4 = z^2$$

where x , y and z are positive integers.

Let $S(N) = \sum (x + y + z)$ where the sum is over all solutions (x, y, z) such that $1 \leq x, y, z \leq N$ and $\gcd(x, y, z) = 1$.

For $N = 100$, there are only two such solutions: $(3, 4, 20)$ and $(10, 3, 41)$. So $S(10^2) = 81$.

You are also given that $S(10^4) = 112851$ (with 26 solutions), and $S(10^7) \equiv 248876211 \pmod{10^9}$.

Find $S(10^{16})$. Give your answer modulo 10^9 .

Problem 765: Trillionaire

Link

Tags: Game Theory, Number Theory, Probability

Starting with 1 gram of gold you play a game. Each round you bet a certain amount of your gold: if you have x grams you can bet b grams for any $0 \leq b \leq x$. You then toss an unfair coin: with a probability of 0.6 you double your bet (so you now have $x + b$), otherwise you lose your bet (so you now have $x - b$).

Choosing your bets to maximize your probability of having at least a trillion (10^{12}) grams of gold after 1000 rounds, what is the probability that you become a trillionaire?

All computations are assumed to be exact (no rounding), but give your answer rounded to 10 digits behind the decimal point.

Problem 766: Sliding Block Puzzle

Link

Tags: Combinatorics, Number Theory

A sliding block puzzle is a puzzle where pieces are confined to a grid and by sliding the pieces a final configuration is reached. In this problem the pieces can only be slid in multiples of one unit in the directions up, down, left, right.

A reachable configuration is any arrangement of the pieces that can be achieved by sliding the pieces from the initial configuration.

Two configurations are identical if the same shape pieces occupy the same position in the grid. So in the case below the red squares are indistinguishable. For this example the number of reachable configurations is 208.

Problem 767: Window into a Matrix II

[Link](#)

Tags: Algebra, Game Theory

A window into a matrix is a contiguous sub matrix.

Consider a $16 \times n$ matrix where every entry is either 0 or 1. Let $B(k, n)$ be the total number of these matrices such that the sum of the entries in every $2 \times k$ window is k .

You are given that $B(2, 4) = 65550$ and $B(3, 9) \equiv 87273560 \pmod{1\,000\,000\,007}$.

Find $B(10^5, 10^{16})$. Give your answer modulo $1\,000\,000\,007$.

Problem 768: Chandelier

[Link](#)

Tags: Combinatorics

A certain type of chandelier contains a circular ring of n evenly spaced candleholders.

If only one candle is fitted, then the chandelier will be imbalanced. However, if a second identical candle is placed in the opposite candleholder (assuming n is even) then perfect balance will be achieved and the chandelier will hang level.

Let $f(n, m)$ be the number of ways of arranging m identical candles in distinct sockets of a chandelier with n candleholders such that the chandelier is perfectly balanced.

For example, $f(4, 2) = 2$: assuming the chandelier's four candleholders are aligned with the compass points, the two valid arrangements are "North & South" and "East & West". Note that these are considered to be different arrangements even though they are related by rotation.

You are given that $f(12, 4) = 15$ and $f(36, 6) = 876$.

Find $f(360, 20)$.

Problem 769: Binary Quadratic Form II

Link

Tags: Algebra, Game Theory, Number Theory

Consider the following binary quadratic form:

$$f(x, y) = x^2 + 5xy + 3y^2$$

A positive integer q has a primitive representation if there exist positive integers x and y such that $q = f(x, y)$ and $\gcd(x, y) = 1$.

We are interested in primitive representations of perfect squares. For example:

$$17^2 = f(1, 9)$$

$$87^2 = f(13, 40) = f(46, 19)$$

Define $C(N)$ as the total number of primitive representations of z^2 for $0 < z \leq N$.

Multiple representations are counted separately, so for example $z = 87$ is counted twice.

You are given $C(10^3) = 142$ and $C(10^6) = 142463$.

Find $C(10^{14})$.

Problem 770: Delphi Flip

Link

Tags: Combinatorics, Game Theory

A and B play a game. A has originally 1 gram of gold and B has an unlimited amount. Each round goes as follows:

- A chooses and displays, x , a nonnegative real number no larger than the amount of gold that A has.
- Either B chooses to TAKE. Then A gives B x grams of gold.
- Or B chooses to GIVE. Then B gives A x grams of gold.

B TAKES n times and GIVES n times after which the game finishes.

Define $g(X)$ to be the smallest value of n so that A can guarantee to have at least X grams of gold at the end of the game. You are given $g(1.7) = 10$.

Find $g(1.9999)$.

Problem 771: Pseudo Geometric Sequences

Link

Tags: Game Theory, Sequences

We define a pseudo-geometric sequence to be a finite sequence $a_0, a_1, \dots, a_n$ of positive integers, satisfying the following conditions:

- $n \geq 4$, i.e. the sequence has at least 5 terms.
- $0a_0a_1 \cdots a_n$, i.e. the sequence is strictly increasing.
- $|a_i^2 - a_{i-1}a_{i+1}| \leq 2$ for $1 \leq i \leq n-1$.

Let $G(N)$ be the number of different pseudo-geometric sequences whose terms do not exceed N .

For example, $G(6) = 4$, as the following 4 sequences give a complete list:

1, 2, 3, 4, 5 1, 2, 3, 4, 6 2, 3, 4, 5, 6 1, 2, 3, 4, 5, 6

Also, $G(10) = 26$, $G(100) = 4710$ and $G(1000) = 496805$.

Find $G(10^{18})$. Give your answer modulo 1 000 000 007.

Problem 772: Balanceable k -bounded Partitions

Link

Tags: Combinatorics

A k -bounded partition of a positive integer N is a way of writing N as a sum of positive integers not exceeding k .

A balanceable partition is a partition that can be further divided into two parts of equal sums.

For example, $3 + 2 + 2 + 2 + 2 + 1$ is a balanceable 3-bounded partition of 12 since $3 + 2 + 1 = 2 + 2 + 2$. Conversely, $3 + 3 + 3 + 1$ is a 3-bounded partition of 10 which is not balanceable.

Let $f(k)$ be the smallest positive integer N all of whose k -bounded partitions are balanceable. For example, $f(3) = 12$ and $f(30) \equiv 179092994 \pmod{1\,000\,000\,007}$.

Find $f(10^8)$. Give your answer modulo 1 000 000 007.

Problem 773: Ruff Numbers

Link

Tags: Number Theory

Let S_k be the set containing 2 and 5 and the first k primes that end in 7. For example, $S_3 = \{2, 5, 7, 17, 37\}$.

Define a k -Ruff number to be one that is not divisible by any element in S_k .

If N_k is the product of the numbers in S_k then define $F(k)$ to be the sum of all k -Ruff numbers less than N_k that have last digit 7. You are given $F(3) = 76101452$.

Find $F(97)$, give your answer modulo 1 000 000 007.

Problem 774: Conjunctive Sequences

Link

Tags: Sequences

Let ' $\&$ ' denote the bitwise AND operation.

For example, $10 \& 12 = 1010_2 \& 1100_2 = 1000_2 = 8$.

We shall call a finite sequence of non-negative integers $(a_1, a_2, \dots, a_n)$ conjunctive if $a_i \& a_{i+1} \neq 0$ for all $i = 1 \dots n - 1$.

Define $c(n, b)$ to be the number of conjunctive sequences of length n in which all terms are $\leq b$.

You are given that $c(3, 4) = 18$, $c(10, 6) = 2496120$, and $c(100, 200) \equiv 268159379 \pmod{998244353}$.

Find $c(123, 123456789)$. Give your answer modulo 998244353.

Problem 775: Saving Paper

Link

Tags: Combinatorics, Graph Theory

When wrapping several cubes in paper, it is more efficient to wrap them all together than to wrap each one individually. For example, with 10 cubes of unit edge length, it would take 30 units of paper to wrap them in the arrangement shown below, but 60 units to wrap them separately.

Problem 776: Digit Sum Division

Link

Tags: Number Theory

For a positive integer n , $d(n)$ is defined to be the sum of the digits of n . For example, $d(12345) = 15$.

$$\text{Let } F(N) = \sum_{n=1}^N \frac{n}{d(n)}.$$

You are given $F(10) = 19$, $F(123) \approx 1.187764610390\text{e}3$ and $F(12345) \approx 4.855801996238\text{e}6$.

Find $F(1234567890123456789)$. Write your answer in scientific notation rounded to twelve significant digits after the decimal point. Use a lowercase e to separate the mantissa and the exponent.

Problem 777: Lissajous Curves

Link

Tags: Number Theory

For coprime positive integers a and b , let $C_{a,b}$ be the curve defined by:

$$\begin{aligned}x &= \cos(at) \\ y &= \cos\left(b\left(t - \frac{\pi}{10}\right)\right)\end{aligned}$$

where t varies between 0 and 2π .

For example, the images below show $C_{2,5}$ (left) and $C_{7,4}$ (right):

Problem 778: Freshman's Product

Link

Tags: Number Theory, Sequences

If a, b are two nonnegative integers with decimal representations $a = (\dots a_2 a_1 a_0)$ and $b = (\dots b_2 b_1 b_0)$ respectively, then the freshman's product of a and b , denoted $a \boxtimes b$, is the integer c with decimal representation $c = (\dots c_2 c_1 c_0)$ such that c_i is the last digit of $a_i \cdot b_i$.

For example, $234 \boxtimes 765 = 480$.

Let $F(R, M)$ be the sum of $x_1 \boxtimes \dots \boxtimes x_R$ for all sequences of integers $(x_1, \dots, x_R)$ with $0 \leq x_i \leq M$.

For example, $F(2, 7) = 204$, and $F(23, 76) \equiv 5870548 \pmod{1\,000\,000\,009}$.

Find $F(234567, 765432)$. Give your answer modulo $1\,000\,000\,009$.

Problem 779: Prime Factor and Exponent

Link

Tags: Number Theory

For a positive integer n , let $p(n)$ be the smallest prime dividing n , and let $\alpha(n)$ be its p -adic order, i.e. the largest integer such that $p(n)^{\alpha(n)}$ divides n .

For a positive integer K , define the function $f_K(n)$ by:

$$f_K(n) = \frac{\alpha(n) - 1}{(p(n))^K}.$$

Also define $\overline{f_K}$ by:

$$\overline{f_K} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=2}^N f_K(n).$$

It can be verified that $\overline{f_1} \approx 0.282419756159$.

Find $\sum_{K=1}^{\infty} \overline{f_K}$. Give your answer rounded to 12 digits after the decimal point.

Problem 780: Toriangles

Link

Tags: Combinatorics, Game Theory, Geometry, Graph Theory

For positive real numbers a, b , an $a \times b$ torus is a rectangle of width a and height b , with left and right sides identified, as well as top and bottom sides identified. In other words, when tracing a path on the rectangle, reaching an edge results in “wrapping round” to the corresponding point on the opposite edge.

A tiling of a torus is a way to dissect it into equilateral triangles of edge length 1. For example, the following three diagrams illustrate respectively a $1 \times \frac{\sqrt{3}}{2}$ torus with two triangles, a $\sqrt{3} \times 1$ torus with four triangles, and an approximately 2.8432×2.1322 torus with fourteen triangles:

Problem 781: Feynman Diagrams

Link

Tags: Graph Theory

Let $F(n)$ be the number of connected graphs with blue edges (directed) and red edges (undirected) containing:

- two vertices of degree 1, one with a single outgoing blue edge and the other with a single incoming blue edge.
- n vertices of degree 3, each of which has an incoming blue edge, a different outgoing blue edge and a red edge.

For example, $F(4) = 5$ because there are 5 graphs with these properties:

Problem 782: Distinct Rows and Columns

Link

Tags: Algebra, Game Theory

The complexity of an $n \times n$ binary matrix is the number of distinct rows and columns.

For example, consider the 3×3 matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$\mathbf{A}$ has complexity 2 because the set of rows and columns is $\{000, 101\}$. $\mathbf{B}$ has complexity 3 because the set of rows and columns is $\{000, 001, 111\}$.

For $0 \leq k \leq n^2$, let $c(n, k)$ be the minimum complexity of an $n \times n$ binary matrix with exactly k ones.

Let

$$C(n) = \sum_{k=0}^{n^2} c(n, k)$$

For example, $C(2) = c(2, 0) + c(2, 1) + c(2, 2) + c(2, 3) + c(2, 4) = 1 + 2 + 2 + 2 + 1 = 8$.

You are given $C(5) = 64$, $C(10) = 274$ and $C(20) = 1150$.

Find $C(10^4)$.

Problem 783: Urns

Link

Tags: Probability

Given n and k two positive integers we begin with an urn that contains kn white balls. We then proceed through n turns where on each turn k black balls are added to the urn and then $2k$ random balls are removed from the urn.

We let $B_t(n, k)$ be the number of black balls that are removed on turn t .

Further define $E(n, k)$ as the expectation of $\sum_{t=1}^n B_t(n, k)^2$.

You are given $E(2, 2) = 9.6$.

Find $E(10^6, 10)$. Round your answer to the nearest whole number.

Problem 784: Reciprocal Pairs

Link

Tags: Number Theory

Let's call a pair of positive integers p, q (pq) reciprocal, if there is a positive integer r such that r equals both the inverse of p modulo q and the inverse of q modulo p .

For example, $(3, 5)$ is one reciprocal pair for $r = 2$.

Let $F(N)$ be the total sum of $p + q$ for all reciprocal pairs (p, q) where $p \leq N$.

$F(5) = 59$ due to these four reciprocal pairs $(3, 5)$, $(4, 11)$, $(5, 7)$ and $(5, 19)$.

You are also given $F(10^2) = 697317$.

Find $F(2 \cdot 10^6)$.

Problem 785: Symmetric Diophantine Equation

Link

Tags: Algebra, Game Theory, Number Theory

Consider the following Diophantine equation:

$$15(x^2 + y^2 + z^2) = 34(xy + yz + zx)$$

where x, y and z are positive integers.

Let $S(N)$ be the sum of all solutions, (x, y, z) , of this equation such that, $1 \leq x \leq y \leq z \leq N$ and $\gcd(x, y, z) = 1$.

For $N = 10^2$, there are three such solutions - $(1, 7, 16), (8, 9, 39), (11, 21, 72)$. So $S(10^2) = 184$.

Find $S(10^9)$.

Problem 786: Billiard

[Link](#)

Tags: Game Theory, Geometry

The following diagram shows a billiard table of a special quadrilateral shape. The four angles A, B, C, D are $120^\circ, 90^\circ, 60^\circ, 90^\circ$ respectively, and the lengths AB and AD are equal.

Problem 787: Bézout's Game

[Link](#)

Tags: Game Theory, Number Theory

Two players play a game with two piles of stones. They take alternating turns. If there are currently a stones in the first pile and b stones in the second, a turn consists of removing $c \geq 0$ stones from the first pile and $d \geq 0$ from the second in such a way that $ad - bc = \pm 1$. The winner is the player who first empties one of the piles.

Note that the game is only playable if the sizes of the two piles are coprime.

A game state (a, b) is a winning position if the next player can guarantee a win with optimal play. Define $H(N)$ to be the number of winning positions (a, b) with $\gcd(a, b) = 1$, $a > 0$, $b > 0$ and $a + b \leq N$. Note the order matters, so for example $(2, 1)$ and $(1, 2)$ are distinct positions.

You are given $H(4) = 5$ and $H(100) = 2043$.

Find $H(10^9)$.

Problem 788: Dominating Numbers

[Link](#)

Tags: Number Theory

A dominating number is a positive integer that has more than half of its digits equal.

For example, 2022 is a dominating number because three of its four digits are equal to 2. But 2021 is not a dominating number.

Let $D(N)$ be how many dominating numbers are less than 10^N . For example, $D(4) = 603$ and $D(10) = 21893256$.

Find $D(2022)$. Give your answer modulo 1 000 000 007.

Problem 789: Minimal Pairing Modulo p

[Link](#)

Tags: Game Theory, Number Theory

Given an odd prime p , put the numbers $1, \dots, p-1$ into $\frac{p-1}{2}$ pairs such that each number appears exactly once. Each pair (a, b) has a cost of $ab \bmod p$. For example, if $p = 5$ the pair $(3, 4)$ has a cost of $12 \bmod 5 = 2$.

The total cost of a pairing is the sum of the costs of its pairs. We say that such pairing is optimal if its total cost is minimal for that p .

For example, if $p = 5$, then there is a unique optimal pairing: $(1, 2), (3, 4)$, with total cost of $2 + 2 = 4$.

The cost product of a pairing is the product of the costs of its pairs. For example, the cost product of the optimal pairing for $p = 5$ is $2 \cdot 2 = 4$.

It turns out that all optimal pairings for $p = 2\,000\,000\,011$ have the same cost product.

Find the value of this product.

Problem 790: Clock Grid

[Link](#)

Tags: Combinatorics, Game Theory, Sequences

There is a grid of length and width 50515093 points. A clock is placed on each grid point. The clocks are all analogue showing a single hour hand initially pointing at 12.

A sequence S_t is created where:

$$S_0 = 290797$$

$$S_t = S_{t-1}^2 \bmod 50515093 \quad t > 0$$

The four numbers $N_t = (S_{4t-4}, S_{4t-3}, S_{4t-2}, S_{4t-1})$ represent a range within the grid, with the first pair of numbers representing the x-bounds and the second pair representing the y-bounds. For example, if $N_t = (3, 9, 47, 20)$, the range would be $3 \leq x \leq 9$ and $20 \leq y \leq 47$, and would include 196 clocks.

For each t ($t > 0$), the clocks within the range represented by N_t are moved to the next hour $\$12 \rightarrow 1 \rightarrow 2 \rightarrow \dots \$$.

We define $C(t)$ to be the sum of the hours that the clock hands are pointing to after timestep t .

You are given $C(0) = 30621295449583788$, $C(1) = 30613048345941659$, $C(10) = 21808930308198471$ and $C(100) = 16190667393984172$.

Find $C(10^5)$.

Problem 791: Average and Variance

Link

Tags: Number Theory

Denote the average of k numbers $x_1, \dots, x_k$ by $\bar{x} = \frac{1}{k} \sum_i x_i$. Their variance is defined as $\frac{1}{k} \sum_i (x_i - \bar{x})^2$.

Let $S(n)$ be the sum of all quadruples of integers (a, b, c, d) satisfying $1 \leq a \leq b \leq c \leq d \leq n$ such that their average is exactly twice their variance.

For $n = 5$, there are 5 such quadruples, namely: $(1, 1, 1, 3)$, $(1, 1, 3, 3)$, $(1, 2, 3, 4)$, $(1, 3, 4, 4)$, $(2, 2, 3, 5)$.

Hence $S(5) = 48$. You are also given $S(10^3) = 37048340$.

Find $S(10^8)$. Give your answer modulo 433494437.

Problem 792: Too Many Twos

Link

Tags: Number Theory

We define $\nu_2(n)$ to be the largest integer r such that 2^r divides n . For example, $\nu_2(24) = 3$.

Define $S(n) = \sum_{k=1}^n (-2)^k \binom{2k}{k}$ and $u(n) = \nu_2(3S(n) + 4)$.

For example, when $n = 4$ then $S(4) = 980$ and $3S(4) + 4 = 2944 = 2^7 \cdot 23$, hence $u(4) = 7$.

You are also given $u(20) = 24$.

Also define $U(N) = \sum_{n=1}^N u(n^3)$. You are given $U(5) = 241$.

Find $U(10^4)$.

Problem 793: Median of Products

[Link](#)

Tags: Game Theory, Probability, Sequences

Let S_i be an integer sequence produced with the following pseudo-random number generator:

- $S_0 = 290797$
- $S_{i+1} = S_i^2 \bmod 50515093$

Let $M(n)$ be the median of the pairwise products $S_i S_j$ for $0 \leq i < j < n$.

You are given $M(3) = 3878983057768$ and $M(103) = 492700616748525$.

Find $M(1\,000\,003)$.

Problem 794: Seventeen Points

[Link](#)

Tags: Game Theory

This problem uses half open interval notation where $[a, b)$ represents $a \leq x < b$.

A real number, x_1 , is chosen in the interval $[0, 1)$.

A second real number, x_2 , is chosen such that each of $[0, \frac{1}{2})$ and $[\frac{1}{2}, 1)$ contains exactly one of (x_1, x_2) .

Continue such that on the n -th step a real number, x_n , is chosen so that each of the intervals $[\frac{k-1}{n}, \frac{k}{n})$ for $k \in \{1, \dots, n\}$ contains exactly one of $(x_1, x_2, \dots, x_n)$.

Define $F(n)$ to be the minimal value of the sum $x_1 + x_2 + \dots + x_n$ of a tuple $(x_1, x_2, \dots, x_n)$ chosen by such a procedure. For example, $F(4) = 1.5$ obtained with $(x_1, x_2, x_3, x_4) = (0, 0.75, 0.5, 0.25)$.

Surprisingly, no more than 17 points can be chosen by this procedure.

Find $F(17)$ and give your answer rounded to 12 decimal places.

Problem 795: Alternating GCD Sum

Link

Tags: Number Theory

For a positive integer n , the function $g(n)$ is defined as

$$g(n) = \sum_{i=1}^n (-1)^i \gcd(n, i^2).$$

For example, $g(4) = -\gcd(4, 1^2) + \gcd(4, 2^2) - \gcd(4, 3^2) + \gcd(4, 4^2) = -1 + 4 - 1 + 4 = 6$.

You are also given $g(1234) = 1233$.

Let $G(N) = \sum_{n=1}^N g(n)$. You are given $G(1234) = 2194708$.

Find $G(12345678)$.

Problem 796: A Grand Shuffle

Link

Tags: Probability

A standard 52 card deck comprises thirteen ranks in four suits. However, modern decks have two additional Jokers, which neither have a suit nor a rank, for a total of 54 cards. If we shuffle such a deck and draw cards without replacement, then we would need, on average, approximately 29.05361725 cards so that we have at least one card for each rank.

Now, assume you have 10 such decks, each with a different back design. We shuffle all 10×54 cards together and draw cards without replacement. What is the expected number of cards needed so every suit, rank and deck design have at least one card?

Give your answer rounded to eight places after the decimal point.

Problem 797: Cyclogenic Polynomials

Link

Tags: Algebra

A monic polynomial is a single-variable polynomial in which the coefficient of highest degree is equal to 1.

Define $\mathcal{F}$ to be the set of all monic polynomials with integer coefficients (including the constant polynomial $p(x) = 1$). A polynomial $p(x) \in \mathcal{F}$ is cyclogenic if there exists $q(x) \in \mathcal{F}$ and a positive integer n such that $p(x)q(x) = x^n - 1$. If n is the smallest such positive integer then $p(x)$ is n -cyclogenic.

Define $P_n(x)$ to be the sum of all n -cyclogenic polynomials. For example, there exist ten 6-cyclogenic polynomials (which divide $x^6 - 1$ and no smaller $x^k - 1$):

$$\begin{array}{ccccccc} x^6 - 1 & & x^4 + x^3 - x - 1 & & x^3 + 2x^2 + 2x + 1 & & x^2 - x + 1 \\ x^5 + x^4 + x^3 + x^2 + x + 1 & & x^4 - x^3 + x - 1 & & x^3 - 2x^2 + 2x - 1 & & \\ x^5 - x^4 + x^3 - x^2 + x - 1 & & x^4 + x^2 + 1 & & x^3 + 1 & & \end{array}$$

giving

$$P_6(x) = x^6 + 2x^5 + 3x^4 + 5x^3 + 2x^2 + 5x.$$

Also define

$$Q_N(x) = \sum_{n=1}^N P_n(x).$$

It's given that $Q_{10}(x) = x^{10} + 3x^9 + 3x^8 + 7x^7 + 8x^6 + 14x^5 + 11x^4 + 18x^3 + 12x^2 + 23x$ and $Q_{10}(2) = 5598$.

Find $Q_{10^7}(2)$. Give your answer modulo 1 000 000 007.

Problem 798: Card Stacking Game

Link

Tags: Game Theory

Two players play a game with a deck of cards which contains s suits with each suit containing n cards numbered from 1 to n .

Before the game starts, a set of cards (which may be empty) is picked from the deck and placed face-up on the table, with no overlap. These are called the visible cards.

The players then make moves in turn.

A move consists of choosing a card X from the rest of the deck and placing it face-up on top of a visible card Y , subject to the following restrictions:

- X and Y must be the same suit;

- the value of X must be larger than the value of Y.

The card X then covers the card Y and replaces Y as a visible card.

The player unable to make a valid move loses and play stops.

Let $C(n, s)$ be the number of different initial sets of cards for which the first player will lose given best play for both players.

For example, $C(3, 2) = 26$ and $C(13, 4) \equiv 540318329 \pmod{1\,000\,000\,007}$.

Find $C(10^7, 10^7)$. Give your answer modulo $1\,000\,000\,007$.

Problem 799: Pentagonal Puzzle

Link

Tags: Sequences

Pentagonal numbers are generated by the formula: $P_n = \frac{1}{2}n(3n - 1)$ giving the sequence:

$$1, 5, 12, 22, 35, 51, 70, 92, \dots$$

Some pentagonal numbers can be expressed as the sum of two other pentagonal numbers.

For example:

$$P_8 = 92 = 22 + 70 = P_4 + P_7$$

3577 is the smallest pentagonal number that can be expressed as the sum of two pentagonal numbers in two different ways

$$\begin{aligned} P_{49} = 3577 &= 3432 + 145 = P_{48} + P_{10} \\ &= 3290 + 287 = P_{47} + P_{14} \end{aligned}$$

107602 is the smallest pentagonal number that can be expressed as the sum of two pentagonal numbers in three different ways.

Find the smallest pentagonal number that can be expressed as the sum of two pentagonal numbers in over 100 different ways.

Problem 800: Hybrid Integers

Link

Tags: Number Theory

An integer of the form $p^q q^p$ with prime numbers $p \neq q$ is called a hybrid-integer.

For example, $800 = 2^5 5^2$ is a hybrid-integer.

We define $C(n)$ to be the number of hybrid-integers less than or equal to n .

You are given $C(800) = 2$ and $C(800^{800}) = 10790$.

Find $C(800800^{800800})$.

Problem 801: $x^y \equiv y^x$

Link

Tags: Algebra, Number Theory

The positive integral solutions of the equation $x^y = y^x$ are $(2, 4)$, $(4, 2)$ and (k, k) for all $k > 0$.

For a given positive integer n , let $f(n)$ be the number of integral values $0 < x, y \leq n^2 - n$ such that

$$x^y \equiv y^x \pmod{n}.$$

For example, $f(5) = 104$ and $f(97) = 1614336$.

Let $S(M, N) = \sum f(p)$ where the sum is taken over all primes p satisfying $M \leq p \leq N$.

You are given $S(1, 10^2) = 7381000$ and $S(1, 10^5) \equiv 701331986 \pmod{993353399}$.

Find $S(10^{16}, 10^{16} + 10^6)$. Give your answer modulo 993353399.

Problem 802: Iterated Composition

Link

Tags: Number Theory

Let $\mathbb{R}^2$ be the set of pairs of real numbers (x, y) . Let $\pi = 3.14159 \dots$.

Consider the function f from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by $f(x, y) = (x^2 - x - y^2, 2xy - y + \pi)$, and its n -th iterated composition $f^{(n)}(x, y) = f(f(\dots f(x, y) \dots))$. For example $f^{(3)}(x, y) = f(f(f(x, y)))$. A pair (x, y) is said to have period n if n is the smallest positive integer such that $f^{(n)}(x, y) = (x, y)$.

Let $P(n)$ denote the sum of x -coordinates of all points having period not exceeding n . Interestingly, $P(n)$ is always an integer. For example, $P(1) = 2$, $P(2) = 2$, $P(3) = 4$.

Find $P(10^7)$ and give your answer modulo 1 020 340 567.

Problem 803: Pseudorandom Sequence

Link

Tags: Combinatorics, Probability, Sequences

Rand48 is a pseudorandom number generator used by some programming languages. It generates a sequence from any given integer $0 \leq a_0 < 2^{48}$ using the rule $a_n = (25214903917 \cdot a_{n-1} + 11) \bmod 2^{48}$.

Let $b_n = \lfloor a_n / 2^{16} \rfloor \bmod 52$. The sequence $b_0, b_1, \dots$ is translated to an infinite string $c = c_0 c_1 \dots$ via the rule:

$0 \rightarrow a, 1 \rightarrow b, \dots, 25 \rightarrow z, 26 \rightarrow A, 27 \rightarrow B, \dots, 51 \rightarrow Z$.

For example, if we choose $a_0 = 123456$, then the string c starts with: “bQYicNGCY...”.

Moreover, starting from index 100, we encounter the substring “RxqLBfWzv” for the first time.

Alternatively, if c starts with “EULERcats...”, then a_0 must be 78580612777175.

Now suppose that the string c starts with “PuzzleOne...”.

Find the starting index of the first occurrence of the substring “LuckyText” in c .

Problem 804: Counting Binary Quadratic Representations

Link

Tags: Number Theory

Let $g(n)$ denote the number of ways a positive integer n can be represented in the form:

$$x^2 + xy + 41y^2$$

where x and y are integers. For example, $g(53) = 4$ due to $(x, y) \in \{(-4, 1), (-3, -1), (3, 1), (4, -1)\}$.

Define $T(N) = \sum_{n=1}^N g(n)$. You are given $T(10^3) = 474$ and $T(10^6) = 492128$.

Find $T(10^{16})$.

Problem 805: Shifted Multiples

Link

Tags: Number Theory

For a positive integer n , let $s(n)$ be the integer obtained by shifting the leftmost digit of the decimal representation of n to the rightmost position.

For example, $s(142857) = 428571$ and $s(10) = 1$.

For a positive rational number r , we define $N(r)$ as the smallest positive integer n such that $s(n) = r \cdot n$.

If no such integer exists, then $N(r)$ is defined as zero.

For example, $N(3) = 142857$, $N(\frac{1}{10}) = 10$ and $N(2) = 0$.

Let $T(M)$ be the sum of $N(u^3/v^3)$ where (u, v) ranges over all ordered pairs of coprime positive integers not exceeding M .

For example, $T(3) \equiv 262429173 \pmod{1\,000\,000\,007}$.

Find $T(200)$. Give your answer modulo $1\,000\,000\,007$.

Problem 806: Nim on Towers of Hanoi

Link

Tags: Game Theory, Probability

This problem combines the game of Nim with the Towers of Hanoi. For a brief introduction to the rules of these games, please refer to Problem 301 and Problem 497, respectively.

The unique shortest solution to the Towers of Hanoi problem with n disks and 3 pegs requires $2^n - 1$ moves. Number the positions in the solution from index 0 (starting position, all disks on the first peg) to index $2^n - 1$ (final position, all disks on the third peg).

Each of these 2^n positions can be considered as the starting configuration for a game of Nim, in which two players take turns to select a peg and remove any positive number of disks from it. The winner is the player who removes the last disk.

We define $f(n)$ to be the sum of the indices of those positions for which, when considered as a Nim game, the first player will lose (assuming an optimal strategy from both players).

For $n = 4$, the indices of losing positions in the shortest solution are 3,6,9 and 12. So we have $f(4) = 30$.

You are given that $f(10) = 67518$.

Find $f(10^5)$. Give your answer modulo 1 000 000 007.

Problem 807: Loops of Ropes

Link

Tags: Combinatorics, Game Theory, Geometry, Probability

Given a circle C and an integer $n > 1$, we perform the following operations.

In step 0, we choose two uniformly random points R_0 and B_0 on C .

In step i ($1 \leq i < n$), we first choose a uniformly random point R_i on C and connect the points R_{i-1} and R_i with a red rope; then choose a uniformly random point B_i on C and connect the points B_{i-1} and B_i with a blue rope.

In step n , we first connect the points R_{n-1} and R_0 with a red rope; then connect the points B_{n-1} and B_0 with a blue rope.

Each rope is straight between its two end points, and lies above all previous ropes.

After step n , we get a loop of red ropes, and a loop of blue ropes.

Sometimes the two loops can be separated, as in the left figure below; sometimes they are “linked”, hence cannot be separated, as in the middle and right figures below.

Problem 808: Reversible Prime Squares

Link

Tags: Number Theory

Both 169 and 961 are the square of a prime. 169 is the reverse of 961.

We call a number a reversible prime square if:

- It is not a palindrome, and
- It is the square of a prime, and
- Its reverse is also the square of a prime.

169 and 961 are not palindromes, so both are reversible prime squares.

Find the sum of the first 50 reversible prime squares.

Problem 809: Rational Recurrence Relation

Link

Tags: Game Theory

The following is a function defined for all positive rational values of x .

$$f(x) = \begin{cases} x & x \text{ is integral} \\ f\left(\frac{1}{1-x}\right) & x1 \\ f\left(\frac{1}{|x|-x} - 1 + f(x-1)\right) & \text{otherwise} \end{cases}$$

For example, $f(3/2) = 3$, $f(1/6) = 65533$ and $f(13/10) = 7625597484985$.

Find $f(22/7)$. Give your answer modulo 10^{15} .

Problem 810: XOR-Primes

Link

Tags: Number Theory

We use $x \oplus y$ for the bitwise XOR of x and y .

Define the XOR-product of x and y , denoted by $x \otimes y$, similar to a long multiplication in base 2, except that the intermediate results are XORed instead of the usual integer addition.

For example, $7 \otimes 3 = 9$, or in base 2, $111_2 \otimes 11_2 = 1001_2$:

$$\begin{array}{r} 111_2 \\ \otimes \quad 11_2 \\ \hline 111_2 \\ \oplus \quad 111_2 \\ \hline 1001_2 \end{array}$$

An XOR-prime is an integer n greater than 1 that is not an XOR-product of two integers greater than 1. The above example shows that 9 is not an XOR-prime. Similarly, $5 = 3 \otimes 3$ is not an XOR-prime. The first few XOR-primes are 2, 3, 7, 11, 13, ... and the 10th XOR-prime is 41.

Find the 5 000 000th XOR-prime.

Problem 811: Bitwise Recursion

Link

Tags: Number Theory

Let $b(n)$ be the largest power of 2 that divides n . For example $b(24) = 8$.

Define the recursive function:

$$\begin{aligned} A(0) &= 1 \\ A(2n) &= 3A(n) + 5A(2n - b(n)) \quad n > 0 \\ A(2n + 1) &= A(n) \end{aligned}$$

and let $H(t, r) = A((2^t + 1)^r)$.

You are given $H(3, 2) = A(81) = 636056$.

Find $H(10^{14} + 31, 62)$. Give your answer modulo 1 000 062 031.

Problem 812: Dynamical Polynomials

Link

Tags: Algebra

A dynamical polynomial is a monic polynomial $f(x)$ with integer coefficients such that $f(x)$ divides $f(x^2 - 2)$.

For example, $f(x) = x^2 - x - 2$ is a dynamical polynomial because $f(x^2 - 2) = x^4 - 5x^2 + 4 = (x^2 + x - 2)f(x)$.

Let $S(n)$ be the number of dynamical polynomials of degree n .

For example, $S(2) = 6$, as there are six dynamical polynomials of degree 2:

$$x^2 - 4x + 4 \quad , \quad x^2 - x - 2 \quad , \quad x^2 - 4 \quad , \quad x^2 - 1 \quad , \quad x^2 + x - 1 \quad , \quad x^2 + 2x + 1$$

Also, $S(5) = 58$ and $S(20) = 122087$.

Find $S(10\,000)$. Give your answer modulo 998244353.

Problem 813: XOR-Powers

Link

Tags: Number Theory

We use $x \oplus y$ to be the bitwise XOR of x and y .

Define the XOR-product of x and y , denoted by $x \otimes y$, similar to a long multiplication in base 2, except that the intermediate results are XORed instead of the usual integer addition.

For example, $11 \otimes 11 = 69$, or in base 2, $1011_2 \otimes 1011_2 = 1000101_2$:

$$\begin{array}{r} 1011_2 \\ \otimes 1011_2 \\ \hline 1011_2 \\ 1011_2 \\ \oplus 1011_2 \\ \hline 1000101_2 \end{array}$$

Further we define $P(n) = 11^{\otimes n} = \overbrace{11 \otimes 11 \otimes \dots \otimes 11}^n$. For example $P(2) = 69$. Find $P(8^{12} \cdot 12^8)$. Give your answer modulo $10^9 + 7$.

Problem 814: Mezzo-forte

[Link](#)

Tags: Geometry

$4n$ people stand in a circle with their heads down. When the bell rings they all raise their heads and either look at the person immediately to their left, the person immediately to their right or the person diametrically opposite. If two people find themselves looking at each other they both scream.

Define $S(n)$ to be the number of ways that exactly half of the people scream. You are given $S(1) = 48$ and $S(10) \equiv 420121075 \pmod{998244353}$.

Find $S(10^3)$. Enter your answer modulo 998244353.

Problem 815: Group by Value

[Link](#)

Tags: Number Theory, Probability

A pack of cards contains $4n$ cards with four identical cards of each value. The pack is shuffled and cards are dealt one at a time and placed in piles of equal value. If the card has the same value as any pile it is placed in that pile. If there is no pile of that value then it begins a new pile. When a pile has four cards of the same value it is removed.

Throughout the process the maximum number of non empty piles is recorded. Let $E(n)$ be its expected value. You are given $E(2) = 1.97142857$ rounded to 8 decimal places.

Find $E(60)$. Give your answer rounded to 8 digits after the decimal point.

Problem 816: Shortest Distance Among Points

Link

Tags: Game Theory, Geometry, Probability

We create an array of points P_n in a two dimensional plane using the following random number generator:

$$s_0 = 290797$$

$$s_{n+1} = s_n^2 \bmod 50515093$$

$$P_n = (s_{2n}, s_{2n+1})$$

Let $d(k)$ be the shortest distance of any two (distinct) points among $P_0, \dots, P_{k-1}$.

E.g. $d(14) = 546446.466846479$.

Find $d(2000000)$. Give your answer rounded to 9 places after the decimal point.

Problem 817: Digits in Squares

Link

Tags: Number Theory

Define $m = M(n, d)$ to be the smallest positive integer such that when m^2 is written in base n it includes the base n digit d . For example, $M(10, 7) = 24$ because if all the squares are written out in base 10 the first time the digit 7 occurs is in $24^2 = 576$. $M(11, 10) = 19$ as $19^2 = 361 = 2A9_{11}$.

Find $\sum_{d=1}^{10^5} M(p, p-d)$ where $p = 10^9 + 7$.

Problem 818: SET

Link

Tags: Game Theory

The SET® card game is played with a pack of 81 distinct cards. Each card has four features (Shape, Color, Number, Shading). Each feature has three different variants (e.g. Color can be red, purple, green).

A SET consists of three different cards such that each feature is either the same on each card or different on each card.

For a collection C_n of n cards, let $S(C_n)$ denote the number of SETs in C_n . Then define $F(n) = \sum_{C_n} S(C_n)^4$ where C_n ranges through all collections of n cards (among the 81 cards). You are given $F(3) = 1080$ and $F(6) = 159690960$. Find $F(12)$.

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Problem 819: Iterative Sampling

[Link](#)

Tags: Number Theory, Probability

Given an n -tuple of numbers another n -tuple is created where each element of the new n -tuple is chosen randomly from the numbers in the previous n -tuple. For example, given $(2, 2, 3)$ the probability that 2 occurs in the first position in the next 3-tuple is $2/3$. The probability of getting all 2's would be $8/27$ while the probability of getting the same 3-tuple (in any order) would be $4/9$.

Let $E(n)$ be the expected number of steps starting with $(1, 2, \dots, n)$ and ending with all numbers being the same.

You are given $E(3) = 27/7$ and $E(5) = 468125/60701 \approx 7.711982$ rounded to 6 digits after the decimal place.

Find $E(10^3)$. Give the answer rounded to 6 digits after the decimal place.

Problem 820: N th Digit of Reciprocals

[Link](#)

Tags: Game Theory, Number Theory

Let $d_n(x)$ be the n th decimal digit of the fractional part of x , or 0 if the fractional part has fewer than n digits.

For example:

- $d_7(1) = d_7(\frac{1}{2}) = d_7(\frac{1}{4}) = d_7(\frac{1}{5}) = 0$
- $d_7(\frac{1}{3}) = 3$ since $\frac{1}{3} = 0.333333333\dots$

- $d_7\left(\frac{1}{6}\right) = 6$ since $\frac{1}{6} = 0.1666666666\dots$
- $d_7\left(\frac{1}{7}\right) = 1$ since $\frac{1}{7} = 0.1428571428\dots$

Let $S(n) = \sum_{k=1}^n d_n\left(\frac{1}{k}\right)$.

You are given:

- $S(7) = 0 + 0 + 3 + 0 + 0 + 6 + 1 = 10$
- $S(100) = 418$

Find $S(10^7)$.

Problem 821: 123-Separable

Link

Tags: Number Theory

A set, S , of integers is called 123-separable if S , $2S$ and $3S$ are disjoint. Here $2S$ and $3S$ are obtained by multiplying all the elements in S by 2 and 3 respectively.

Define $F(n)$ to be the maximum number of elements of

$$(S \cup 2S \cup 3S) \cap \{1, 2, 3, \dots, n\}$$

where S ranges over all 123-separable sets.

For example, $F(6) = 5$ can be achieved with either $S = \{1, 4, 5\}$ or $S = \{1, 5, 6\}$.

You are also given $F(20) = 19$.

Find $F(10^{16})$.

Problem 822: Square the Smallest

Link

Tags: Number Theory

A list initially contains the numbers $2, 3, \dots, n$.

At each round, the smallest number in the list is replaced by its square. If there is more than one such number, then only one of them is replaced.

For example, below are the first three rounds for $n = 5$:

$$[2, 3, 4, 5] \xrightarrow{(1)} [4, 3, 4, 5] \xrightarrow{(2)} [4, 9, 4, 5] \xrightarrow{(3)} [16, 9, 4, 5].$$

Let $S(n, m)$ be the sum of all numbers in the list after m rounds.

For example, $S(5, 3) = 16 + 9 + 4 + 5 = 34$. Also $S(10, 100) \equiv 845339386 \pmod{1234567891}$.

Find $S(10^4, 10^{16})$. Give your answer modulo 1234567891.

Problem 823: Factor Shuffle

[Link](#)

Tags: Number Theory

A list initially contains the numbers $2, 3, \dots, n$.

At each round, every number in the list is divided by its smallest prime factor. Then the product of these smallest prime factors is added to the list as a new number. Finally, all numbers that become 1 are removed from the list.

For example, below are the first three rounds for $n = 5$:

$$[2, 3, 4, 5] \xrightarrow{(1)} [2, 60] \xrightarrow{(2)} [30, 4] \xrightarrow{(3)} [15, 2, 4].$$

Let $S(n, m)$ be the sum of all numbers in the list after m rounds.

For example, $S(5, 3) = 15 + 2 + 4 = 21$. Also $S(10, 100) = 257$.

Find $S(10^4, 10^{16})$. Give your answer modulo 1234567891.

Problem 824: Chess Sliders

[Link](#)

Tags: Graph Theory

A Slider is a chess piece that can move one square left or right.

This problem uses a cylindrical chess board where the left hand edge of the board is connected to the right hand edge. This means that a Slider that is on the left hand edge of the chess board can move to the right hand edge of the same row and vice versa.

Let $L(N, K)$ be the number of ways K non-attacking Sliders can be placed on an $N \times N$ cylindrical chess-board.

For example, $L(2, 2) = 4$ and $L(6, 12) = 4204761$.

Find $L(10^9, 10^{15}) \pmod{(10^7 + 19)^2}$.

Problem 825: Chasing Game

[Link](#)

Tags: Game Theory, Geometry, Number Theory, Probability

Two cars are on a circular track of total length $2n$, facing the same direction, initially distance n apart.

They move in turn. At each turn, the moving car will advance a distance of 1, 2 or 3, with equal probabilities.

The chase ends when the moving car reaches or goes beyond the position of the other car. The moving car is declared the winner.

Let $S(n)$ be the difference between the winning probabilities of the two cars.

For example, when $n = 2$, the winning probabilities of the two cars are $\frac{9}{11}$ and $\frac{2}{11}$, and thus $S(2) = \frac{7}{11}$.

Let $T(N) = \sum_{n=2}^N S(n)$.

You are given that $T(10) = 2.38235282$ rounded to 8 digits after the decimal point.

Find $T(10^{14})$, rounded to 8 digits after the decimal point.

Problem 826: Birds on a Wire

[Link](#)

Tags: Game Theory, Number Theory, Probability

Consider a wire of length 1 unit between two posts. Every morning n birds land on it randomly with every point on the wire equally likely to host a bird. The interval from each bird to its closest neighbour is then painted.

Define $F(n)$ to be the expected length of the wire that is painted. You are given $F(3) = 0.5$.

Find the average of $F(n)$ where n ranges through all odd prime less than a million. Give your answer rounded to 10 places after the decimal point.

Problem 827: Pythagorean Triple Occurrence

[Link](#)

Tags: Geometry

Define $Q(n)$ to be the smallest number that occurs in exactly n Pythagorean triples (a, b, c) where abc .

For example, 15 is the smallest number occurring in exactly 5 Pythagorean triples:

$$(9, 12, \mathbf{15}) \quad (8, \mathbf{15}, 17) \quad (\mathbf{15}, 20, 25) \quad (\mathbf{15}, 36, 39) \quad (\mathbf{15}, 112, 113)$$

and so $Q(5) = 15$.

You are also given $Q(10) = 48$ and $Q(10^3) = 8064000$.

Find $\sum_{k=1}^{18} Q(10^k)$. Give your answer modulo 409120391.

Problem 828: Numbers Challenge

Link

Tags: Game Theory

It is a common recreational problem to make a target number using a selection of other numbers. In this problem you will be given six numbers and a target number.

For example, given the six numbers 2, 3, 4, 6, 7, 25, and a target of 211, one possible solution is:

$$211 = (3 + 6) \times 25(4 \times 7) \div 2$$

This uses all six numbers. However, it is not necessary to do so. Another solution that does not use the 7 is:

$$211 = (252) \times (6 + 3) + 4$$

Define the score of a solution to be the sum of the numbers used. In the above example problem, the two given solutions have scores 47 and 40 respectively. It turns out that this problem has no solutions with score less than 40.

When combining numbers, the following rules must be observed:

- Each available number may be used at most once.
- Only the four basic arithmetic operations are permitted: $+$, $-$, $\times$, $\div$.
- All intermediate values must be positive integers, so for example $(3 \div 2)$ is never permitted as a subexpression (even if the final answer is an integer).

The attached file number-challenges.txt contains 200 problems, one per line in the format:

211:2,3,4,6,7,25

where the number before the colon is the target and the remaining comma-separated numbers are those available to be used.

Numbering the problems 1, 2, ..., 200, we let s_n be the minimum score of the solution to the n th problem. For example, $s_1 = 40$, as the first problem in the file is the example given above. Note that not all problems have a solution; in such cases we take $s_n = 0$.

Find $\sum_{n=1}^{200} 3^n s_n$. Give your answer modulo 1005075251.

Problem 829: Integral Fusion

[Link](#)

Tags: Graph Theory, Number Theory

Given any integer n a binary factor tree $T(n)$ is defined to be:

- A tree with the single node n when n is prime.
- A binary tree that has root node n , left subtree $T(a)$ and right subtree $T(b)$, when n is not prime. Here a and b are positive integers such that $n = ab$, $a \leq b$ and $b - a$ is the smallest.

For example $T(20)$:

We define $M(n)$ to be the smallest number that has a factor tree identical in shape to the factor tree for $n!!$, the double factorial of n .

For example, consider $9!! = 9 \times 7 \times 5 \times 3 \times 1 = 945$. The factor tree for 945 is shown below together with the factor tree for 72 which is the smallest number that has a factor tree of the same shape. Hence $M(9) = 72$.

Find $\sum_{n=2}^{31} M(n)$.

Problem 830: Binomials and Powers

[Link](#)

Tags: Number Theory

Let $S(n) = \sum_{k=0}^n \binom{n}{k} k^n$.

You are given, $S(10) = 142469423360$.

Find $S(10^{18})$. Submit your answer modulo $83^3 89^3 97^3$.

Problem 831: Triple Product

Link

Tags: Combinatorics, Game Theory, Number Theory

Let $g(m)$ be the integer defined by the following double sum of products of binomial coefficients:

$$\sum_{j=0}^m \sum_{i=0}^j (-1)^{j-i} \binom{m}{j} \binom{j}{i} \binom{j+5+6i}{j+5}.$$

You are given that $g(10) = 127278262644918$. Its first (most significant) five digits are 12727.

Find the first ten digits of $g(142857)$ when written in base 7.

Problem 832: Mex Sequence

Link

Tags: Game Theory

In this problem $\oplus$ is used to represent the bitwise exclusive or of two numbers.

Starting with blank paper repeatedly do the following:

- Write down the smallest positive integer a which is currently not on the paper;
- Find the smallest positive integer b such that neither b nor $(a \oplus b)$ is currently on the paper. Then write down both b and $(a \oplus b)$.

After the first round $\{1, 2, 3\}$ will be written on the paper. In the second round $a = 4$ and because $(4 \oplus 5)$, $(4 \oplus 6)$ and $(4 \oplus 7)$ are all already written b must be 8.

After n rounds there will be $3n$ numbers on the paper. Their sum is denoted by $M(n)$.

For example, $M(10) = 642$ and $M(1000) = 5432148$.

Find $M(10^{18})$. Give your answer modulo 1 000 000 007.

Problem 833: Square Triangle Products

Link

Tags: Geometry

Triangle numbers T_k are integers of the form $\frac{k(k+1)}{2}$.

A few triangle numbers happen to be perfect squares like $T_1 = 1$ and $T_8 = 36$, but more can be found when considering the product of two triangle numbers. For example, $T_2 \cdot T_{24} = 3 \cdot 300 = 30^2$.

Let $S(n)$ be the sum of c for all integers triples (a, b, c) with $0 < c \leq n$, $c^2 = T_a \cdot T_b$ and $0 < a < b$. For example, $S(100) = \sqrt{T_1 T_8} + \sqrt{T_2 T_{24}} + \sqrt{T_1 T_{49}} + \sqrt{T_3 T_{48}} = 6 + 30 + 35 + 84 = 155$.

You are given $S(10^5) = 1479802$ and $S(10^9) = 241614948794$.

Find $S(10^{35})$. Give your answer modulo 136101521.

Problem 834: Add and Divide

Link

Tags: Probability, Sequences

A sequence is created by starting with a positive integer n and incrementing by $(n + m)$ at the m^{th} step. If $n = 10$, the resulting sequence will be 21, 33, 46, 60, 75, 91, 108, 126, ...

Let $S(n)$ be the set of indices m , for which the m^{th} term in the sequence is divisible by $(n + m)$.

For example, $S(10) = \{5, 8, 20, 35, 80\}$.

Define $T(n)$ to be the sum of the indices in $S(n)$. For example, $T(10) = 148$ and $T(10^2) = 21828$.

Let $U(N) = \sum_{n=3}^N T(n)$. You are given, $U(10^2) = 612572$.

Find $U(1234567)$.

Problem 835: Supernatural Triangles

Link

Tags: Geometry

A Pythagorean triangle is called supernatural if two of its three sides are consecutive integers.

Let $S(N)$ be the sum of the perimeters of all distinct supernatural triangles with perimeters less than or equal to N .

For example, $S(100) = 258$ and $S(10000) = 172004$.

Find $S(10^{10})$. Give your answer modulo 1234567891.

Problem 836: A Bold Proposition

Link

Tags: Number Theory

Let A be an affine plane over a radically integral local field F with residual characteristic p .

We consider an open oriented line section U of A with normalized Haar measure m .

Define $f(m, p)$ as the maximal possible discriminant of the jacobian associated to the orthogonal kernel embedding of U into A .

Find $f(20230401, 57)$. Give as your answer the concatenation of the first letters of each bolded word.

Problem 837: Amidakuji

Link

Tags: Combinatorics, Game Theory, Probability

Amidakuji (Japanese:) is a method for producing a random permutation of a set of objects.

In the beginning, a number of parallel vertical lines are drawn, one for each object. Then a specified number of horizontal rungs are added, each lower than any previous rungs. Each rung is drawn as a line segment spanning a randomly select pair of adjacent vertical lines.

For example, the following diagram depicts an Amidakuji with three objects (A , B , C) and six rungs:

Problem 838: Not Coprime

[Link](#)

Tags: Number Theory

Let $f(N)$ be the smallest positive integer that is not coprime to any positive integer $n \leq N$ whose least significant digit is 3.

For example $f(40)$ equals to $897 = 3 \cdot 13 \cdot 23$ since it is not coprime to any of 3, 13, 23, 33. By taking the natural logarithm (log to base e) we obtain $\ln f(40) = \ln 897 \approx 6.799056$ when rounded to six digits after the decimal point.

You are also given $\ln f(2800) \approx 715.019337$.

Find $f(10^6)$. Enter its natural logarithm rounded to six digits after the decimal point.

Problem 839: Beans in Bowls

[Link](#)

Tags: Sequences

The sequence S_n is defined by $S_0 = 290797$ and $S_n = S_{n-1}^2 \bmod 50515093$ for $n > 0$.

There are N bowls indexed $0, 1, \dots, N-1$. Initially there are S_n beans in bowl n .

At each step, the smallest index n is found such that bowl n has strictly more beans than bowl $n+1$. Then one bean is moved from bowl n to bowl $n+1$.

Let $B(N)$ be the number of steps needed to sort the bowls into non-descending order.

For example, $B(5) = 0$, $B(6) = 14263289$ and $B(100) = 3284417556$.

Find $B(10^7)$.

Problem 840: Sum of Products

[Link](#)

Tags: Combinatorics, Number Theory

A partition of n is a set of positive integers for which the sum equals n .

The partitions of 5 are:

$\{5\}$, $\{1, 4\}$, $\{2, 3\}$, $\{1, 1, 3\}$, $\{1, 2, 2\}$, $\{1, 1, 1, 2\}$ and $\{1, 1, 1, 1, 1\}$.

Further we define the function $D(p)$ as:

$$\begin{aligned} D(1) &= 1 \\ D(p) &= 1, \text{ for any prime } p \\ D(pq) &= D(p)q + pD(q), \text{ for any positive integers } p, q. \end{aligned}$$

Now let $\{a_1, a_2, \dots, a_k\}$ be a partition of n .

We assign to this particular partition the value:

$$P = \prod_{j=1}^k D(a_j).$$

$G(n)$ is the sum of P for all partitions of n .

We can verify that $G(10) = 164$.

We also define:

$$S(N) = \sum_{n=1}^N G(n).$$

You are given $S(10) = 396$.

Find $S(5 \times 10^4) \pmod{999676999}$.

Problem 841: Regular Star Polygons

Link

Tags: Geometry, Graph Theory, Number Theory

The regular star polygon $\{p/q\}$, for coprime integers p, q with $p > 2q$, is a polygon formed from p edges of equal length and equal internal angles, such that tracing the complete polygon wraps q times around the centre. For example, $\{8/3\}$ is illustrated below:

Problem 842: Irregular Star Polygons

Link

Tags: Geometry

Given n equally spaced points on a circle, we define an n -star polygon as an n -gon having those n points as vertices. Two n -star polygons differing by a rotation/reflection are considered different.

For example, there are twelve 5-star polygons shown below.

For an n -star polygon S , let $I(S)$ be the number of its self intersection points.

Let $T(n)$ be the sum of $I(S)$ over all n -star polygons S .

For the example above $T(5) = 20$ because in total there are 20 self intersection points.

Some star polygons may have intersection points made from more than two lines. These are only counted once. For example, S , shown below is one of the sixty 6-star polygons. This one has $I(S) = 4$.

You are also given that $T(8) = 14640$.

Find $\sum_{n=3}^{60} T(n)$. Give your answer modulo $(10^9 + 7)$.

Problem 843: Periodic Circles

Link

Tags: Geometry

This problem involves an iterative procedure that begins with a circle of $n \geq 3$ integers. At each step every number is simultaneously replaced with the absolute difference of its two neighbours.

For any initial values, the procedure eventually becomes periodic.

Let $S(N)$ be the sum of all possible periods for $3 \leq n \leq N$. For example, $S(6) = 6$, because the possible periods for $3 \leq n \leq 6$ are 1, 2, 3. Specifically, $n = 3$ and $n = 4$ can each have period 1 only, while $n = 5$ can have period 1 or 3, and $n = 6$ can have period 1 or 2.

You are also given $S(30) = 20381$.

Find $S(100)$.

Problem 844: k -Markov Numbers

Link

Tags: Number Theory

Consider positive integer solutions to

$$a^2 + b^2 + c^2 = 3abc$$

For example, $(1, 5, 13)$ is a solution. We define a 3-Markov number to be any part of a solution, so 1, 5 and 13 are all 3-Markov numbers. Adding distinct 3-Markov numbers $\leq 10^3$ would give 2797.

Now we define a k -Markov number to be a positive integer that is part of a solution to:

$$\sum_{i=1}^k x_i^2 = k \prod_{i=1}^k x_i, \quad x_i \text{ are positive integers}$$

Let $M_k(N)$ be the sum of k -Markov numbers $\leq N$. Hence $M_3(10^3) = 2797$, also $M_8(10^8) = 131493335$.

Define $S(K, N) = \sum_{k=3}^K M_k(N)$. You are given $S(4, 10^2) = 229$ and $S(10, 10^8) = 2383369980$.

Find $S(10^{18}, 10^{18})$. Give your answer modulo 1 405 695 061.

Problem 845: Prime Digit Sum

Link

Tags: Number Theory

Let $D(n)$ be the n -th positive integer that has the sum of its digits a prime.

For example, $D(61) = 157$ and $D(10^8) = 403539364$.

Find $D(10^{16})$.

Problem 846: Magic Bracelets

Link

Tags: Game Theory, Geometry, Number Theory

A bracelet is made by connecting at least three numbered beads in a circle. Each bead can only display 1, 2, or any number of the form p^k or $2p^k$ for odd prime p .

In addition a magic bracelet must satisfy the following two conditions:

- no two beads display the same number
 - the product of the numbers of any two adjacent beads is of the form $x^2 + 1$
-

Problem 847: Jack's Bean

Link

Tags: Combinatorics, Game Theory, Number Theory

Jack has three plates in front of him. The giant has N beans that he distributes to the three plates. All the beans look the same, but one of them is a magic bean. Jack doesn't know which one it is, but the giant knows.

Jack can ask the giant questions of the form: "Does this subset of the beans contain the magic bean?" In each question Jack may choose any subset of beans from a single plate, and the giant will respond truthfully.

If the three plates contain a , b and c beans respectively, we let $h(a, b, c)$ be the minimal number of questions Jack needs to ask in order to guarantee he locates the magic bean. For example, $h(1, 2, 3) = 3$ and $h(2, 3, 3) = 4$.

Let $H(N)$ be the sum of $h(a, b, c)$ over all triples of non-negative integers a, b, c with $1 \leq a + b + c \leq N$.

You are given: $H(6) = 203$ and $H(20) = 7718$.

A repunit, R_n , is a number made up with n digits all '1'. For example, $R_3 = 111$ and $H(R_3) = 1634144$.

Find $H(R_{19})$. Give your answer modulo 1 000 000 007.

Problem 848: Guessing with Sets

Link

Tags: Combinatorics, Game Theory, Number Theory, Probability

Two players play a game. At the start of the game each player secretly chooses an integer; the first player from $1, \dots, n$ and the second player from $1, \dots, m$. Then they take alternate turns, starting with the first player. The player, whose turn it is, displays a set of numbers and the other player tells whether their secret number is in the set or not. The player to correctly guess a set with a single number is the winner and the game ends.

Let $p(m, n)$ be the winning probability of the first player assuming both players play optimally. For example $p(1, n) = 1$ and $p(m, 1) = 1/m$.

You are also given $p(7, 5) \approx 0.51428571$.

Find $\sum_{i=0}^{20} \sum_{j=0}^{20} p(7^i, 5^j)$ and give your answer rounded to 8 digits after the decimal point.

Problem 849: The Tournament

Link

Tags: Game Theory

In a tournament there are n teams and each team plays each other team twice. A team gets two points for a win, one point for a draw and no points for a loss.

With two teams there are three possible outcomes for the total points. $(4, 0)$ where a team wins twice, $(3, 1)$ where a team wins and draws, and $(2, 2)$ where either there are two draws or a team wins one game and loses the other. Here we do not distinguish the teams and so $(3, 1)$ and $(1, 3)$ are considered identical.

Let $F(n)$ be the total number of possible final outcomes with n teams, so that $F(2) = 3$.

You are also given $F(7) = 32923$.

Find $F(100)$. Give your answer modulo $10^9 + 7$.

Problem 850: Fractions of Powers

Link

Tags: Number Theory

Any positive real number x can be decomposed into integer and fractional parts $\lfloor x \rfloor + \{x\}$, where $\lfloor x \rfloor$ (the floor function) is an integer, and $0 \leq \{x\} < 1$.

For positive integers k and n , define the function

$$f_k(n) = \sum_{i=1}^n \left\{ \frac{i^k}{n} \right\}$$

For example, $f_5(10) = 4.5$ and $f_7(1234) = 616.5$.

Let

$$S(N) = \sum_{\substack{k=1 \\ k \text{ odd}}}^N \sum_{n=1}^N f_k(n)$$

You are given that $S(10) = 100.5$ and $S(10^3) = 123687804$.

Find $\lfloor S(33557799775533) \rfloor$. Give your answer modulo 977676779.

Problem 851: SOP and POS

Link

Tags: Geometry

Let n be a positive integer and let E_n be the set of n -tuples of strictly positive integers.

For $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$ two elements of E_n , we define:

- the Sum Of Products of u and v , denoted by $\langle u, v \rangle$, as the sum $\sum_{i=1}^n u_i v_i$;
- the Product Of Sums of u and v , denoted by $u \star v$, as the product $\prod_{i=1}^n (u_i + v_i)$.

Let $R_n(M)$ be the sum of $u \star v$ over all ordered pairs (u, v) in E_n such that $\langle u, v \rangle = M$.

For example: $R_1(10) = 36$, $R_2(100) = 1873044$, $R_2(100!) \equiv 446575636 \pmod{10^9 + 7}$.

Find $R_6(10000!)$. Give your answer modulo $10^9 + 7$.

Problem 852: Coins in a Box

Link

Tags: Game Theory, Number Theory, Probability

This game has a box of N unfair coins and N fair coins. Fair coins have probability 50% of landing heads while unfair coins have probability 75% of landing heads.

The player begins with a score of 0 which may become negative during play.

At each round the player randomly picks a coin from the box and guesses its type: fair or unfair. Before guessing they may toss the coin any number of times; however, each toss subtracts 1 from their score. The decision to stop tossing and make a guess can be made at any time. After guessing the player's score is increased by 20 if they are right and decreased by 50 if they are wrong. Then the coin type is revealed to the player and the coin is discarded.

After $2N$ rounds the box will be empty and the game is over. Let $S(N)$ be the expected score of the player at the end of the game assuming that they play optimally in order to maximize their expected score.

You are given $S(1) = 20.558591$ rounded to 6 digits after the decimal point.

Find $S(50)$. Give your answer rounded to 6 digits after the decimal point.

Problem 853: Pisano Periods 1

[Link](#)

Tags: Sequences

For every positive integer n the Fibonacci sequence modulo n is periodic. The period depends on the value of n . This period is called the Pisano period for n , often shortened to $\pi(n)$.

There are three values of n for which $\pi(n)$ equals 18: 19, 38 and 76. The sum of those smaller than 50 is 57.

Find the sum of the values of n smaller than 1 000 000 000 for which $\pi(n)$ equals 120.

Problem 854: Pisano Periods 2

[Link](#)

Tags: Sequences

For every positive integer n the Fibonacci sequence modulo n is periodic. The period depends on the value of n . This period is called the Pisano period for n , often shortened to $\pi(n)$.

Define $M(p)$ as the largest integer n such that $\pi(n) = p$, and define $M(p) = 1$ if there is no such n .

For example, there are three values of n for which $\pi(n)$ equals 18: 19, 38, 76. Therefore $M(18) = 76$.

Let the product function $P(n)$ be:

$$P(n) = \prod_{p=1}^n M(p).$$

You are given: $P(10) = 264$.

Find $P(1\,000\,000) \bmod 1\,234\,567\,891$.

Problem 855: Delphi Paper

[Link](#)

Tags: Combinatorics, Game Theory, Geometry, Graph Theory, Number Theory, Probability

Given two positive integers a, b , Alex and Bianca play a game in ab rounds. They begin with a square piece of paper of side length 1.

In each round Alex divides the current rectangular piece of paper into $a \times b$ pieces using $a - 1$ horizontal cuts and $b - 1$ vertical ones. The cuts do not need to be evenly spaced. Moreover, a piece can have zero width/height when a cut coincides with another cut or the edge of the paper. The pieces are then numbered $1, 2, \dots, ab$ starting from the left top corner, moving from left to right and starting from the left of the next row when a row is finished.

Then Bianca chooses one of the pieces for the game to continue on. However, Bianca must not choose a piece with a number she has already chosen during the game.

Bianca wants to minimize the area of the final piece of paper while Alex wants to maximize it. Let $S(a, b)$ be the area of the final piece assuming optimal play.

For example, $S(2, 2) = 1/36$ and $S(2, 3) = 1/1800 \approx 5.5555555556e-4$.

Find $S(5, 8)$. Give your answer in scientific notation rounded to ten significant digits after the decimal point. Use a lowercase e to separate the mantissa and the exponent.

Problem 856: Waiting for a Pair

Link

Tags: Probability

A standard 52-card deck comprises 13 ranks in four suits. A pair is a set of two cards of the same rank.

Cards are drawn, without replacement, from a well shuffled 52-card deck waiting for consecutive cards that form a pair. For example, the probability of finding a pair in the first two draws is $\frac{1}{17}$.

Cards are drawn until either such a pair is found or the pack is exhausted waiting for one. In the latter case we say that all 52 cards were drawn.

Find the expected number of cards that were drawn. Give your answer rounded to eight places after the decimal point.

Problem 857: Beautiful Graphs

Link

Tags: Game Theory, Geometry, Graph Theory

A graph is made up of vertices and coloured edges. Between every two distinct vertices there must be exactly one of the following:

- A red directed edge one way, and a blue directed edge the other way
- A green undirected edge
- A brown undirected edge

Such a graph is called beautiful if

- A cycle of edges contains a red edge if and only if it also contains a blue edge
- No triangle of edges is made up of entirely green or entirely brown edges

Below are four distinct examples of beautiful graphs on three vertices:

Below are four examples of graphs that are not beautiful:

Let $G(n)$ be the number of beautiful graphs on the labelled vertices: $1, 2, \dots, n$. You are given $G(3) = 24$, $G(4) = 186$ and $G(15) = 12472315010483328$.

Find $G(10^7)$. Give your answer modulo $10^9 + 7$.

Problem 858: LCM

[Link](#)

Tags: Combinatorics, Number Theory

Define $G(N) = \sum_S \text{lcm}(S)$ where S ranges through all subsets of $\{1, \dots, N\}$ and lcm denotes the lowest common multiple. Note that the lcm of the empty set is 1.

You are given $G(5) = 528$ and $G(20) = 8463108648960$.

Find $G(800)$. Give your answer modulo $10^9 + 7$.

Problem 859: Cookie Game

[Link](#)

Tags: Combinatorics, Game Theory

Odd and Even are playing a game with N cookies.

The game begins with the N cookies divided into one or more piles, not necessarily of the same size. They then make moves in turn, starting with Odd.

Odd's turn: Odd may choose any pile with an odd number of cookies, eat one and divide the remaining (if any) into two equal piles.

Even's turn: Even may choose any pile with an even number of cookies, eat two of them and divide the remaining (if any) into two equal piles.

The player that does not have a valid move loses the game.

Let $C(N)$ be the number of ways that N cookies can be divided so that Even has a winning strategy.

For example, $C(5) = 2$ because there are two winning configurations for Even: a single pile containing all five cookies; three piles containing one, two and two cookies.

You are also given $C(16) = 64$.

Find $C(300)$.

Problem 860: Gold and Silver Coin Game

Link

Tags: Combinatorics, Game Theory, Probability

Gary and Sally play a game using gold and silver coins arranged into a number of vertical stacks, alternating turns. On Gary's turn he chooses a gold coin and removes it from the game along with any other coins sitting on top. Sally does the same on her turn by removing a silver coin. The first player unable to make a move loses.

An arrangement is called fair if the person moving first, whether it be Gary or Sally, will lose the game if both play optimally.

Define $F(n)$ to be the number of fair arrangements of n stacks, all of size 2. Different orderings of the stacks are to be counted separately, so $F(2) = 4$ due to the following four arrangements:

Problem 861: Products of Bi-Unitary Divisors

Link

Tags: Number Theory

A unitary divisor of a positive integer n is a divisor d of n such that $\gcd(d, \frac{n}{d}) = 1$.

A bi-unitary divisor of n is a divisor d for which 1 is the only unitary divisor of d that is also a unitary divisor of $\frac{n}{d}$.

For example, 2 is a bi-unitary divisor of 8, because the unitary divisors of 2 are $\{1, 2\}$, and the unitary divisors of $8/2$ are $\{1, 4\}$, with 1 being the only unitary divisor in common.

The bi-unitary divisors of 240 are $\{1, 2, 3, 5, 6, 8, 10, 15, 16, 24, 30, 40, 48, 80, 120, 240\}$.

Let $P(n)$ be the product of all bi-unitary divisors of n . Define $Q_k(N)$ as the number of positive integers $1n \leq N$ such that $P(n) = n^k$. For example, $Q_2(10^2) = 51$ and $Q_6(10^6) = 6189$.

Find $\sum_{k=2}^{10} Q_k(10^{12})$.

Problem 862: Larger Digit Permutation

Link

Tags: Combinatorics, Number Theory

For a positive integer n define $T(n)$ to be the number of strictly larger integers which can be formed by permuting the digits of n .

Leading zeros are not allowed and so for $n = 2302$ the total list of permutations would be:

2023, 2032, 2203, 2230, **2302**, 2320, 3022, 3202, 3220

Problem 863: Different Dice

Link

Tags: Game Theory, Probability, Sequences

Using only a six-sided fair dice and a five-sided fair dice, we would like to emulate an n -sided fair dice.

For example, one way to emulate a 28-sided dice is to follow this procedure:

- Roll both dice, obtaining integers $1 \leq p \leq 6$ and $1 \leq q \leq 5$.
- Combine them using $r = 5(p - 1) + q$ to obtain an integer $1 \leq r \leq 30$.
- If $r \leq 28$, return the value r and stop.
- Otherwise (r being 29 or 30), roll both dice again, obtaining integers $1 \leq s \leq 6$ and $1 \leq t \leq 5$.
- Compute $u = 30(r - 29) + 5(s - 1) + t$ to obtain an integer $1 \leq u \leq 60$.
- If $u > 4$, return the value $((u - 5) \bmod 28) + 1$ and stop.
- Otherwise (with $1 \leq u \leq 4$), roll the six-sided dice twice, obtaining integers $1 \leq v \leq 6$ and $1 \leq w \leq 6$.
- Compute $x = 36(u - 1) + 6(v - 1) + w$ to obtain an integer $1 \leq x \leq 144$.
- If $x > 4$, return the value $((x - 5) \bmod 28) + 1$ and stop.

- Otherwise (with $1 \leq x \leq 4$), assign $u := x$ and go back to step 7.

The expected number of dice rolls in following this procedure is 2.142476 (rounded to 6 decimal places). Note that rolling both dice at the same time is still counted as two dice rolls.

There exist other more complex procedures for emulating a 28-sided dice that entail a smaller average number of dice rolls. However, the above procedure has the attractive property that the sequence of dice rolled is predetermined: regardless of the outcome, it follows (D5,D6,D5,D6,D6,D6,D6,...), truncated wherever the process stops. In fact, amongst procedures for $n = 28$ with this restriction, this one is optimal in the sense of minimising the expected number of rolls needed.

Different values of n will in general use different predetermined sequences. For example, for $n = 8$, the sequence (D5,D5,D5,...) gives an optimal procedure, taking 2.083333... dice rolls on average.

Define $R(n)$ to be the expected number of dice rolls for an optimal procedure for emulating an n -sided dice using only a five-sided and a six-sided dice, considering only those procedures where the sequence of dice rolled is predetermined. So, $R(8) \approx 2.083333$ and $R(28) \approx 2.142476$.

Let $S(n) = \sum_{k=2}^n R(k)$. You are given that $S(30) \approx 56.054622$.

Find $S(1000)$. Give your answer rounded to 6 decimal places.

Problem 864: Square + 1 = Squarefree

Link

Tags: Number Theory

Let $C(n)$ be the number of squarefree integers of the form $x^2 + 1$ such that $1 \leq x \leq n$.

For example, $C(10) = 9$ and $C(1000) = 895$.

Find $C(123567101113)$.

Problem 865: Triplicate Numbers

Link

Tags: Number Theory

A triplicate number is a positive integer such that, after repeatedly removing three consecutive identical digits from it, all its digits can be removed.

For example, the integer 122555211 is a triplicate number:

$$122\color{red}{555}211 \rightarrow 1\color{red}{222}11 \rightarrow \color{red}{111} \rightarrow .$$

On the other hand, neither 663633 nor 9990 are triplicate numbers.

Let $T(n)$ be how many triplicate numbers are less than 10^n .

For example, $T(6) = 261$ and $T(30) = 5576195181577716$.

Find $T(10^4)$. Give your answer modulo 998244353.

Problem 866: Tidying Up B

[Link](#)

Tags: Probability

A small child has a “number caterpillar” consisting of N jigsaw pieces, each with one number on it, which, when connected together in a line, reveal the numbers 1 to N in order.

Every night, the child’s father has to pick up the pieces of the caterpillar that have been scattered across the play room. He picks up the pieces at random and places them in the correct order.

As the caterpillar is built up in this way, it forms distinct segments that gradually merge together.

Any time the father places a new piece in its correct position, a segment of length k is formed and he writes down the k th hexagonal number $k \cdot (2k - 1)$. Once all pieces have been placed and the full caterpillar constructed he calculates the product of all the numbers written down. Interestingly, the expected value of this product is always an integer. For example if $N = 4$ then the expected value is 994.

Find the expected value of the product for a caterpillar of $N = 100$ pieces. Give your answer modulo 987654319.

Problem 867: Tiling Dodecagon

[Link](#)

Tags: Geometry

There are 5 ways to tile a regular dodecagon of side 1 with regular polygons of side 1.

Let $T(n)$ be the number of ways to tile a regular dodecagon of side n with regular polygons of side 1. Then $T(1) = 5$. You are also given $T(2) = 48$.

Find $T(10)$. Give your answer modulo $10^9 + 7$.

Problem 868: Belfry Maths

Link

Tags: Combinatorics

There is a method that is used by Bell ringers to generate all variations of the order that bells are rung.

The same method can be used to create all permutations of a set of letters. Consider the letters to be permuted initially in order from smallest to largest. At each step swap the largest letter with the letter on its left or right whichever generates a permutation that has not yet been seen. If neither gives a new permutation then try the next largest letter and so on. This procedure continues until all permutations have been generated.

For example, 3 swaps are required to reach the permutation CBA when starting with ABC.

The swaps are $ABC \rightarrow ACB \rightarrow CAB \rightarrow CBA$.

Also 59 swaps are required to reach BELFRY when starting with these letters in alphabetical order.

Find the number of swaps that are required to reach NOWPICKBELFRYMATHS when starting with these letters in alphabetical order.

Problem 869: Prime Guessing

Link

Tags: Game Theory, Number Theory, Probability

A prime is drawn uniformly from all primes not exceeding N . The prime is written in binary notation, and a player tries to guess it bit-by-bit starting at the least significant bit. The player scores one point for each bit they guess correctly. Immediately after each guess, the player is informed whether their guess was correct, and also whether it was the last bit in the number - in which case the game is over.

Let $E(N)$ be the expected number of points assuming that the player always guesses to maximize their score. For example, $E(10) = 2$, achievable by always guessing "1". You are also given $E(30) = 2.9$.

Find $E(10^8)$. Give your answer rounded to eight digits after the decimal point.

Problem 870: Stone Game IV

Link

Tags: Game Theory, Number Theory

Two players play a game with a single pile of stones of initial size n . They take stones from the pile in turn, according to the following rules which depend on a fixed real number $r > 0$:

- In the first turn, the first player may take k stones with $1 \leq kn$.
- If a player takes m stones in a turn, then in the next turn the opponent may take k stones with $1 \leq k \leq \lfloor r \cdot m \rfloor$.

Whoever cannot make a legal move loses the game.

Let $L(r)$ be the set of initial pile sizes n for which the second player has a winning strategy. For example, $L(0.5) = \{1\}$, $L(1) = \{1, 2, 4, 8, 16, \dots\}$, $L(2) = \{1, 2, 3, 5, 8, \dots\}$.

A real number q_0 is a transition value if $L(s)$ is different from $L(t)$ for all $s < q < t$.

Let $T(i)$ be the i -th transition value. For example, $T(1) = 1$, $T(2) = 2$, $T(22) \approx 6.3043478261$.

Find $T(123456)$ and give your answer rounded to 10 digits after the decimal point.

Problem 871: Drifting Subsets

Link

Tags: Combinatorics

Let f be a function from a finite set S to itself. A drifting subset for f is a subset A of S such that the number of elements in the union $A \cup f(A)$ is equal to twice the number of elements of A .

We write $D(f)$ for the maximal number of elements among all drifting subsets for f .

For a positive integer n , define f_n as the function from $\{0, 1, \dots, n-1\}$ to itself sending x to $x^3 + x + 1 \bmod n$.

You are given $D(f_5) = 1$ and $D(f_{10}) = 3$.

Find $\sum_{i=1}^{100} D(f_{10^5+i})$.

Problem 872: Recursive Tree

Link

Tags: Combinatorics, Game Theory, Graph Theory, Sequences

A sequence of rooted trees T_n is constructed such that T_n has n nodes numbered 1 to n .

The sequence starts at T_1 , a tree with a single node as a root with the number 1.

For $n > 1$, T_n is constructed from T_{n-1} using the following procedure:

- Trace a path from the root of T_{n-1} to a leaf by following the largest-numbered child at each node.
- Remove all edges along the traced path, disconnecting all nodes along it from their parents.
- Connect all orphaned nodes directly to a new node numbered n , which becomes the root of T_n .

For example, the following figure shows T_6 and T_7 . The path traced through T_6 during the construction of T_7 is coloured red.

Problem 873: Words with Gaps

Link

Tags: Number Theory

Let $W(p, q, r)$ be the number of words that can be formed using the letter A p times, the letter B q times and the letter C r times with the condition that every A is separated from every B by at least two Cs. For example, CACACCBB is a valid word for $W(2, 2, 4)$ but ACBCACBC is not.

You are given $W(2, 2, 4) = 32$ and $W(4, 4, 44) = 13908607644$.

Find $W(10^6, 10^7, 10^8)$. Give your answer modulo 1 000 000 007.

Problem 874: Maximal Prime Score

Link

Tags: Number Theory

Let $p(t)$ denote the $(t + 1)$ th prime number. So that $p(0) = 2$, $p(1) = 3$, etc.

We define the prime score of a list of nonnegative integers $[a_1, \dots, a_n]$ as the sum $\sum_{i=1}^n p(a_i)$.

Let $M(k, n)$ be the maximal prime score among all lists $[a_1, \dots, a_n]$ such that:

- $0 \leq a_i < k$ for each i ;
- the sum $\sum_{i=1}^n a_i$ is a multiple of k .

For example, $M(2, 5) = 14$ as $[0, 1, 1, 1, 1]$ attains a maximal prime score of 14.

Find $M(7000, p(7000))$.

Problem 875: Quadruple Congruence

Link

Tags: Number Theory

For a positive integer n we define $q(n)$ to be the number of solutions to:

$$a_1^2 + a_2^2 + a_3^2 + a_4^2 \equiv b_1^2 + b_2^2 + b_3^2 + b_4^2 \pmod{n}$$

where $0 \leq a_i, b_i < n$. For example, $q(4) = 18432$.

Define $Q(n) = \sum_{i=1}^n q(i)$. You are given $Q(10) = 18573381$.

Find $Q(12345678)$. Give your answer modulo 1001961001.

Problem 876: Triplet Tricks

Link

Tags: Game Theory, Sequences

Starting with three numbers a, b, c , at each step do one of the three operations:

- change a to $2(b + c) - a$;
- change b to $2(c + a) - b$;
- change c to $2(a + b) - c$;

Define $f(a, b, c)$ to be the minimum number of steps required for one number to become zero. If this is not possible then $f(a, b, c) = 0$.

For example, $f(6, 10, 35) = 3$:

$$(6, 10, 35) \rightarrow (6, 10, -3) \rightarrow (8, 10, -3) \rightarrow (8, 0, -3).$$

However, $f(6, 10, 36) = 0$ as no series of operations leads to a zero number.

Also define $F(a, b) = \sum_{c=1}^{\infty} f(a, b, c)$. You are given $F(6, 10) = 17$ and $F(36, 100) = 179$.

Find $\sum_{k=1}^{18} F(6^k, 10^k)$.

Problem 877: XOR-Equation A

Link

Tags: Algebra

We use $x \oplus y$ for the bitwise XOR of x and y .

Define the XOR-product of x and y , denoted by $x \otimes y$, similar to a long multiplication in base 2, except that the intermediate results are XORed instead of the usual integer addition.

For example, $7 \otimes 3 = 9$, or in base 2, $111_2 \otimes 11_2 = 1001_2$:

$$\begin{array}{r} 111_2 \\ \otimes \quad 11_2 \\ \hline 111_2 \\ \oplus \quad 111_2 \\ \hline 1001_2 \end{array}$$

We consider the equation:

$$(a \otimes a) \oplus (2 \otimes a \otimes b) \oplus (b \otimes b) = 5$$

For example, $(a, b) = (3, 6)$ is a solution.

Let $X(N)$ be the XOR of the b values for all solutions to this equation satisfying $0 \leq a \leq b \leq N$. You are given $X(10) = 5$.

Find $X(10^{18})$.

Problem 878: XOR-Equation B

Link

Tags: Algebra

We use $x \oplus y$ for the bitwise XOR of x and y .

Define the XOR-product of x and y , denoted by $x \otimes y$, similar to a long multiplication in base 2, except that the intermediate results are XORed instead of the usual integer addition.

For example, $7 \otimes 3 = 9$, or in base 2, $111_2 \otimes 11_2 = 1001_2$:

$$\begin{array}{r} 111_2 \\ \otimes \quad 11_2 \\ \hline 111_2 \\ \oplus \quad 111_2 \\ \hline 1001_2 \end{array}$$

We consider the equation:

$$(a \otimes a) \oplus (2 \otimes a \otimes b) \oplus (b \otimes b) = k.$$

For example, $(a, b) = (3, 6)$ is a solution to this equation for $k = 5$.

Let $G(N, m)$ be the number of solutions to those equations with $k \leq m$ and $0 \leq a \leq b \leq N$.

You are given $G(1000, 100) = 398$.

Find $G(10^{17}, 1\,000\,000)$.

Problem 879: Touch-screen Password

Link

Tags: Combinatorics, Number Theory, Sequences

A touch-screen device can be unlocked with a “password” consisting of a sequence of two or more distinct spots that the user selects from a rectangular grid of spots on the screen. The user enters their sequence by touching the first spot, then tracing a straight line segment to the next spot, and so on until the end of the sequence. The user’s finger remains in contact with the screen throughout, and may only move in straight line segments from spot to spot.

If the finger traces a straight line that passes over an intermediate spot, then that is treated as two line segments with the intermediate spot included in the

password sequence. For example, on a 3×3 grid labelled with digits 1 to 9 (shown below), tracing $1 - 9$ is interpreted as $1 - 5 - 9$.

Once a spot has been selected it disappears from the screen. Thereafter, the spot may not be used as an endpoint of future line segments, and it is ignored by any future line segments which happen to pass through it. For example, tracing $1 - 9 - 3 - 7$ (which crosses the 5 spot twice) will give the password $1 - 5 - 9 - 6 - 3 - 7$.

There are 389488 different passwords that can be formed on a 3×3 grid.

Find the number of different passwords that can be formed on a 4×4 grid.

Problem 880: Nested Radicals

Link

Tags: Number Theory

(x, y) is called a nested radical pair if x and y are non-zero integers such that $\frac{x}{y}$ is not a cube of a rational number, and there exist integers a , b and c such that:

$$\sqrt{\sqrt[3]{x} + \sqrt[3]{y}} = \sqrt[3]{a} + \sqrt[3]{b} + \sqrt[3]{c}$$

For example, both $(-4, 125)$ and $(5, 5324)$ are nested radical pairs:

$$\begin{aligned}\sqrt{\sqrt[3]{-4} + \sqrt[3]{125}} &= \sqrt[3]{-1} + \sqrt[3]{2} + \sqrt[3]{4} \\ \sqrt{\sqrt[3]{5} + \sqrt[3]{5324}} &= \sqrt[3]{-2} + \sqrt[3]{20} + \sqrt[3]{25}\end{aligned}$$

Let $H(N)$ be the sum of $|x| + |y|$ for all the nested radical pairs (x, y) where $|x| \leq |y| \leq N$.

For example, $H(10^3) = 2535$.

Find $H(10^{15})$. Give your answer modulo $1031^3 + 2$.

Problem 881: Divisor Graph Width

Link

Tags: Combinatorics, Geometry, Graph Theory, Number Theory

For a positive integer n create a graph using its divisors as vertices. An edge is drawn between two vertices ab if their quotient b/a is prime. The graph can be

arranged into levels where vertex n is at level 0 and vertices that are a distance k from n are on level k . Define $g(n)$ to be the maximum number of vertices in a single level.

The example above shows that $g(45) = 2$. You are also given $g(5040) = 12$.

Find the smallest number, n , such that $g(n) \geq 10^4$.

Problem 882: Removing Bits

Link

Tags: Combinatorics, Game Theory

Dr. One and Dr. Zero are playing the following partisan game.

The game begins with one 1, two 2's, three 3's, $\dots$, n n 's. Starting with Dr. One, they make moves in turn.

Dr. One chooses a number and changes it by removing a 1 from its binary expansion.

Dr. Zero chooses a number and changes it by removing a 0 from its binary expansion.

The player that is unable to move loses.

Note that leading zeros are not allowed in any binary expansion; in particular nobody can make a move on the number 0.

They soon realize that Dr. Zero can never win the game. In order to make it more interesting, Dr. Zero is allowed to "skip the turn" several times, i.e. passing the turn back to Dr. One without making a move.

For example, when $n = 2$, Dr. Zero can win the game if allowed to skip 2 turns. A sample game:

$$[1, 2, 2] \xrightarrow{\text{Dr. One}} [1, 0, 2] \xrightarrow{\text{Dr. Zero}} [1, 0, 1] \xrightarrow{\text{Dr. One}} [1, 0, 0] \xrightarrow[\text{skip}]{\text{Dr. Zero}} [1, 0, 0] \xrightarrow{\text{Dr. One}} [0, 0, 0] \xrightarrow[\text{skip}]{\text{Dr. Zero}} [0, 0, 0].$$

Let $S(n)$ be the minimal number of skips needed so that Dr. Zero has a winning strategy.

For example, $S(2) = 2$, $S(5) = 17$, $S(10) = 64$.

Find $S(10^5)$.

Problem 883: Remarkable Triangles

Link

Tags: Combinatorics, Geometry

In this problem we consider triangles drawn on a hexagonal lattice, where each lattice point in the plane has six neighbouring points equally spaced around it, all distance 1 away.

We call a triangle remarkable if

- All three vertices and its incentre lie on lattice points
- At least one of its angles is 60°

Above are four examples of remarkable triangles, with 60° angles illustrated in red. Triangles A and B have inradius 1; C has inradius $\sqrt{3}$; D has inradius 2.

Define $T(r)$ to be the number of remarkable triangles with inradius $\leq r$. Rotations and reflections, such as triangles A and B above, are counted separately; however direct translations are not. That is, the same triangle drawn in different positions of the lattice is only counted once.

You are given $T(0.5) = 2$, $T(2) = 44$, and $T(10) = 1302$.

Find $T(10^6)$.

Problem 884: Removing Cubes

Link

Tags: Number Theory

Starting from a positive integer n , at each step we subtract from n the largest perfect cube not exceeding n , until n becomes 0.

For example, with $n = 100$ the procedure ends in 4 steps:

$$100 \xrightarrow{-4^3} 36 \xrightarrow{-3^3} 9 \xrightarrow{-2^3} 1 \xrightarrow{-1^3} 0.$$

Let $D(n)$ denote the number of steps of the procedure. Thus $D(100) = 4$.

Let $S(N)$ denote the sum of $D(n)$ for all positive integers n strictly less than N .

For example, $S(100) = 512$.

Find $S(10^{17})$.

Problem 885: Sorted Digits

[Link](#)

Tags: Number Theory

For a positive integer d , let $f(d)$ be the number created by sorting the digits of d in ascending order, removing any zeros. For example, $f(3403) = 334$.

Let $S(n)$ be the sum of $f(d)$ for all positive integers d of n digits or less. You are given $S(1) = 45$ and $S(5) = 1543545675$.

Find $S(18)$. Give your answer modulo 1123455689.

Problem 886: Coprime Permutations

[Link](#)

Tags: Combinatorics, Number Theory

A permutation of $\{2, 3, \dots, n\}$ is a rearrangement of these numbers. A coprime permutation is a rearrangement such that all pairs of adjacent numbers are coprime.

Let $P(n)$ be the number of coprime permutations of $\{2, 3, \dots, n\}$.

For example, $P(4) = 2$ as there are two coprime permutations, $(2, 3, 4)$ and $(4, 3, 2)$. You are also given $P(10) = 576$.

Find $P(34)$ and give your answer modulo 83 456 729.

Problem 887: Bounded Binary Search

[Link](#)

Tags: Number Theory

Consider the problem of determining a secret number from a set $\{1, \dots, N\}$ by repeatedly choosing a number y and asking “Is the secret number greater than y ?”.

If $N = 1$ then no questions need to be asked. If $N = 2$ then only one question needs to be asked. If $N = 64$ then six questions need to be asked. However, in the latter case if the secret number is 1 then six questions still need to be asked. We want to restrict the number of questions asked for small values.

Let $Q(N, d)$ be the least number of questions needed for a strategy that can find any secret number from the set $\{1, \dots, N\}$ where no more than $x + d$ questions are needed to find the secret value x .

It can be proved that $Q(N, 0) = N - 1$. You are also given $Q(7, 1) = 3$ and $Q(777, 2) = 10$.

Find $\sum_{d=0}^7 \sum_{N=1}^{7^{10}} Q(N, d)$.

Problem 888: 1249 Nim

Link

Tags: Combinatorics, Game Theory

Two players play a game with a number of piles of stones, alternating turns. Each turn a player can choose to remove 1, 2, 4, or 9 stones from a single pile; or alternatively they can choose to split a pile containing two or more stones into two non-empty piles. The winner is the player who removes the last stone.

A collection of piles is called a losing position if the player to move cannot force a win with optimal play. Define $S(N, m)$ to be the number of distinct losing positions arising from m piles of stones where each pile contains from 1 to N stones. Two positions are considered equivalent if they consist of the same pile sizes. That is, the order of the piles does not matter.

You are given $S(12, 4) = 204$ and $S(124, 9) = 2259208528408$.

Find $S(12491249, 1249)$. Give your answer modulo 912491249.

Problem 889: Rational Blancmange

Link

Tags: Geometry

Recall the blancmange function from Problem 226: $T(x) = \sum_{n=0}^{\infty} \frac{s(2^n x)}{2^n}$, where $s(x)$ is the distance from x to the nearest integer.

For positive integers k, t, r , we write

$$F(k, t, r) = (2^{2k} - 1)T\left(\frac{(2^t + 1)^r}{2^k + 1}\right).$$

It can be shown that $F(k, t, r)$ is always an integer.

For example, $F(3, 1, 1) = 42$, $F(13, 3, 3) = 23093880$ and $F(103, 13, 6) \equiv 878922518 \pmod{1\,000\,062\,031}$.

Find $F(10^{18} + 31, 10^{14} + 31, 62)$. Give your answer modulo 1 000 062 031.

Problem 890: Binary Partitions

Link

Tags: Combinatorics

Let $p(n)$ be the number of ways to write n as the sum of powers of two, ignoring order.

For example, $p(7) = 6$, the partitions being

$$\begin{aligned} 7 &= 1 + 1 + 1 + 1 + 1 + 1 + 1 \\ &= 1 + 1 + 1 + 1 + 1 + 2 \\ &= 1 + 1 + 1 + 2 + 2 \\ &= 1 + 1 + 1 + 4 \\ &= 1 + 2 + 2 + 2 \\ &= 1 + 2 + 4 \end{aligned}$$

You are also given $p(7^7) \equiv 144548435 \pmod{10^9 + 7}$.

Find $p(7^{777})$. Give your answer modulo $10^9 + 7$.

Problem 891: Ambiguous Clock

Link

Tags: Game Theory, Geometry, Probability

A round clock only has three hands: hour, minute, second. All hands look identical and move continuously. Moreover, there is no number or reference mark so that the “upright position” is unknown. The clock functions the same as a normal 12-hour analogue clock.

Despite the inconvenient design, for most time it is possible to tell the correct time (within a 12-hour cycle) from the clock, just by measuring accurately the angles between the hands. For example, if all three hands coincide, then the time must be 12:00:00.

Nevertheless, there are several moments where the clock shows an ambiguous reading. For example, the following moment could be either 1:30:00 or 7:30:00 (with the clock rotated 180°). Thus both 1:30:00 and 7:30:00 are ambiguous moments.

Note that even if two hands perfectly coincide, we can still see them as two distinct hands in the same position. Thus for example 3:00:00 and 9:00:00 are not ambiguous moments.

Problem 892: Zebra Circles

[Link](#)

Tags: Geometry

Consider a circle where $2n$ distinct points have been marked on its circumference.

A cutting C consists of connecting the $2n$ points with n line segments, so that no two line segments intersect, including on their end points. The n line segments then cut the circle into $n + 1$ pieces. Each piece is painted either black or white, so that adjacent pieces are opposite colours. Let $d(C)$ be the absolute difference between the numbers of black and white pieces under the cutting C .

Let $D(n)$ be the sum of $d(C)$ over all different cuttings C . For example, there are five different cuttings with $n = 3$.

Problem 893: Matchsticks

[Link](#)

Tags: Game Theory, Number Theory

Define $M(n)$ to be the minimum number of matchsticks needed to represent the number n .

A number can be represented in digit form or as an expression involving addition and/or multiplication. Also order of operations must be followed, that is multiplication binding tighter than addition. Any other symbols or operations, such as brackets, subtraction, division or exponentiation, are not allowed.

The valid digits and symbols are shown below:

Problem 894: Spiral of Circles

[Link](#)

Tags: Game Theory, Geometry, Number Theory

Consider a unit circle C_0 on the plane that does not enclose the origin. For $k \geq 1$, a circle C_k is created by scaling and rotating C_{k-1} with respect to the origin. That is, both the radius and the distance to the origin are scaled by the same factor, and the centre of rotation is the origin. The scaling factor is positive and strictly less than one. Both it and the rotation angle remain constant for each k .

It is given that C_0 is externally tangent to C_1 , C_7 and C_8 , as shown in the diagram below, and no two circles overlap.

Problem 895: Gold & Silver Coin Game II

[Link](#)

Tags: Combinatorics, Game Theory, Probability

Gary and Sally play a game using gold and silver coins arranged into a number of vertical stacks, alternating turns. On Gary's turn he chooses a gold coin and removes it from the game along with any other coins sitting on top. Sally does the same on her turn by removing a silver coin. The first player unable to make a move loses.

An arrangement is called fair if the person moving first, whether it be Gary or Sally, will lose the game if both play optimally.

An arrangement is called balanced if the number of gold and silver coins are equal.

Define $G(m)$ to be the number of fair and balanced arrangements consisting of three non-empty stacks, each not exceeding m in size. Different orderings of the stacks are to be counted separately, so $G(2) = 6$ due to the following six arrangements:

Problem 896: Divisible Ranges

[Link](#)

Tags: Combinatorics, Number Theory

A contiguous range of positive integers is called a divisible range if all the integers in the range can be arranged in a row such that the n -th term is a multiple of n .

For example, the range [6..9] is a divisible range because we can arrange the numbers as 7, 6, 9, 8.

In fact, it is the 4th divisible range of length 4, the first three being [1..4], [2..5], [3..6].

Find the 36th divisible range of length 36.

Give as answer the smallest number in the range.

Problem 897: Maximal n -gon in a region

[Link](#)

Tags: Geometry, Number Theory

Let $G(n)$ denote the largest possible area of an n -gon contained in the region $\{(x, y) \in \mathbb{R}^2 : x^4 \leq y \leq 1\}$.

For example, $G(3) = 1$ and $G(5) \approx 1.477309771$.

Find $G(101)$ rounded to nine digits after the decimal point.

Problem 898: Claire Voyant

Link

Tags: Game Theory, Number Theory, Probability

Claire Voyant is a teacher playing a game with a class of students. A fair coin is tossed on the table. All the students can see the outcome of the toss, but Claire cannot. Each student then tells Claire whether the outcome is head or tail. The students may lie, but Claire knows the probability that each individual student lies. Moreover, the students lie independently. After that, Claire attempts to guess the outcome using an optimal strategy.

For example, for a class of four students with lying probabilities 20%, 40%, 60%, 80%, Claire guesses correctly with probability 0.832.

Find the probability that Claire guesses correctly for a class of 51 students each lying with a probability of 25%, 26%, ..., 75% respectively.

Give your answer rounded to 10 digits after the decimal point.

Problem 899: DistribuNim I

Link

Tags: Game Theory

Two players play a game with two piles of stones. The players alternately take stones from one or both piles, subject to:

- the total number of stones taken is equal to the size of the smallest pile before the move;
- the move cannot take all the stones from a pile.

The player that is unable to move loses.

For example, if the piles are of sizes 3 and 5 then there are three possible moves.

$$(3, 5) \xrightarrow{(2,1)} (1, 4) \qquad (3, 5) \xrightarrow{(1,2)} (2, 3) \qquad (3, 5) \xrightarrow{(0,3)} (3, 2)$$

Let $L(n)$ be the number of ordered pairs (a, b) with $1 \leq a, b \leq n$ such that the initial game position with piles of sizes a and b is losing for the first player assuming optimal play.

You are given $L(7) = 21$ and $L(7^2) = 221$.

Find $L(7^{17})$.

Problem 900: DistribuNim II

Link

Tags: Game Theory

Two players play a game with at least two piles of stones. The players alternately take stones from one or more piles, subject to:

- the total number of stones taken is equal to the size of the smallest pile before the move;
- the move cannot take all the stones from a pile.

The player that is unable to move loses.

For example, if the piles are of sizes 2, 2 and 4 then there are four possible moves.

$$(2, 2, 4) \xrightarrow{(1,1,0)} (1, 1, 4) \quad (2, 2, 4) \xrightarrow{(1,0,1)} (1, 2, 3) \quad (2, 2, 4) \xrightarrow{(0,1,1)} (2, 1, 3) \quad (2, 2, 4) \xrightarrow{(0,0,2)} (2, 2, 2)$$

Let $t(n)$ be the smallest nonnegative integer k such that the position with n piles of n stones and a single pile of $n + k$ stones is losing for the first player assuming optimal play. For example, $t(1) = t(2) = 0$ and $t(3) = 2$.

Define $S(N) = \sum_{n=1}^{2^N} t(n)$. You are given $S(10) = 361522$.

Find $S(10^4)$. Give your answer modulo 900497239.

Problem 901: Well Drilling

Link

Tags: Combinatorics, Game Theory, Geometry, Probability

A driller drills for water. At each iteration the driller chooses a depth d (a positive real number), drills to this depth and then checks if water was found. If so, the process terminates. Otherwise, a new depth is chosen and a new drilling starts from the ground level in a new location nearby.

Drilling to depth d takes exactly d hours. The groundwater depth is constant in the relevant area and its distribution is known to be an exponential random

variable with expected value of 1. In other words, the probability that the groundwater is deeper than d is e^{-d} .

Assuming an optimal strategy, find the minimal expected drilling time in hours required to find water. Give your answer rounded to 9 places after the decimal point.

Problem 902: Permutation Powers

Link

Tags: Combinatorics, Game Theory, Graph Theory

A permutation π of $\{1, \dots, n\}$ can be represented in one-line notation as $\pi(1), \dots, \pi(n)$. If all $n!$ permutations are written in lexicographic order then $\text{rank}(\pi)$ is the position of π in this 1-based list.

For example, $\text{rank}(2, 1, 3) = 3$ because the six permutations of $\{1, 2, 3\}$ in lexicographic order are:

$$1, 2, 3 \quad 1, 3, 2 \quad 2, 1, 3 \quad 2, 3, 1 \quad 3, 1, 2 \quad 3, 2, 1$$

For a positive integer m , we define the following permutation of $\{1, \dots, n\}$ with $n = \frac{m(m+1)}{2}$:

$$\begin{aligned} \sigma(i) &= \begin{cases} \frac{k(k-1)}{2} + 1 & \text{if } i = \frac{k(k+1)}{2} \text{ for } k \in \{1, \dots, m\}; \\ i + 1 & \text{otherwise;} \end{cases} \\ \tau(i) &= ((10^9 + 7)i \bmod n) + 1 \\ \pi(i) &= \tau^{-1}(\sigma(\tau(i))) \end{aligned}$$

where τ^{-1} is the inverse permutation of τ .

Define $P(m) = \sum_{k=1}^{m!} \text{rank}(\pi^k)$, where π^k is the permutation arising from applying π k times.

For example, $P(2) = 4$, $P(3) = 780$ and $P(4) = 38810300$.

Find $P(100)$. Give your answer modulo $(10^9 + 7)$.

Problem 903: Total Permutation Powers

Link

Tags: Combinatorics, Graph Theory

A permutation π of $\{1, \dots, n\}$ can be represented in one-line notation as $\pi(1), \dots, \pi(n)$. If all $n!$ permutations are written in lexicographic order then $\text{rank}(\pi)$ is the position of π in this 1-based list.

For example, $\text{rank}(2, 1, 3) = 3$ because the six permutations of $\{1, 2, 3\}$ in lexicographic order are:

1, 2, 3 1, 3, 2 2, 1, 3 2, 3, 1 3, 1, 2 3, 2, 1

Let $Q(n)$ be the sum $\sum_{\pi} \sum_{i=1}^{n!} \text{rank}(\pi^i)$, where π ranges over all permutations of $\{1, \dots, n\}$, and π^i is the permutation arising from applying π i times.

For example, $Q(2) = 5$, $Q(3) = 88$, $Q(6) = 133103808$ and $Q(10) \equiv 468421536 \pmod{10^9 + 7}$.

Find $Q(10^6)$. Give your answer modulo $(10^9 + 7)$.

Problem 904: Pythagorean Angle

Link

Tags: Geometry

Given a right-angled triangle with integer sides, the smaller angle formed by the two medians drawn on the the two perpendicular sides is denoted by θ .

Problem 905: Now I Know

Link

Tags: Game Theory

Three epistemologists, known as A, B, and C, are in a room, each wearing a hat with a number on it. They have been informed beforehand that all three numbers are positive and that one of the numbers is the sum of the other two.

Once in the room, they can see the numbers on each other's hats but not on their own. Starting with A and proceeding cyclically, each epistemologist must either honestly state "I don't know my number" or announce "Now I know my number!" which terminates the game.

For instance, if their numbers are $A = 2, B = 1, C = 1$ then A declares "Now I know" at the first turn. If their numbers are $A = 2, B = 7, C = 5$ then "I don't know" is heard four times before B finally declares "Now I know" at the fifth turn.

Let $F(A, B, C)$ be the number of turns it takes until an epistemologist declares “Now I know”, including the turn this declaration is made. So $F(2, 1, 1) = 1$ and $F(2, 7, 5) = 5$.

Find $\sum_{a=1}^7 \sum_{b=1}^{19} F(a^b, b^a, a^b + b^a)$.

Problem 906: A Collective Decision

Link

Tags: Combinatorics, Probability

Three friends attempt to collectively choose one of n options, labeled $1, \dots, n$, based upon their individual preferences. They choose option i if for every alternative option j at least two of the three friends prefer i over j . If no such option i exists they fail to reach an agreement.

Define $P(n)$ to be the probability the three friends successfully reach an agreement and choose one option, where each of the friends’ individual order of preference is given by a (possibly different) random permutation of $1, \dots, n$.

You are given $P(3) = 17/18$ and $P(10) \approx 0.6760292265$.

Find $P(20\,000)$. Give your answer rounded to ten places after the decimal point.

Problem 907: Stacking Cups

Link

Tags: Combinatorics, Game Theory

An infant’s toy consists of n cups, labelled $C_1, \dots, C_n$ in increasing order of size.

The cups may be stacked in various combinations and orientations to form towers. The cups are shaped such that the following means of stacking are possible:

- Nesting: C_k may sit snugly inside C_{k+1} .
- Base-to-base: C_{k+2} or C_{k-2} may sit, right-way-up, on top of an up-side-down C_k , with their bottoms fitting together snugly.
- Rim-to-rim: C_{k+2} or C_{k-2} may sit, up-side-down, on top of a right-way-up C_k , with their tops fitting together snugly.
- For the purposes of this problem, it is not permitted to stack both C_{k+2} and C_{k-2} rim-to-rim on top of C_k , despite the schematic diagrams appearing to allow it:

Define $S(n)$ to be the number of ways to build a single tower using all n cups according to the above rules.

You are given $S(4) = 12$, $S(8) = 58$, and $S(20) = 5560$.

Find $S(10^7)$, giving your answer modulo 1 000 000 007.

Problem 908: Clock Sequence II

[Link](#)

Tags: Sequences

A clock sequence is a periodic sequence of positive integers that can be broken into contiguous segments such that the sum of the n -th segment is equal to n .

For example, the sequence

$$1\ 2\ 3\ 4\ 3\ 2\ 1\ 2\ 3\ 4\ 3\ 2\ 1\ 2\ 3\ 4\ 3\ 2\ 1\ \dots$$

is a clock sequence with period 6, as it can be broken into

$$1\left|2\right|3\left|4\right|3\ 2\left|1\ 2\ 3\right|4\ 3\left|2\ 1\ 2\ 3\right|4\ 3\ 2\left|1\ 2\ 3\ 4\right|3\ 2\ 1\ 2\ 3\left|\dots\right.$$

Let $C(N)$ be the number of different clock sequences with period at most N .

For example, $C(3) = 3$, $C(4) = 7$ and $C(10) = 561$.

Find $C(10^4) \bmod 1111211113$.

Problem 909: L-expressions I

[Link](#)

Tags: Game Theory, Number Theory

An L-expression is defined as any one of the following:

- a natural number;
- the symbol A ;
- the symbol Z ;
- the symbol S ;
- a pair of L-expressions u, v , which is written as $u(v)$.

An L-expression can be transformed according to the following rules:

- $A(x) \rightarrow x + 1$ for any natural number x ;

- $Z(u)(v) \rightarrow v$ for any L-expressions u, v ;
- $S(u)(v)(w) \rightarrow v(u(v)(w))$ for any L-expressions u, v, w .

For example, after applying all possible rules, the L-expression $S(Z)(A)(0)$ is transformed to the number 1:

$$S(Z)(A)(0) \rightarrow A(Z(A)(0)) \rightarrow A(0) \rightarrow 1.$$

Similarly, the L-expression $S(S)(S(S))(S(Z))(A)(0)$ is transformed to the number 6 after applying all possible rules.

Find the result of the L-expression $S(S)(S(S))(S(S))(S(Z))(A)(0)$ after applying all possible rules. Give the last nine digits as your answer.

Note: it can be proved that the L-expression in question can only be transformed a finite number of times, and the final result does not depend on the order of the transformations.

Problem 910: L-expressions II

Link

Tags: Game Theory, Number Theory

An L-expression is defined as any one of the following:

- a natural number;
- the symbol A ;
- the symbol Z ;
- the symbol S ;
- a pair of L-expressions u, v , which is written as $u(v)$.

An L-expression can be transformed according to the following rules:

- $A(x) \rightarrow x + 1$ for any natural number x ;
- $Z(u)(v) \rightarrow v$ for any L-expressions u, v ;
- $S(u)(v)(w) \rightarrow v(u(v)(w))$ for any L-expressions u, v, w .

For example, after applying all possible rules, the L-expression $S(Z)(A)(0)$ is transformed to the number 1:

$$S(Z)(A)(0) \rightarrow A(Z(A)(0)) \rightarrow A(0) \rightarrow 1.$$

Similarly, the L-expression $S(S)(S(S))(S(Z))(A)(0)$ is transformed to the number 6 after applying all possible rules.

Define the following L-expressions:

- $C_0 = Z$;
- $C_i = S(C_{i-1})$ for $i \geq 1$;
- $D_i = C_i(S)(S)$.

For natural numbers a, b, c, d, e , let $F(a, b, c, d, e)$ denote the result of the L-expression $D_a(D_b)(D_c)(C_d)(A)(e)$ after applying all possible rules.

Find the last nine digits of $F(12, 345678, 9012345, 678, 90)$.

Note: it can be proved that the L-expression in question can only be transformed a finite number of times, and the final result does not depend on the order of the transformations.

Problem 911: Khinchin Exceptions

Link

Tags: Number Theory, Sequences

An irrational number x can be uniquely expressed as a continued fraction $[a_0; a_1, a_2, a_3, \dots]$:

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \ddots}}}$$

where a_0 is an integer and $a_1, a_2, a_3, \dots$ are positive integers.

Define $k_j(x)$ to be the geometric mean of $a_1, a_2, \dots, a_j$. That is, $k_j(x) = (a_1 a_2 \cdots a_j)^{1/j}$. Also define $k_\infty(x) = \lim_{j \rightarrow \infty} k_j(x)$.

Khinchin proved that almost all irrational numbers x have the same value of $k_\infty(x) \approx 2.685452\dots$ known as Khinchin's constant. However, there are some exceptions to this rule.

For $n \geq 0$ define

$$\rho_n = \sum_{i=0}^{\infty} \frac{2^n}{2^{2^i}}$$

For example ρ_2 , with continued fraction beginning $[3; 3, 1, 3, 4, 3, 1, 3, \dots]$, has $k_\infty(\rho_2) \approx 2.059767$.

Find the geometric mean of $k_\infty(\rho_n)$ for $0 \leq n \leq 50$, giving your answer rounded to six digits after the decimal point.

Problem 912: Where are the Odds?

Link

Tags: Number Theory

Let s_n be the n -th positive integer that does not contain three consecutive ones in its binary representation.

For example, $s_1 = 1$ and $s_7 = 8$.

Define $F(N)$ to be the sum of n^2 for all $n \leq N$ where s_n is odd. You are given $F(10) = 199$.

Find $F(10^{16})$ giving your answer modulo $10^9 + 7$.

Problem 913: Row-major vs Column-major

Link

Tags: Algebra, Combinatorics, Game Theory

The numbers from 1 to 12 can be arranged into a 3×4 matrix in either row-major or column-major order:

$$R = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{pmatrix}, C = \begin{pmatrix} 1 & 4 & 7 & 10 \\ 2 & 5 & 8 & 11 \\ 3 & 6 & 9 & 12 \end{pmatrix}$$

By swapping two entries at a time, at least 8 swaps are needed to transform R to C .

Let $S(n, m)$ be the minimal number of swaps needed to transform an $n \times m$ matrix of 1 to nm from row-major order to column-major order. Thus $S(3, 4) = 8$.

You are given that the sum of $S(n, m)$ for $2 \leq n \leq m \leq 100$ is 12578833.

Find the sum of $S(n^4, m^4)$ for $2 \leq n \leq m \leq 100$.

Problem 914: Triangles inside Circles

Link

Tags: Geometry

For a given integer R consider all primitive Pythagorean triangles that can fit inside, without touching, a circle with radius R . Define $F(R)$ to be the largest inradius of those triangles. You are given $F(100) = 36$.

Find $F(10^{18})$.

Problem 915: Giant GCDs

Link

Tags: Number Theory, Sequences

The function $s(n)$ is defined recursively for positive integers by $s(1) = 1$ and $s(n+1) = (s(n) - 1)^3 + 2$ for $n \geq 1$.

The sequence begins: $s(1) = 1, s(2) = 2, s(3) = 3, s(4) = 10, \dots$

For positive integers N , define

$$T(N) = \sum_{a=1}^N \sum_{b=1}^N \gcd(s(a), s(b)).$$

You are given $T(3) = 12$, $T(4) \equiv 24881925$ and $T(100) \equiv 14416749$ both modulo 123456789.

Find $T(10^8)$. Give your answer modulo 123456789.

Problem 916: Restricted Permutations

Link

Tags: Combinatorics, Sequences

Let $P(n)$ be the number of permutations of $\{1, 2, 3, \dots, 2n\}$ such that:

1. There is no ascending subsequence with more than $n + 1$ elements, and
2. There is no descending subsequence with more than two elements.

Note that subsequences need not be contiguous. For example, the permutation $(4, 1, 3, 2)$ is not counted because it has a descending subsequence of three elements: $(4, 3, 2)$. You are given $P(2) = 13$ and $P(10) \equiv 45265702 \pmod{10^9 + 7}$.

Find $P(10^8)$ and give your answer modulo $10^9 + 7$.

Problem 917: Minimal Path Using Additive Cost

Link

Tags: Algebra, Combinatorics, Game Theory, Sequences

The sequence s_n is defined by $s_1 = 102022661$ and $s_n = s_{n-1}^2 \pmod{998388889}$ for $n > 1$.

Let $a_n = s_{2n-1}$ and $b_n = s_{2n}$ for $n = 1, 2, \dots$

Define an $N \times N$ matrix whose values are $M_{i,j} = a_i + b_j$.

Let $A(N)$ be the minimal path sum from $M_{1,1}$ (top left) to $M_{N,N}$ (bottom right), where each step is either right or down.

You are given $A(1) = 966774091$, $A(2) = 2388327490$ and $A(10) = 13389278727$.

Find $A(10^7)$.

Problem 918: Recursive Sequence Summation

Link

Tags: Sequences

The sequence a_n is defined by $a_1 = 1$, and then recursively for $n \geq 1$:

$$\begin{aligned}a_{2n} &= 2a_n \\ a_{2n+1} &= a_n - 3a_{n+1}\end{aligned}$$

The first ten terms are 1, 2, -5, 4, 17, -10, -17, 8, -47, 34.

Define $S(N) = \sum_{n=1}^N a_n$. You are given $S(10) = -13$.

Find $S(10^{12})$.

Problem 919: Fortunate Triangles

Link

Tags: Geometry, Graph Theory

We call a triangle fortunate if it has integral sides and at least one of its vertices has the property that the distance from it to the triangle's orthocentre is exactly half the distance from the same vertex to the triangle's circumcentre.

Triangle ABC above is an example of a fortunate triangle with sides (6, 7, 8). The distance from the vertex C to the circumcentre O is ≈ 4.131182 , while the distance from C to the orthocentre H is half that, at ≈ 2.065591 .

Define $S(P)$ to be the sum of $a + b + c$ over all fortunate triangles with sides $a \leq b \leq c$ and perimeter not exceeding P .

For example $S(10) = 24$, arising from three triangles with sides (1, 2, 2), (2, 3, 4), and (2, 4, 4). You are also given $S(100) = 3331$.

Find $S(10^7)$.

Problem 920: Tau Numbers

Link

Tags: Number Theory

For a positive integer n we define $\tau(n)$ to be the count of the divisors of n . For example, the divisors of 12 are $\{1, 2, 3, 4, 6, 12\}$ and so $\tau(12) = 6$.

A positive integer n is a tau number if it is divisible by $\tau(n)$. For example $\tau(12) = 6$ and 6 divides 12 so 12 is a tau number.

Let $m(k)$ be the smallest tau number x such that $\tau(x) = k$. For example, $m(8) = 24$, $m(12) = 60$ and $m(16) = 384$.

Further define $M(n)$ to be the sum of all $m(k)$ whose values do not exceed 10^n . You are given $M(3) = 3189$.

Find $M(16)$.

Problem 921: Golden Recurrence

Link

Tags: Game Theory, Sequences

Consider the following recurrence relation:

$$a_0 = \frac{\sqrt{5} + 1}{2}$$
$$a_{n+1} = \frac{a_n(a_n^4 + 10a_n^2 + 5)}{5a_n^4 + 10a_n^2 + 1}$$

Note that a_0 is the golden ratio.

a_n can always be written in the form $\frac{p_n\sqrt{5} + 1}{q_n}$, where p_n and q_n are positive integers.

Let $s(n) = p_n^5 + q_n^5$. So, $s(0) = 1^5 + 2^5 = 33$.

The Fibonacci sequence is defined as: $F_1 = 1$, $F_2 = 1$, $F_n = F_{n-1} + F_{n-2}$ for $n > 2$.

Define $S(m) = \sum_{i=2}^m s(F_i)$.

Find $S(1618034)$. Submit your answer modulo 398874989.

Problem 922: Young's Game A

[Link](#)

Tags: Combinatorics

A Young diagram is a finite collection of (equally-sized) squares in a grid-like arrangement of rows and columns, such that

- the left-most squares of all rows are aligned vertically;
- the top squares of all columns are aligned horizontally;
- the rows are non-increasing in size as we move top to bottom;
- the columns are non-increasing in size as we move left to right.

Two examples of Young diagrams are shown below.

Problem 923: Young's Game B

[Link](#)

Tags: Combinatorics

A Young diagram is a finite collection of (equally-sized) squares in a grid-like arrangement of rows and columns, such that

- the left-most squares of all rows are aligned vertically;
- the top squares of all columns are aligned horizontally;
- the rows are non-increasing in size as we move top to bottom;
- the columns are non-increasing in size as we move left to right.

Two examples of Young diagrams are shown below.

Problem 924: Larger Digit Permutation II

[Link](#)

Tags: Number Theory

Let $B(n)$ be the smallest number larger than n that can be formed by rearranging digits of n , or 0 if no such number exists. For example, $B(245) = 254$ and $B(542) = 0$.

Define $a_0 = 0$ and $a_n = a_{n-1}^2 + 2$ for $n > 0$. Let $U(N) = \sum_{n=1}^N B(a_n)$. You are given $U(10) \equiv 543870437 \pmod{10^9 + 7}$.

Find $U(10^{16})$. Give your answer modulo $10^9 + 7$.

Problem 925: Larger Digit Permutation III

Link

Tags: Number Theory

Let $B(n)$ be the smallest number larger than n that can be formed by rearranging digits of n , or 0 if no such number exists. For example, $B(245) = 254$ and $B(542) = 0$.

Define $T(N) = \sum_{n=1}^N B(n^2)$. You are given $T(10) = 270$ and $T(100) = 335316$.

Find $T(10^{16})$. Give your answer modulo $10^9 + 7$.

Problem 926: Total Roundness

Link

Tags: Number Theory

A round number is a number that ends with one or more zeros in a given base.

Let us define the roundness of a number n in base b as the number of zeros at the end of the base b representation of n .

For example, 20 has roundness 2 in base 2, because the base 2 representation of 20 is 10100, which ends with 2 zeros.

Also define $R(n)$, the total roundness of a number n , as the sum of the roundness of n in base b for all $b > 1$.

For example, 20 has roundness 2 in base 2 and roundness 1 in base 4, 5, 10, 20, hence we get $R(20) = 6$.

You are also given $R(10!) = 312$.

Find $R(10\,000\,000!)$. Give your answer modulo $10^9 + 7$.

Problem 927: Prime-ary Tree

Link

Tags: Combinatorics, Graph Theory, Number Theory

A full k -ary tree is a tree with a single root node, such that every node is either a leaf or has exactly k ordered children. The height of a k -ary tree is the number of edges in the longest path from the root to a leaf.

For instance, there is one full 3-ary tree of height 0, one full 3-ary tree of height 1, and seven full 3-ary trees of height 2. These seven are shown below.

For integers n and k with $n \geq 0$ and $k \geq 2$, define $t_k(n)$ to be the number of full k -ary trees of height n or less.

Thus, $t_3(0) = 1$, $t_3(1) = 2$, and $t_3(2) = 9$. Also, $t_2(0) = 1$, $t_2(1) = 2$, and $t_2(2) = 5$.

Define S_k to be the set of positive integers m such that m divides $t_k(n)$ for some integer $n \geq 0$. For instance, the above values show that 1, 2, and 5 are in S_2 and 1, 2, 3, and 9 are in S_3 .

Let $S = \bigcap_p S_p$ where the intersection is taken over all primes p . Finally, define $R(N)$ to be the sum of all elements of S not exceeding N . You are given that $R(20) = 18$ and $R(1000) = 2089$.

Find $R(10^7)$.

Problem 928: Cribbage

Link

Tags: Combinatorics, Game Theory

This problem is based on (but not identical to) the scoring for the card game Cribbage.

Consider a normal pack of 52 cards. A Hand is a selection of one or more of these cards.

For each Hand the Hand score is the sum of the values of the cards in the Hand where the value of Aces is 1 and the value of court cards (Jack, Queen, King) is 10.

The Cribbage score is obtained for a Hand by adding together the scores for:

- Pairs. A pair is two cards of the same rank. Every pair is worth 2 points.
- Runs. A run is a set of at least 3 cards whose ranks are consecutive, e.g. 9, 10, Jack. Note that Ace is never high, so Queen, King, Ace is not a valid run. The number of points for each run is the size of the run. All locally maximum runs are counted. For example, 2, 3, 4, 5, 7, 8, 9 the two runs of 2, 3, 4, 5 and 7, 8, 9 are counted but not 2, 3, 4 or 3, 4, 5.

- Fifteens. A fifteen is a combination of cards that has value adding to 15. Every fifteen is worth 2 points. For this purpose the value of the cards is the same as in the Hand Score.

For example, $(5\spadesuit, 5\clubsuit, 5\diamond, K\heartsuit)$ has a Cribbage score of 14 as there are four ways that fifteen can be made and also three pairs can be made.

The example $(A\diamond, A\heartsuit, 2\clubsuit, 3\heartsuit, 4\clubsuit, 5\spadesuit)$ has a Cribbage score of 16: two runs of five worth 10 points, two ways of getting fifteen worth 4 points and one pair worth 2 points. In this example the Hand score is equal to the Cribbage score.

Find the number of Hands in a normal pack of cards where the Hand score is equal to the Cribbage score.

Problem 929: Odd-Run Compositions

Link

Tags: Sequences

A composition of n is a sequence of positive integers which sum to n . Such a sequence can be split into runs, where a run is a maximal contiguous subsequence of equal terms.

For example, $2, 2, 1, 1, 1, 3, 2, 2$ is a composition of 14 consisting of four runs:

$2, 2 \quad 1, 1, 1 \quad 3 \quad 2, 2$

Let $F(n)$ be the number of compositions of n where every run has odd length.

For example, $F(5) = 10$:

$5 \qquad 4, 1 \quad 3, 2 \quad 2, 3 \quad 2, 1, 2$
 $2, 1, 1, 1 \quad 1, 4 \quad 1, 3, 1 \quad 1, 1, 1, 2 \quad 1, 1, 1, 1, 1$

Find $F(10^5)$. Give your answer modulo 1111124111.

Problem 930: The Gathering

Link

Tags: Combinatorics, Game Theory, Geometry, Number Theory, Probability

Given $n \geq 2$ bowls arranged in a circle, $m \geq 2$ balls are distributed amongst them.

Initially the balls are distributed randomly: for each ball, a bowl is chosen equiprobably and independently of the other balls. After this is done, we start the following process:

- Choose one of the m balls equiprobably at random.
- Choose a direction to move - either clockwise or anticlockwise - again equiprobably at random.
- Move the chosen ball to the neighbouring bowl in the chosen direction.
- Return to step 1.

This process stops when all the m balls are located in the same bowl. Note that this may be after zero steps, if the balls happen to have been initially distributed all in the same bowl.

Let $F(n, m)$ be the expected number of times we move a ball before the process stops. For example, $F(2, 2) = \frac{1}{2}$, $F(3, 2) = \frac{4}{3}$, $F(2, 3) = \frac{9}{4}$, and $F(4, 5) = \frac{6875}{24}$.

Let $G(N, M) = \sum_{n=2}^N \sum_{m=2}^M F(n, m)$. For example, $G(3, 3) = \frac{137}{12}$ and $G(4, 5) = \frac{6277}{12}$. You are also given that $G(6, 6) \approx 1.681521567954e4$ in scientific format with 12 significant digits after the decimal point.

Find $G(12, 12)$. Give your answer in scientific format with 12 significant digits after the decimal point.

Problem 931: Totient Graph

[Link](#)

Tags: Graph Theory, Number Theory

For a positive integer n construct a graph using all the divisors of n as the vertices. An edge is drawn between a and b if a is divisible by b and a/b is prime, and is given weight $\phi(a) - \phi(b)$, where ϕ is the Euler totient function. Define $t(n)$ to be the total weight of this graph.

The example below shows that $t(45) = 52$

Let $T(N) = \sum_{n=1}^N t(n)$. You are given $T(10) = 26$ and $T(10^2) = 5282$.

Find $T(10^{12})$. Give your answer modulo 715827883.

Problem 932: 2025

[Link](#)

Tags: Number Theory

For the year 2025

$$2025 = (20 + 25)^2$$

Given positive integers a and b , the concatenation ab we call a 2025-number if $ab = (a + b)^2$.

Other examples are 3025 and 81.

Note 9801 is not a 2025-number because the concatenation of 98 and 1 is 981.

Let $T(n)$ be the sum of all 2025-numbers with n digits or less. You are given $T(4) = 5131$.

Find $T(16)$.

Problem 933: Paper Cutting

[Link](#)

Tags: Game Theory, Geometry

Starting with one piece of integer-sized rectangle paper, two players make moves in turn.

A valid move consists of choosing one piece of paper and cutting it both horizontally and vertically, so that it becomes four pieces of smaller rectangle papers, all of which are integer-sized.

The player that does not have a valid move loses the game.

Let $C(w, h)$ be the number of winning moves for the first player, when the original paper has size $w \times h$. For example, $C(5, 3) = 4$, with the four winning moves shown below.

Also write $D(W, H) = \sum_{w=2}^W \sum_{h=2}^H C(w, h)$. You are given that $D(12, 123) = 327398$.

Find $D(123, 1234567)$.

Problem 934: Unlucky Primes

[Link](#)

Tags: Number Theory

We define the unlucky prime of a number n , denoted $u(n)$, as the smallest prime number p such that the remainder of n divided by p (i.e. $n \bmod p$) is not a multiple of seven.

For example, $u(14) = 3$, $u(147) = 2$ and $u(1470) = 13$.

Let $U(N)$ be the sum $\sum_{n=1}^N u(n)$.

You are given $U(1470) = 4293$.

Find $U(10^{17})$.

Problem 935: Rolling Square

Link

Tags: Number Theory

A square of side length $b < 1$ is rolling around the inside of a larger square of side length 1, always touching the larger square but without sliding.

Initially the two squares share a common corner. At each step, the small square rotates clockwise about a corner that touches the large square, until another of its corners touches the large square. Here is an illustration of the first three steps for $b = \frac{5}{13}$.

For some values of b , the small square may return to its initial position after several steps. For example, when $b = \frac{1}{2}$, this happens in 4 steps; and for $b = \frac{5}{13}$ it happens in 24 steps.

Let $F(N)$ be the number of different values of b for which the small square first returns to its initial position within at most N steps. For example, $F(6) = 4$, with the corresponding b values:

$$\frac{1}{2}, \quad 2 - \sqrt{2}, \quad 2 + \sqrt{2} - \sqrt{2 + 4\sqrt{2}}, \quad 8 - 5\sqrt{2} + 4\sqrt{3} - 3\sqrt{6},$$

the first three in 4 steps and the last one in 6 steps. Note that it does not matter whether the small square returns to its original orientation.

Also $F(100) = 805$.

Find $F(10^8)$.

Problem 936: Peerless Trees

Link

Tags: Graph Theory

A peerless tree is a tree with no edge between two vertices of the same degree. Let $P(n)$ be the number of peerless trees on n unlabelled vertices.

There are six of these trees on seven unlabelled vertices, $P(7) = 6$, shown below.

Define $S(N) = \sum_{n=3}^N P(n)$. You are given $S(10) = 74$.

Find $S(50)$.

Problem 937: Equiproduct Partition

Link

Tags: Combinatorics, Game Theory, Number Theory

Let $\theta = \sqrt{-2}$.

Define T to be the set of numbers of the form $a + b\theta$, where a and b are integers and either $a\theta$, or $a = 0$ and $b\theta$. For a set $S \subseteq T$ and element $z \in T$, define $p(S, z)$ to be the number of ways of choosing two distinct elements from S with product either z or $-z$.

For example if $S = \{1, 2, 4\}$ and $z = 4$, there is only one valid pair of elements with product ± 4 , namely 1 and 4. Thus, in this case $p(S, z) = 1$.

For another example, if $S = \{1, \theta, 1+\theta, 2-\theta\}$ and $z = 2-\theta$, we have $1 \cdot (2-\theta) = z$ and $\theta \cdot (1+\theta) = -z$, giving $p(S, z) = 2$.

Let A and B be two sets satisfying the following conditions:

- $1 \in A$
- $A \cap B = \emptyset$
- $A \cup B = T$
- $p(A, z) = p(B, z)$ for all $z \in T$

Remarkably, these four conditions uniquely determine the sets A and B .

Let F_n be the set of the first n factorials: $F_n = \{1!, 2!, \dots, n!\}$, and define $G(n)$ to be the sum of all elements of $F_n \cap A$.

You are given $G(4) = 25$, $G(7) = 745$, and $G(100) \equiv 709772949 \pmod{10^9 + 7}$.

Find $G(10^8)$ and give your answer modulo $10^9 + 7$.

Problem 938: Exhausting a Colour

Link

Tags: Number Theory, Probability

A deck of cards contains R red cards and B black cards.

A card is chosen uniformly randomly from the deck and removed. A second card is then chosen uniformly randomly from the cards remaining and removed.

- If both cards are red, they are discarded.
- If both cards are black, they are both put back in the deck.
- If they are different colours, the red card is put back in the deck and the black card is discarded.

Play ends when all the remaining cards in the deck are the same colour and let $P(R, B)$ be the probability that this colour is black.

You are given $P(2, 2) = 0.4666666667$, $P(10, 9) = 0.4118903397$ and $P(34, 25) = 0.3665688069$.

Find $P(24690, 12345)$. Give your answer with 10 digits after the decimal point.

Problem 939: Partisan Nim

[Link](#)

Tags: Combinatorics, Game Theory

Two players A and B are playing a variant of Nim.

At the beginning, there are several piles of stones. Each pile is either at the side of A or at the side of B. The piles are unordered.

They make moves in turn. At a player's turn, the player can

- either choose a pile on the opponent's side and remove one stone from that pile;
- or choose a pile on their own side and remove the whole pile.

The winner is the player who removes the last stone.

Let $E(N)$ be the number of initial settings with at most N stones such that, whoever plays first, A always has a winning strategy.

For example $E(4) = 9$; the settings are:

Nr. Piles at the side of A Piles at the side of B 1 4 none 2 1, 3 none 3 2, 2 none
4 1, 1, 2 none 5 3 1 6 1, 2 1 7 2 1, 1 8 3 none 9 2 none

Problem 940: Two-Dimensional Recurrence

[Link](#)

Tags: Sequences

The Fibonacci sequence (f_i) is the unique sequence such that

- $f_0 = 0$
- $f_1 = 1$
- $f_{i+1} = f_i + f_{i-1}$

Similarly, there is a unique function $A(m, n)$ such that

- $A(0, 0) = 0$
- $A(0, 1) = 1$
- $A(m + 1, n) = A(m, n + 1) + A(m, n)$
- $A(m + 1, n + 1) = 2A(m + 1, n) + A(m, n)$

Define $S(k) = \sum_{i=2}^k \sum_{j=2}^k A(f_i, f_j)$. For example

$$\begin{aligned} S(3) &= A(1, 1) + A(1, 2) + A(2, 1) + A(2, 2) \\ &= 2 + 5 + 7 + 16 \\ &= 30 \end{aligned}$$

You are also given $S(5) = 10396$.

Find $S(50)$, giving your answer modulo 1123581313.

Problem 941: de Bruijn's Combination Lock

Link

Tags: Combinatorics, Graph Theory, Number Theory, Sequences

de Bruijn has a digital combination lock with k buttons numbered 0 to $k - 1$ where $k \leq 10$.

The lock opens when the last n buttons pressed match the preset combination.

Unfortunately he has forgotten the combination. He creates a sequence of these digits which contains every possible combination of length n . Then by pressing the buttons in this order he is sure to open the lock.

Consider all sequences of shortest possible length that contains every possible combination of the digits.

Denote by $C(k, n)$ the lexicographically smallest of these.

For example, $C(3, 2) = 0010211220$.

Define the sequence a_n by $a_0 = 0$ and

$$a_n = (920461a_{n-1} + 800217387569) \bmod 10^{12} \text{ for } n > 0$$

Interpret each a_n as a 12-digit combination, adding leading zeros for any a_n with less than 12 digits.

Given a positive integer N , we are interested in the order the combinations $a_1, \dots, a_N$ appear in $C(10, 12)$.

Denote by p_n the place, numbered $1, \dots, N$, in which a_n appears out of $a_1, \dots, a_N$.

Define $F(N) = \sum_{n=1}^N p_n a_n$.

For example, the combination $a_1 = 800217387569$ is entered before $a_2 = 696996536878$. Therefore:

$$F(2) = 1 \cdot 800217387569 + 2 \cdot 696996536878 = 2194210461325$$

You are also given $F(10) = 32698850376317$.

Find $F(10^7)$. Give your answer modulo 1234567891.

Problem 942: Mersenne's Square Root

Link

Tags: Game Theory, Number Theory

Given a natural number q , let $p = 2^q - 1$ be the q -th Mersenne number.

Let $R(q)$ be the minimal square root of q modulo p , if one exists. In other words, $R(q)$ is the smallest positive integer x such that $x^2 - q$ is divisible by p .

For example, $R(5) = 6$ and $R(17) = 47569$.

Find $R(74\,207\,281)$. Give your answer modulo $10^9 + 7$.

Note: $2^{74207281} - 1$ is prime.

Problem 943: Self Describing Sequences

Link

Tags: Sequences

Given two unequal positive integers a and b , we define a self-describing sequence consisting of alternating runs of a s and b s. The first element is a and the sequence of run lengths is the original sequence.

For $a = 2, b = 3$, the sequence is:

2, 2, 3, 3, 2, 2, 2, 3, 3, 3, 2, 2, 3, 3, 2, 2, 3, 3, 3, 2, 2, 2, 3, 3, 3, ...

The sequence begins with two 2s and two 3s, then three 2s and three 3s, so the run lengths 2, 2, 3, 3, ... are given by the original sequence.

Let $T(a, b, N)$ be the sum of the first N elements of the sequence. You are given $T(2, 3, 10) = 25$, $T(4, 2, 10^4) = 30004$, $T(5, 8, 10^6) = 6499871$.

Find $\sum T(a, b, 22332223332233)$ for $2 \leq a \leq 223$, $2 \leq b \leq 223$ and $a \neq b$. Give your answer modulo 2233222333.

Problem 944: Sum of Elevisors

Link

Tags: Combinatorics, Number Theory

Given a set E of positive integers, an element x of E is called an element divisor (elevisor) of E if x divides another element of E .

The sum of all elevisors of E is denoted $\text{sev}(E)$.

For example, $\text{sev}(\{1, 2, 5, 6\}) = 1 + 2 = 3$.

Let $S(n)$ be the sum of $\text{sev}(E)$ for all subsets E of $\{1, 2, \dots, n\}$.

You are given $S(10) = 4927$.

Find $S(10^{14}) \bmod 1234567891$.

Problem 945: XOR-Equation C

Link

Tags: Algebra

We use $x \oplus y$ for the bitwise XOR of x and y .

Define the XOR-product of x and y , denoted by $x \otimes y$, similar to a long multiplication in base 2, except that the intermediate results are XORed instead of the usual integer addition.

For example, $7 \otimes 3 = 9$, or in base 2, $111_2 \otimes 11_2 = 1001_2$:

$$\begin{array}{r} 111_2 \\ \otimes 11_2 \\ \hline 111_2 \\ \oplus 111_2 \\ \hline 1001_2 \end{array}$$

We consider the equation:

$$(a \otimes a) \oplus (2 \otimes a \otimes b) \oplus (b \otimes b) = c \otimes c$$

For example, $(a, b, c) = (1, 2, 1)$ is a solution to this equation, and so is $(1, 8, 13)$.

Let $F(N)$ be the number of solutions to this equation satisfying $0 \leq a \leq b \leq N$. You are given $F(10) = 21$.

Find $F(10^7)$.

Problem 946: Continued Fraction Fraction

Link

Tags: Number Theory, Sequences

Given the representation of a continued fraction

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \ddots}}} = [a_0; a_1, a_2, a_3, \dots]$$

α is a real number with continued fraction representation: $\alpha = [2; 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 2, 1,$ where the number of 1's between each of the 2's are consecutive prime numbers.

β is another real number defined as

$$\beta = \frac{2\alpha + 3}{3\alpha + 2}$$

The first ten coefficients of the continued fraction of β are $[0; 1, 5, 6, 16, 9, 1, 10, 16, 11]$ with sum 75.

Find the sum of the first 10^8 coefficients of the continued fraction of β .

Problem 947: Fibonacci Residues

[Link](#)

Tags: Sequences

The (a, b, m) -sequence, where $0 \leq a, bm$, is defined as

Problem 948: Left vs Right

[Link](#)

Tags: Game Theory

Left and Right play a game with a word consisting of L's and R's, alternating turns. On Left's turn, Left can remove any positive number of letters, but not all the letters, from the left side of the word. Right does the same on Right's turn except that Right removes letters from the right side. The game continues until only one letter remains: if it is an 'L' then Left wins; if it is an 'R' then Right wins.

Let $F(n)$ be the number of words of length n where the player moving first, whether it's Left or Right, will win the game if both play optimally.

You are given $F(3) = 4$ and $F(8) = 181$.

Find $F(60)$.

Problem 949: Left vs Right II

[Link](#)

Tags: Game Theory

Left and Right play a game with a number of words, each consisting of L's and R's, alternating turns. On Left's turn, for each word, Left can remove any number of letters (possibly zero), but not all the letters, from the left side of the word. However, at least one letter must be removed from at least one word. Right does the same on Right's turn except that Right removes letters from the right side of each word. The game continues until each word is reduced to a single letter. If there are more L's than R's remaining then Left wins; otherwise if there are more R's than L's then Right wins. In this problem we only consider games with an odd number of words, thus making ties impossible.

Let $G(n, k)$ be the number of ways of choosing k words of length n , for which Right has a winning strategy when Left plays first. Different orderings of the same set of words are to be counted separately.

It can be seen that $G(2, 3) = 14$ due to the following solutions (and their reorderings):

(LL, RR, RR) : 3 orderings
 (LR, LR, LR) : 1 ordering
 (LR, LR, RR) : 3 orderings
 (LR, RR, RR) : 3 orderings
 (RL, RR, RR) : 3 orderings
 (RR, RR, RR) : 1 ordering

You are also given $G(4, 3) = 496$ and $G(8, 5) = 26359197010$.

Find $G(20, 7)$ giving your answer modulo 1001001011.

Problem 950: Pirate Treasure

Link

Tags: Number Theory, Probability

A band of pirates has come into a hoard of treasure, and must decide how to distribute it amongst themselves. The treasure consists of identical, indivisible gold coins.

According to pirate law, the distribution of treasure must proceed as follows:

- The most senior pirate proposes a distribution of the coins.
- All pirates, including the most senior, vote on whether to accept the distribution.
- If at least half of the pirates vote to accept, the distribution stands.
- Otherwise, the most senior pirate must walk the plank, and the process resumes from step 1 with the next most senior pirate proposing another distribution.

The happiness of a pirate is equal to $-\infty$ if he doesn't survive; otherwise, it is equal to $c + p \cdot w$, where c is the number of coins that pirate receives in the distribution, w is the total number of pirates who were made to walk the plank, and p is the bloodthirstiness of the pirate.

The pirates have a number of characteristics:

- Greed: to maximise their happiness.
- Ruthlessness: incapable of cooperation, making promises or maintaining any kind of reputation.
- Shrewdness: perfectly rational and logical.

Consider the happiness $c(n, C, p) + p \cdot w(n, C, p)$ of the most senior surviving pirate in the situation where n pirates, all with equal bloodthirstiness p , have found C coins. For example, $c(5, 5, \frac{1}{10}) = 3$ and $w(5, 5, \frac{1}{10}) = 0$ because it can be shown that if the most senior pirate proposes a distribution of 3, 0, 1, 0, 1 coins to the pirates (in decreasing order of seniority), the three pirates receiving coins will all vote to accept. On the other hand, $c(5, 1, \frac{1}{10}) = 0$ and $w(5, 1, \frac{1}{10}) = 1$: the most senior pirate cannot survive with any proposal, and then the second most senior pirate must give the only coin to another pirate in order to survive.

Define $T(N, C, p) = \sum_{n=1}^N (c(n, C, p) + w(n, C, p))$. You are given that $T(30, 3, \frac{1}{\sqrt{3}}) = 190$, $T(50, 3, \frac{1}{\sqrt{31}}) = 385$, and $T(10^3, 101, \frac{1}{\sqrt{101}}) = 142427$.

Find $\sum_{k=1}^6 T(10^{16}, 10^k + 1, \frac{1}{\sqrt{10^k + 1}})$. Give the last 9 digits as your answer.

Problem 951: A Game of Chance

Link

Tags: Game Theory, Probability

Two players play a game using a deck of $2n$ cards: n red and n black. Initially the deck is shuffled into one of the $\binom{2n}{n}$ possible starting configurations. Play then proceeds, alternating turns, where a player follows two steps on each turn:

- Remove the top card from the deck, taking note of its colour.
- If there is a next card and it is the same colour as the previous card they toss a fair coin. If the coin lands on heads they remove that card as well; otherwise leave it on top of the deck.

The player who removes the final card from the deck wins the game.

Some starting configurations give an advantage to one of the players; while some starting configurations are fair, in which both players have exactly 50% chance to win the game. For example, if $n = 2$ there are four starting configurations which are fair: RRBB, BBRR, RBBR, BRRB. The remaining two, RBRB and BRBR, result in a guaranteed win for the second player.

Define $F(n)$ to be the number of starting configurations which are fair. Therefore $F(2) = 4$. You are also given $F(8) = 11892$.

Find $F(26)$.

Problem 952: Order Modulo Factorial

[Link](#)

Tags: Game Theory, Number Theory

Given a prime p and a positive integer np , let $R(p, n)$ be the multiplicative order of p modulo $n!$.

In other words, $R(p, n)$ is the minimal positive integer r such that

$$p^r \equiv 1 \pmod{n!}$$

For example, $R(7, 4) = 2$ and $R(10^9 + 7, 12) = 17280$.

Find $R(10^9 + 7, 10^7)$. Give your answer modulo $10^9 + 7$.

Problem 953: Factorisation Nim

[Link](#)

Tags: Game Theory, Number Theory

In the classical game of Nim two players take turns removing stones from piles. A player may remove any positive number of stones from a single pile. If there are no remaining stones, the next player to move loses.

In Factorisation Nim the initial position of the game is chosen according to the prime factorisation of a given natural number n by setting a pile for each prime factor, including multiplicity. For example, if $n = 12 = 2 \times 2 \times 3$ the game starts with three piles: two piles with two stones and one pile with three stones.

It can be verified that the first player to move loses for $n = 1$ and for $n = 70$, assuming both players play optimally.

Let $S(N)$ be the sum of n for $1 \leq n \leq N$ such that the first player to move loses, assuming both players play optimally. You are given $S(10) = 14$ and $S(100) = 455$.

Find $S(10^{14})$. Give your answer modulo $10^9 + 7$.

Problem 954: Heptaphobia

[Link](#)

Tags: Number Theory

A positive integer is called heptaphobic if it is not divisible by seven and no number divisible by seven can be produced by swapping two of its digits. Note that leading zeros are not allowed before or after the swap.

For example, 17 and 1305 are heptaphobic, but 14 and 132 are not because 14 and 231 are divisible by seven.

Let $C(N)$ count heptaphobic numbers smaller than N . You are given $C(100) = 74$ and $C(10^4) = 3737$.

Find $C(10^{13})$.

Problem 955: Finding Triangles

Link

Tags: Geometry, Sequences

A sequence $(a_n)_{n \geq 0}$ starts with $a_0 = 3$ and for each $n \geq 0$,

- if a_n is a triangle number, then $a_{n+1} = a_n + 1$;
- otherwise, $a_{n+1} = 2a_n - a_{n-1} + 1$.

The sequence begins:

3, 4, **6**, 7, 9, 12, 16, **21**, 22, 24, 27, 31, **36**, 37, 39, 42, . . .

where triangle numbers are marked red.

The 10th triangle number in the sequence is $a_{2964} = 1439056$.

Find the index n such that a_n is the 70th triangle number in the sequence.

Problem 956: Super Duper Sum

Link

Tags: Combinatorics, Number Theory

The total number of prime factors of n , counted with multiplicity, is denoted $\Omega(n)$.

For example, $\Omega(12) = 3$, counting the factor 2 twice, and the factor 3 once.

Define $D(n, m)$ to be the sum of all divisors d of n where $\Omega(d)$ is divisible by m .

For example, $D(24, 3) = 1 + 8 + 12 = 21$.

The superfactorial of n , often written as $n\$$, is defined as the product of the first n factorials:

$$n\$ = 1! \times 2! \times \cdots \times n!$$

The superduperfactorial of n , we write as $n\star$, is defined as the product of the first n superfactorials:

$$n\star = 1\$ \times 2\$ \times \cdots \times n\$$$

You are given $D(6\star, 6) = 6368195719791280$.

Find $D(1\,000\star, 1\,000)$. Give your answer modulo 999 999 001.

Problem 957: Point Genesis

Link

Tags: Number Theory

There is a plane on which all points are initially white, except three red points and two blue points.

On each day, every line passing through a red point and a blue point is constructed. Then every white point, where two different such lines meet, turns blue.

Let $g(n)$ be the maximal possible number of blue points after n days.

For example, $g(1) = 8$ and $g(2) = 28$.

Find $g(16)$.

Problem 958: Euclid's Labour

Link

Tags: Game Theory, Number Theory

The Euclidean algorithm can be used to find the greatest common divisor of two positive integers. At each step of the algorithm the smaller number is subtracted from the larger one. The algorithm terminates when the numbers are equal, which is then the greatest common divisor of the original numbers.

For two numbers n and m , let $d(n, m)$ be the number of subtraction steps used by the Euclidean algorithm for computing the greatest common divisor of n and m .

For a number n , let $f(n)$ be the positive number m coprime to n that minimizes $d(n, m)$. If more than one number attains the minimum, the minimal m is chosen.

For example, at least four steps are needed for computing the GCD of 7 and any positive number m coprime to 7. This number of steps is obtained by $m = 2, 3, 4, 5$, yielding $f(7) = 2$. You are also given $f(89) = 34$ and $f(8191) = 1856$.

Find $f(10^{12} + 39)$.

Problem 959: Asymmetric Random Walk

Link

Tags: Number Theory, Probability

A frog is placed on the number line. Every step the frog jumps either a units to the left or b units to the right, both with $1/2$ probability.

Define $f(a, b)$ as the limit $\lim_{n \rightarrow \infty} \frac{c_n}{n}$ where c_n is the expected number of unique numbers visited in the first n steps. You are given $f(1, 1) = 0$ and $f(1, 2) \approx 0.427050983$.

Find $f(89, 97)$. Give your answer rounded to nine digits after the decimal point.

Problem 960: Stone Game Solitaire

Link

Tags: Combinatorics, Game Theory, Sequences

There are n distinct piles of stones, each of size $n - 1$. Starting with an initial score of 0, the following procedure is repeated:

- Choose any two piles and remove exactly n stones in total from the two piles.
- If the number of stones removed from the two piles were a and b , add $\min(a, b)$ to the score.

If all piles are eventually emptied, the current score is confirmed as final. However, if one gets “stuck” and cannot empty all piles, the current score is discarded, resulting in a final score of 0.

Three example sequences of turns are illustrated below for $n = 4$, with each tuple representing pile sizes as one proceeds, and with additions to the score indicated above the arrows.

$$\begin{aligned}(3, 3, 3, 3) &\xrightarrow{+1} (0, 3, 2, 3) \xrightarrow{+1} (0, 3, 1, 0) \xrightarrow{+1} (0, 0, 0, 0) & : \quad \text{final score} &= 3 \\(3, 3, 3, 3) &\xrightarrow{+1} (3, 0, 3, 2) \xrightarrow{+2} (1, 0, 3, 0) \xrightarrow{+1} (0, 0, 0, 0) & : \quad \text{final score} &= 4 \\(3, 3, 3, 3) &\xrightarrow{+2} (1, 3, 1, 3) \xrightarrow{+1} (1, 2, 1, 0) \rightarrow \text{stuck!} & : \quad \text{final score} &= 0\end{aligned}$$

Define $F(n)$ to be the sum of the final scores achieved for every sequence of turns which successfully empty all piles.

You are given $F(3) = 12$, $F(4) = 360$, and $F(8) = 16785941760$.

Find $F(100)$. Give your answer modulo $10^9 + 7$.

Problem 961: Removing Digits

Link

Tags: Game Theory, Number Theory

This game starts with a positive integer. Two players take turns to remove a single digit from that integer. After the digit is removed any resulting leading zeros are removed.

For example, removing a digit from 105 results in either 5, 10 or 15.

The winner is the person who removes the last nonzero digit.

Define $W(N)$ to be how many positive integers less than N for which the first player can guarantee a win given optimal play. You are given $W(100) = 18$ and $W(10^4) = 1656$.

Find $W(10^{18})$.

Problem 962: Angular Bisector and Tangent 2

Link

Tags: Geometry

Given is an integer sided triangle ABC with $BC \leq AC \leq AB$. k is the angular bisector of angle ACB . m is the tangent at C to the circumscribed circle of ABC . n is a line parallel to m through B .

The intersection of n and k is called E .

Problem 963: Removing Trits

Link

Tags: Game Theory

NOTE: This problem is related to Problem 882. It is recommended to solve that problem before doing this one.

Two players are playing a game. When the game starts, each player holds a paper with two positive integers written on it.

They make moves in turn. At a player's turn, the player can do one of the following:

- pick a number on the player's own paper and change it by removing a 0 from its ternary expansion;
- pick a number on the opponent's paper and change it by removing a 1 from its ternary expansion;
- pick a number on either paper and change it by removing a 2 from its ternary expansion.

The player that is unable to make a move loses.

Leading zeros are not allowed in any ternary expansion; in particular nobody can make a move on the number 0.

An initial setting is called fair if whichever player moves first will lose the game if both play optimally.

For example, if initially the integers on the paper of the first player are 1, 5 and those on the paper of the second player are 2, 4, then this is a fair initial setting, which we can denote as $(1, 5 \mid 2, 4)$.

Note that the order of the two integers on a paper does not matter, but the order of the two papers matter.

Thus $(5, 1 \mid 4, 2)$ is considered the same as $(1, 5 \mid 2, 4)$, while $(2, 4 \mid 1, 5)$ is a different initial setting.

Let $F(N)$ be the number of fair initial settings where each initial number does not exceed N .

For example, $F(5) = 21$.

Find $F(10^5)$.

Problem 964: Musical Chairs Revisited

Link

Tags: Combinatorics, Game Theory, Geometry, Number Theory, Probability

A group of $k(k-1)/2 + 1$ children play a game of k rounds.

At the beginning, they are all seated on chairs arranged in a circle.

During the i -th round:

- The music starts playing and i children are randomly selected, with all combinations being equally likely. The selected children stand up and dance around.
- When the music stops, these i children sit back down randomly in the i available chairs, with all permutations being equally likely.

Let $P(k)$ be the probability that every child ends up sitting exactly one chair to the right of their original chair when the game ends (at the end of the k -th round).

You are given $P(3) \approx 1.388888889\text{e}-2$.

Find $P(7)$. Give your answer in scientific notation rounded to ten significant digits after the decimal point. Use a lowercase e to separate the mantissa and the exponent.

Problem 965: Expected Minimal Fractional Value

[Link](#)

Tags: Game Theory, Number Theory, Probability

Let $\{x\}$ denote the fractional part of a real number x .

Define $f_N(x)$ to be the minimal value of $\{nx\}$ for integer n satisfying $0 < n \leq N$.

Further define $F(N)$ to be the expected value of $f_N(x)$ when x is sampled uniformly in $[0, 1]$.

You are given $F(1) = \frac{1}{2}$, $F(4) = \frac{1}{4}$ and $F(10) \approx 0.131944444444$.

Find $F(10^4)$ and give your answer rounded to 13 digits after the decimal point.

Problem 966: Triangle Circle Intersection

[Link](#)

Tags: Geometry, Number Theory

Let $I(a, b, c)$ be the largest possible area of intersection between a triangle of side lengths a, b, c and a circle which has the same area as the triangle.

For example $I(3, 4, 5) \approx 4.593049$ and $I(3, 4, 6) \approx 3.552564$.

Find the sum of $I(a, b, c)$ for integers a, b, c such that $1 \leq a \leq b \leq ca + b$ and $a + b + c \leq 200$.

Give your answer rounded to two digits after the decimal point.

Problem 967: B -Trivisible Numbers

[Link](#)

Tags: Number Theory

A positive integer n is considered B -trivisible if the sum of all different prime factors of n which are not larger than B is divisible by 3.

For example, $175 = 5^2 \cdot 7$ is 10-trivisible because $5 + 7 = 12$ which is divisible by 3. Similarly, 175 is 4-trivisible because all primes dividing 175 are larger than 4, and the empty summation 0 is divisible by 3.

On the other hand, 175 is not 6-trivisible because the sum of relevant primes is 5 which is not divisible by 3.

Let $F(N, B)$ be the number of B -trivisible integers not larger than N .

For example, $F(10, 4) = 5$, the 4-trivisible numbers being 1, 3, 5, 7, 9.

You are also given $F(10, 10) = 3$ and $F(100, 10) = 41$.

Find $F(10^{18}, 120)$.

Problem 968: 5D Summation

Link

Tags: Sequences

Define

$$P(X_{a,b}, X_{a,c}, X_{a,d}, X_{a,e}, X_{b,c}, X_{b,d}, X_{b,e}, X_{c,d}, X_{c,e}, X_{d,e})$$

as the sum of $2^a 3^b 5^c 7^d 11^e$ over all quintuples of non-negative integers (a, b, c, d, e) such that the sum of each two of the five variables is restricted by a given value. In other words, $a + b \leq X_{a,b}$, $a + d \leq X_{a,d}$, $b + e \leq X_{b,e}$ etc.

For example, $P(2, 2, 2, 2, 2, 2, 2, 2, 2, 2) = 7120$ and $P(1, 2, 3, 4, 5, 6, 7, 8, 9, 10) \equiv 799809376 \pmod{10^9 + 7}$.

Define a sequence A as follows:

- $A_0 = 1$, $A_1 = 7$;
- $A_n = (7A_{n-1} + A_{n-2}^2) \pmod{10^9 + 7}$ for $n \geq 2$.

Also define $Q(n) = P(A_{10n}, A_{10n+1}, A_{10n+2}, \dots, A_{10n+9})$.

Find $\sum_{0 \leq n < 100} Q(n)$. Give your answer modulo $10^9 + 7$.

Problem 969: Kangaroo Hopping

Link

Tags: Algebra, Geometry, Probability

Starting at zero, a kangaroo hops along the real number line in the positive direction. Each successive hop takes the kangaroo forward a uniformly random distance between 0 and 1. Let $H(n)$ be the expected number of hops needed for the kangaroo to pass n on the real line.

If we write $\alpha = H(1)$, then for all positive integers n , $H(n)$ can be expressed as a polynomial function of α with rational coefficients. For example $H(3) = \alpha^3 - 2\alpha^2 + \frac{1}{2}\alpha$. Define $S(n)$ to be the sum of all integer coefficients in this polynomial form of $H(n)$. Therefore $S(1) = 1$ and $S(3) = 1 + (-2) = -1$.

You are also given $\sum_{n=1}^{10} S(n) = 43$.

Find $\sum_{n=1}^{10^{18}} S(n)$. Give your answer modulo $10^9 + 7$.

Problem 970: Kangaroo Hopping over Sixes

[Link](#)

Tags: Geometry, Number Theory, Probability

Starting at zero, a kangaroo hops along the real number line in the positive direction. Each successive hop takes the kangaroo forward a uniformly random distance between 0 and 1. Let $H(n)$ be the expected number of hops needed for the kangaroo to pass n on the real line.

For example, $H(2) \approx 4.67077427047$. The first eight digits after the decimal point that are different from six are 70774270.

Similarly, $H(3) \approx 6.6665656395558899$. Here the first eight digits after the decimal point that are different from six are 55395558.

Find $H(10^6)$ and give as your answer the first eight digits after the decimal point that are different from six.

Problem 971: Modular Polynomial Composition

[Link](#)

Tags: Number Theory

Let p be a prime of the form $5k - 4$ and define $f_p(x) = (x^k + x) \bmod p$.

Let $C(p)$ be the number of values $0 \leq x < p$ such that $f_p^{(m)}(x) = x$ for some positive integer m , that is, x can be obtained by iteratively applying f_p on itself starting at x .

For example, $C(11) = 7$, due to $x = 0, 1, 2, 3, 8, 9, 10$.

Let $S(N)$ be the sum of $C(p)$ for all primes of the form $5k - 4$ not exceeding N .
For example $S(100) = 127$.

Find $S(10^8)$.

Problem 972: Hyperbolic Plane

Link

Tags: Game Theory

The hyperbolic plane can be represented by the open unit disc, namely the set of points (x, y) in $\mathbb{R}^2$ with $x^2 + y^2 < 1$.

A geodesic is defined as either a diameter of the open unit disc or a circular arc contained within the disc that is orthogonal to the boundary of the disc.

The following diagram shows the hyperbolic plane with two geodesics; one is a diameter and the other is a circular arc.

Problem 973: Random Dealings

Link

Tags: Combinatorics, Game Theory, Probability

A game is played with n cards. At the start the cards are dealt out onto a table to get n piles of size one.

Each round proceeds as follows:

- Select a pile at random and pick it up.
- Randomly choose a pile from the table and add the top card of the picked-up pile to it.
- Redistribute any remaining cards from the picked-up pile by dealing them into new single-card piles.

The game ends when all cards are in a single pile.

At the end of each round a score is obtained by bitwise-XORing the size of each pile. The score is summed across the rounds. Let $X(n)$ be the expected total score at the end of the game.

You are given $X(2) = 2$, $X(4) = 14$ and $X(10) = 1418$.

Find $X(10^4)$. Give your answer modulo $10^9 + 7$.

Problem 974: Very Odd Numbers

Link

Tags: Number Theory

A very odd number is a number which contains only odd digits and is divisible by 105. Further each odd digit occurs an odd number of times.

Define $\Theta(n)$ be the n th very odd number, then $\Theta(1) = 1117935$ and $\Theta(10^3) = 11137955115$.

Find $\Theta(10^{16})$.
